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Volume - 16

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Chapter - 1

Genomic Identification and Classification of Phytoplasma

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Chapter - 1

Genomic Identification and Classification of Phytoplasma

Shoeb Ahmad, Syed Saad, Shveta Malhotra, Azahar Sajjad and Akil Ahmad Khan

Abstract

Phytoplasmas are versatile, lacking a cell wall, bacteria that can induce severe illnesses in a diverse range of plant species. Swift detection and accurate identification of phytoplasmas accountable for new plant diseases are essential for halting the illnesses' further dissemination and minimizing financial losses. The taxonomy of phytoplasma, which includes identification, nomenclature, and categorization, has not progressed as much as that of culturable bacteria. This is mainly because phytoplasma cannot be cultured in isolation, making it difficult to study their phenotypic traits. Nevertheless, the swift growth of molecular techniques and the introduction of high-throughput genome sequencing have greatly improved the taxonomy of phytoplasma based on nucleotide sequences. This article focuses on the significant events and milestones that have influenced the development of the current phytoplasma taxonomy. This study also explains the differences and links between two systems that work at the same time: the "*Candidatus* phytoplasma" species/nomenclature system and the group/subgroup categorization system. Both systems are essential as they fulfill distinct functions. In addition, this article also addresses controversial topics with the naming of phytoplasmas, particularly those related to the application of recently updated rules for describing '*Candidatus* Phytoplasma' species. In summary, a concise overview of the difficulties and potential future directions of phytoplasma taxonomy is provided.

Keywords: Phytoplasma, bacterial taxonomy, whole genome-based average nucleotide identity (ANI), iPhyClassifier

Introduction

Phytoplasmas are intracellular, pleomorphic, wall less bacteria that are capable of inducing catastrophic diseases in an extensive array of plant species. It is critical to promptly diagnose and accurately identify phytoplasmas that are accountable for emerging plant diseases in order to

mitigate economic losses and prevent the diseases' further transmission. The field of phytoplasma taxonomy, which encompasses identification, nomenclature, and classification, has fallen behind that of cultivable bacteria. This is primarily because there is no axenic phytoplasma culture, which renders phenotypic characteristics inaccessible. Nucleotide sequence-based phytoplasma taxonomy has been significantly improved, however, due to the swift advancement of molecular techniques and the introduction of high-throughput genome sequencing. The article emphasizes the significant occurrences and turning points that influenced the present phytoplasma taxonomy. It also explains the differences and links between two classification systems that are used at the same time for "*Candidatus* phytoplasma": the species/nomenclature system and the group/subgroup classification system. Both systems are essential despite fulfilling distinct functions (Wei, W., & Zhao, Y., 2022). Bacterial taxonomy has witnessed notable advancements in recent years, as evidenced by the shift from the 16SrRNA gene to the whole genome sequence in culturable bacteria (Hugenholtz, P., *et al.*, 2021). Furthermore, efforts have been made to enhance the taxonomy of uncultivable phytoplasma so that it can more effectively correspond with culturable bacteria. Recent developments in phytoplasma taxonomy, including nomenclature, classification, and identification, are examined in this article.

Phytoplasmas, previously known as mycoplasma-like organisms (MLOs), are small prokaryotes that lack a cell wall and are linked to diseases in hundreds of plant species (Lee *et al.*, 2000; Bertaccini, 2007; Hogenhout *et al.*, 2008). Since these different plant pathogens were first identified more than forty years ago, attempts to grow them on media without cells have failed. This makes it hard to classify phytoplasmas using the usual methods for grown prokaryotes. The field of mollicute systematics has recently embraced a polyphasic taxonomy system. This system, developed within the past twenty years, relies on multiple criteria, including phenotype, genotype, and phylogenetics, to classify members of the Mollicutes. Various studies carried out by Weisburg *et al.* (1989), Murray *et al.* (1990), Vandamme *et al.* (1996), and Razin *et al.* (1998) have supported this strategy. Phylogenetic studies using 16SrRNA and other housekeeping genes have definitively classified phytoplasmas as members of the Mollicutes class. Several studies by Lim and Sears (1989), Namba *et al.* (1993), Gundersen *et al.* (1994), Seemüller *et al.* (1998), Lee *et al.* (2000, 2006b), Zhao *et al.* (2005), Martini *et al.* (2007), and Hodgetts *et al.* (2008) have supported this classification. Due to the lack of easily identifiable physical traits, it was unavoidable that

the classification of phytoplasmas would mainly rely on molecular characteristics and phylogeny. Over the past twenty years, molecular-based investigations have demonstrated greater accuracy and reliability compared to the traditional biological criteria used for identifying phytoplasmas (Lee *et al.*, 2000). The development of PCR-based assays in the late 1980s and early 1990s greatly improved the detection and categorization of phytoplasmas by offering a more sensitive method for their detection. For additional information, see the references in Lee *et al.*, 2000. The 16SrRNA gene sequence, which is highly conserved, has been widely employed as the principal molecular tool for phytoplasma categorization. The 19 groups they found were called 16SrRNA groups (16Sr groups). They did this by using either real-life RFLP analysis of PCR-amplified 16SrDNA sequences or computer-based RFLP analysis. Wei *et al.* reported these findings in 2007, as well as Lee *et al.* in 1998 and 2000. This method led to the subsequent classification of numerous 16Sr subgroups. Gundersen *et al.* (1994) advocated that each group should include at least one species of phytoplasma. Currently, differences in 16SrDNA sequences primarily determine the classification of phytoplasmas. According to IRPCM in 2004, a specific criterion of 2.5% dissimilarity served as a guideline for choosing a new species. Due to the high level of conservation seen in the 16SrRNA gene, this rule may leave out many phytoplasma strains that are unique in their environment or biology. Some of these excluded strains may really deserve classification as a new taxonomic group. Speciation requires the inclusion of additional distinct biological features, such as insect vectors and plant hosts, together with other molecular requirements. In the past ten years, epidemiological studies have shown that different strains of phytoplasma, which are highly similar according to studies of 16SrRNA gene sequences, have been responsible for causing comparable diseases in different cultivars of a certain crop farmed in different parts of the world. In order to enhance the progress of disease control strategies, it is crucial to understand the association between different phytoplasma strains and their distinct ecological habitats. Frequently, it is challenging to distinguish these strains based on the examination of 16SrRNA gene sequences. This highlights the necessity of finding new indicators to distinguish more precisely between closely related phytoplasma strains. Over the past ten years, additional conserved genes or particular regions of genomic DNA have been utilized as supplementary molecular markers to more precisely distinguish closely related strains. This chapter aims to provide a concise overview of the current advancements made in this field and suggest potential genes that can be used to categorize phytoplasmas at various taxonomic levels. The

comments will discuss the possible utilization of several gene-based approaches to establish a taxonomic classification at the species or strain level.

16SrRNA Gene-based system for classification of phytoplasmas

Various molecular markers have been used to distinguish and categorize phytoplasmas. The 16SrRNA gene is extensively employed as a marker in the phytoplasma research community and has great utility in the initial categorization of phytoplasmas. Different sets of universal or generic oligonucleotide primers have been made based on the 16SrRNA gene, the 16S-23S intergenic spacer region, and partial 23SrRNA gene sequences. These primer pairs make it possible to copy over 1200 base pairs, which gives us almost full 16SrRNA gene sequences for all phytoplasmas found in different plants and insect vectors. The design of these primer pairs has been described in studies by Lee *et al.* (1993), Namba *et al.* (1993), Schneider *et al.* (1993), Gundersen and Lee (1996), and Smart *et al.* (1996). Schneider *et al.* (1993) and Lee *et al.* (1993) utilized restriction fragment length polymorphism (RFLP) analysis to classify phytoplasmas. This involved analyzing PCR-amplified 16SrRNA gene sequences using specific restriction enzymes. Lee *et al.* (1993, 1998, 2000) developed a thorough categorization system for phytoplasmas using RFLP tests with 17 restriction enzymes.

Diseases caused by phytoplasma have been documented in over a hundred countries, according to the sequences accessible in the GenBank database. The extent to which phytoplasma maladies affect various geographic regions is contingent upon the insect vector's polyphagous feeding behavior and the host range of the phytoplasma (Foissac and Wilson, 2009). Periodic reviews provide a comprehensive understanding of phytoplasma studies, including their taxonomy, etiology, transmission, and interactions with insect and plant hosts (Rao *et al.*, 2017; Duduk *et al.*, 2018; Bertaccini *et al.*, 2022; Wei and Zhao, 2022). Phytoplasma research has experienced significant growth over the past two decades.

This review provides a comprehensive overview of the historical development and present state of phytoplasma taxonomy. It specifically examines the various classification systems that have been implemented, evaluating their merits and demerits; challenges associated with published phytoplasma species; recent endeavors to establish a classification system for '*Candidatus*' bacteria; and other pertinent matters pertaining to phytoplasma.

The classification of major groups was determined by analyzing the similarity coefficients of collective restriction fragment length polymorphism (RFLP) patterns of a 1.2-kilobase polymerase chain reaction (PCR) amplicon, as described by Lee *et al.* in 1998. The similarity coefficients of restriction fragment length polymorphism (RFLP) patterns between two separate groups were 90% or below. The process of dividing a certain group into smaller subgroups was determined by analyzing the restriction sites inside this amplicon. If an unidentified phytoplasma strain possessed distinct restriction sites from all the established members of the group, a new subgroup would be designated. The scheme has been regularly revised (Lee *et al.*, 1998, 2000, 2006a; Montano *et al.*, 2001; Arocha *et al.*, 2005; Al-Saady *et al.*, 2008; and references in Lee *et al.*, 2004a, b; Zhao *et al.*, 2009a). Currently, it consists of 19 prominent phytoplasma groupings and over 50 subgroups. The grouping closely aligns with the phylogenetic tree formed by the examination of 16SrRNA sequences. Gundersen *et al.* (1994) suggested that each group should represent a minimum of one phytoplasma species. Wei *et al.* (2007, 2008) and Zhao *et al.* (2009b) have improved and enhanced the approach using virtual RFLP patterns. To do this, they used a computer program to simulate RFLP analysis on large collections of phytoplasma 16SrRNA gene sequences that were logged and saved in GenBank. Presently, the system consists of 29 groups and 89 subgroups. Every subgroup is characterized by distinct collective RFLP patterns. The approach, along with the illustrated restriction fragment length polymorphism (RFLP) patterns of all typical strains available in print or online, has effectively compiled the most extensive catalog of reference phytoplasma strains. To find an unknown phytoplasma strain, one can either do a physical (for initial identification) or computer-simulated virtual restriction fragment length polymorphism (RFLP) analysis of the 16S ribosomal RNA (rRNA) gene sequence and compare it to these patterns. Performing virtual RFLP analysis necessitates the sequencing of the 16S rRNA gene. Practically, RFLP analysis is the preferred method for the first identification of unknown phytoplasmas linked to a novel disease, especially when a large number of samples need to be examined or when there is no access to sequencing technology. The updated plan has the most complete classification system for phytoplasmas and has provided reliable molecular markers for quickly identifying different phytoplasma types. The 16S rRNA gene-based system is good for classifying phytoplasma because it has a high level of conservation, which makes it easy to design universal oligonucleotide primers, and there are a lot of sequences in the GenBank database, which lets scientists do full phylogenetic studies. However, the

16SrRNA gene is not enough to accurately tell the difference between closely related but different phytoplasma strains because its characteristics have been preserved for a long time. Multiple biologically relevant strains were present in certain subgroups, as was apparent.

The use of housekeeping genes in phylogenetic analysis proved and improved classification systems based on 16SrRNA genes. Adding more distinct branches allowed for a better separation of strains. At the same time, none of the studies looked at how well the marker genes worked by comparing them to other genes in the phytoplasma genome, like the 16SrRNA gene, which is used in a multi-locus classification system. This way, the researchers could see how quickly the genes changed. So, it was important to look at how well these marker genes worked at putting phytoplasma strains into groups at the genome level, taking into account how they evolved (Martini *et al.*, 2019). The phylogenetic history of an entire organism may not reflect the evolutionary history of a single gene due to the exclusion of horizontal gene transfer and unrecognized paralogy. The suitability of published markers, which are typically described for specific groups or species, may not extend to another group of phytoplasma species. Cho *et al.* (2020) suggested the use of the five-gene marker MLSA for the Aster Yellow (16SrI, AY) group of phytoplasmas as a supplement to the 16SrRNA gene phylogeny in order to improve phytoplasma taxonomy. The aforementioned marker genes DegV family protein (degV), TIGR00282 family metallo phosphoesterase, Preprotein translocase SecY (secY), and RluA family pseudo-uridine synthase-were successful in differentiating the strains of phytoplasmas belonging to the AY (16SrI) group into three distinct Operational Taxonomic Units (OTUs). In this instance, the authors deduced the five marker genes with the highest densities of informative genome sites and the highest substitution rates. These particular genes were observed to exhibit minimal deviation in comparison to previously reported shared single-copy genes. The ones chosen by Cho *et al.* (2020) were chosen because they are located in separate and unique parts of the genome that show unlikely homologous recombination between the same OTUs.

16S-23SrRNA Intergenic Spacer Region (ISR)

The intergenic spacer section of the phytoplasma 16S-23SrRNA, which is approximately 232 base pairs long, includes a segment that encodes the tRNA^{Ile}, a molecule that is extremely conserved. Nevertheless, the sequences that border the tDNA^{Ile} and continue towards the 16SrDNA and 23SrDNA differ across different phytoplasmas. The ISR can be utilized as an effective method for distinguishing between distinct phytoplasma groups and

subgroups. In general, the ISR and the 16SrRNA gene sequence are both good at telling the difference between different phytoplasma lineages (Smart *et al.*, 1996). Due to the low number of relevant characters included in its relatively brief sequence, ISR is unable to effectively distinguish between all the 16Sr subgroups. The full 16S rRNA gene and ISR sequence, on the other hand, have been studied together and have been useful for telling different strain types within a 16Sr subgroup apart. This has been demonstrated in several examples by Griffiths *et al.* (1999), Marcone *et al.* (2000), Padovan *et al.* (2000), and Andersen *et al.* (2006).

Tuf Gene-based system

The *tuf* gene, which encodes the elongation factor EF-Tu, is a highly conserved gene that is commonly employed for distinguishing and categorizing phytoplasmas. In 1997, Schneider *et al.* developed primer pairs suitable for amplifying *tuf* gene sequences from a wide range of phytoplasma families. The *tuf* gene, similar to the 16SrRNA gene, has been identified as a possible indicator for categorizing phytoplasma groups. Between 87.8% and 97.0% of the nucleotide sequences were the same between the aster yellows (AY), peach X-disease, and stolbur (STOL) phytoplasma families. Phytoplasma groups and subgroups can be distinguished by doing RFLP studies with multiple restriction enzymes. The ability to distinguish between different lineages of phytoplasmas is slightly less effective compared to the 16SrRNA gene (Schneider *et al.*, 1997; Marcone *et al.*, 2000). Still, the *tuf* gene has been shown to be useful for telling the difference between different ecological strains or variants within 16SrRNA subgroups (Langer and Maixner, 2004). Multiple strain variants were identified within 16XII-A and 16XII-B by analyzing *tuf* gene sequences (Langer and Maixner, 2004; Streten and Gibb, 2005; Andersen *et al.*, 2006; Pacifico *et al.*, 2007; Riolo *et al.*, 2007; Iriti *et al.*, 2008).

Ribosomal protein gene-based system

There is more variation in ribosomal protein (rp) genes than in 16SrRNA genes, and these genes have more phylogenetically significant traits. This significantly improves the ability to differentiate between different phytoplasma strains. Prior research on distinguishing phytoplasma strains in groups 16SrI and 16SrV revealed that analyzing rp gene sequences not only effectively categorized subgroups that align with 16Sr subgroups but also identified separate strains (lineages) within certain subgroups. These lineages could not be distinguished by analyzing 16S rRNA gene sequences alone (Martini *et al.*, 2002; Lee *et al.*, 2004a). The majority of the newly

discovered lineages possess unique biological characteristics. The phytoplasma that causes maize bushy stunt (MBS) is part of the 16SrI-B subgroup. This subgroup has specific host plants and vectors. This subgroup is distinct from other members of the 16SrI group, making it a separate and unique subgroup. Furthermore, subgroup 16SrV-C can be more precisely classified into many rp subgroups using RFLP studies with specific important restriction enzymes, as demonstrated by Martini *et al.* (2002) and Lee *et al.* (2004). A new study by Martini *et al.* (2007) made a detailed phylogenetic tree by looking at two ribosomal protein genes, rplV (rpl22) and rpsC (rps3), from 46 different phytoplasma strains. These strains belonged to 12 different 16Sr groups. When comparing the two trees, the one made with the rp gene showed the relationships between the different phytoplasma strains more clearly and precisely than the one made with the 16S rRNA gene. It also revealed more distinct subclades and lineages within the phytoplasma group. Three species of '*Ca. Phytoplasma*' (namely '*Ca. Phytoplasma mali*', '*Ca. Phytoplasma pyri*', and '*Ca. Phytoplasma prunorum*') were found to have a 16SrDNA sequence similarity of 98.9-99.1%. Additionally, they shared a RP gene sequence similarity of 94.3-94.6%. These species were easily distinguished by analyzing their RP gene sequences. When rp genes were compared, the mean sequence similarity between two given 16S phytoplasma groups was 50.4% to 83.5%. When 16SrRNA genes were compared, it was 85.0 % to 96.9%. Due to the increased sequence variety, rp genes are more effective as a molecular tool for distinguishing various phytoplasma strains.

SecYGene-based system

The secY gene, which codes for a subunit of a protein translocase, serves as an additional molecular identifier that aids in the more precise distinction of phytoplasma strains. The sequence variability of the secY gene is comparable to that of the rp genes. The mean sequence similarities of the secY gene between two specified 16Sr phytoplasma groups vary from 57.4% to 76.0% (Lee *et al.*, unpublished). The classification of SecY subgroups within groups 16SrI and 16SrV phytoplasmas was determined by analyzing the RFLP patterns of secY gene sequences. These subgroups were found to align with the subgroups identified using rp gene sequences (Lee *et al.*, 2004a, b, 2006b; Martini *et al.*, 2007). Nevertheless, the resolving capability of secY is marginally superior to that of rp gene sequences, owing to its more informative features. Ongoing efforts are being made to fully characterize the majority of phytoplasma groups and their representative strains (Lee *et al.*, unpublished). The secY gene, similar to the rp gene, may serve as a promising potential marker for phytoplasma strain classification.

SecA gene and other genes

The gene *secA*, which encodes a component of a protein translocase, was recently utilized to classify phytoplasmas (Hodgetts *et al.*, 2008). A 480-base-pair segment of the gene sequence was amplified using PCR from different phytoplasma strains that belong to 12 16Sr groups. The sequence similarity between the two specified 16Sr groups varied from 69.7% to 84.4%. The *secA* gene's resolving strength as a phylogenetic parameter for distinguishing phytoplasmas is comparable to that of the *rp* and *secY* genes. Besides *nusA* (Shao *et al.*, 2006), folate genes (*folP* and *folK*) (Davis *et al.*, 2003), and *dnaB*, other genes have been used and shown to be useful for telling strains in groups 16SrI and 16SrXII apart. Additional possible genes that could be useful for classifying phytoplasmas include *dnaA*, which encodes the chromosomal replication initiator protein; *polC*, which encodes the DNA polymerase III alpha subunit; and *dnaE*, which also encodes the DNA polymerase III alpha subunit. The genetic sequence variability of these genes is comparable to, or slightly higher than, that of the *secY* or *rp* genes.

Perspective of multiple gene-based systems

Phytoplasmas are insect-transmitted plant diseases and are capable of growth in both vector and plant hosts (Lee *et al.*, 2000). The susceptibility of various insect and plant species to phytoplasmas differs depending on the specific phytoplasma strains involved, including strains within a particular 16Sr subgroup. As a result, the selection pressures that insect and/or plant hosts apply to the phytoplasmas they are associated with have over time resulted in the evolution and/or isolation of distinct phytoplasma populations or strains. Geographic isolation, which results from the different distributions of plant and vector species in various regions, may also have an impact on the process. The interactions among phytoplasmas, vectors, and plant hosts, known as three-way interactions, play a significant role in the intricate nature of phytoplasma ecology. These phytoplasma strains that are ecologically isolated frequently have distinct biological characteristics, including their selectivity for certain plant and vector species, as well as the symptoms they cause in the affected plants. It frequently happens that various but closely related groups of phytoplasma can cause a specific disease, such as grapevine yellows. Researchers from various fields have studied these groups (Angelini *et al.*, 2001; Leyva-López *et al.*, 2002; Martini *et al.*, 2002; Langer and Maixner, 2004; Lee *et al.*, 2006b; Botti and Bertaccini, 2007). Certain groups may be more prevalent in specific grapevine varieties or geographic areas. Conversely, strains that are closely related, such as '*Ca. Phytoplasma asteris*' strains, have the potential to cause

various diseases and elicit distinct symptoms (Lee *et al.*, 2004b). In order to facilitate epidemiological investigations, it is crucial to identify and describe the various ecological strains that may play a role in the disease. The fact that the 16S rRNA gene-based system isn't very good at telling closely related strains apart makes it clear that other molecular markers for regular phytoplasma classification need to be added to this system right away. Several alternative molecular markers, in addition to the 16SrRNA gene, have been discovered that have significantly enhanced the ability to differentiate these ecological strains. The new multiple-gene systems are expected to offer improved molecular criteria for more precise differentiation of species and strains. The Phytoplasma /Spiroplasma Working Team of the International Research Program for Comparative Mycology (IRPCM) suggested using a certain threshold value of 97.5% similarity in the 16SrRNA gene sequence to tell the difference between two different species of "*Ca. Phytoplasma*." There are no precise threshold values available to determine the classification of new species based on strains that have a similarity of more than 97.5%. Similarly, there are no clear guidelines to differentiate closely related strains that have distinct ecological characteristics. To find these closely related but different biological or ecological strains, you can use molecular markers we've talked about in earlier sections along with other well-known phylogenetic factors that are more powerful than the 16SrRNA gene. Using the 16SrRNA gene along with one or more variable genes or DNA fragments, like 16SrRNA plus *secY*, 16SrRNA plus *rp*, or 16SrRNA plus *secA*, has been shown to effectively tell two strains that are very similar apart (Lee *et al.*, 2004a, b, 2006b; Martini *et al.*, 2007; Hodgetts *et al.*, 2008). Adding the 16SrRNA plus ISR (Internal Spacer Region) made it easier to tell closely related bacteria apart, which wasn't possible with just the 16SrRNA gene alone (Langer and Maixner, 2004). Due to the increased variety in the sequence, these sets of markers could aid in the more precise identification of closely related strains, including derived variations from a specific strain. Regarding the 16SrRNA gene, it is possible to define arbitrary threshold values of sequence similarities for assigning bacteria to various taxonomic rankings for these supplementary markers. A new study by Arnaud *et al.* (2007) used a technique called multi-locus sequence typing to tell the difference between three groups of *flavescence dorée* phytoplasma strains and group 16SrV phytoplasmas that infect European grapevines and alders. This method involved analyzing the gene sequences of *secY*, *map*, and *uvrB-degV*. Four phytoplasma genomes have been successfully annotated thanks to advances in genome sequencing and bioinformatics (Oshima *et al.*, 2004; Bai *et al.*,

2006; Kube *et al.*, 2008; Tran-Nguyen *et al.*, 2008). Additional studies for sequencing phytoplasma genomes are now in progress. Comparing these genomes will give us a lot of different genes. We will then use these genes to find markers that will help us define and describe phytoplasmas at different levels, such as strain, species, genus, and population. However, the development of a comprehensive gene-based system for classifying all types of phytoplasmas will not be possible until we have a nearly complete sequencing database for these prospective molecular markers. In order to distinguish between different strains, it is important to have an up-to-date database of sequence information (>1000 bp) for molecular markers in GenBank. This database should be easily accessible to the phytoplasma research community.

Issues in phytoplasma taxonomy

Phytoplasmas are insect-transmitted plant diseases and are capable of growth in both vector and plant hosts (Lee *et al.*, 2000). The susceptibility of various insect and plant species to phytoplasmas differs depending on the specific phytoplasma strains involved, including strains within a particular 16Sr subgroup. As a result, the selection pressures that insect and/or plant hosts apply to the phytoplasmas they are associated with have, over time, resulted in the evolution and/or isolation of distinct phytoplasma populations or strains. Geographic isolation, which results from the different distributions of plant and vector species in various regions, may also have an impact on the process. The interactions among phytoplasmas, vectors, and plant hosts, known as three-way interactions, play a significant role in the intricate nature of phytoplasma ecology. These phytoplasma strains that are ecologically isolated frequently have distinct biological characteristics, including their selectivity for certain plant and vector species, as well as the symptoms they cause in the affected plants. It frequently happens that various but closely related groups of phytoplasma can cause a specific disease, such as grapevine yellows. Researchers from various fields have studied these groups (Angelini *et al.*, 2001; Leyva-López *et al.*, 2002; Martini *et al.*, 2002; Langer and Maixner, 2004; Lee *et al.*, 2006b; Botti and Bertaccini, 2007). Certain groups may be more prevalent in specific grapevine varieties or geographic areas. Conversely, strains that are closely related, such as 'Ca. *Phytoplasma asteris*' strains, have the potential to cause various diseases and elicit distinct symptoms (Lee *et al.*, 2004b). In order to facilitate epidemiological investigations, it is crucial to identify and describe the various ecological strains that may play a role in the disease. The fact that the 16S rRNA gene-based system isn't very good at telling closely

related strains apart makes it clear that other molecular markers for regular phytoplasma classification need to be added to this system right away. Besides the 16S rRNA gene, other molecular markers have been found that make it much easier to tell these ecological strains apart. The new multiple-gene systems are expected to offer improved molecular criteria for more precise differentiation of species and strains. The Phytoplasma/Spiroplasma Working Team of the International Research Program for Comparative Mycology (IRPCM) suggested that to tell the difference between two species of "*Ca. Phytoplasma*," the 16SrRNA gene sequence should be 97.5% similar. There are no precise threshold values available to determine the classification of new species based on strains that have a similarity of more than 97.5%. Similarly, there are no clear guidelines to differentiate closely related strains that have distinct ecological characteristics. To find these closely related but different biological or ecological strains, you can use molecular markers we've talked about in earlier sections along with other well-known phylogenetic factors that are more powerful than the 16SrRNA gene. It has been shown that the 16S rRNA gene plus one or more variable genes or DNA fragments, such as 16SrRNA plus *secY*, 16SrRNA plus *rp*, or 16SrRNA plus *secA*, can tell two strains that are very similar apart (Lee *et al.*, 2004a, b, 2006b; Martini *et al.*, 2007; Hodgetts *et al.*, 2008). Adding the 16SrRNA plus ISR (Internal Spacer Region) made it easier to tell closely related bacteria apart, which wasn't possible with just the 16SrRNA gene alone (Langer and Maixner, 2004). Due to the increased variety in the sequence, these sets of markers could aid in the more precise identification of closely related strains, including derived variations from a specific strain. With these extra markers for the 16S rRNA gene, it is possible to set any number of sequence similarity thresholds that will put bacteria in different taxonomic groups. Arnaud *et al.* (2007) did a new study that used a method called multi-locus sequence typing to tell the difference between three groups of *flavescence dorée* phytoplasma strains and group 16SrV phytoplasmas that infect grapevines and alders in Europe. This method involved analyzing the gene sequences of *secY*, *map*, and *uvrB-degV*. Four phytoplasma genomes have been successfully annotated thanks to advances in genome sequencing and bioinformatics (Oshima *et al.*, 2004; Bai *et al.*, 2006; Kube *et al.*, 2008; Tran-Nguyen *et al.*, 2008). Additional studies for sequencing phytoplasma genomes are now in progress. Comparing these genomes will give us a lot of different genes. We will then use these genes to find markers that will help us define and describe phytoplasmas at different levels, such as strain, species, genus, and population. However, the development of a comprehensive gene-based system for classifying all types

of phytoplasmas will not be possible until we have a nearly complete sequencing database for these prospective molecular markers. In order to distinguish between different strains, it is important to have an up-to-date database of sequence information (>1000 bp) for molecular markers in GenBank. This database should be easily accessible to the phytoplasma research community.

References

1. Al-Saady NA, Khan AJ, Calari A, Al-Subhi AM, Bertaccini A. *Candidatus Phytoplasma omanense*, associated with witches' broom of *Cassia italica* (Mill.) Spreng in Oman. *Int J Syst Evol Microbiol*. 2008;58:461-466.
2. Andersen MT, Newcomb RD, Liefting LW, Beever RE. Phylogenetic analysis of *Candidatus Phytoplasma australiense* reveals distinct populations in New Zealand. *Phytopathology*. 2006;96:838-845.
3. Angelini E, Clair D, Borgo M, Bertaccini A, Boudon-Padieu E. Flavescence dorée in France and Italy. Occurrence of closely related phytoplasma isolates and their near relationships to Palatinate grapevine yellows and an alder yellows phytoplasma. *Vitis*. 2001;40:79-84.
4. Arnaud G, Malembic-Maher S, Salar P, Bonnet P, Maixner M, Marcone C, *et al*. Multilocus sequence typing confirms the close genetic interrelatedness of three distinct Flavescence dorée phytoplasma strain clusters and group 16SrV phytoplasmas infecting grapevine and alder in Europe. *Appl Environ Microbiol*. 2007;73:4001-4010.
5. Arocha Y, López M, Piñol B, Fernández M, Picornell B, Almeida R, *et al*. *Candidatus Phytoplasma graminis* and *Candidatus Phytoplasma caricae*, two novel phytoplasmas associated with disease of sugarcane, weeds and papaya in Cuba. *Int J Syst Evol Microbiol*. 2005;55:2451-2463.
6. Bai X, Zhang J, Ewing A, Miller SA, Radek AJ, Shevchenko DV, *et al*. Living with genome instability, the adaptation of phytoplasmas to diverse environments of their insect and plant hosts. *J Bacteriol*. 2006;188:3682-3696.
7. Bertaccini A. Phytoplasmas, diversity, taxonomy, and epidemiology. *Front Biosci*. 2007;12:673-689.
8. Bertaccini A, Arocha-Rosete Y, Contaldo N, Duduk B, Fiore N, Montano HG, *et al*. Revision of the '*Candidatus Phytoplasma*' species

- description guidelines. *Int J Syst Evol Microbiol.* 2022;72(4). doi:10.1099/ijsem.0.005353.
9. Botti S, Bertaccini A. Grapevine yellows in northern Italy, molecular identification of *Flavescence dorée* phytoplasma strains and of Bois Noir phytoplasmas. *J Appl Microbiol.* 2007;103:2325-2330.
 10. Brown D, Whitcomb RF, Bradbury JM. Revised minimal standard for description of new species of the class Mollicutes (division Tenericutes). *Int J Syst Evol Microbiol.* 2007;57:2703-2719.
 11. Cho S-T, Kung H-J, Huang W, Hogenhout SA, Kuo C-H. Species boundaries and molecular markers for the classification of 16S rDNA phytoplasmas inferred by genome analysis. *Front Microbiol.* 2020;11. doi:10.3389/fmicb.2020.01531.
 12. Davis RE, Jomantiene R, Zhao Y, Dally EL. Folate biosynthesis pseudogenes, yfolP and yfolK, and an O-sialoglycoprotein endopeptidase gene homolog in the phytoplasma genome. *DNA Cell Biol.* 2003;22:697-706.
 13. Duduk B, Stepanović J, Yadav A, Rao GP. Phytoplasmas in Weeds and Wild Plants. In: Rao GP, editor. *Phytoplasmas: Plant Pathogenic Bacteria-I*. Springer Singapore; 2018. p. 313-345.
 14. Foissac X, Wilson MR. Current and possible future distributions of phytoplasma diseases and their vectors. In: Weintraub PG, Jones P, editors. *Phytoplasmas: genomes, plant hosts and vectors*. CABI; 2009. p. 309-324.
 15. Fox GE, Wisotzkey JD, Jurtshuk P. How close is close? 16S rRNA sequence identity may not be sufficient to guarantee species identity. *Int J Syst Bacteriol.* 1992;42:166-170.
 16. Griffiths HM, Sinclair WA, Boudon-Padiou E, Daire X, Lee I-M, Sfalanga A, *et al.* Phytoplasmas associated with elm yellows. Molecular variability and differentiation from related organisms. *Plant Dis.* 1999;83:1101-1104.
 17. Gundersen DE, Lee I-M. Ultrasensitive detection of phytoplasmas by nested-PCR assays using two universal primer pairs. *Phytopathol Mediterr.* 1996;35:144-151.
 18. Gundersen DE, Lee IM, Rehner SA, Davis RE, Kingsbury DT. Phylogeny of mycoplasma like organisms (phytoplasmas), a basis for their classification. *J Bacteriol.* 1994;176:5244-5254.

19. Hodgetts J, Boonham N, Mumford R, Harrison N, Dickinson M. Phytoplasma phylogenetics based on analysis of secA and 23S rRNA gene sequences for improved resolution of candidate species of 'Candidatus Phytoplasma'. *Int J Syst Evol Microbiol.* 2008;58:1826-1837.
20. Hogenhout SA, Oshima K, Ammar el-D, Kakizawa S, Kingdom HN, Namba S. Phytoplasmas, bacteria that manipulate plants and insects. *Mol Plant Pathol.* 2008;9:403-423.
21. Hugenholtz P, Chuvochina M, Oren A, Parks DH, Soo RM. Prokaryotic taxonomy and nomenclature in the age of big sequence data. *ISME J.* 2021;15:1879-1892.
22. Iriti M, Quaglino F, Maffi D, Casatti P, Bianco PA, Faoro F. *Solanum malacoxylon*, a new natural host of stolbur phytoplasma. *J Phytopathol.* 2008;156:8-14.
23. IRPCM Phytoplasma/Spiroplasma Working Team-Phytoplasma Taxonomy Group. 'Candidatus Phytoplasma', a taxon for the wall-less, non-helical prokaryotes that colonize plant phloem and insects. *Int J Syst. Evol. Microbiol.* 2004;54:1243-1255.
24. Kube M, Schneider B, Kuhl H, Dandekar T, Heitmann K, Migdoll AM, *et al.* The linear chromosome of the plant-pathogenic mycoplasma 'Candidatus Phytoplasma mali'. *BMC Genomics.* 2008;9:306.
25. Langer M, Maixner M. Molecular characterization of grapevine yellows associated with phytoplasmas of stolbur group based on RFLP-analysis of non-ribosomal DNA. *Vitis.* 2004;43:185-189.
26. Lee I-M, Hammond RW, Davis RE, Gundersen DE. Universal amplification and analysis of pathogen 16S rDNA for classification and identification of mycoplasma like organisms. *Phytopathology.* 1993;83:834-842.
27. Lee I-M, Gundersen-Rindal DE, Davis RE, Bartoszyk IM. Revised classification scheme of phytoplasmas based on RFLP analyses of 16SrRNA and ribosomal protein gene sequences. *Int. J Syst. Bacteriol.* 1998;48:1153-1169.
28. Lee IM, Davis RE, Gundersen-Rindal DE. Phytoplasma, phytopathogenic mollicutes. *Annu. Rev Microbiol.* 2000;54:221-255.
29. Lee IM, Martini M, Marcone C, Zhu SF. Classification of phytoplasma strains in the elm yellows group (16SrV) and proposition of 'Candidatus

- Phytoplasma ulmi' for the phytoplasma associated with elm yellows. Int J Syst Evol Microbiol. 2004;54:337-347.
30. Lee IM, Gundersen DE, Davis RE, Bottner KD, Marcone C, Seemüller E. 'Candidatus Phytoplasma asteris', a novel phytoplasma taxon associated with aster yellows and related diseases. Int. J Syst. Evol. Microbiol. 2004;54:1037-1048.
 31. Lee IM, Bottner KD, Secor G, Rivera-Varas V. 'Candidatus Phytoplasma americanum', a phytoplasma associated with a purple top wilt disease complex. Int. J Syst. Evol. Microbiol. 2006;56:1593-1597.
 32. Lee I-M, Zhao Y, Bottner KD. SecY gene sequence analysis for finer differentiation of diverse strains in the aster yellows phytoplasma group. Mol Cell Probes. 2006;20:87-91.
 33. Leyva-López NE, Ochoa-Sánchez JC, Leal-Klevezas DS, Martinez-Soriano JP. Multiple phytoplasmas associated with potato diseases in Mexico. Can J Microbiol. 2002;48:1062-1068.
 34. Lim PO, Sears BB. 16S rRNA sequence indicates that plant-pathogenic mycoplasma-like organisms are evolutionarily distinct from animal mycoplasmas. J Bacteriol. 1989;171:5901-5906.
 35. Marcone C, Lee IM, Davis RE, Ragozzino A, Seemüller E. Classification of aster yellows-group phytoplasmas based on combined analyses of rRNA and tuf gene sequences. Int. J Syst. Evol. Microbiol. 2000;50:1703-1713.
 36. Martini M, Botti S, Marcone C, Marzachi C, Casati P, Bianco PA, *et al.* Genetic variability among Flavescence dorée phytoplasmas from different origins in Italy and France. Mol Cell Probes. 2002;16:197-208.
 37. Martini M, Lee I-M, Bottner KD, Zhao Y, Botti S, Bertaccini A, *et al.* Ribosomal protein gene-based phylogeny for finer differentiation and classification of phytoplasmas. Int J Syst Evol Microbiol. 2007;57:2037-2051.
 38. Martini M, Quaglino F, Bertaccini A. Multilocus Genetic Characterization of Phytoplasmas. In: Bertaccini A, Weintraub PG, Rao GP, Mori N, editors. Phytoplasmas: Plant Pathogenic Bacteria-III. Springer Singapore; 2019. p. 161-200.
 39. Montano HG, Davis RE, Dally EL, Hogenhout S, Pimentel P, Brioso PST. 'Candidatus Phytoplasma brasiliense', a new phytoplasma taxon associated with hibiscus witches'-broom disease. Int. J Syst. Evol. Bacteriol. 2001;51:1109-1118.

40. Murray RGE, Schleifer KH. Taxonomic notes, a proposal for recording the properties of putative taxa of prokaryotes. *Int J Syst Bacteriol.* 1994;44:174-176.
41. Murray RGE, Brenner DJ, Colwell RR, De Vos P, Goodfellow M, Grimont PAD, *et al.* Report of the ad hoc committee on approaches to taxonomy within the Proteobacteria. *Int J Syst Bacteriol.* 1990;40:213-215.
42. Namba S, Kato S, Iwanami S, Shiozawa H, *et al.* Phylogenetic diversity of phytopathogenic organisms. *Int J Syst Bacteriol.* 1993;43:461-467.
43. Oshima K, Kakizawa S, Nishigawa H, Jung HY, Wei W, Suzuki S, *et al.* Reductive evolution suggested from the complete genome sequence of a plant pathogenic phytoplasma. *Nat Genet.* 2004;36:27-29.
44. Pacifico D, Foissac X, Veratti F, Marzachi C. Genetic diversity of Italian and French 'bois noir' phytoplasma isolates. *Bull Insectol.* 2007;60(2):345-346.
45. Padovan A, Gibb K, Persley D. Association of 'Candidatus Phytoplasma australiense' with green petal and lethal yellows diseases in strawberry. *Plant Pathol.* 2000;49:362-369.
46. Rao GP, Madhupriya, Thorat V, Manimekalai R, Tiwari AK, Yadav A. A century progress of research on phytoplasma diseases in India. *Phytopathogenic Mollicutes.* 2017;7(1):1. <https://doi.org/10.5958/2249-4677.2017.00001.9>
47. Razin S, Yogeve D, Naot Y. Molecular biology and pathology of mycoplasmas. *Microbiol Mol Biol Rev.* 1998;62:1094-1156.
48. Riolo P, Landi L, Nardi S, Isidoro N. Relationships among *Hyalostyles obsoletus*, its herbaceous host plants and 'bois noir' phytoplasma strains in vineyard ecosystems in the Marche region (central-eastern Italy). *Bull Insectol.* 2007;60(2):353-354.
49. Schneider B, Aherns U, Kirkpatrick BC, Seemüller E. Classification of plant-pathogenic mycoplasma-like organisms using restriction-site analysis of PCR-amplified 16S rDNA. *J Gen Microbiol.* 1993;139:519-527.
50. Schneider B, Gibb KS, Seemüller E. Sequence and RFLP analysis of the elongation factor Tu gene used in differentiation and classification of phytoplasmas. *Microbiology.* 1997;143:3381-3389.

51. Seemüller E, Marcone C, Lauer U, Ragozzino A, Gó'schl M. Current status of molecular classification of the phytoplasmas. *J Plant Pathol.* 1998;80:3-26.
52. Shao J, Jomantiene R, Dally EL, Zhao Y, Lee I-M, Nuss DL, *et al.* Phylogeny and characterization of phytoplasma lNusA and use of the nusA gene in detection of group 16SrI strains. *J Plant Pathol.* 2006;88:193-201.
53. Smart CD, Schneider B, Blomquist CL, Guerra LJ, Harrison NA, *et al.* Phytoplasma-specific PCR primers based on sequences of 16S-23S rRNA spacer region. *Appl Environ Microbiol.* 1996;62:2988-2993.
54. Stackebrandt E. Forces shaping bacterial systematics. *Microbe.* 2007;2:283-288.
55. Stackebrandt E, Goebel BM. Taxonomic note, a place for DNA-DNA reassociation and 16SrRNA sequence analysis in the present species definition in bacteriology. *Int J Syst Bacteriol.* 1994;44:846-849.
56. Stackebrandt E, Frederiksen W, Garrity GM, Grimont PA, Kampfer P, Maiden MC, *et al.* Report of the ad hoc committee for the re-evaluation of the species definition in bacteriology. *Int J Syst Evol Microbiol.* 2002;52:1043-1047.
57. Streten C, Gibb KS. Genetic variation in *Candidatus Phytoplasma australiense*. *Plant Pathol.* 2005;54:8-14.
58. Tran-Nguyen LT, Kube M, Schneider B, Reinhardt R, Gibb KB. Comparative genome analysis of 'Candidatus *Phytoplasma australiense*' (subgroup tuf-Australia I; rp-A) and 'Ca. *Phytoplasma asteris*' strains OY-M and AY-WB. *J Bacteriol.* 2008;190:3979-3991.
59. Vandamme P, Pot B, Gillis M, Devos P, Kersters K, Swings J. Polyphasic taxonomy, a consensus approach to bacterial systematics. *Microbiol Rev.* 1996;60:407-438.
60. Wei W, Davis RE, Lee I-M, Zhao Y. Computer-simulated RFLP analysis of 16S rRNA genes, identification of ten new phytoplasma groups. *Int J Syst Evol Microbiol.* 2007;57:1855-1867.
61. Wei W, Lee I-M, Davis RE, Suo X, Zhao Y. Automated RFLP pattern comparison and similarity coefficient calculation for rapid delineation of new and distinct phytoplasma 16Sr subgroup lineages. *Int. J Syst. Evol. Microbiol.* 2008;58:2368-2377.

62. Wei W, Zhao Y. Phytoplasma taxonomy: Nomenclature classification, and identification. *Biology*. 2022;11(8):11-19.
<https://doi.org/10.3390/biology11081119>.
63. Weisburg WG, Tully JG, Rose DL, Petzel JP, Oyaizu H, *et al.* A phylogenetic analysis of the mycoplasmas, basis for their classification. *J Bacteriol*. 1989;171:6455-6467.
64. Woese CR. Bacterial evolution. *Microbiol Rev*. 1987;51:221-271.
65. Woese CR. Interpreting the universal phylogenetic tree. *Proc. Natl. Acad. Sci USA*. 2000;97:8392-8396.
66. Woese CR, Maniloff J, Zablen LB. Phylogenetic analysis of the mycoplasmas. *Proc Natl Acad. Sci USA*. 1980;77:494-498.
67. Zhao Y, Davis RE, Lee IM. Phylogenetic positions of ‘Candidatus Phytoplasma asteris’ and *Spiroplasma kunkelii* as inferred from multiple sets of concatenated core housekeeping proteins. *Int. J Syst. Evol. Microbiol*. 2005;55:2131-2141.
68. Zhao Y, Sun Q, Wei W, Davis RE, Wu W, Liu Q. ‘Candidatus Phytoplasma tamaricis’, a novel taxon discovered in witches’-broom diseased salt cedar (*Tamarix chinensis* Lour.). *Int. J Syst. Evol. Microbiol.*, 2009, 59.
69. Zhao Y, Wei W, Lee IM, Shao J, Suo X, Davis RE. Construction of an interactive online phytoplasma classification tool, iPhyClassifier, and its application in analysis of the peach X-disease phytoplasma group (16SrIII). *Int. J Syst. Evol. Microbiol.*, 2009, 59.

Chapter - 2

Next Generation Sequencing Applications in ChIP-seq, BS-seq and Metagenomics and Molecular Biology of Viral Metagenomics

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Chapter - 2

Next Generation Sequencing Applications in ChIP-seq, BS-seq and Metagenomics and Molecular Biology of Viral Metagenomics

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Abstract

A method for characterisation of DNA-binding proteins, histone modifications, or nucleosomes on a genome-wide scale is chromatin immunoprecipitation followed by sequencing, or ChIP-seq. The remarkable advancements in next-generation sequencing technology have made ChIP-seq more productive than its array-based predecessor, ChIP-chip, in terms of resolution, noise reduction, and coverage. ChIP-seq has become an essential tool for researching gene regulation and epigenetic mechanisms. One epigenetic alteration that plays a significant part in numerous biological functions and diseases is DNA methylation. Recently, bisulfite sequencing (BS-seq) has become the preferred method for profiling DNA methylation due to its increased resolution, accuracy, and genome coverage. However, the need for flexible, powerful and computationally efficient statistical methods is required for more development. The pace of metagenomic investigations has increased globally, thanks to advancements in next-generation sequencing methods. Crop plants and microbial interactions is an important aspect of agriculture. Therefore, examining these interactions via the metagenomics lens might provide a whole new significance for crop development and health. An important source of the microbial community for agricultural soil is the rhizosphere. A special emphasis has been provided on viral-plant metagenomics. Various studies have been carried out on the microflora present in the rhizosphere in various crops such as sorghum, rice, wheat, legumes, and chickpeas.

Keywords: ChIP-seq, Bisulfite sequencing, viral metagenomics

Introduction

Applications of Next Generation Sequencing have been successful and have expanded to include Protein-DNA Interaction Analysis (ChIP-seq),

Epigenome Research, DNA Methylation Pattern Analysis, Bisulfite Sequencing Research, and Metagenomic Research ^[1]. The many sequencing facility types that are available are shown. The creation of resistant cultivars has shown to be a practical and cost-effective method of managing crop development and control of diseases. Food security and crop protection are interdependent, and in order to guarantee that there is enough food for everyone, it is crucial for the development of crops resistant to biotic and abiotic stresses by means of correct breeding procedures ^[2]. Because breeding requires a lot of labour, integrated disease control is frequently limited to chemical treatment, which defeats the entire goal of this environmentally beneficial strategy. However, as breeding techniques have advanced, it has been easier to remain ahead of disease adaptation. A plant breeder's ultimate goal should be to determine the phenotypes of climate-resilient lines and the SNPs linked to trait of interest.

Researchers may now delve deeply into the nooks and crannies of the crop genomes thanks to the improvements made in research made possible by the various genome genome sequence projects. The fundamental method relies on NGS quality control and pre-processing data, read mapping using certain algorithms and reference sequences, quality control checks, data visualisation, and functional annotations. Following each stage, quality control processes are typically implemented in order to identify the precise genomic areas linked to the trait of interest. Sequencing the target (DNA/RNA), shearing into smaller fragments using nebulization, sonication, enzyme-mediated cleavage, and acoustic shearing are some general aspects of the initial stage of NGS data interpretation. The size selection step in the designated target range comes after this fragmentation stage. An RNA fragment needs to be transformed with specialised reverse transcriptase into complementary DNA. Adapters should next be immobilised to a solid support and ligated to the DNA/RNA fragments. These adapters are unique to the NGS platform and are necessary for sequences that call for an NGS instrument to recognise sequencing reads ^[3, 4].

There are four parts to the adapters. PCR-specific primer sequences are amplified for sequencing template enrichment, universal sequencing primer sequences are used to initiate reactions in each fragment, anchoring sequences are used to secure DNA fragments to solid supports, and indexing barcodes are used to distinguish between multiple samples when sequenced together. The creation of sequencing reads, or contigs, and sequencing read alignment come next. Nucleotides with a reversible terminator and fluorescent labelling are used in the core Illumina sequencing technique. The terminator moiety

stops nascent DNA strands from expanding for a brief period of time. The terminator moiety is broken once the integrated nucleotide is optically detected using a particular fluorescent label. This allows new strand synthesis to start for the next cycle of nucleotide incorporation. To identify nucleotide inclusion in millions of sequencing operations simultaneously, distinct fluorescent signals are generated to mark deoxy ATP, deoxy CTP, deoxy GTP, and deoxy TTP. Various fluorescence signals can be used to identify each nucleotide [5, 6].

IDT has developed sophisticated indexing techniques in partnership with Illumina. The xGen UDI-UMI indexes, which are especially made for multiplexing solutions to yield faster results, may have single indexing, unique dual indexing (UDI), and unique molecular identifiers (UMI) in these adapters [7]. The design of the adaptor is as follows:

- i) Unique/Combinatorial dual indexes (UDI/CDI).
- ii) Index lengths of 8 to 10 nucleotides (CDI or UDI).
- iii) Multiplex capacity up to 1.5 kb unique indexes.
- iv) Indexing primers intended for enzymatic library normalisations.

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Sample indexing enables numerous samples to be multiplexed in a flow cell, also known as a chip, for sequencing in a single attempt. Each sample index, which typically consists of 8-10 bases, is unique to a particular sample library and is used to demultiplex data in order to assign each sequence read to the correct sample. In addition to maintaining colour balance for Illumina instruments with two and four colours, primer indexing eliminates GC bias. By using unique dual indexes (UDIs), errors caused by index-hopping-a process related to primer multiplexing are reduced. Through the use of unique molecular identifiers (UMI), molecular barcoding boosts throughput and provides dependable error correction. UMIs are brief sequences that share the same insert and UMI tag and add a distinct barcode to every molecule in a sample library. UMIs improves variant detection and lower the percentage of false-positive variant calls. UMIs, such as the xGen UDI-UMI Adapters, which are available with a UDI design that identifies low-frequency variants, may also be contained in adapters [7].

In addition to the genetic components found by researching DNA sequences alone, understanding the dynamics of DNA methylation and their influence has expanded and improved our knowledge on various biological processes and disease incidence and escaping mechanisms. In recent years, there has been a rapid evolution of technologies for assessing DNA methylation. Next-generation sequencing (BS-seq) is a new technology that examines the methylation levels at single-nucleotide precision. It involves treating DNA with sodium bisulfite. Enhanced accuracy, decreased background bias, and increased spatial resolution characterise BS-seq in comparison to other high-throughput technologies like methylation microarray and capture-based sequencing like MeDIP-seq. As a result, it has swiftly emerged as the method of choice for researching DNA methylation [8, 9].

ChIP-seq and BS-seq

ChIP sequencing (ChIP-Seq), which combines sequencing with chromatin immunoprecipitation (ChIP) tests, is an effective technique for locating DNA binding sites throughout the genome that transcription factors and other proteins bind to. Genome-wide identification and characterization of histone modification, enhancer analysis, and genome-wide chromatin annotation may be carried out through ChIP-seq to help analysis of dynamics that contribute to cell identity, development, and transition stages [10]. A recent ChIP-seq study revealed homeobox gene OshOX24-binding elements that modulate plethora of target genes like ERF1, MYB1, LTREs and SORLIP2 in response to heat stress. Regulation of transcription factor FoxO in brown plant hopper enabled modulation of fecundity by regulation of expression of mTOR/S6K and vitellogenin resulting in increased progeny and infectivity of hopper in rice as revealed by ChIP-seq [11].

The flow-summary for steps involved in ChIP-seq Data Analysis workflow involve obtained standard data quality followed by read mapping, peak calling, peak visualization, followed by differential binding assay, function assay and DNA binding motif assay and profiling of histone modification topography (Transcription factor binding regions, eQTL discovery, genome-wide association study). In a ChIP-seq study, antibodies specific to DNA regions of modified histones must be designed and prior to the workflow. Recently reported standard protocols are available for the ChIP-seq methodology [12, 13]. In past decade, ChIP-chip analysis performed in microarrays chips for deciphering several pathways in rice through protein location analysis or variations in expression levels of genes [14]. However, they are unable to produce more reliable and accurate data and unable to offer better

resolution. Presently, with the cutting costs of NGS technologies word wide, the ChIP-seq methodology is gaining more popularity than microarray technologies. Modern-day ChIP-seq is a different procedure altogether and has been reported for genome-wide analysis of rice transcription factor NAC involved in abiotic stress resistance ^[15].

In a recent experiment, activity of enhancers of reference rice epigenomes was validated by qRT-PCR. ChIP-seq and FAIRE-seq data analysis viz. 3K4m3, H3K27ac and RNAPII library preparation in MAC2 peak calling software as has been used for Transposase Accessible Chromatin (ATAC-seq) in acoe *Hofstenia miamia* for identification of open chromatin representing putative regulatory elements ^[16]. Narrow peak calling algorithm settings were used for the analysis of H3K9me2 for detection of transcription factor binding at nearby regions and broad peak calling algorithm settings were used for H3K9me2, H3K4me1, and H3K27me3 library by wide peak mode for deciphering signature histone modifications. Quality of the peak dataset was accessed through FriP (Fraction of Reads in Peak) ^[17]. FAIRE-seq peaks are also identified using MACS2 software by ATAC-seq of acinar cells ^[18]. Later FAIRE-seq and ChIP-seq data visualization was performed in IGV browser. Signal levels were normalized to FPKM values (FPKM abbvr. Fragments per Kilobase of transcript per Million mapped reads). Spearman's correlation coefficient was used for the normalization of signals between two biological replicates of eChIP-seq or FAIRE-seq libraries by the creation of bins ^[19].

DNA methylation may be studied by means of Whole Genome Bisulfite Sequencing (WGBS). The use of bisulfite treatment of DNA before usual sequencing helps in the analysis of pattern for methylation. Treating DNA with bisulfite convert cytosine residues for uracil, yet causes no modifications to 5-methylcytosine. In a study, the libraries were evaluated using Fast-QC, and filtration of adapter and low-quality reads were performed in Trimnomatic ^[20]. Methylated genome reads are mapped to reference *Oryza sativa* indica genomes built by combination of PacBio read and Illumina paired-end sequencing data of indica cultivars MH63/ZS97 ^[21]. For mapping of reads BatMeth 2.0 is used ^[22].

For bisulphite treated DNA sequencing standard procedure include bisulfite conversion of DNA by means of EZ-DNA Methylation Gold kit and bisulfite treated DNA libraries are constructed using Illumina True Seq DNA sample preparation kit and after following required protocol is sequenced in Illumina X platform. The DNA methylation sequencing depth may be further analysed for differential methylation containing regions and characterization of chromatin states using multivariate Hidden Markov Model ^[23, 22].

The quantity of methylated and total readings at each CpG site can be used to summarise the BS-seq data. A beta-binomial distribution is commonly used to predict the methylation read counts, taking into consideration both sampling and biological variability. At each CpG site, a simple way to call DML (Defined Mutant Library) under general design is to use a beta-binomial generalised linear model (GLM). RADMeth and BiSeq are the two approaches that are now available for BS-seq under generic design. The beta-binomial GLM with the ‘logit’ link function is directly applied by RADMeth. Prior to using a beta regression with a ‘probit’ link on the smoothed methylation levels; BiSeq first carries out a local smoothing inside each sample. The primary goal of BiSeq's hierarchical testing process is Differentially Methylated Region (DMR) identification. However, the RADMeth and BiSeq experience poor computational performance, secondly GLM procedure being numerically unstable at numerous covariates in numerous instances of separation or quasi-separation. A software for direct DMR detection is GetisDMR that is based on statistical inference for direct DMR detection. Another new designed method of data analysis for data under linear model with fixed effects has been discovered ^[9].

The prevailing flowchart for integrative analysis of ChIP-Seq and DNA methylation sequencing (BS-seq) involved is depicted (Fig. 1).

Integrated approach of ChIP-seq, BS-seq approach for gene and epigenome regulation analysis

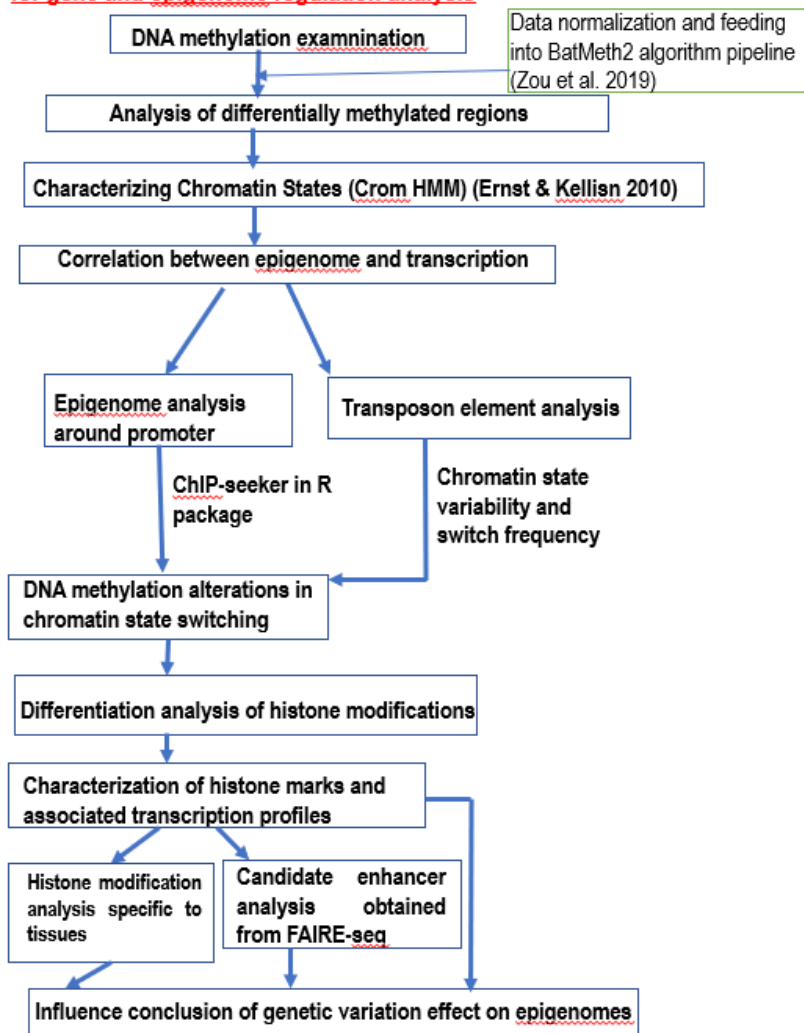


Fig 1: ChIP-seq and BS-seq procedure flow diagram

Combining the ChIP-seq, BS-Seq and RNA-seq approach can further pinpoint the definitive causal genomic regions for a phenotype. RNA-seq analysis of rice and mapping of sequencing reads to *Oryza* genome with annotation from combined gene obtained from NCBI databases using Cuffcompare method in Cufflinks simultaneously using Cuffdiff^[24]. Quality reads are obtained from raw paired end reads using Trim Galore and aligned to *Oryza sativa* Nipponbare reference genome version IGRSP-1.0 reference

genome using TopHat2 for read alignment, Bowtie2 for read alignment of long reference genome sequence reads and estimation of sequencing errors [25].

Long non-coding RNAs are predicted using Cuffmerge and Cuffcompare and following mapping to reference genome lncRNAs from transcripts coding for proteins are predicted by tools viz. CPC2, COME. Differential expression analysis may be performed in DESeq R package followed by KEGG pathway enrichment mediated by Hypergenomic Distribution test may also be performed in R package. Further differentiation between alternative splicing analysis may be performed from indexed BAM files and merged with reference read sequence in.gtf format using rMATS [26] and alternative splicing events and novel introns and novel exons may be predicted. An overview of this protocol may be obtained from studying post-transcriptional regulation in response to cadmium stress in rice [27]. Bisulfite sequencing has been performed in rice cultivars [28] and various other crops [29, 30, 31].

Alteration in rice DNA methylation patterns that affect gene expressions in virus-associated biotic stress has been tracked. Sequencing was performed in Hiseq 4000 platform. Reads mapped for reference genome with WALT [32, 33]. DNA methylation levels were determined in MethPipe pipeline [34, 35]. Whole genome Bisulfite sequencing and epigenetic variation have been investigated in cereal methylomes [36]. DNA methylation specific changes associated with gene expression when exposed to cadmium have also been studied and reported [37]. The data from ChIP-seq, BS-seq, FAIRE-seq, RNA-seq and ncRNA-seq obtained from rice has been put together towards the development of RiceENCODE database that proclaims to be the encyclopedia of DNA elements [38]. The effect of SNPs variations on the effector genes may be determined by SnpEff [39]. The ChIP-seq assay based on requirement may be validated by mass spectrometric techniques [40].

Metagenomic sequencing

A great number of microorganisms including bacteria, fungi, archaea, oomycetes and virus inhabits plants. These micro-organisms that are associated to plants might be pathogenic or non-pathogenic which might be beneficial, neutral, mutualistic or commensal. Organisms demonstrating beneficial association with the plant is termed endophytes [41]. These endophytes don't cause any harmful effects to the plants. Various endophytic microbes are specific to their colonization plant species. They demonstrate mutualistic relationship and are beneficial for improvement of plant growth and health [42]. Recent findings reveal presence of pathogenic species [43, 44].

Endophytes colonize tissues by undergoing horizontal transmission and through vertical seedling methods from the rhizospheric soil [45, 46]. Endophytes create a variety of chemicals that aid in plant growth promotion and improved environmental adaption in an effort to maintain stable symbiosis. This chapter provides a thorough overview of metagenomics, which is used to research microorganisms linked with plants. It focuses on endophytes, which exploit information from genome sequences to their benefit. The prevailing flowchart for integrative analysis of Metagenomic analysis flow-diagram is depicted (Fig. 2).

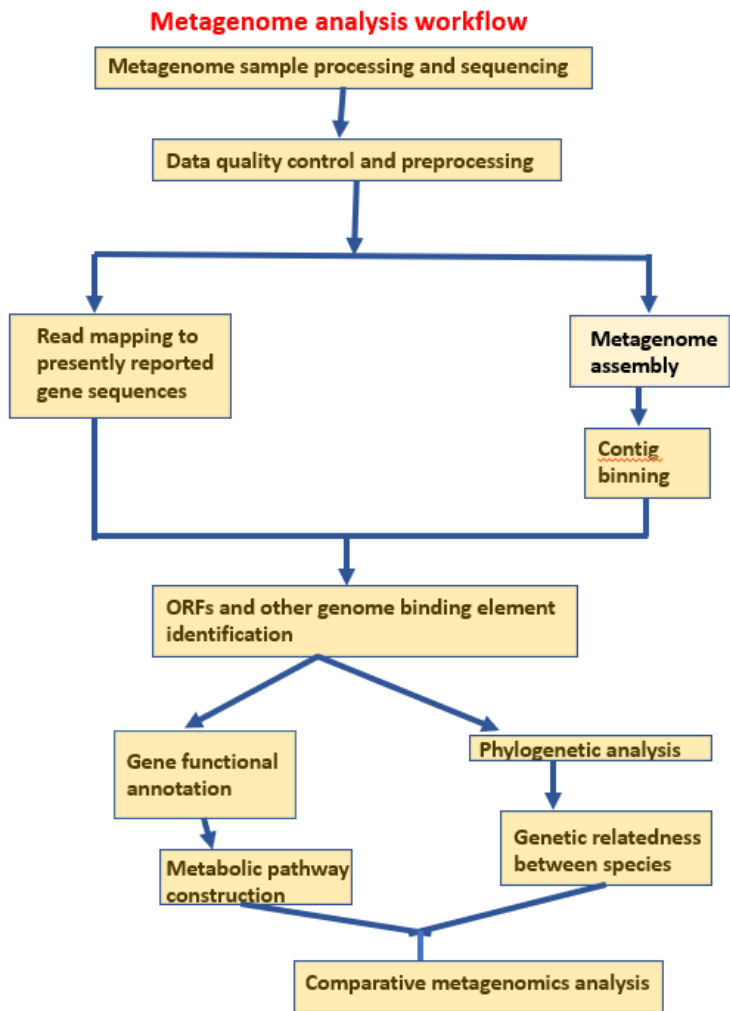


Fig 2: Metagenomic analysis procedure flow diagram

The metagenomic data analysis helps to create taxonomic relationships of microbial communities or prediction of complex genomic elements like CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) in microbes. Metagenomic samples may be collected from chemically or ecologically characterized habitats and are purified and made free of contaminants before proceeding from library preparation. The rRNA genes are the most conserved in DNA in all organisms. Remarkable similarities have been observed in distantly related organisms. Signature sequences from distantly related organisms may be utilized for alignment and mapping leading to the determination of taxonomy, phylogeny, and estimation rates of species divergence. The 16s rRNA is usually used for the evolutionary relationship study and also for identification of the plant-soil microbial interactions. The biotic influence may occur in terms of activities of human and grazing animals causing an impact on rhizosphere microbes that increases solubilization and uptake of important macronutrient and improve soil fertility that positively impacts yield, growth and productivity ^[47]. On the other hand, abiotic influences like geo-climatic changes, soil quality, application of various elements heavy metals, and salts in response to which plants release root exudates that suppress pathogenic effect, biotic-abiotic stress resistance, and reduced penetration of toxic matter. The microbial environment at the zone of the rhizosphere induces chemotaxis, recognition, adherence, growth, colonization, and biocontrol for microbes. The plant microbiome induces the uptake of hormones, nutrients, enzymes, antibiotics, and the release of effectors for stress resistance ^[48].

The sequencing may be carried out in MiSeq platform that generates reads up to 300 base pairs or Illumina Next-seq 550 platform. Sequence processing may be performed by trimming of adapters and read removal using BBmap v37.66 and DIAMOND v0.9.118 ^[49]. Next, denovo assembly of Illumina reads are performed in SOAP denovo. Scaffolds need to be organized into read binning based on k-mer abundance in sequence assembly based on new technologies and algorithms ^[50]. Gene prediction of bacteria, fungi, and viruses is performed in updated versions of August and Metagenemark or any other available updated tools ^[51, 52]. Next pathway annotation may be performed in KEGG. This procedure has been utilized for the evaluation of the relationship in-between microbiota composition and quality in Chinese Rice Wine (Hong *et al.* 2016). The SOAPaligner may be used for quality control and mapping of reads. Operational taxonomic units (OTU) may be created with Python script of Qiime and Perl scripts. OTUs may be assigned and diversity analysis may be carried out on Qiime pipeline. Taxonomy and functional gene assignments may be prepared against NCBI SEED.DB by

using Megan 6.0 with weighted lowest common ancestor algorithm by use of top percent setting of 3.0. Next correlation and regression analysis of microbial effect on plant biomass was studied [53, 51]. Characterization study of leaf microbiome interaction through whole genome sequencing data of 3000 rice genome project resulted in key revelations about microbial assembly on rice leaf as a donor for modulation of rice consortium for crop improvement [54].

In another, metagenomic study revealed microbes responsible for generation of electricity in sediment-type microbial fuel cells in rice paddy fields [55]. Bioremediation-related study of rice rhizosphere revealed key insights into spatial variation and dynamics [56]. Metagenomics of nitrogenase reductase collected from four geographical locations across the United States at four locations Alaska, Hawaii, Utah, and Florida has been metagenomically studied using FrancBot that lead to new perspectives to meet the need for metagenomics studies [57]. The rice dispensable genome that is responsible for phenotypic variation and genome evolution was assembled using *denovo*-metagenome-like assembly and low coverage population sequencing data were analysed and candidate genes associated with phenotypic variation were investigated [58]. Identification of various methanotroph species was based on 16rRNA gene amplicon obtained Metagenomics signatures of Rhizobiome community in growth stage-specific at various temporal dimensions were carried out. These studies revealed the presence of several ammonia-oxidizing Archeae as good bacteria predominant and *Nitrosospora* and *Nitrospira* [59, 60].

Microbial diversity analysis involving 16s rRNA amplicon sequencing of rhizosphere was performed from double cropping rice and rice-shrimp farming predominating in Vietnam. The study revealed dominant presence of Actinobacteriota, Acidobacteriota, Chloroflexi, Firmicutes and Myxococcota family members [61]. Metagenomic analysis study of root endophytic microbial community using 16s rRNA amplicon sequencing revealed dominance of Proteobacteria, Cyanobacteria, Actinomycetes, Firmicutes, and species related to nitrogen fixation in abundance by QIIME2 statistical analysis [62]. Metagenomic analysis revealed novel cellulases that are that are efficient in saccharification of ionic-liquid pretreated rice straw in small scale [63]. In a fascinating experiment gnotobiotically grown aromatic rice reveals presence of endophytic bacteria *Sphingomonas* sp. that helps in the promotion of growth and aroma increase in rice. Operational taxonomic units were picked and taxonomy was assigned through Greengenes database in QIIME [64]. In another similar experiment, the deciphering of core microbiome present in rice leaf endospheric region through metagenomics and microbial analysis of

aromatic, non-aromatic rice genotype collected from various geographical locations viz. Pusa Basmati-1 and BPT collected from Palampur, Almora and Hazaribagh belonging to three geographical zones were used in the study. The presence of bacterial community belonging to Proteobacteria, Bacterioidetes, Firmicutes, and Actinobacteria were found to be abundant. Sparse network analysis was employed to demonstrate intra-microbial interaction among microbial communities. The observations obtained could enable the development microbiome-assisted crop management courtesy microbiome reengineering in the coming times ^[65]. Thus, the metagenomics studies may unravel a whole new domain associated with rice and rice microbiome-associated biotic and biotic stress responses towards the development of climate-resilient rice genotypes in the future. The type of metagenomic studies may be extended to other crops as well.

Challenges while enriching virus and extraction of nucleic acid

The relative abundance of viral readings will vary depending on whether total nucleic acid extraction or viral particle enrichment procedures are used, as discussed ^[66]. It is crucial to understand the bias that the method of choice will bring before metagenomic sequencing. For instance, we anticipate variations in the sensitivity of dsRNA-based and Viron-associated nucleic acids (VANA) methods. Because VANA procedures enhance virus particles, they will not be able to recover viruses that are not encapsidated, have low titers, or have unstable particle shape. DNA viruses and single-stranded RNA (ssRNA) can both be found using dsRNA-based methods. For DNA viruses, overlapping transcripts from ORFs in opposite orientations can hybridise to form dsRNA, whereas for ssRNA viruses, dsRNA is a replication intermediate ^[67, 68]. Nevertheless, low levels of dsRNA can be produced by some ssRNA viruses ^[69], and successful detection of DNA viruses depends on how the viral transcripts are organised. As a result, many viruses without dsRNA genomes may be difficult to identify using this method. In order to conduct plant virome analysis in complex pools that are typical of the most common plant species in both managed and cultivated habitats, Ma ^[70] compared the two viral sequence enrichment techniques, dsRNA and VANA.

Furthermore, a unique bioinformatic method was applied, and OTUs were identified after the clustering of conserved viral domains in order to estimate virus richness ^[71]. In order to examine virome diversity, 200 individual plant samples were gathered in the spring of 2016 from six distinct locations in southwest France. High viral diversity, including the existence of novel dsRNA viruses, was found in all sites when analysis was conducted using either the VANA or dsRNA metagenomic techniques. In general, the same

viral families were recovered by both methods; however, the dsRNA strategy could not effectively recover DNA viruses. Conversely, a low detection efficiency of virus families with lower titers and/or fewer stable particles was noted using the VANA technique. Consequently, a dsRNA-based method was advised for researchers who were primarily interested in RNA viruses since it offered a more thorough description of the examined (phyto)viromes ^[72].

Also, the creation of superior nucleic acid extracts is an essential step in the study of plant viruses through metagenomics. Plant samples frequently contain polyphenols, polysaccharides, high concentrations of RNases, and fibrous tissues. Fibrous tissues, including lignin (wood), are difficult to break apart and remove, which makes it difficult for wild plant species to extract high-quality nucleic acids. Depending on the tissue type and the presence of secondary metabolites and inhibitors, numerous modifications have been reported to nucleic acid isolation methods, including adjustments to tissue storage, maceration, preparation, and extraction, in order to obtain sufficient, high-purity nucleic acids for non-model plant species ^[73].

Challenges in high throughput sequencing and bioinformatics

It is difficult to analyse virus metagenomics data in weeds and wild plants from a conceptual and practical standpoint. Illumina sequencing is now the HTS technique that is frequently utilised for viral metagenomics. The Illumina HiSeq and NextSeq produce short paired end reads, measuring 2×75 -150 bp, which are obtained from the brief fragments generated throughout the library preparation procedure. While read volume is much lower with Illumina MiSeq, yet read length can be doubled up to 2×300 bp paired end reads. Using a range of procedures and software tools, longer virus genome sequences can be built from these short sequencing reads. Viral genomes may, in certain circumstances, be incomplete after assembly (for example, having partial genomes or missing genome segments), or the assembly process may produce chimeric sequences (where reads from various genetic origins within a sample are similar and thus mistakenly assembled into one sequence). In plant virus metagenomics investigations, metagenomics is frequently conducted on pooled samples, which increases the likelihood of chimera assemblies which is a problem. In subsequent research, the issue of chimeras might be resolved by utilising HTS techniques that produce longer reads, including PacBio SMRT, Oxford Nanopore sequencing, or HiFi sequencing. Plant viruses have short genomes, ranging from 4 to 30 kb ^[74]. This means that whole genome portions can be captured in a single run using long read sequencing. Long read sequencing has been applied to numerous plant viruses like geminiviruses ^[75], tombusviridae ^[76] and potyviruses, such as Wheat

streak mosaic virus ^[77], Plum pox virus ^[78], and Yam mild mosaic virus ^[79]. All of the host plants used in these investigations, however, were crop plants, with the exception of the geminiviruses identified in the Fabaceae family shrub *Medicago arborea* ^[75]. Many DNA and RNA HTS library preparation techniques are employed in these investigations, and the choice of library preparation influences the sensitivity of viral identification. Long-read sequencing often yielded reliable viral detection.

While plant virus metagenomics can benefit from long read sequencing, there are several disadvantages. First being, since sequencing adapters are made to bind to poly-A tailed RNA, poly-A tailed RNA must be used as input in all library preparation procedures for RNA sequencing. Nonetheless, a large number of plant virus species lack a poly-A tail. This restriction can be overcome by including a poly-A tailing step before sequencing; however, it must be noted that this reaction will also impact any host RNA in the sample and is not specific to viral RNA ^[80]. Secondly, barcoding is not supported by many RNA-based library preparation techniques, which reduces the number of samples that can be sequenced at a high throughput, thereby reducing the quality. The limitations mentioned above do not apply to DNA viruses. The demand for a higher quantity and purity of the nucleic acid input compared to other sequencing technologies like Illumina is a major barrier to the large-scale adoption of long read Nanopore sequencing, and this is more challenging to achieve for wild plant species. It is unclear if Nanopore sequencing is easily adaptable to bigger metagenomic surveys of plant viruses in weeds and wild plants, as the majority of long-read plant viral metagenomics have been carried out in crops.

Due to their diverse evolutionary histories, viruses lack conserved genomic sections, whereas other microbial groups, like bacteria or fungi, can be regularly identified by a conserved gene (16S or ITS, respectively). Viral genome segments are identified by sequence similarity to other viruses, as viruses lack a universal marker. This makes it difficult to distinguish viral metagenomes from weeds in a number of ways. Firstly, there is now a bias towards plant viruses that are found in crops, and virus genome databases are not fully complete. This implies that viruses that are only found in wild plants can go unnoticed because their sequences differ from those of known viruses. "Dark matter" refers to the sizable fraction of viral metagenomics data that lacks detectable sequence similarity with any known biological entity ^[72, 81]. Numerous researchers have observed that even when specifically enriching for viral nucleic acids linked to viruses using VANA or dsRNA purification, many viral sequences are detected that are unclear of their viral nature, difficult to assign to species because of their novelty or short length ^[71, 81].

This widespread ambiguity highlights the need for developing novel and inventive algorithms to identify viruses that not only rely on sequence similarity but also make use of conserved, informative genome regions or domains for viral sequence clustering or the functional protein motifs that viruses employ to successfully infect host cells ^[82, 83]. However, there are numerous obstacles to the taxonomic assignment of the viral contigs found from HTS data, which have been well examined ^[82]. The most popular method, BLAST-based annotation, is often unreliable at the species and genus level and still has flaws at the family level, especially considering the sheer number of unique agents found in metagenomics research ^[81, 82]. These issues prompt consideration of whether and how to include metagenomic sequence data in the ICTV taxonomy. In an attempt to rectify the situation, Simmonds *et al.* ^[82] suggested ways to classify viruses, such as generating new virus species, classifying new species and genera into already-existing families, and streamlining the virus classification process. These issues are not specific to viruses, and efforts made with other microbes may serve as sources of inspiration. Numerous bacterial and archaeal species identified through metagenomics, for instance, are not cultivable. A taxonomy classification convention has been established for these candidate species, and recommendations have been made for improving the identification of the significance of these genome sequences ^[84, 85]. Finally, non-specialized teams face a practical barrier when analysing virus metagenomics data. Large-scale computing hardware for storing and processing datasets, together with knowledge of genome assembly analytical processes and software, are needed when working with HTS-data. However, metagenomics will become more accessible to a wider range of researchers due to the decline in its fundamental operating expenses and the continued development of straightforward, dependable, and user-friendly bioinformatics tools for data analysis and sequence mapping. While high-throughput sequencing (HTS) is a valuable tool for virus detection, confirmation testing can be necessary if the sequencing depth and coverage of the virus are poor.

Conclusion

Owing to understanding epigenetic modification and prediction is crucial for treating abnormalities, we discussed about how CHiP-seq and BS-seq can be applied in different situations where methylation patterns and histone modifications are obtained in the chromatin. This aids in the identification of epigenotypes and the alteration of epigenetic patterns in chloroplasts and mitochondrion. Due to the significance of polyploid plants and the established role that epigenetic inheritance plays in polyploidy, software and prediction

algorithms for the identification of epigenetic inheritance in polyploid plants must be created. Gene expression patterns and the prediction of evolutionary relationships are two examples of the newly updated capabilities that the databases need. The use of metagenomics techniques has great promise in the identification of bacteria, archaea, fungi, and viruses, as well as in the study of their emergence and epidemiology. Metagenomic research on plant viruses has revealed that viruses are frequently found in weeds and wild plants. These results demonstrate the remarkable amount of viral biodiversity that wild plants harbour, as well as the effectiveness of HTS techniques for virus discovery. However, there are still a number of drawbacks to HTS approaches and obstacles to overcome in this area of research. Decisions made about plant virome enrichment and sequencing protocols will impact the types of viruses and viral genetic variation that can be identified. This can be accomplished by expanding the use of plant virus metagenomics research, which can identify new or unknown viruses and detect viruses either by themselves or in combination with other infections.

References

1. Zhou X, Bai X, Xing Y. A Rice Genetic Improvement Boom by Next-generation Sequencing. *Current Issues in Molecular Biology*. 2018;27(1):109-126.
2. Béné C, Barange M, Subasinghe R, Pinstrup-Andersen P, Merino G, Hemre GI, *et al.* Feeding 9 billion by 2050-Putting fish back on the menu. *Food Security*. 2015;7:261-274.
3. Zhang H, Zhang Y, Xu W, Li R, Zhang D, Yang L. Development and performance evaluation of whole-genome sequencing with paired-end and mate-pair strategies in molecular characterization of GM crops: One GM rice 114-7-2 line as an example. *Food Chemistry: Molecular Sciences*. 2022;4:100-061.
4. Patel N, Ferns BR, Nastouli E, Kozlakidis Z, Kellam P, Morris S. Cost analysis of standard Sanger sequencing versus next generation sequencing in the ICONIC study. *The Lancet*. 2016;388:S86.
5. Head SR, Komori HK, LaMere SA, Whisenant T, Van Nieuwerburgh F, Salomon DR, *et al.* Library construction for next-generation sequencing: overviews and challenges. *Biotechniques*. 2014;56(2):61-77.
6. Wheeler DA, Srinivasan M, Egholm M, Shen Y, Chen L, McGuire A, *et al.* The complete genome of an individual by massively parallel DNA sequencing. *Nature*. 2008;452(7189):872-876.

7. MacConaill LE, Burns RT, Nag A, Coleman HA, Slevin MK, Giorda K, *et al.* Unique, dual-indexed sequencing adapters with UMIs effectively eliminate index cross-talk and significantly improve sensitivity of massively parallel sequencing. *BMC genomics*. 2018;19:1-10.
8. Piao Y, Xu W, Park KH, Ryu KH, Xiang R. Comprehensive evaluation of differential methylation analysis methods for bisulfite sequencing data. *International Journal of Environmental Research and Public Health*. 2021;18(15):79-75.
9. Yongseok Park, Hao Wu. Differential methylation analysis for BS-seq data under general experimental design. *Bioinformatics*. 2016;32(10):1446-1453.
10. Ferreira LJ, Ravasco S, Figueiredo DD, Peterhänsel C, Saibo NJ, Santos AP, *et al.* Deciphering Histone Modifications in Rice by Chromatin Immunoprecipitation (ChIP): Applications to Study the Impact of Stress Imposition. In *advances in international rice research 2017*.
11. Dong Y, Chen W, Kang K, Pang R, Dong Y, Liu K, *et al.* FoxO directly regulates the expression of TOR/S6K and vitellogenin to modulate the fecundity of the brown planthopper. *Science China Life Sciences*. 2021;64(1):133-143.
12. Nakato R, Sakata T. Methods for ChIP-seq analysis: a practical workflow and advanced applications. *Methods*. 2021;187: 44-53.
13. Tu X, Mejía-Guerra MK, Valdes Franco JA, Tzeng D, Chu PY, Shen W, *et al.* Reconstructing the maize leaf regulatory network using ChIP-seq data of 104 transcription factors. *Nature communications*. 2020;11(1):50-89.
14. Kim S, Mitchell TK. The application of ChIP-chip analysis in the rice blast pathogen. *Fungal Genomics: Methods and Protocols*; 2011. p. 121-31.
15. Chung PJ, Jung H, Do Choi Y, Kim JK. Genome-wide analyses of direct target genes of four rice NAC-domain transcription factors involved in drought tolerance. *BMC genomics*. 2018;19(1):1-7.
16. Gehrke AR, Srivastava M. Assessing chromatin accessibility during WBR in acoels. In *Whole-Body Regeneration: Methods and Protocols*; 2022 Jan. p. 549-561.
17. Landt SG, Marinov GK, Kundaje A, Kheradpour P, Pauli F, Batzoglou S, Bernstein BE, Bickel P, Brown JB, Cayting P, Chen Y. ChIP-seq

- guidelines and practices of the ENCODE and modENCODE consortia. *Genome research*. 2012;22(9):1813-1831.
18. Ackermann AM, Wang Z, Schug J, Naji A, Kaestner KH. Integration of ATAC-seq and RNA-seq identifies human alpha cell and beta cell signature genes. *Molecular metabolism*. 2016;5(3):233-244.
 19. Wang Y, Wang K, Lu YY, Sun F. Improving contig binning of metagenomic data using d 2 S d _2^ S oligonucleotide frequency dissimilarity. *BMC bioinformatics*. 2017 Dec;18:1-4.
 20. Wang Y, Wang K, Lu YY, Sun F. Improving contig binning of metagenomic data using d 2 S d _2^ S oligonucleotide frequency dissimilarity. *BMC bioinformatics*. 2017;18:1-4.
 21. Zhang J, Chen LL, Sun S, Kudrna D, Copetti D, Li W, *et al*. Building two indica rice reference genomes with PacBio long-read and Illumina paired-end sequencing data. *Scientific data*. 2016;3(1):1-8.
 22. Zhou Q, Lim JQ, Sung WK, Li G. An integrated package for bisulfite DNA methylation data analysis with Indel-sensitive mapping. *BMC bioinformatics*. 2019;20(1):1-1.
 23. Zhao L, Xie L, Zhang Q, Ouyang W, Deng L, Guan P, *et al*. Integrative analysis of reference epigenomes in 20 rice varieties. *Nature communications*. 2020;11(1):1-6.
 24. Ghosh S, Chan CK. Analysis of RNA-Seq data using TopHat and Cufflinks. *Plant Bioinformatics: Methods and Protocols*; 2016. p. 339-361.
 25. Kim D, Pertea G, Trapnell C, Pimentel H, Kelley R, Salzberg SL. TopHat2: accurate alignment of transcriptomes in the presence of insertions, deletions and gene fusions. *Genome biology*. 2013;14:1-3.
 26. Shen S, Park JW, Lu ZX, Lin L, Henry MD, Wu YN, *et al*. rMATS: robust and flexible detection of differential alternative splicing from replicate RNA-Seq data. *Proceedings of the National Academy of Sciences*. 2014;111(51): E5593-E5601.
 27. He F, Liu Q, Zheng L, Cui Y, Shen Z, Zheng L. RNA-Seq analysis of rice roots reveals the involvement of post-transcriptional regulation in response to cadmium stress. *Frontiers in plant science*. 2015;16:11-36.
 28. Rajkumar MS, Shankar R, Garg R, Jain M. Bisulphite sequencing reveals dynamic DNA methylation under desiccation and salinity stresses in rice cultivars. *Genomics*. 2020;112(5):3537-3548.

29. Kou S, Gu Q, Duan L, Liu G, Yuan P, Li H, *et al.* Genome-wide bisulphite sequencing uncovered the contribution of DNA methylation to rice short-term drought memory formation. *Journal of Plant Growth Regulation*. 2022;41(7):2903-2917.
30. Li L, He Y, Zhang X, Zhang H, Sun Z, Li J, Hong G. Alterations of rice (*Oryza sativa* L.) DNA methylation patterns associated with gene expression in response to Rice black streaked dwarf virus. *International journal of molecular sciences*. 2020;21(16):5753.
31. Zhang J, Liu Y, Xia EH, Yao QY, Liu XD, Gao LZ. Autotetraploid rice methylome analysis reveals methylation variation of transposable elements and their effects on gene expression. *Proceedings of the National Academy of Sciences*. 2015;112(50):E7022-E7029.
32. Li R, Hu F, Li B, Zhang Y, Chen M, Fan T, Wang T. Whole genome bisulfite sequencing methylome analysis of mulberry (*Morus alba*) reveals epigenome modifications in response to drought stress. *Scientific reports*. 2020;10(1):1-7.
33. Chen H, Smith AD, Chen T. WALT: fast and accurate read mapping for bisulfite sequencing. *Bioinformatics*. 2016;32(22):3507-3509.
34. Song Q, Decato B, Kessler M, Fang F, Qu J, Garvin T, Zhou M, Smith A. The Smithlab DNA Methylation Data Analysis Pipeline (MethPipe). 2021
35. Song Q, Decato B, Hong EE, Zhou M, Fang F, Qu J, *et al.* A reference methylome database and analysis pipeline to facilitate integrative and comparative epigenomics. *PloS one*. 2013;8(12):e81148.
36. Kawakatsu T. Whole-genome bisulfite sequencing and epigenetic variation in cereal methylomes. *Cereal Genomics: Methods and Protocols*. 2020: 119-128.
37. Feng SJ, Liu XS, Tao H, Tan SK, Chu SS, Oono Y, *et al.* Variation of DNA methylation patterns associated with gene expression in rice (*Oryza sativa*) exposed to cadmium. *Plant, Cell & Environment*. 2016;39(12):2629-2649.
38. Xie L, Liu M, Zhao L, Cao K, Wang P, Xu W, *et al.* RiceENCODE: A comprehensive epigenomic database as a rice Encyclopedia of DNA Elements. *Molecular Plant*. 2021; 14(10):1604-1606.
39. Cingolani P, Platts A, Wang LL, Coon M, Nguyen T, Wang L, *et al.* A program for annotating and predicting the effects of single nucleotide

- polymorphisms, SnpEff: SNPs in the genome of *Drosophila melanogaster* strain w1118;iso-2;iso-3. fly. 2012;6(2):80-92.
40. Lloyd SM, Bao X. Pinpointing the Genomic Localizations of Chromatin-Associated Proteins: The Yesterday, Today, and Tomorrow of ChIP-seq. *Current protocols in cell biology*. 2019;84(1):e89.
 41. Huang Y, Kuang Z, Wang W, Cao L. Exploring potential bacterial and fungal biocontrol agents transmitted from seeds to sprouts of wheat. *Biological control*. 2016;98:27-33.
 42. De Mandal S, Sonali, Singh S, Hussain K, Hussain T. Plant–microbe association for mutual benefits for plant growth and soil health. *Current trends in microbial biotechnology for sustainable agriculture*. 2021:95-121.
 43. Brader G, Compant S, Vescio K, Mitter B, Trognitz F, Ma LJ, Sessitsch A. Ecology and genomic insights into plant-pathogenic and plant-nonpathogenic endophytes. *Annual Review of Phytopathology*. 2017;55:61-83.
 44. Brader G, Corretto E, Sessitsch A. Metagenomics of plant microbiomes. *Functional metagenomics: tools and applications*. 2017:179-200.
 45. Wu C, Zhang X, Fan Y, Ye J, Dong L, Wang Y, Ren Y, Yong H, Liu R, Wang A. Vertical transfer and functional characterization of cotton seed core microbiome. *Frontiers in Microbiology*. 2024;14:1323342.
 46. Abdelfattah A, Wisniewski M, Schena L, Tack AJ. Experimental evidence of microbial inheritance in plants and transmission routes from seed to phyllosphere and root. *Environmental Microbiology*. 2021;23(4):2199-2214.
 47. Jang SW, Yoou MH, Hong WJ, Kim YJ, Lee EJ, Jung KH. Re-Analysis of 16S Amplicon Sequencing Data Reveals Soil Microbial Population Shifts in Rice Fields under Drought Condition. *Rice*. 2020;13(1):1-7.
 48. Vishwakarma K, Kumar N, Shandilya C, Mohapatra S, Bhayana S, Varma A. Revisiting plant–microbe interactions and microbial consortia application for enhancing sustainable agriculture: A review. *Frontiers in Microbiology*. 2020;11:3195.
 49. Buchfink, B.;Xie, C.;Huson, D.H. Fast and sensitive protein alignment using DIAMOND. *Nat. Methods* 2015, 12, 59–60.
 50. Ghurye J, Pop M. Modern technologies and algorithms for scaffolding assembled genomes. *PLoS computational biology*. 2019;15(6):e1006994.

51. Wang, X. (2016). Next-Generation Sequencing Data Analysis (1st ed.). CRC Press. 2016;<https://doi.org/10.1201/b19532>
52. Hyatt D, LoCascio PF, Hauser LJ, Uberbacher EC. Gene and translation initiation site prediction in metagenomic sequences. *Bioinformatics*. 2012;28(17):2223-2230.
53. Fernández-Baca CP, Rivers AR, Maul JE, Kim W, Poudel R, McClung AM, *et al*. Rice Plant-Soil Microbiome Interactions Driven by Root and Shoot Biomass. *Diversity*. 2021;13(3):125.
54. Roman-Reyna V, Pinili D, Borja FN, Quibod IL, Groen SC, Alexandrov N, *et al*. Characterization of the leaf microbiome from whole-genome sequencing data of the 3000 rice genomes project. *Rice*. 2020;13(1):1-8.
55. Kouzuma A, Kaku N, Watanabe K. Microbial electricity generation in rice paddy fields: recent advances and perspectives in rhizosphere microbial fuel cells. *Applied Microbiology and Biotechnology*. 2014;98(23):9521-9526. doi: 10.1007/s00253-014-6138-0.
56. Ma B, Lyu XF, Zha T, Gong J, He Y, Xu JM. Reconstructed metagenomes reveal changes of microbial functional profiling during PAH s degradation along a rice (*Oryza sativa*) rhizosphere gradient. *Journal of applied microbiology*. 2015;118(4):890-900.
57. Wang Q, Quensen JF, Fish JA, Kwon Lee T, Sun Y, Tiedje JM, *et al*. Ecological patterns of nifH genes in four terrestrial climatic zones explored with targeted metagenomics using FrameBot, a new informatics tool. *MBio*. 2013;4(5):e00592-13.
58. Yao W, Li G, Zhao H, Wang G, Lian X, Xie W. Exploring the rice dispensable genome using a metagenome-like assembly strategy. *Genome biology*. 2015;16(1):1-20.
59. Zhao Z, Chu C, Zhou D, Wang Q, Wu S, Zheng X, Song K, Lv W. Soil bacterial community composition in rice-fish integrated farming systems with different planting years. *Scientific reports*. 2021;11(1):1-10.
60. Imchen M, Kumavath R, Vaz A, Góes-Neto A, Barh D, Ghosh P, *et al*. 16S rRNA gene amplicon based metagenomic signatures of rhizobiome community in rice field during various growth stages. *Frontiers in microbiology*. 2019;10:2103.
61. Do TX, Huynh VP, Le LA, Nguyen TV, Nguyen-Pham AT, Bui-Thi MD, *et al*. Microbial diversity analysis using 16S rRNA Gene amplicon sequencing of rhizosphere soils from double-cropping rice and rice-

- shrimp farming systems in Soc Trang, Vietnam. *Microbiology Resource Announcements*. 2021;10(44):e00595-21.
62. Sengupta S, Ganguli S, Singh PK. Metagenome analysis of the root endophytic microbial community of Indian rice (*O. sativa* L.). *Genomics data*. 2017;12:41-43.
63. Maleki M, Ariaeenejad S, Salekdeh GH. Efficient saccharification of ionic liquid-pretreated rice straw in a one-pot system using novel metagenomics derived cellulases. *Bioresource Technology*. 2022;345:126-536.
64. Krishnamoorthy A, Gupta A, Sar P, Maiti MK. Metagenomics of two gnotobiotically grown aromatic rice cultivars reveals genotype-dependent and tissue-specific colonization of endophytic bacterial communities attributing multiple plant growth promoting traits. *World Journal of Microbiology and Biotechnology*. 2021;37(4):1-6.
65. Kumar M, Kumar A, Sahu KP, Patel A, Reddy B, Sheoran N, *et al*. Deciphering core-microbiome of rice leaf endosphere: revelation by metagenomic and microbiological analysis of aromatic and non-aromatic genotypes grown in three geographical zones. *Microbiological Research*. 2021;246:126-704.
66. Maclot F, Candresse T, Filloux D, Malmstrom CM, Roumagnac P, Van Der Vlugt R, *et al*. Illuminating an ecological blackbox: using high throughput sequencing to characterize the plant virome across scales. *Frontiers in Microbiology*. 2020;11:578-064.
67. Colby C, Jurale C, Kates JR. Mechanism of synthesis of vaccinia virus double-stranded ribonucleic acid in vivo and in vitro. *Journal of virology*. 1971;7:71-76.
68. Son KN, Liang Z, Lipton HL. Double-stranded RNA is detected by immunofluorescence analysis in RNA and DNA virus infections, including those by negative-stranded RNA viruses. *Journal of virology*. 2015;89(18):9383-9392.
69. Weber F, Wagner V, Rasmussen SB, Hartmann R, Paludan SR. Double-stranded RNA is produced by positive-strand RNA viruses and DNA viruses but not in detectable amounts by negative-strand RNA viruses. *Journal of virology*. 2006;80(10):5059-5064.
70. Ma Y. Wild Plant Species Associated Viromes: Towards Improved Characterization Strategies and Variability in Various Ecological Environments. In *Human Health and Pathology*; English NNT:

71. Bernardo P, Charles-Dominique T, Barakat M, Ortet P, Fernandez E, Filloux D, Hartnady P, *et al.* Geometagenomics illuminates the impact of agriculture on the distribution and prevalence of plant viruses at the ecosystem scale. *The ISME journal*. 2018;12(1):173-184.
72. Ma Y, Marais A, Lefebvre M, Theil S, Svanella-Dumas L, Faure C, *et al.* Phytoviroome analysis of wild plant populations: comparison of double-stranded RNA and virion-associated nucleic acid metagenomic approaches. *Journal of Virology*. 2019;94(1):10-128.
73. Yockteng R, Almeida AM, Yee S, Andre T, Hill C, Specht CD. A method for extracting high-quality RNA from diverse plants for next-generation sequencing and gene expression analyses. *Applications in plant sciences*. 2013;1(12):1300-070.
74. Holmes EC. *The Evolution and Emergence of RNA Viruses*; Oxford University Press: Oxford, UK; 2009. ISBN 9780199211128.
75. Ben Chehida S, Filloux D, Fernandez E, Moubset O, Hoareau M, Julian C, *et al.* Nanopore sequencing is a credible alternative to recover complete genomes of geminiviruses. *Microorganisms*. 2021;9(5):903.
76. Adams IP, Braidwood LA, Stomeo F, Phiri N, Uwumukiza B, Feyissa B, *et al.* Characterizing maize viruses associated with maize lethal necrosis symptoms in Sub Saharan Africa. *BioRxiv*; 2017. p. 161-489.
77. Fellers JP, Webb C, Fellers MC, Shoup Rupp J, De Wolf E. Wheat virus identification within infected tissue using nano-pore sequencing technology. *Plant disease*. 2019;103(9):2199-2203.
78. Phannareth T, Nunziata SO, Stulberg MJ, Galvez ME, Rivera Y. Comparison of nano-pore sequencing protocols and real-time analysis for phytopathogen diagnostics. *Plant Health Progress*. 2020;22(1):31-36.
79. Filloux D, Fernandez E, Loire E, Claude L, Galzi S, Candresse T, *et al.* Nanopore-based detection and characterization of yam viruses. *Scientific reports*. 2018;8(1):178-79.
80. Wongsurawat T, Jenjaroenpun P, Taylor MK, Lee J, Tolardo AL, Parvathareddy J, *et al.* Rapid sequencing of multiple RNA viruses in their native form. *Frontiers in microbiology*. 2019;10:260.
81. Roossinck MJ, Martin DP, Roumagnac P. *Plant Virus Metagenomics: Advances in Virus Discovery*. *Phytopathology*. 2015;105:716-727.

82. Simmonds P, Adams MJ, Benkő M, Breitbart M, Brister JR, Carstens EB, *et al.* Virus taxonomy in the age of metagenomics. *Nature Reviews Microbiology*. 2017;15(3):161-168.
83. Sobhy HA. A review of functional motifs utilized by viruses. *Proteomes*. 2016;4(1):3.
84. Konstantinidis KT, Rosselló-Móra R, Amann R. Uncultivated microbes in need of their own taxonomy. *The ISME journal*. 2017;11(11):2399-2406.
85. Chuvochina M, Rinke C, Parks DH, Rappé MS, Tyson GW, Yilmaz P, *et al.* The importance of designating type material for uncultured taxa. *Systematic and applied microbiology*. 2019;42(1):15-21.

Chapter - 3

Biodiversity: Concept, Types, Importance, Threats and Conservation

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Chapter - 3

Biodiversity: Concept, Types, Importance, Threats and Conservation

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Abstract

Biodiversity forms the basis of life. Agricultural supplies, purification of water, flood and drought control, nutrient flow, and climate regulation are the critical ecosystem services which are essential for the continuation of life. These services are essential to promote both economic development and human well-being. It provides the entire population oxygen to breathe, the food for living and the water for drinking for all living things. Wetlands clean the polluted water, plants and trees slow down global warming by collecting carbon and bacteria and fungi breakdown decaying organisms to provide fertiliser for the soil. In spite of the beneficial effects of biodiversity, threats to species and ecosystems are increasing extremely quickly and are primarily the result of human mismanagement of natural resources, which is frequently increased by careless economic policies, pollution and faulty practices in addition to climate change. This brief study describes the extent of biodiversity while emphasising its enormous benefits to humanity and its inherent and anthropogenic values, which support the necessity for their preservation and sustainable usage.

Keywords: Biodiversity, threats to biodiversity, *in situ*, *ex situ* conservation, gene bank

Introduction

Edward Wilson, a sociobiologist, coined the term "biodiversity" in 1988. He became to be known as the "Father of Biodiversity." According to Edward Wilson "Biodiversity is the sum of all types of biological organization's diversity." It can also be defined as the variety and variability of life on Earth. The term "biodiversity" refers to a measure of genetic, species-level and ecosystem-level variation. It is the sum of total of life forms at all levels of organization in biological system.

Biodiversity is a broad phrase that refers to the degree of variation in the natural world, both in terms of frequency and number. It is typically explained by the diversity of plants, animals and microorganisms, the genes they carry and the eco-systems they produce. The biodiversity that we see today is the result of billions of years of evolution, affected to a greater and larger amount by both natural processes and human activities. It builds the web of life, of which we are completely dependent for essential component. There are currently about 2.1 million species recognised, mostly small animals like insects. Scientists estimate that there are roughly 13 million species, contrary to UNEP's estimate of just 9.0 million species. (Mora *et al.*, 2011).

Biodiversity is not spread uniformly throughout the Earth. In tropics it is the richest. The equator often has the highest levels of terrestrial biodiversity, according to Gaston (2000), which is assumed to be a result of the area's warm climate and high primary productivity (Field *et al.*, 2009). Marine biodiversity appears to be most plentiful along the mid-latitudinal band of all oceans, particularly along the Western Pacific coasts where sea surface temperature is highest. Biodiversity has been growing throughout time, but it is anticipated to slow down in the coming years since it tends to congregate in hotspots.

Types of biodiversity/levels of biodiversity

Biodiversity or Biological diversity comprises of Genetic diversity, Species diversity and Ecosystem diversity.

Genetic diversity

It refers to the variation of genes within the species which stores as immense amount of genetic information. It is defined as the sum of the genetic information, contained in the genes of

individuals of plants, animals and microorganisms that inhabit the earth. There is genetic diversity among members of a species. For instance, cattle come in a wide range of colours, milk yields, sizes, disease resistance, the chromosomal architecture, complete genes or individual alleles may all exhibit genetic diversity. It results in enhanced species adaption to the altered environment. Genetic diversity leads to the emergence of new species. Any species needs it to preserve reproductive health, disease resistance and the capacity to adapt to shifting environmental conditions. It permits a population to respond to natural selection and adapt to its surroundings. Variability in the environment frequently leads to an increase in genetic diversity within a species. New breeds of domesticated animals, plants and crops have made genetic variety.

Species diversity

According to Mayr (1963), species are collections of naturally reproducing populations that are either interbreeding or have the ability to do so. Each species has a unique function in the environment that make them different units of diversity. In nature, there is a wide variation in the number and types of species as well as the number of individuals within each species, which promotes more diversity. The diverse species found within a region are referred to as species diversity. Variability can be found between species in a population or within a single species. Both species richness (the number of species per unit area) and evenness or equitability (the consistency in the number of individuals within a species) are used to measure species diversity. The total number of species is the standard unit of measurement for measuring species diversity.

Ecosystem diversity

Ecological equilibrium should not be altered because the current diversity has evolved over millions of years of evolution. Ecosystem diversity is the variety of habitats, biological communities, ecological processes and internal variations that exist within a single ecosystem. The various biological communities that the species inhabit are meant by this. For quantifying biodiversity at different spatial scales, Whittaker (1972) defined three terms: Alpha, Beta and Gamma Diversity.

Alpha diversity

Alpha diversity, also known as species richness. It is the diversity that exists within a specific area or ecosystem and is typically quantified by the number of species present there. It speaks of the typical species variety in a habitat or location. A local metric is alpha diversity. Alpha diversity, to put it in order words, is a community's diversity.

Beta diversity

The ratio of local or alpha diversity to regional diversity is known as beta diversity. This is the species diversity between two habitats or geographic areas. Beta diversity refers to the diversity inside communities.

Gamma diversity

It is a blend of alpha and beta diversity and represents the overall variety of a terrain. Gamma diversity (or) Landscape diversity is the diversity of habitats across the entire landscape or geological area. It is a way to gauge how diverse a vast area is overall.

Loss of biodiversity/threats to biodiversity

The loss of biodiversity is caused by a variety of circumstances, including pollution, habitat loss, hunting, the introduction of exotic species, overexploitation of desired species, climate change, etc. (Figure 1).

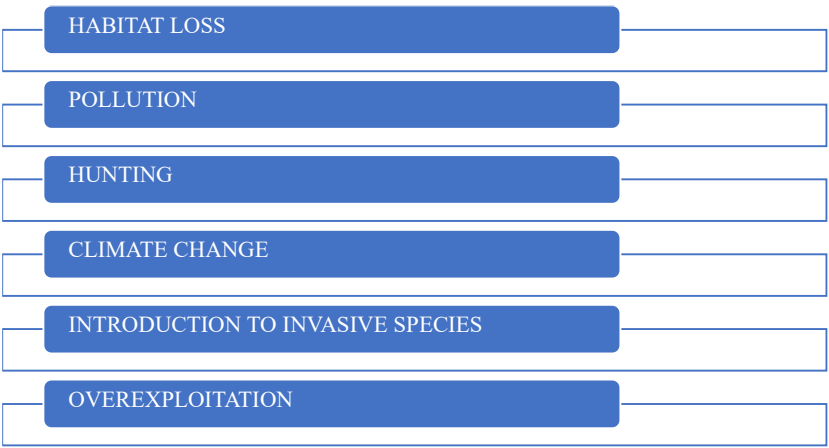


Fig 1: Principal causes for the loss of variety of life on earth

Loss of habitat

One of the biggest dangers to biodiversity is this. Loss of habitat is directly related to the stress humans are putting on the planet. A key contributing factor in the extinction of many species is habitat loss, degradation and fragmentation. Along with total loss, pollution-induced habitat deterioration has put many species survival at risk. Large habitats are fragmented into small pieces for various human activities, which negatively affects mammals, birds, and some species with migratory tendencies and causes population decreases. Every species on earth has unique dietary and ecological requirements. The greater the sensitivity of a species to habitat loss due to agricultural land, cattle and towns, the more specialised are the demands and localised the habitat. The deterioration, damage and fragmentation of habitat are the three predominant categories of habitat loss (Klappenbach L *et al.*, 2020).

Examples of habitat degradation includes harvesting fossil fuels, filling marshes and mowing fields so to create residential or commercial areas and cutting down trees to prepare fields for agricultural use (Martins IS *et al.*, 2018). These factors are responsible for the extinction of thousands of species on planet which ultimately leads to a decrease of biodiversity or the extinction of species. The term "habitat destruction" refers to the systematic destruction

of a species' natural environment to the point where it is no longer able to support both the species and its native eco-systems. The destruction and fragmentation of habitats, which is the main contributor to the loss of biodiversity, results from the conversion of natural ecosystems into agricultural land, urban areas, and infrastructural development.

The World Health Organisation (WHO) estimates that traditional medicines, primarily those derived from plants, are used by about 80% of the population in underdeveloped nations. As a result of habitat loss and degradation brought on by overharvesting, many plants become endangered. For example, the Amazon rain forest (it is so huge that it is called the 'lungs of the planet') harbouring probably millions of species is being cut and cleared for cultivating soya beans or for conversion of grasslands for raising beef cattle. Habitat destruction renders entire habitats functionally unable to support the species present in the habitat (Ayoade *et al.*, 2009; Agarwal *et al.*, 2011).

Human destruction of habitats has accelerated greatly in the latter half of the twentieth century. Natural habitats are often destroyed through human activity for the purpose of harvesting natural resources for industry production and urbanization (Agarwal *et al.*, 2014). Development of agricultural practices, reduced resources such as food, water, air quality, mining, pollution, logging, catastrophic fishing activities, activities related to urbanization, and the interruption of processes related to ecosystem are the predominant elements of degradation of habitat. The habitat degradation affects both the species dwelling in the habitat and the humans both. The erosion, depletion of nutrients and desertification cause the further loss of the degraded land (Carrete M *et al.*, 2009).

Another significant problem brought on by human developmental activities is habitat fragmentation. Large natural regions are reduced to smaller pieces of land by humans for the sake of development and to satisfy their never-ending urge to grow, even at the expense of extinction of their own species. These fragmented landscapes reduce animal and plant species habitats, isolate animal populations, and reduce genetic diversity. (Bright PW, 1993).

Wild animals are hunted for their goods, such as hides and skins, which are used commercially. Indian hunters target rhinos for their horns, tigers for their bones and skin, musk deer for their scent (which has therapeutic value), elephants for their ivory, gharials and crocodiles for their skin, and jackals for the fur trade in Kashmir. The whale hunt is one of the most well-known

commercial snares. Nine Indian animal species have been put on the CITES (Convention on International commerce in Endangered Species of Wild Fauna and Flora) list as being highly threatened by international commerce.

Pollution

Living organisms have developed over an extended period, but due to polluted planet they strive for existence. Air, soil and water pollution can harm species by degrading their habitats, physically harming them, or increasing their vulnerability to diseases or predation. Some pollutants, such as pesticides and heavy metals, can be passed through the food chain, therefore contaminating many levels of the ecosystem. Thus, human activities influence the natural environment which produces negative effects that in turn alter the flow of energy, the chemical and physical constitution of the environment etc.

Air pollution

Animal's respiratory systems are affected by air pollution, which has a negative impact on their health as well as their ability to lay eggs and their behaviour (Sanderfoot OV *et al.*, 2017). Both directly and indirectly, air pollutants like soot, dust, ammonia, or carbon dioxide can affect biodiversity. According to Mohalleem SV *et al.* (2005), Animals ability to reproduce is also affected by air pollution. It is challenging to assess the indirect effects of air pollution on animals since they are challenging to analyse in a controlled environment. The massive emission of greenhouse gases including nitrous oxide, carbon dioxide, and methane is rapidly changing the earth's climate. Acid rain is other pollutant, and it causes enhanced build-up of the mucus in fish gills leading to suffocation. Soil prone to acid rain has diminished activity of microbes (Pennanen T *et al.*, 1998).

Water pollution

Water pollution has a negative impact on biodiversity. Chemical fertilisers are added to the soil to increase crop productivity; these fertilizers mainly contain nitrogen and phosphorus. Phosphorus and nitrogen leach out of the soil and into the ground water bodies. Thus these fertilizers cause eutrophication in the water bodies. Eutrophication brings reduction in oxygen levels which is harmful to biodiversity. Because of the insufficient dissolved oxygen in the water, fish and other aquatic species perish. Similar effect is caused by fertilisers, pesticides in water bodies. Given that fertilisers are not washed away and animals in water bodies have trouble reproducing, pesticides have a negative impact on non-flowing waterbodies such lakes and ponds (Bhateria R *et al.*, 2016). Heavy metals affect the behaviour as well as the survival rates of aquatic animals specially fish. Further, events such as oil

spills greatly impact the wildlife specially in the deeper oceans. Oil spills cause disruption of the animal senses, suffocation, impair the vital organs of the organisms, reduction in growth rates and induce the higher mortality of the larvae (Brussaard CP *et al.*, 2016). Like the oil spills, plastic stays in the environment for longer time-period and hence influence the wildlife. Microplastics in the environment also impact the survival rate of larvae, diminished food consumption and gradually weight loss in aquatic animals (Wright SL *et al.*, 2013).

Soil pollution

Soil pollution is another factor adversely affecting biodiversity. Heavy metal-contaminated soil adversely affects the microorganisms necessary for the survival of living things. Plants absorb the excess heavy metals that are present in the soil because they are difficult to break down (Singh V *et al.*, 2015). A major threat to biodiversity is the overuse of fertilisers, herbicides, and antibiotics in agriculture. The pH and nutrient content of the soil are altered by these agricultural contaminants, such as nitrogen from fertilisers (Klein AM *et al.*, 2007). It has a significant impact on biodiversity.

Use of Artificial lighting and other forms of light pollution also harm biodiversity. The streetlights, building lights, car headlights and other late-night lighting sources have a particular negative impact on nocturnal species like bats. In environments with artificial lighting, bats feeding activity has diminished and their exit from roosts has been seen to be delayed. The light pollution has a significant impact on the behaviour of the moths as well. Moths are both pollinators of numerous species and prey for other species. (Macgregor CJ *et al.*, 2017).

Noise pollution also has tremendous negative impact on biodiversity. As per the studies it has found that the birds in noisier areas begin laying less eggs as it masks the important territorial calls birds make (Wouter H *et al.*, 2011). The machinery noise in Brazil at a mining site was found to impact wildlife. Species numbers declined at sites nearer the mine and were higher farther away (Duarte MHL *et al.*, 2015).

Hunting

The main factor that led to the loss of so many species holding important positions in the food chain was hunting. As a result, the numerous species in the area are negatively impacted since, in contrast to the normal scenario, they experience food scarcity or full food inaccessibility. The main cause of biodiversity loss is hunting. Hunting activities place a heavy load on animals, causing a significant decline in species and disrupting and inefficient

ecosystems. Overhunting can result in the eradication of large mammals in healthy habitat in most of such circumstances, forcing a change in forest structure. (Symes *et al.*, 2018).

Because of this, not only the particular species become prone to extinction but also the other species dependent on that species. Overhunting of a species to meet high demand for meat or animal byproducts for sport. Industrialised hunting does not take into consideration the effects of species deterioration on the rest of the ecosystem and causes depletion in populations. For example, Industrialised fishing has led to the depletion of highly-demanded species like tuna, whales and salmon to meet the world's demand. Unsustainable methods of fishing, like bottom trawling, have also destroyed sea-floor habitats, which are important nursery areas for many species. This has affected the marine ecosystem by increasing the populations of predators at the expense of their prey.

Climate changes

Around the world, species and ecosystems are threatened by climate change. Climate change and biodiversity have a close relationship. The fast climate change disrupts the ecological systems since the climate has continually changed along with ecological groups and species evolving and going extinct. The changing environments brought on by global warming and the resultant changes in climate patterns make it difficult for species to carry out their normal duties or adapt to new circumstances.

The distribution of species is largely determined by climate. Due to climate change, many plants and animals are not be able to adjust. As a result, some of the species and ecosystems are likely to be eliminated by climate change. For example, agricultural production likely will show regional variation in gains and losses, depending upon crop and climate. Based on the degree of threats faced by the species, species are categories into different conservation category i.e. Extinct, Endangered, Vulnerable and Risk. Therefore, it is very important to protect and conserve the existing biodiversity for the socio-economic development and to maintain ecological balance. Every species plays an important role in the environment. Climate change, beside other components such as habitat loss, land degradation, hunting, overexploitation of certain species etc., is turning up as a great threat to biodiversity on earth (Abrahms B *et al.*, 2017).

Natural disasters

Natural disasters such as volcanoes, wildfires, floods, storms, draughts, diseases, tsunamis, etc. have also caused a significant loss of biodiversity (Rondeau D *et al.*, 2020).

Due to flooding large amounts of soil nutrients are carried away (Chapin FS *et al.*, 2020). Animals and flora are harmed as a result of these natural disaster. Similar to forest fires, earthquakes also severely interrupt life of creatures, which in turn has an impact on biodiversity. Flora and fauna are frequently destroyed in the surrounding areas by volcanoes. Occasionally, epidemics kill off the vast bulk of the population. Since the pathogen is typically species-specific, epidemics in nature are typically confined to a certain group of animals or plants. (Isbell F *et al.*, 2010).

This raises serious concerns, especially as the amount of atmospheric CO₂ rises and causes global warming. The majority of species develop within a fairly small physiological range, therefore nature maintains a range of tolerance for ecological stability. Changes might be gradual or abrupt, and if the top or lower limit is surpassed, a species will end up extinct. The biodiversity and environment have already been significantly impacted by recent climatic changes, such as the warmer temperatures in some areas. They have impacted the frequency of pest and disease outbreaks along with the distribution of species, population levels, and the timing of reproduction or migration events. Many species existing in specific geographical areas could become extinct because of projected climate changes by 2050. By the end of the century, climate change and its impacts may become the main reason for the biodiversity loss.

Introduction of non-native species

When non-native species also known as “alien species” or “exotic species” are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive and cause decline or extinction of indigenous species. These species introduced in an ecosystem will cause changes in the ecosystem. These species have been a major threat to biological diversity worldwide as the introduced species have often flourished at the cost of the local species. Species originating from a particular area when introduced into new natural environments, it can lead to imbalance in biodiversity. As these invasive species cause harm to native species by competing for food and shelter, spreading diseases unknown to them, causing genetic changes through interbreeding with native species, and disrupting various aspects of their food chains and physical environment. For example, water hyacinth was introduced by the British for beautification in India. However, it has evolved into an invasive species, clogging rivers, lakes and other bodies of water, preventing aquatic life from growing and surviving. Similarly there are many instances when introduction of exotic species has caused extensive damage to natural biotic community of the ecosystem. For example, the introduction of Nile

perch from north in Lake Victoria, Africa's largest lake, has driven almost half of the 400 original fish species of the lake to near extinction. The introduction of such invasive species is the tremendous threat to biodiversity crisis.

Invasive species is the species, which is not native to the ecosystem, arrives or is introduced mostly via humans in the new ecosystem. Such species are detrimental as they effect the ecosystem vastly as compared to any other species. Most of the new species introduced in the ecosystem do not become invasive, but few of them turn into invasive species and adversely affect the ecosystem (Linders TEW *et al.*, 2019). As ecosystems have evolved to maintain a relative stability of species populations, non-native species introduced to new environments can outcompete native species for resources, prey on them or transmit diseases.

Overexploitation

Many plants and animals are on the verge of extinction because of their overexploitation. Overexploitation is the collection of species from their natural habitat at higher estimates than the native communities can retrieve. Presently, nearly one third of the earth's vertebrates facing extinction are vulnerable due to overexploitation. Extreme fishing and hunting are the examples of overexploitation. Passenger pigeons which were once very commonly seen are the sufferers of overhunting. Similarly, various creatures both plants and animals are collected to be used as pets, trophies, or souvenirs. Such collections are illegal and usually called as poaching (Hammerchlag N *et al.*, 2017).

Over-exploitation of biological resources results when individuals of a particular species are taken at a higher rate than can be sustained by the natural reproductive capacity of the population being harvested. This can be through hunting, fishing, trade, food gathering etc.

Overexploitation of plants for their medicinal values have resulted in their loss from native habitats. Many plants such as *Drosera* sp., *Gnetum* sp., *pitcher plants*, *Psilotum* sp., *Napenthes khasiana* etc. are hunted and collected for academic tutoring and laboratory experiments. Similarly, various orchids have also been overexploited (Bodeker *et al.*, 2014).

Overexploitation remains a serious threat to many species, such as marine fish and invertebrates, trees and animals hunted for meat. The grazing pressure on most of the high-altitude grasslands of the Uttarakhand state both from migrant and local communities, is the extensive extraction of medicinal herbs in these areas resulting in their over exploitation (Rawat, 1998). Most industrial fisheries are either fully or overexploited, while destructive fishing

techniques harm estuaries and wetlands. The trade in wild plants and animals and their derivatives is poorly documented but is estimated at nearly \$160 billion annually. It ranges from live animals for the food and pet trade to ornamental plants and timber. Because the trade in wild animals and plants crosses national borders, the effort to regulate it requires international cooperation to safeguard certain species from overexploitation.

Biodiversity conservation

There are numerous ways to preserve Biodiversity. Conservation techniques can be grouped into two categories: *In situ* and *Ex situ*. *Ex situ* (off-site) conservation refers to the preservation of genetic resources away from the natural ecosystem in which they are found, whereas *in situ* (on-site) conservation refers to the preservation of genetic resources inside the ecosystem in which they are found.

***In situ* conservation**

In-situ conservation provides flow of gene which in turn preserves species in dynamic connections with their habitat (Edwards and Kelbessa, 1999). Populations, species and ecosystems are all dynamic and change across in time and location. *In situ* conservation fosters genetic diversity and enhances the species ability to adapt to shifting environmental conditions by allowing evolutionary and ecological processes to take place. Therefore, natural ecosystems are the greatest place to conserve biodiversity (WCMC, 1992; Heywood and Baste, 1995). The *in-situ* conservation technique enables the cost-effective and extensive conservation of a significant quantity of genetic variety, species diversity, and ecosystem diversity (Wolf, 1999). *In situ* conservation often has low costs and research requirements. *In situ* conservation methods include the different protected area systems and on-farm conservation in which cultivated plants and domesticated animals are conserved in the agroecosystems where they have been developed and utilized.

Protected area system: The creation and administration of protected areas is the most common method for preserving biodiversity. Geographically defined regions that are governed and maintained to accomplish certain conservation goals are known as protected areas (Miller *et al.*, 1995). In order to maintain a specific ecosystem or species, protected areas need to be large enough (IUCN, 1990; Jeffries, 1997; Wolf, 1999). Additionally, there is a need to have protected area systems that are representative of major landforms and ecosystems. Protected areas play a great role in the conservation of biodiversity. They have artistic, scientific, educational, social, economic, and

environmental values. The effectiveness of management and long-term protection are essential for the sustainability of *in situ* conservation in protected areas (IUCN, 1990, 1992; Miller *et al.*, 1995; Jeffries, 1997; Wolf, 1999).

It is essential to provide sufficient human and financial resources to ensure that protected areas are successful in conserving biodiversity. Additionally, the local population must be involved in decision-making and conservation efforts for protected areas to be developed and managed. The creation of suitable incentives from international, regional, and national authorities for resource users both inside and outside of the protected areas is necessary for the conservation of biodiversity in protected areas. To maintain the survival of protected areas, it is crucial to offer incentives in a variety of ways (such as tax exemptions and subsidies) to the surrounding population.

The advantages of biodiversity conservation and sustainable use inside protected areas must be distributed according to the right systems. For instance, local participation in conservation efforts and the distribution of benefits from conservation areas have been proven to be the most effective approaches in Cameroon and Zimbabwean (Jeffries, 1997). However, protected areas have mostly failed to achieve their conservation goals in many nations due to the exclusion of resident's involvement and a lack of appropriate systems to offer incentives (Jeffries, 1997; Zegeye, 2004). Government organisations and protected areas are managing protected areas for several nations with high tropical biodiversity, but they lack the knowledge and expertise essential to effectively prevent biodiversity loss (Rao and Ginsberg, 2010). Protected area managers frequently do not have easy access, knowledge and tools used in conservation research. There is an urgent need to transfer the advances in conservation science to individuals and institutions in biodiversity-rich countries.

On-farm conservation involves the preservation of crops and their wild relatives, animals, and the agroecosystems in which they thrive. Home gardens, crop fields, agroforestry systems, and grazing grounds are all examples of agroecosystems. Agricultural biodiversity benefits food production and agriculture. *In situ* conservation on farms is critical for preserving agrobiodiversity. Agriculture is critical to the survival and variety of domesticated plants and animals (McNeely *et al.*, 1995). Farmers across the world keep a huge variety of crop and livestock kinds. Farmers indigenous knowledge, skills and practises are critical to the conservation and management of agricultural biodiversity. They are superior possibilities for developing the scientific foundation for on-farm conservation of

agrobiodiversity (Deribe *et al.*, 2002). Farmers indigenous knowledge and practises in germplasm selection, storage and sharing are important components in the conservation of agricultural biodiversity through community gene banks. The spread of contemporary agricultural systems, in which a few enhanced varieties have largely replaced numerous farmer varieties, has resulted in the loss of agricultural biodiversity.

***Ex situ* conservation**

Threatened species can only be saved via *ex situ* conservation if *in situ* conservation is not possible for a variety of reasons (Jeffries, 1997; Wolf, 1999). Furthermore, *ex situ* conservation provides material for study and ecological restoration. *Ex situ* conservation disrupts evolutionary and ecological processes. It reduces species genetic variety and adaptation to the changing environmental conditions. Furthermore, *ex situ* conservation requires much higher costs and research requirements than *in situ* conservation. Botanic gardens, zoo and gene banks are examples of *ex situ* conservation strategies.

Botanic gardens are institutions that preserve recorded collections of living plants for the purpose of scientific research, conservation and education (Wyse Jackson, 1999). The world's plant diversity has been preserved in large part because of botanical gardens. Living collections or seedbanks include many of the most endangered plant species in the world, serving as an insurance policy for the survival of biodiversity worldwide (Waylen, 2006). Wild species that are endangered in their natural habitat are highly valued by botanical gardens (Heywood, 1999; Wyse Jackson, 1998).

In recent years, many botanical gardens around the world have been established and expanded to serve as complete botanical resource hubs (Wyse Jackson, 1998). They remain outstanding organisations for managing the gene banks of wild species to preserve rare species. Many botanical gardens also work to preserve plants that are essential to agriculture and food production as well as those that are used for a range of other commercial endeavours. Botanic gardens around the world contain more than 4 million living individual plant collections, as well as 250,000 seedbank accessions, representing around 80,000 species of living plant (almost 30% are vascular plant) (Hawksworth, 1995; Wyse Jackson, 1999). Numerous botanic gardens all over the world have developed effective seedbanks in order to preserve the genetic diversity of plants. To maintain the genetic richness of wild plants, numerous botanic gardens throughout the world have created efficient seedbanks. The Royal Botanic Gardens at Kew, UK's Millennium Seed Bank

(MSB), is a large investment in the long-term viability of a range of plants (Smith *et al.*, 2007).

Botanical gardens serve as living classrooms. Through scientific research, they promote and engage in the study of plants and biological variety in general. Molecular biology, biotechnology, population genetics, ecology, horticulture, plant breeding, taxonomy/systematics, evolution, conservation biology, climate change, and environmental education are among the fields of study. Numerous botanic gardens maintain substantial collections of plants used in agriculture, medicine, gardening, forestry, habitat management and restoration, tourism, and a range of other industries, as well as perform study on these plants. So, it is crucial to give the resources needed so that botanical gardens can play significant roles in the evaluation and conservation of biodiversity.

Zoos

Various organisations which maintain wild or exotic animals are referred to as zoos. Zoos are now more crucial than ever for the preservation of endangered wildlife. Furthermore, they are essential for both recreational and instructional purposes. Zoos, however, are typically underfunded and poorly developed. Therefore, appropriate human, financial, and physical resources must be made available so that zoos can play a significant part in the conservation of endangered wild species.

Field gene banks

Plant species that do not generate seeds and spread vegetatively are called refractory seed, such seeds that cannot be stored at low temperatures, need field gene banks to be conserved.

In most cases, field gene banks can keep these hard to preserve species as living collections. On the other hand, conservation in field gene banks calls for accurate knowledge of the ecological needs of the concerned species. It requires a large enough region with site characteristics that are comparable to those of the original population but with less environmental load or pressure (Wolf, 1999).

Gene bank

The preservation of genetic material or germplasm depends on gene banks. In addition to seeds, pollen, spores, semen (sperm), eggs, embryos, cells and tissues, gene banks can also store other kinds of genetic material. Currently, DNA sequences are also kept in dedicated banks (Smith *et al.*, 2007). It is crucial to use a representative sample in each case. Approximately

6 million plant accessions are stored in gene banks around the world (Engels and Engelmann, 2002; Frison *et al.*, 2002). The storage of genetic material in gene banks enables research and restoration. Utilising *in vitro* conservation techniques, such as tissue culture storage (micropropagation) and cryopreservation (storage of germplasm in latent state), which is necessary for the preservation of genetic material. Biomaterial has been preserved using cryopreservation using liquid nitrogen (-196 °C).

The two main goals of the projects are:

- i) To gather and maintain the wild species which yield seeds.
- ii) To foster global collaboration in bilateral research, training, and capacity-building in order to support and advance the seed conservation effort.

The primary emphasis is on both endemic and endangered species. For the reemergence of threatened species and restoration of ecological systems, an indigenous plant germplasm collection is going to be needed for the development of novel agricultural products, the adaptation of existing ones, the production of medicinal plants and other uses. Gene banks will also serve as a source of plants and genetic variety.

- Address the underlying causes of biodiversity loss.

Conclusion

The range of degree of variation within the natural world are referred to as "biodiversity" in this context. It could be evaluated on several levels. The biological diversity measurements that are most frequently used are genetic, species and ecosystem diversity. Four separate methods-alpha, beta, gamma and delta diversity are used to measure biodiversity at various spatial scales. A part of biodiversity is cultural variety. Biodiversity is linked to values in the fields of education, science, healthcare, science, finances, society, and the arts. However, a rapid loss of diversity is currently being driven by environmental and more importantly, anthropogenic procedures. The risk to biodiversity are overexploitation, growing urbanisation, industrial development, pollution, agricultural growth, increased frequency of wildfires, invasive species, genetically engineered organisms (GMOs) and global climate change. It is crucial to preserve biodiversity to ensure intra and intergenerational equity. Reforestation, zoological gardens, botanical gardens, national parks, biosphere reserves, germplasm banks and adoption of breeding techniques, tissue culture techniques and social forestry to reduce pressure on the exploitation of forest resources are some of the current measures of biodiversity conservation.

References

1. Abrahms B, Dipietro D, Graffis A, Hollander A. Managing biodiversity under climate change: challenges, frameworks, and tools for adaptation. *Biodiversity Conservation*. 2017;26:2277-2293. Available from: <https://bit.ly/35UjMWk>
2. Agarwal NK, Singh G, Rawat US. Present status and threats to the Ichthyofaunal diversity of a snow fed river Nandakini in central Himalaya (Garhwal), India. In: Rawat US, Semwal VP, editors. *Uttarakhand Disaster: Contemporary issue of Climate Change and Development with Holistic Approach*. Dehradun, India: Winsar Publication; 2014. p. 173-182.
3. Agarwal NK, Singh G, Singh H. Present status of Ichthyofaunal diversity of Garhwal Himalayan river Bhilangna and its tributaries with reference to changing environment. *Environ Conserv J*. 2011;12(3):101-108.
4. Ayoade AA, Agarwal NK, Chandola-Saklani A. Changes in physico-chemical features and plankton of two regulated high altitude rivers in Garhwal Himalaya, India. *Eur. J Sci. Res*. 2009;27(1):77-92.
5. Bhateria R, Jain D. Water quality assessment of lake water: a review. *Sustain Water Resour Manag*. 2016;2:161-173. Available from: <https://bit.ly/39Kan56>
6. Bright PW. Habitat fragmentation-problems and predictions for British mammals. *Mammal Rev*. 1993;23:101-112. Available from: <https://bit.ly/39LPNRQ>
7. Brussaard CP, Peperzak L, Beggah S, Wick LY, Wuerz B, *et al*. Immediate ecotoxicological effects of short-lived oil spills on marine biota. *Nat Commun*. 2016;7:112-06. Available from: <https://bit.ly/39RPWmR>
8. Carrete M, Tella JL, Blanco G, Bertellotti M. Effects of habitat degradation on the abundance, richness, and diversity of raptors across Neotropical biomes. *Biol. Conserv*. 2009;142:2002-2011. Available from: <https://bit.ly/3bTZIr2>
9. Chapin FS, Diaz S. Interactions between changing climate and biodiversity: shaping humanity's future. *Proc. Natl. Acad. Sci. USA*. 2020;117:6295-6296. Available from: <https://bit.ly/39LQwm2>
10. Duarte MHL, Sousa-Lima RS, Young RJ, Farina A, Vasconcelos M, *et al*. The impact of noise from open-cast mining on Atlantic forest biophony. *Biol Conserv*. 2015;191:623-631. Available from: <https://bit.ly/39KHVQx>

11. Field R, Hawkins BA, Cornell HV, Currie DJ, Diniz JAF, Guégan JF, *et al.* Spatial species-richness gradients across scales: a meta-analysis. *J Biogeogr.* 2009;36(1):132-147.
12. Frison EA, Mitteau M, Sharrock S. Sharing responsibilities for *ex situ* germplasm management. *Plant Genet Resour Newsl.* 2002;131:7-15.
13. Gaston KJ. Global patterns in biodiversity. *Nature.* 2000;405(6783):220-227.
14. Hammerschlag N, Gallagher JA. Extinction risk and conservation of the earth's national animal symbols. *Bioscience.* 2017;67:744-749. Available from: <https://bit.ly/391vdxm>
15. Isbell F, Reich PB, Tilman D, Hobbie SE, Polasky S. Nutrient enrichment, biodiversity loss, and consequent declines in ecosystem productivity. *Proc. Natl. Acad. Sci. USA.* 2013;110:11911-11916. Available from: <https://bit.ly/3o3vGU6>
16. Klappenbach L. Habitat loss, fragmentation, and destruction. ThoughtCo. 2020. Available from: <https://bit.ly/3iFsUUh>
17. Klein AM, Vaissière BE, Cane JH, Steffan-Dewenter I, Cunningham SA. Importance of pollinators in changing landscapes for world crops. *Proc. Biol. Sci.* 2007;274:303-313. Available from: <https://bit.ly/3qCDmyw>
18. Linders TEW, Schaffner U, Eschen R, Abebe A, Choge SK, *et al.* Direct and indirect effects of invasive species: biodiversity loss is a major mechanism by which an invasive tree affects ecosystem functioning. *J Ecol.* 2019;107:2660-2672. Available from: <https://bit.ly/3qARzMb>
19. Macgregor CJ, Evans DM, Fox R, Pocock MJ. The dark side of street lighting impacts on moths and evidence for the disruption of nocturnal pollen transport. *Glob Chang Biol.* 2017;23:697-707. Available from: <https://bit.ly/2Y2wmhT>
20. Martins IS, Rosa IMD, Pereira HM. Projecting impacts of global land-use scenarios on biodiversity change across scales and species groups. 5th European Congress of Conservation Biology, 2018. Available from: <https://bit.ly/3bW0BQ4>
21. Mohallem SV, De Araújo Lobo DJ, Pesquero CR, Assunção JV, De Andre PA, *et al.* Decreased fertility in mice exposed to environmental air pollution in the city of Sao Paulo. *Environ Res.* 2005;98:196-202. Available from: <https://bit.ly/3sQpOkW>

22. Mora C, Tittensor DP, Adl S, Simpson AG, Worm B. How many species are there on Earth and in the ocean? *PLoS Biol.* 2011;9(8). Available from: <https://doi.org/10.1371/journal.pbio.1001127>
23. Pennanen T, Fritze H, Vanhala P, Kiikkila O, Neuvonen S. Structure of a microbial community in soil after prolonged addition of low levels of simulated acid rain. *Appl Environ Microbiol.* 1998;64:2173-2180. Available from: <https://bit.ly/39TMBUm>
24. Singh V, Tripathi BN, Sharma V. Interaction of Mg with heavy metals (Cu, Cd) in *T. aestivum* with special reference to oxidative and proline metabolism. *J Plant Res.* 2015;129:487-497. Available from: <https://bit.ly/391ttVc>
25. Symes WS, Edwards DP, Miettinen J, Rheindt FE, Carrasco LR. Combined impacts of deforestation and wildlife trade on tropical biodiversity are severely underestimated. *Nat Commun.* 2018;9:4052. Available from: <https://go.nature.com/35WtyaC>
26. Rondeau D, Perry B, Grimard F. The consequences of COVID-19 and other disasters for wildlife and biodiversity. *Environ Resource Econ.* 2020;76:945-961. Available from: <https://bit.ly/3qCaYwe>
27. Sanderfoot OV, Holloway T. Air pollution impacts on avian species via inhalation exposure and associated outcomes. *Environ Res Lett.* 2017;12:083002. Available from: <https://bit.ly/35TKOxm>
28. Wouter H, Holleman LJM, Lessells CM, Slabbekoorn H. Negative impact of traffic noise on avian reproductive success. *J Appl Ecol.* 2011;48:210-219. Available from: <https://bit.ly/35WQrL5>
29. Wright SL, Rowe D, Thompson RC, Galloway TS. Microplastic ingestion decreases energy reserves in marine worms. *Curr Biol.* 2013;23. Available from: <https://bit.ly/3oZbO5>

Chapter - 4

Isolation and Screening of Bacteria from Fermented Food Products for Potential Probiotics

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Chapter - 4

Isolation and Screening of Bacteria from Fermented Food Products for Potential Probiotics

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Abstract

Probiotics are beneficial bacteria that promote a healthy gut microbiota balance and play a significant role in maintaining human health. One of the first sources of probiotics in processed food products have been discovered is fermentation. They are created by microorganisms as a result of the biotransformation of basic materials into finished goods. Commercially-produced fermented foods often serve as carriers for probiotic bacteria. Isolating promising probiotic strains has been done for millennia and continues to be of interest because to its industrial and therapeutic usefulness. In this study, we isolated atleast four different bacteria from traditional fermented food products of northeast India. Pure cultures were maintained in glycerol stocks for further investigation. Preliminary studies were done to evaluate the morphological characteristics of the four isolates including gram staining, catalase activity test, and CFU/ml.

Keywords: Probiotics, Lactobacillus, fermented foods, CFU/ml, catalase, gram staining

Introduction

Commercially-produced fermented foods often serve as carriers for probiotic bacteria. One of the first sources of probiotics in processed food products have been discovered is fermentation. They are created by microorganisms as a result of the biotransformation of basic materials into finished goods. Numerous food biotransformation processes depend on the fermentation of ethanol by the yeast *Saccharomyces cerevisiae* or the fermentation of lactic acid by a class of bacteria known as lactic acid bacteria (Papadimitriou *et al.*, 2016). The living bacteria found in fermented foods, in particular, have been linked to gastrointestinal and systemic health advantages, according to numerous studies and researchers (Sanders *et al.*,

2018). Each fermented meal relates to a distinct microbial community, which boosts the amounts of protein, vitamins, vital amino acids, and fatty acids. Food fermentation, food preservation, and feed products all use LAB (Liang *et al.*, 2010). The global market for probiotic products has grown quickly in recent years, as has public knowledge of the health advantages of probiotic microorganisms, most notably lactic acid bacteria (LAB) (Rosenstiel and Stange, 2010; Hill *et al.*, 2014; Hansen *et al.*, 2018; Jackson *et al.*, 2019; Fallico *et al.*, 2020; Kiefer *et al.*, 2020).

The use of fermented fish as a dish in traditional culinary practices stretches back to ancient times in various communities throughout the world (Prajapati *et al.*, 2003). Lactic acid bacteria (LAB) are used in fermented vegetables and yoghurt to convert mono- and disaccharide carbohydrates into lactic acid. In contrast, natto and tempeh (fermented soybeans) are fermented using *fungus* and *Bacillus* microbes (Marco *et al.*, 2017). Traditional fermented mustard leaf dish that is consumed in Northeast India used green mustard which is produced spontaneously by a mixed microbial community dominated by LAB (Swain *et al.*, 2014; Wang *et al.*, 2014).

Probiotics are living microorganisms that, when consumed in adequate amounts, provide health benefits to the host. These beneficial bacteria promote a healthy gut microbiota balance and play a significant role in maintaining human health. There are various types of probiotics, with the most common belonging to the *Lactobacillus* and *Bifidobacterium* genera. These strains are often used in commercial probiotic products and have been extensively studied for their health-promoting properties. However, other types of bacteria and yeasts, such as *Saccharomyces boulardii*, *Streptococcus thermophilus*, and certain strains of *Escherichia coli*, can also exhibit probiotic properties.

LAB or lactic acid bacteria are Gram-positive, catalase-negative bacteria that may create lactic acid as a by-product of carbohydrate fermentation. As they utilise a different metabolic route, the genus *Bifidobacterium* is included among them more conventionally than phylogenetically. *Streptococcus thermophilus* and *Lactococcus lactis*, two of the most economically significant lactic acid bacteria, are two other species that play a vital role in the food business, notably dairy products, but are not strictly called probiotics (Felis *et al.*, 2007).

Even fermented foods that are not subjected to thermal processing may contain low levels of live or viable microorganisms due to unfavorable environmental conditions that reduce microbial populations over time.

Microbiologists have recently been interested in the isolation and characterization of bacteriocins, since they may be a better option to replace antibiotics and synthetic preservatives (Indira *et al.*, 2018). Isolating promising probiotic strains has been done for millennia and continues to be of interest because to its industrial and therapeutic usefulness (Reuben *et al.*, 2020)

In this study, our primary goal is to isolate pure cultures of bacteria living in different fermented food products. Further identification and characterization of these bacteria for their probiotic potential will also be carried out in the near future.

Materials and Methods

Sample collection

Bacterial samples were collected from different fermented food products such as soya bean, fish, yogurt and mustard leaf. These traditional food products were made locally and purchased from markets. Samples were serially diluted before plating on MRS agar (Table 1) and incubated for 24 hours at 37 °C. Single colonies were selected for preparation of glycerol stocks at stored at freezing temperature for further use.

Gram staining

Bacterial smear was made on glass slide and air dried followed by heat fixing. After this, crystal violet was added and rinsed after 1 minute. Few drops of Grams iodine were added on the slide and kept for 1 minute. After washing with distilled water the smear was decolorized with 95% ethyl alcohol for 15-25 secs. Again after rinsing with water safranin was added and rinsed again after 1 minute. Finally, the slide was allowed to air dry and the smear was observed under microscope.

Serial dilution for colony count

Pure culture of the sample was pipette out and centrifuged for 3 mins and the supernatant was discarded. It was then washed with PBS buffer and serial dilution was ranging from 10^{-1} to 10^{-6} was made. For colony counting, samples were spreaded on MRS agar plate from each serial dilution tubes and incubated at 37°C overnight. Colonies forming units (CFU) was estimated with the formula $CFU/ml = (Nos\ of\ colonies \times Dilution\ factor) / Volume$

Table 1: Composition of MRS (De Man, Rogosa and Sharpe)

Ingredients	Grams/litre
Universal peptone	
Meat extract	10.0
Yeast extract	5.0
D(+)-Glucose	5.0
Dipotassium hydrogen phosphate	20.0
Diammonium hydrogen citrate	2.0
Sodium acetate	2.0
Magnesium sulfate	5.0
Manganous sulfate	0.1
Agar	0.05
Final pH 6.5 +/- 0.2 at 25 °C	12.0

Catalase test

The catalase test aids in the identification of the catalase enzyme in bacteria. An isolated colony was placed on a microscope slide using a sterile inoculating loop. A drop of 3% hydrogen peroxide H_2O_2 was placed onto the slide containing the culture, the formation of bubbles on the slide indicates catalase-positive and no bubbles indicate catalase-negative. The results were observed and recorded.

Results and Discussions

Traditional fermented foods have been found to include a variety of probiotic compounds that encourages and attracts probiotic growth. It is essential to have a fundamental understanding of the relationship between food, beneficial bacteria, and human health in order to enhance food quality and prevent a variety of ailments. The ability of people in the North-Eastern states of India to produce a variety of fermented foods has obviously provided a significant opportunity for the development of various food industries related to fermented products, quality improvements, and enhanced wider-scale marketing. Probiotics have gained a great deal of attention in recent years due to their potential ability to support the gut health, immune function, and mental well-being.

Different fermented food products which are locally available in Mizoram were collected and employed for the isolation of bacteria as shown in Figure 1 & 2. Following serial dilution method, the samples were diluted and poured on Petri plates containing media, where growth was observed after 18 hours of incubation at 37 °C. Thereafter, colonies from different

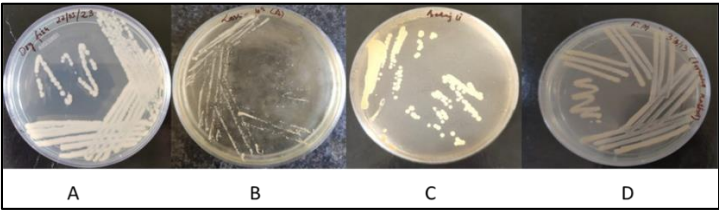
fermented food products were obtained. To produce pure colonies, the isolated bacterial strains were streaked and purified using the streak plate method. The initial morphological investigation was then conducted using the gram staining method. Gram staining is essential for phenotypic characterization of microorganisms. The staining process distinguishes bacteria based on their cell wall structure. Gram-positive cells contain a thick peptidoglycan coating and appear blue to purple when stained. Gram-negative cells are distinguished by a thin peptidoglycan coating that stains red to pink (Smith *et al.*, 2005). In our study all the bacterial isolates exhibited gram positive characteristic but vary in their shapes. While the isolates from fermented mustard leaves and Lassi are cocci shaped, colonies from fermented dry fish and soya bean exhibited bacilli and rod-shaped in appearance.

The preliminary observations of morphological attributes for bacterial isolates (Fermented mustard leaves) in comparison with known strain of *Lactobacillus plantarum* is shown in Table 2. Both exhibited gram positive characteristic, while our isolate appears to be coccus and round shape *L. plantarum* shows rod shape in morphology.

The catalase assay was also conducted to check the activity of catalase enzyme for its hydrogen peroxide scavenging capacity. In all the three samples except for the isolate from fermented mustard, all were found to be catalase negative (data not shown).



Fig 1: Samples collected from locally made fermented food products for bacterial isolation



Colonies forming units (CFU/ml) estimated after serially dilution of cultures ranging from 10^{-1} to 10^{-6} . Number of colonies counted at dilution of 10^{-5} was plotted as shown in Figure 3.

Table 2: Preliminary characterisation of bacteria isolated from fermented mustard in comparison with known *Lactobacillus plantarum* strain.

Morphological characteristics	Bacteria from fermented mustard	<i>Lactobacillus plantarum</i>
Shape	circular/coccus	Rod
Gram staining	Purple/blue	Purple/blue
Gram +/-	Gram positive	Gram positive
Elevation	Convex	Flat
Form	Round	Rod
Margin	Entire	Not entire

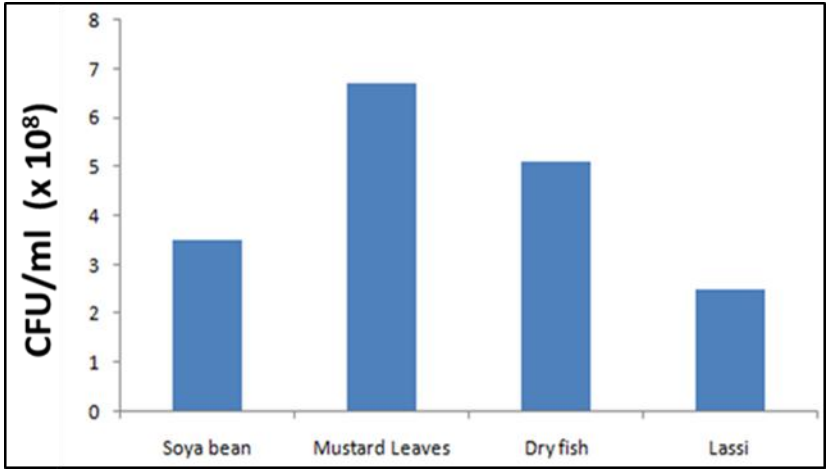


Fig 3: CFU/ml estimated at a dilution of 10^{-5} for different samples

Conclusion

From our study, we have successfully isolated pure cultures of four bacterial species from fermented food products which are all gram positive, catalase negative (except for one). Our future goal is to identify these bacteria by 16S DNA sequencing and further characterization with biochemical assays for their potential role in probiotic applications.

References

- Papadimitriou K, Alegría Á, Bron PA, De Angelis M, Gobbetti M, Kleerebezem M, *et al.* Stress physiology of lactic acid bacteria. Microbiol Mol Biol Rev. 2016 Sep;80(3):837-90.

2. Sanders ME, Merenstein D, Merrifield CA, Hutkins R. Probiotics for human use. *Nutr Bull.* 2018 Sep;43(3):212-25.
3. Liang TW, Wu YY, Huang TY, Wang CY, Yen YH, Liu CP, *et al.* Conversion of squid pen by a novel strain *Lactobacillus paracasei* subsp. *paracasei* TKU010, and its application in antimicrobial and antioxidants activity. *J Gen Appl. Microbiol.* 2010;56(6):481-9.
4. Rosenstiel P, Stange EF. Probiotics and intestinal diseases. *Ann Nutr Metab.* 2010 Sep;57:27.
5. Hill C, Guarner F, Reid G, Gibson GR, Merenstein DJ, Pot B, *et al.* Activity of cecropin P1 and FA-LL-37 against urogenital microflora. *Nat Rev Gastroenterol Hepatol.* 2014;11(8):506.
6. Hansen SJ, Morovic W, DeMeules M, Stahl B, Sindelar CW. Absolute enumeration of probiotic strains *Lactobacillus acidophilus* NCFM® and *Bifidobacterium animalis* subsp. *lactis* BI-04® via chip-based digital PCR. *Front Microbiol.* 2018 Apr;9:342-603.
7. Jackson SA, Schoeni JL, Vegge C, Pane M, Stahl B, Bradley M, *et al.* Improving end-user trust in the quality of commercial probiotic products. *Front Microbiol.* 2019 Apr;10:739.
8. Fallico V, Rea M, Stanton C, Ilestam N, McKinney J. Next-generation multi-parameter flow cytometry assay improves the assessment of oxidative stress in probiotics. *Food Microbiol.* 2020 Oct;91:103-501.
9. Kiefer A, Tang P, Arndt S, Fallico V, Wong C. Optimization of viability treatment essential for accurate droplet digital PCR enumeration of probiotics. *Front Microbiol.* 2020 Jul;11:569-374.
10. Prajapati JB, Nair BM. The history of fermented foods. In: *Handbook of fermented functional foods*; 2003 Mar. p. 1-25.
11. Marco ML, Heeney D, Binda S, Cifelli CJ, Cotter PD, Foligné B, *et al.* Health benefits of fermented foods: microbiota and beyond. *Curr Opin Biotechnol.* 2017 Apr;44:94-102.
12. Swain MR, Anandharaj M, Ray RC, Rani RP. Fermented fruits and vegetables of Asia: a potential source of probiotics. *Biotechnol Res Int.*; 2014.
13. Wang SC, Chang CK, Chan SC, Shieh JS, Chiu CK, Duh PD. Effects of lactic acid bacteria isolated from fermented mustard on lowering cholesterol. *Asian Pac J Trop Biomed.* 2014 Jul;4(7):523-8.

14. Indira M, Venkateswarulu TC, Prabhakar KV, Peele KA, Krupanidhi S. Isolation and characterization of bacteriocin producing *Enterococcus casseliflavus* and its antagonistic effect on *Pseudomonas aeruginosa*. Karbala Int. J Mod Sci. 2018 Dec;4(4):361-8.
15. Reuben RC, Roy PC, Sarkar SL, Alam AR, Jahid IK. Characterization and evaluation of lactic acid bacteria from indigenous raw milk for potential probiotic properties. J Dairy Sci. 2020 Feb;103(2):1223-37.
16. Felis GE, Dellaglio F. Taxonomy of lactobacilli and bifidobacteria. Curr Issues Intest Microbiol. 2007 Sep;8(2):44.
17. Smith AC, Hussey MA. Gram Stain Protocols, American Society for Microbiology; 2005.

Chapter - 5

Photo-Biotechnology a Tool to Improve Agronomic Traits in Crops

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Chapter - 5

Photo-Biotechnology a Tool to Improve Agronomic Traits in Crops

Manoj Kumar Duppala, Shruthi Nagaral K., T. John Ratnakar and Kanavi M.S.P.

Abstract

Light is a critical environmental factor that influences growth and development in plants and also involves complex signaling networks which are tightly regulated by genetic and environmental factors. Sunlight is not only crucial as the ultimate source of energy fueling photosynthesis but it also provides important information about the environment, influencing plant growth and development. Being sessile, during developmental processes plants are affected by light intensity, quality and day length. Importantly, such light mediated adaptive responses are termed as photomorphogenesis, include traits that are important for plant survival. The Growth and developmental plasticity in response to light signals is conferred by specialized information-transducing photoreceptors. Plants have evolved specific photoreceptors that capture information from sunlight. They are red (R)/far-red (FR) light-absorbing phytochromes and the blue/UV-A light-absorbing cryptochromes and phototropins. The three major phytochromes exist in angiosperms, phyA, phyB and phyC, encoded by the genes PHYA, PHYB and PHYC. Two additional phytochromes exist in dicotyledonous plants, phyD and phyE. Genetic engineering of phytochrome genes has provided a potential means to control vegetative growth and reproductive development. The cryptochromes and phytochromes control growth and developmental responses to variation in the wavelength, intensity and diurnal duration of the irradiation whereas the phototropins function primarily in controlling directional (phototropic) growth in response to directional light and/ or intracellular chloroplast movement in response to light intensity. The combined activities of these molecules enable the plant to detect and respond to the presence, absence, color, intensity, directionality and diurnal duration of the light signals. Photobiotechnology refers to the transgenic expression of phytochrome or cryptochrome transgenes or variants of such transgenes. Several studies have indicated that crop cultivars can be improved by

modulating expression photoreceptor genes. The studies suggests that red and far-red light perceiving PHYA overexpressing rice line showed increase in grain yield with reduction in plant height. Important agronomic traits such as flowering time, fruit quality and photosynthetic pigments are controlled by the cryptochrome (mutants and overexpression analysis) in tomato seedling. Several other improved traits includes grass quality, plant architecture, chlorophyll biosynthesis have been reported. Despite significant progress, gaps in our knowledge exist regarding the regulation of photoreceptors signaling and its role in plant development. Transcriptome profiling coupled with proteomic analyses at different stage of plant development under different environmental conditions can contribute to understanding of photoreceptor signalling.

Keywords: Photo biotechnology, light, photomorphogenesis, phytochromes, cryptochrome

Introduction

Plants harness light energy for photosynthesis, a process that fixes carbon dioxide to synthesize organic molecules essential for life. Plants also recognize light as information to optimize their morphology and metabolism for efficient photosynthesis under a fluctuating light environment in nature. Accordingly, various growth and developmental processes in plants are affected by light intensity and quality and day length. Importantly, such light mediated adaptive responses, termed photomorphogenesis. Photomorphogenesis refers to the biological process by which plants undergo growth and development patterns respond to light signals/spectrum for Ex., during de-etiolation plants switch transitioning from skotomorphogenesis to photomorphogenesis (independent of photosynthesis) (fig. 1). After seed germination in the soil, etiolated growth enables the germinated seedling to grow toward the soil surface seeking illumination. Upon exposure to light, the seedling undergoes photomorphogenesis characterized by de-etiolation, chlorophyll synthesis and development of chloroplast all of which enable the seedling to establish itself as an independent autotroph. Quality as well as quantity of incident light influence the developmental and growth of plants. Light functions as a signal to regulate various physiological responses in plants. Photomorphogenic responses are initiated by several classes of photoreceptors. Photo sensory phosphoprotein, which absorb different light waves involved in variety of functions like growth, greening and photosynthesis. Plants detect light of various wavelengths using a wide range of photoreceptors. When these photoreceptors are triggered by light, they trigger specialized transduction

pathways that alter the plant's growth and development in response to the surrounding conditions. Photomorphogenesis refers to the biological process through which plants undergo growth and development in reaction to light stimuli. The process is facilitated by a complex network of photoreceptors, with phytochromes playing a crucial part. Phytochrome-mediated photomorphogenic responses encompass intricate light-induced physiological reactions including as germination, de-etiolation, shadow avoidance, circadian rhythm regulation, and flowering (Taiz *et al.*, 2015). Light receptors like phytochrome (phy), cryptochrome and phototropin help plants perceive the duration, wavelength, and intensity of incident light and adapt to the changing environment.

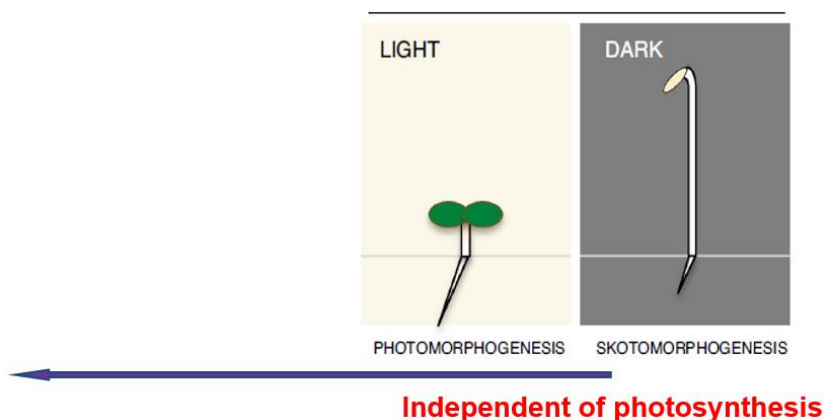


Fig 1: Photomorphogenesis responses during seed germination

Types of photoreceptors in plants

- a) **Cryptochrome:** Blue light reception (600-750nm).
- b) **Phytochrome:** Red and far-red light reception (390-500nm).
- c) **Phototropin:** Mediates phototropism (320-390nm).

Manipulating the expression of photoreceptors or variants of such transgene. Ever since the green revolution occurred between the 1940s and late 1960s, crop improvement programs across the globe have strived to develop high-yielding crop cultivars. Among the major agronomic traits link to the success of the green revolution was dwarfism. N.E. Borlaug, the father of the green revolution, noticed that even though taller wheat varieties were efficiently able to compete for sunlight, they were fragile and could not bear the weight of the extra grain. Since many photoreceptors-regulated processes determine the overall suitability and productivity of crops. There has been a notable level of interest in modulating photoreceptors activity to attain

improved crop yields. Numerous studies have established the fundamental role of photoreceptors in regulating a wide range of complex developmental processes in plants, such as seed germination, de-etiolation, chloroplast biogenesis, and flowering. Photo-biotechnology offers a promising approach for studying the influence of these photoreceptors on plant development. With the knowledge on photoreceptors and their method of functioning to regulate important plant processes, photoreceptors could be engineered to express desirable traits.

Phytochromes

A ubiquitous group of photoreceptors that are sensitive to red and far-red light. Crucial controller of the process by which plants develop and grow in response to light. Phytochromes are found in all green plants, as well as in cyanobacteria, non-photosynthetic bacteria and fungus. These photoreceptors have two thermally stable conformations, which rely on the presence of light: an inactive form that absorbs light at 660 nmR (Pr) and an active form that absorbs light at 730 nm FR (Pfr).

Pr form

- Has maximum absorption at a wavelength of 666 nm
- Is the form synthesized in dark-grown seedlings.
- Red light absorption by Pr results in its conversion to the Pfr form.

Pfr form:

- Has maximum absorption at a wavelength of 730 nm
- Pfr is the active form that triggers biological responses.
- Pfr converts to Pr when absorbing far red light.
- Pfr can also spontaneously revert to the Pr form in the dark over time = dark reversion; Pfr is also sensitive to proteinases.
- The proportion of Pfr and Pr in red light is 85% Pfr and 15% Pr.
- Pr absorbs minimal far red light, resulting in a 97% Pr to 3% Pfr balance.
- Photosensory phosphoproteins phytochromes are essential for plant light-induced development. Functional investigations of phytochromes have shown their unique roles in plant life. Gene expression and plant development are influenced by phytochrome signalling.

The protein family phytochrome has a tiny covalently bonded pigment molecule: Phytochrome proteins exist as a dimer composed of two identical 124 k Da polypeptides, each of which is bonded to a pigment molecule. Every polypeptide consists of three segments, sometimes known as domains, which share amino acid sequences that are identical to domains seen in other sensor proteins.

- The pigment is termed chromophore. This linear tetrapyrrole.
- The structure of the chromophore changes slightly when it absorbs light. This changes protein conformation to the response-initiating form.
- Dark-grown seedlings have 50X higher phytochrome levels than light-grown plants. Highest levels occur toward the plant's peak.
- Molecular genetics has identified many genes for this protein in plants. All of these proteins share a chromophore but have different light sensitivities.
- Phytochromes provide diverse biological responses to red light. Multigene families code phytochromes.

Five phytochrome genes in dicots: *Arabidopsis* has a family of five members (PhyA-PhyE); Monocots like rice have three members: phyA, phyB, and phyC.

Phytochrome signaling-transduction pathways

- Direct transcription factors.
- Phytochrome kinase activity.
- Regulation of phytochrome by itself (auto-phosphorylation).
- cGMP and calcium dependent pathway.

Phytochromobilin is synthesized inside plastids and is derived from heme via a pathway that branches from the chlorophyll biosynthetic pathway. After synthesized in the plastid it is exported from plastid to the cytosol where it autocatalytically attaches to the PHY apoprotein through a thioether linkage to a cysteine residue (fig. 2), then it follows light induced conformation changes within the phy molecules, direct transcription activation, phytochrome kinase activity and regulation of phytochrome by itself (fig. 3). With the knowledge on photoreceptors and their method of functioning to regulate important plant processes, photoreceptors could be engineered to express desirable traits. For example, Light-regulated *Arabidopsis* phytochrome overexpression A rice gene changes plant

architecture and boosts grain yield (Garg *et al.*, 2006). Overexpression of *Arabidopsis thaliana* PHYTOCHROME A (PHYA) gene in a commercially important Indica rice variety (*Oryza sativa* L. Pusa Basmati-1) was studied. The light-regulated and tissue-specific rice *rbcS* promoter drove transgenic expression. Several homozygous sixth generation (T5) transgenic lines were described and found to accumulate PHYA protein in the light. In far-red and red light, PHYA-overexpressing lines inhibited coleoptile expansion more than non-transgenic seedlings (fig. 4a). Mature transgenic rice plants had significantly lower plant height, internode length, and diameter (including cell size and quantity) (fig. 4b) and generated more panicles per plant. Rice grain yield was 6-21% greater in three PHYA-overexpressing lines than in non-transgenic plants under greenhouse conditions. These findings show that altering light signal-transduction pathways can reduce lodging in basmati/aromatic rice and boost grain productivity.

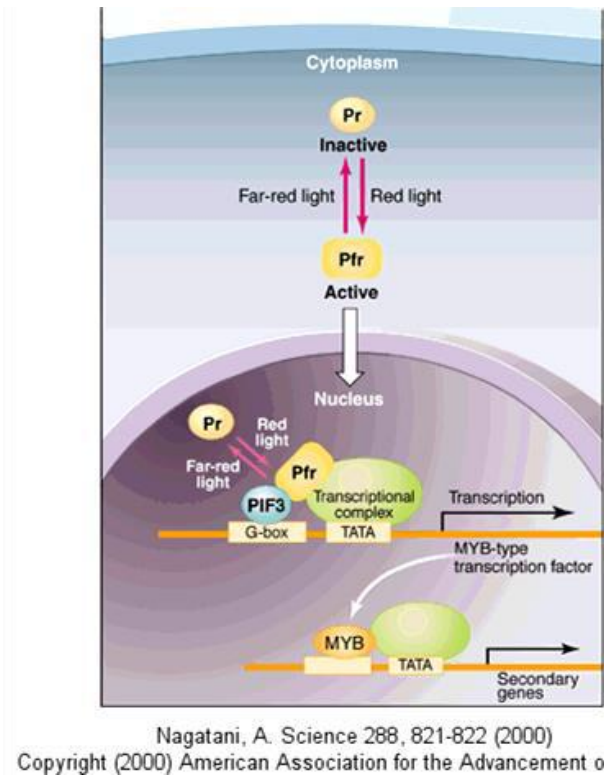
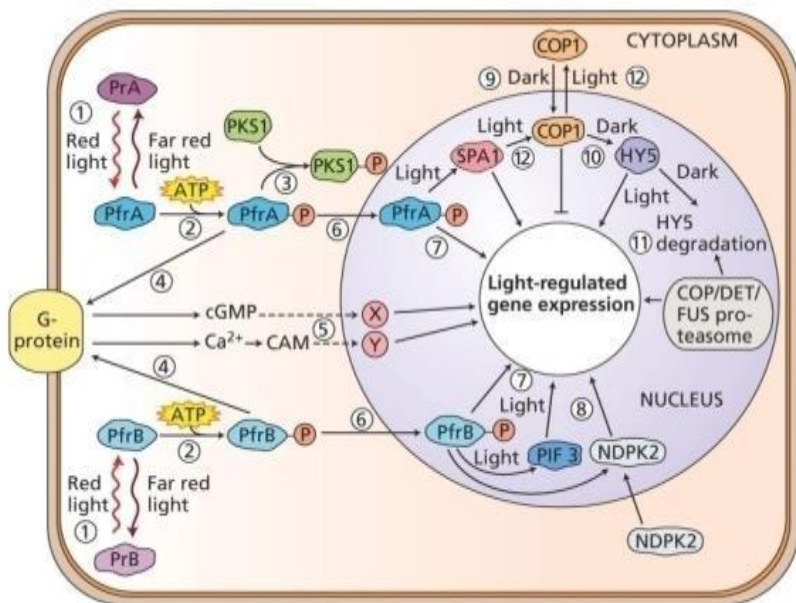


Fig 2: Phytochrome synthesis in the plastid and interaction through direct transcription factor



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Fig 3: Phytochrome synthesis through kinase activity and autophosphorylation in the nucleus

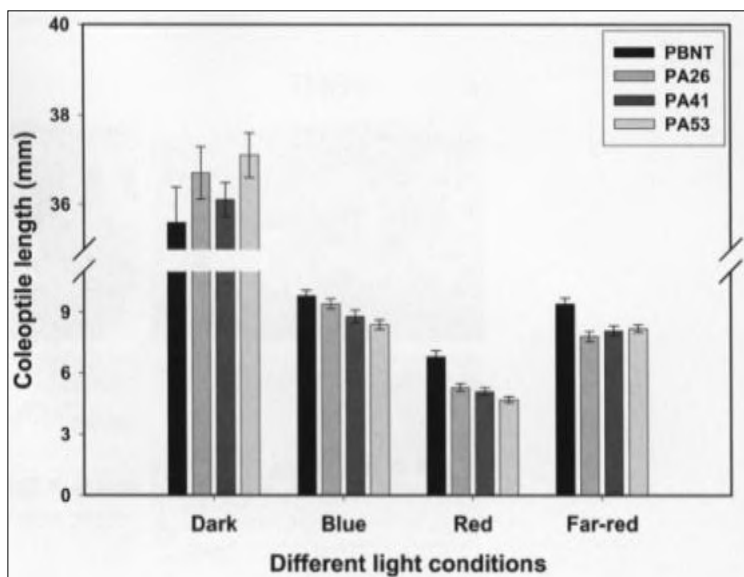


Fig 4a: Photomorphogenesis of rice seedling

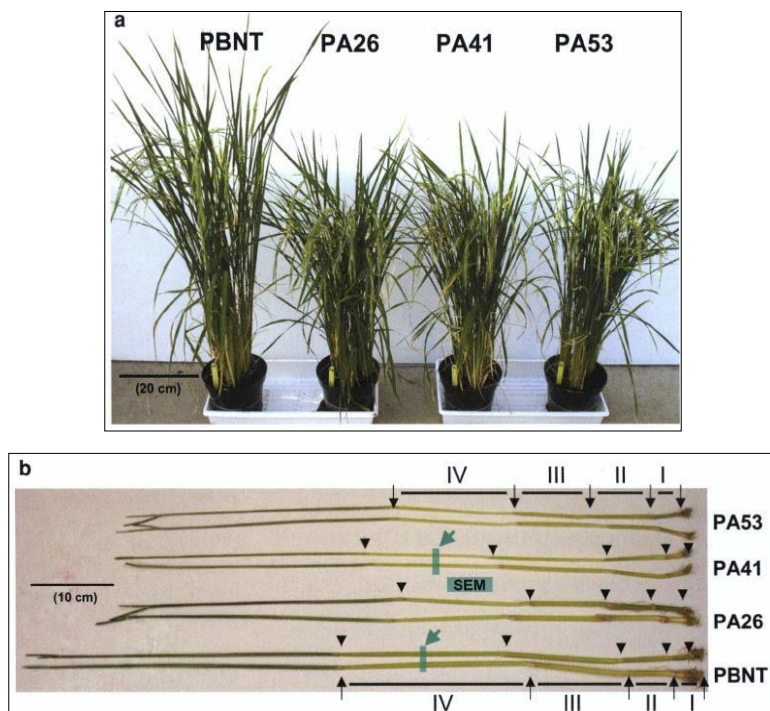


Fig 4b: Differences in (top) plant architecture and (bottom) internode length between non-transgenic and *PHYA* overexpressing transgenic lines

Rice line	PBNT	PA26	PA41	PA53
<i>PHYA</i> transgene copy number	0	2	1	1
Plant height (cm)	92 ± 4	76 ± 6	73 ± 5	68 ± 5
1000-grain weight (gm)	21.2 ± 0.5	21.9 ± 0.4	22.3 ± 0.5	21.7 ± 0.4
Panicle number per plant	23 ± 5.4	32 ± 4.9	27 ± 5.0	27 ± 6.5
Grain yield per plant (gm)	39.2 ± 8.1	47.4 ± 7.0	41.7 ± 8.3	43.4 ± 9.5
Relative grain yield (%)	100	121	106	111

Fig 4c: Yield related characteristics

Phytochrome (PhyA and PhyB) for various qualities in genetic engineering. Abiotic stress tolerance and phytochrome B. Liu *et al.* (2012) reported phytochrome B (*phyB*) mutants exhibit improved drought tolerance compared to wild type (WT) rice (*Oryza sativa* L. cv. Nipponbare). Whereas the leaves of *phyB* mutants displayed larger epidermal cells than WT leaves, resulting in reduced stomatal density. *phyB* deficiency promoted the expression of both putative *ERECTA* family genes and *EXPANSIN* family genes involved in cell expansion in leaves, thus causing greater epidermal cell expansion in the *phyB* mutants. In *phyB* mutants, reduced stomatal density reduced transpiration per unit leaf area and the *phyB* mutant also

showed increased water use efficiency. Rice (*Oryza sativa*) phytochrome B (*osphyB*) mutant has greater tolerance to salt stress than its parent japonica rice (cv. Dongjin) (Kwon *et al.*, 2018). *osphyB* mutant showed a higher survival rate by maintaining significantly lower Na⁺ to K⁺ ratio. There was an upregulation of salt stress-associated genes, including transcription factors and high-affinity K⁺ transporters in the *osphyB* mutant. Therefore together the decreased Na⁺ to K⁺ ratio and enhances cell membrane integrity through upregulation of salt stress-associated genes, resulting in improved tolerance to salt stress.

Phytochrome A shade tolerance and turf quality in creeping bentgrass and zoysiagrass

PHYA, a hyperactive mutant gene (S599A-PHYA), in which a phosphorylation site involved in light-signal attenuation was removed, was transformed into the turfgrasses (Ganesan *et al.*, 2012). Given the economic value of turfgrasses, photo-biotechnology has been used to produce transgenic turfgrass cultivars with improved phenotypic and agronomic traits.

- Shade-tolerant zoysiagrass requiring less irrigation and fertilizer.
- Shade-and herbicide-tolerant zoysiagrass.
- Transgenic zoysia grass with enhanced seed production.
- Transgenic zoysiagrass with improved greenness and increased abiotic stress tolerance.
- Phytochrome B Promotes Rice Chlorophyll Biosynthesis and Chloroplast Development (Jie *et al.*, 2013).

Rice phytochromes are encoded by *PHYA*, *PHYB*, and *PHYC* genes. The photomorphogenesis of phytochrome mutants and wild type (WT) has been used to investigate how phytochromes regulate rice de-etiolation, blooming period, and fertility. The chlorophyll concentrations of wild type and *phyA*, *phyB*, and *phyAphyB* mutants grown under white or red light were compared in this study. *PhyB* senses R to positively regulate chlorophyll production, while *phyA* only appears in the *phyB*-deficient mutant. Analysis of chlorophyll biosynthesis gene expression showed that phytochromes regulate protochlorophyll oxidoreductase A (*PORA*) expression. The studies also showed that *phyB* perceives R to regulate chloroplast development by influencing chloroplast numbers, grana, and membrane systems.

Cryptochromes: Cryptochromes are blue light photoreceptors that contain flavin. They were initially identified in plants but are also present in bacteria and animals. These possess a similar structure to DNA photolyase, but do not exhibit photolyase activity. Cryptochrome 2 (cry2) in *Arabidopsis* largely regulates blooming time and controls photomorphogenesis in response to low-intensity light.

Function of cryptochromes

- Photosensory receptors mediating light regulation of growth and development in plants.
- Flavoproteins that are sensitive to blue light.
- Cryptochromes are involved in the circadian rhythms of plants.

Cryptochrome pathways: CRY-mediate blue light control of gene expression via at least two mechanism (Liu *et al.*, 2011) (fig. 5)

- 1) Light dependent transcription (CIB dependent CRY2 regulation of transcription).
- 2) Light dependent-suppression of proteolysis (CRY-SPA1/COP1 pathway).

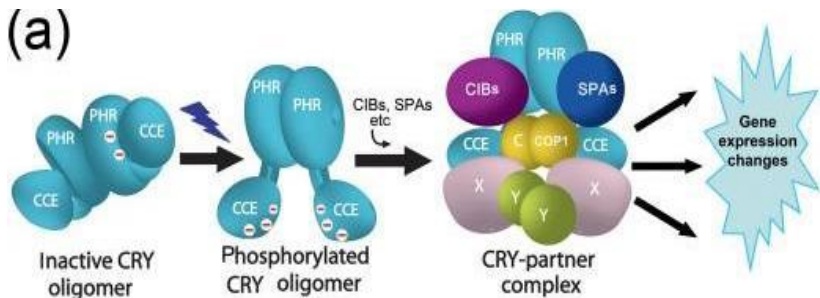


Fig 5a: CRY-mediate blue light control of gene expression

1. **Light dependent transcription (CIB dependent CRY2 regulation of transcription):** CRY-interacting basic helix-loop-helix is the initial protein that interacts with CRY2 and is dependent on blue light. CIB plays a positive role in promoting the initiation of flowers, and this effect is dependent on CRY2. CIB interacts with the chromatin of the promoter DNA of the FLOWERING LOCUS T (FT), which is responsible for encoding a mobile transcriptional regulator. This regulator moves from leaves to the apical meristem and activates the transcription of genes related to floral meristem identity.

2. **Light dependent-suppression of proteolysis (CRY-SPA1/COP1 pathway):** CRY not only directly regulates transcription by interacting with transcription factors, but also indirectly modulates gene expression through post-transcriptional mechanisms by interacting with the SPA1/COP1 complex. CRY1 is responsible for inhibiting the degradation of the bZIP transcription factor long HYPOCOTYL 5 under far-red light, which controls the transcription of genes necessary for de-etiolation responses. This suppression is dependent on COP1. CRY2 facilitates the inhibition of the degradation of CONSTANS (CO), a key transcriptional regulator of floral initiation, by COP1 in response to blue light. CO functions as a positive regulator of blooming in long day plants by stimulating the transcription of the florigen gene (FT). COP1 directly interacts with CO, which enhances the process of ubiquitination of CO leading to its destruction.

In the absence of blue light, COP1 facilitates the decomposition of CO. CRY1 and CRY2 engage in a blue light-dependent interaction with the COP1 interacting protein SPA1. SPA1 functions in a genetic pathway that occurs after CRY1 and CRY2, and is responsible for the inhibition of COP1-mediated degradation of HY5 and CO in response to blue light. SPA1 engages in physical interactions with COP1 in a manner that is reliant on light, and it acts as a positive regulator of COP1. The connection between CRY and SPA1 inhibited the interaction between SPA1 and COP1 in both yeast and plants. In this case, CRY may function as a competitive inhibitor of COP1. Light inhibits the interaction between SPA1 and COP1.

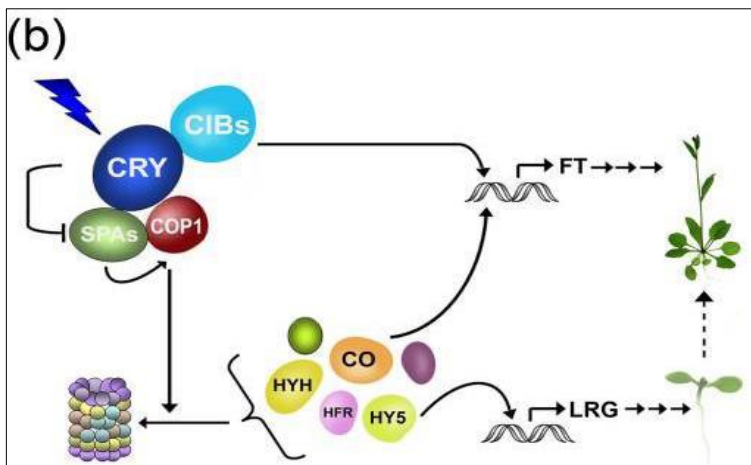


Fig 5b: CRY2-CIBs and CRY-SPA1/COP1 pathway

In their 2018 study, Fantini *et al.* conducted a phenotypic and metabolomic analysis of tomato plants with changed CRY1 and CRY2 expression due to mutations. The researchers specifically investigated the impact of CRY1a and CRY2 on the development of tomato seeds and seedlings. This study aimed to modify the expression of cryptochrome genes by inducing local lesions in the genomes of a tomato cry2 knock-out mutant. Additionally, the study examined the introgression of this mutant into the indeterminate Moneymaker background, as well as the phenotypes of cry1a/cry2 single and double mutants. The cry1a/cry2 mutant exhibited phenotypic characteristics, including elongated hypocotyls (fig. 6a). Both cry1a and cry2 regulate the chlorophyll levels in hypocotyls and cotyledons. Specifically, cry1a is responsible for controlling chlorophyll levels in hypocotyls, while cry2 controls chlorophyll levels in cotyledons. In cry1a/cry2 mutants, there is an increase in chlorophyll concentration in the cotyledons and a decrease in the hypocotyls (fig. 6b).

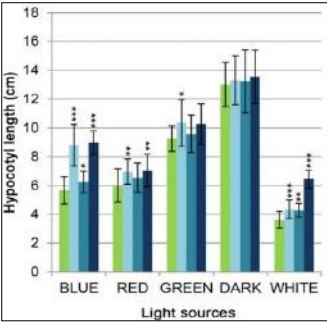


Fig 6a: Tomato wild-type and mutant phenotypes (7-day seedlings)

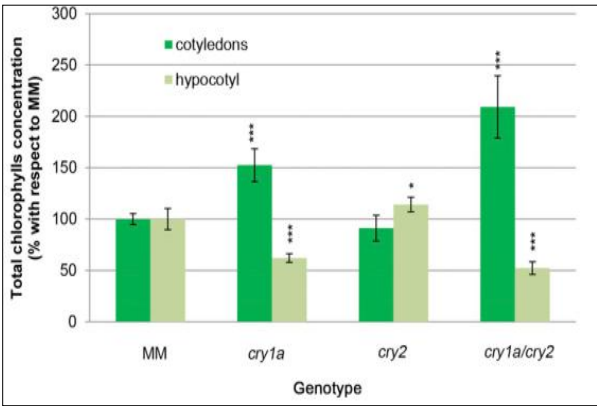


Fig 6b: Relative chlorophyll concentration in cotyledon and hypocotyl (7-day-old seedling, 2μmol m-2s-1 continuous blue light)

Both cry1a and cry2 regulate the concentrations of photosynthetic pigments in leaves, however, cry2 has a more significant function in determining fruit pigmentation. The accumulation of metabolites from the sterol, tocopherol, quinone, and sugar classes varies between cry1a and cry2 leaves and fruits (Giliberto *et al.*, 2005). Both cry1a and cry2 play a significant role in regulating flowering. While single mutants do not exhibit accelerated flowering time, it is evident in the double mutant. The results indicate that cryptochromes play a crucial function in regulating tomato growth and physiology. Manipulating these photoreceptors is a potent method for influencing significant agronomic characteristics, such as the timing of blooming and the quality of fruits.

Phototropins: Phototropins are serine/threonine protein kinases activated by blue-light/UVA radiation. Arabidopsis has two phototropins (phot1 and phot2) with shared as well as specific functions. Phototropins play a key role in the perception of light gradients within plant tissues.

Phototropin promotes

1. Phototropism

Phototropins are localized in or associated with the plasma membrane contains two light sensing light-oxygen voltage (LOV) domain i.e., LOV1 and LOV2. In dark, one FMN molecule is non-covalently bound to each LOV domain. Upon light illumination, the FMN molecule become covalently bound to a cysteine residue in the phototropin molecule forming a cysteine-flavin covalent adduct. The LOV2 domain is primarily responsible for kinase activation in response to blue light (fig. 7). For example Sharma *et al.*, 2014 investigated phot1 mutant tomato plant, which is caused by replacement of arginine to histidine amino acid mutation in the A' α region. When this mutant seedling along with its wild type was subjected to phenotypic characterization phot1 mutant (*Nsp1*) did not show curvature (fig. 7A, indicated in red arrow) whereas wild type showed phototropic movement. *Nps1* seedlings show no phototropic response to 0.1 or 1 $\mu\text{mol m}^{-2} \text{s}^{-2}$ 3-h BL exposure. The white arrow indicates the direction of light (fig. 7B).

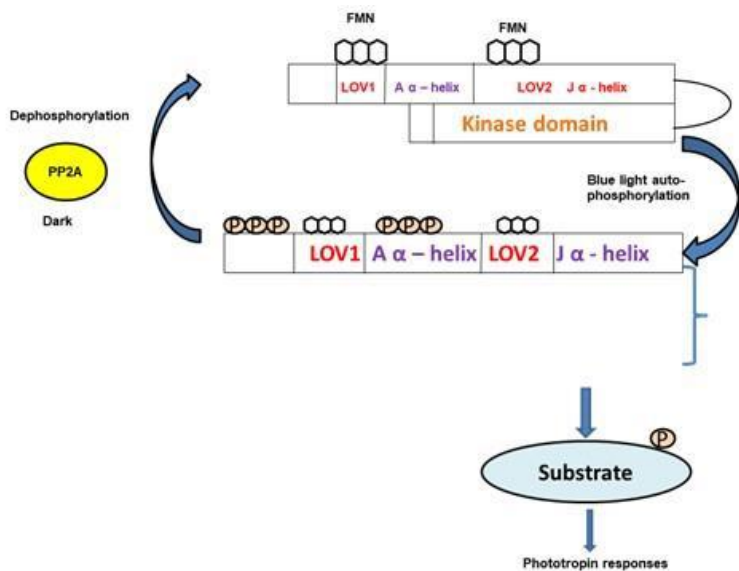
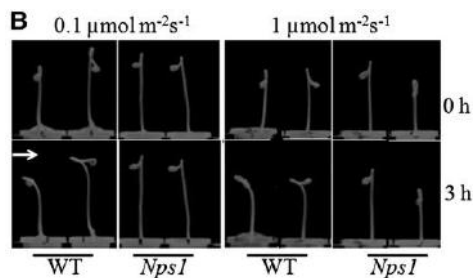


Fig 7: Blue light induced auto-phosphorylation of phototropin



(a)



(b)

Fig 8: Phototropism (A) Non-phototropic migration of tomato. (B) *Nps1* seedlings had no phototropic response to 0.1 or 1 $\mu\text{mol m}^{-2}\text{s}^{-1}$ 3-h BL exposure

Under strong illumination, the chloroplast move to the lateral walls that are parallel to the incident light (avoidance), thus minimizing light absorption and avoiding photo damage (fig. 9b). Phot2 is implicated in the chloroplast avoidance mechanism, although both Phot1 and Phot2 are engaged in both chloroplast accumulation in *Arabidopsis thaliana*. For example,

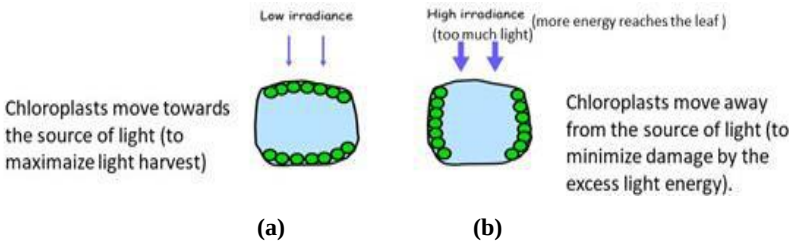
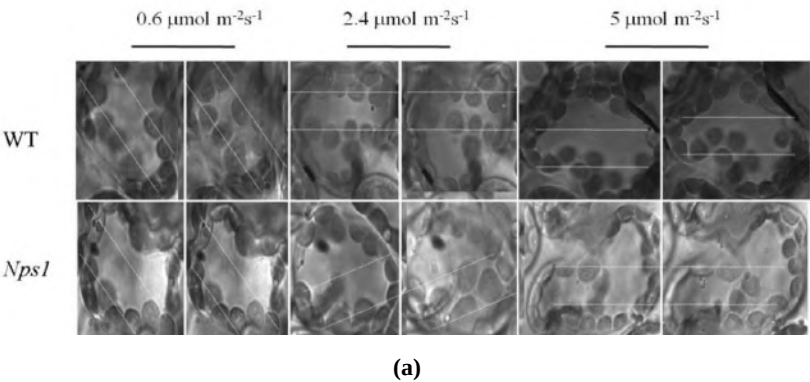
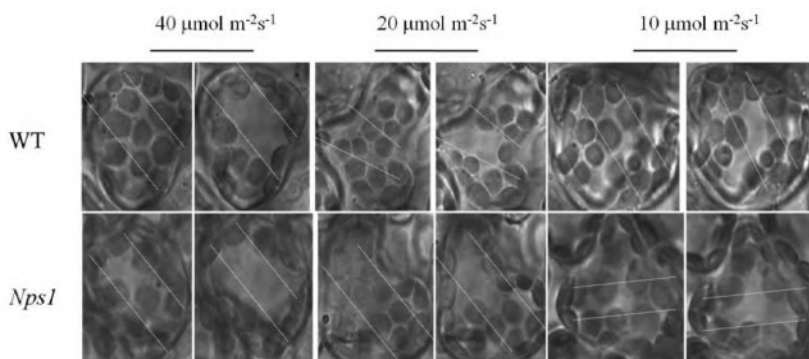


Fig 9: Blue light regulates chloroplast movement in leaf mesophyll cells. (a) Chloroplast accumulation. (b) Chloroplast avoidance

Sharma *et al.*, 2014 observed the migration of chloroplasts in tomato plants, both in the wild type and in a mutant lacking the phot1 gene, in response to blue light (BL). Phot1 is responsible for the chloroplast accumulation response to a wide range of blue light intensities, ranging from 0.1 to 100 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (as shown in figure 10a). On the other hand, Phot2 triggers chloroplast accumulation at light intensities between 2 and 16 $\mu\text{mol m}^{-2} \text{s}^{-1}$, but causes avoidance at intensities higher than 32 $\mu\text{mol m}^{-2} \text{s}^{-1}$. The Nps1 mutant exhibited the chloroplast accumulation response, which was specifically regulated by phot2. In order to confirm whether Nps1 maintains its usual chloroplast avoidance, the leaves were exposed to a range of high-intensity blue light (10-40 $\mu\text{mol m}^{-2} \text{s}^{-1}$) after chloroplast accumulation. In line with the active functioning of phot2, the chloroplasts that had built up in Nps1 moved away from the area that was exposed to radiation (fig.10b).





(b)

Fig 10: Characterization of chloroplast movement in tomato *Nps1* (*phot1* mutant)

Shang *et al.*, 2018 documented the Chloroplast avoidance response of *G. hirsutum* to different intensities of blue light irradiation: 10 (BL10), 30 (BL30), 100 (BL100), or 200 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (BL200). Leaves from cotton plants that were 4 weeks old and had been kept in the dark were removed and exposed to different intensities of blue light over a period of 3 hours using a 5mm opening. Under intense blue light, the chloroplasts relocated to the anticlinal walls, resulting in a pale green appearance in the irradiated area (fig.11a). GhPHOT2-1 is an orthologous protein that is well conserved and shares functional similarities with AtPHOT2. It has the ability to facilitate the same reactions as AtPHOT2 in Arabidopsis. In order to investigate the function of GhPHOT2-1 in the chloroplast avoidance response, we cloned the full length cDNA of GhPHOT2-1 and fused it in-frame with a green fluorescent protein (GFP) gene to create a 35S::The GhPHOT2-1:GFP construct, together with the homozygous transgenic lines in the *phot2* background, were used for subsequent chloroplast avoidance investigation. The leaves were exposed to strong blue light irradiation at a rate of 100 $\mu\text{mol m}^{-2} \text{s}^{-1}$ for a duration of 1 hour. In the irradiated areas of the *phot2* or *phot1 phot2* mutants, no noticeable chloroplast avoidance response was seen. Additionally, the chloroplasts were evenly distributed across the mesophyll cells of these mutants, as shown in figure 11B. Nevertheless, the implementation of the 35S::Introduction of the GhPHOT2-1:GFP construct into *phot2* resulted in the recovery of its chloroplast avoidance response under high light conditions (fig. 11B). We detected the accumulation of chloroplasts at the anticlinal membranes in the leaves of the transgenic lines (C1 & C9) that were exposed to high blue light intensity of 100 $\mu\text{mol m}^{-2} \text{s}^{-1}$, as shown in Figure 11B. The results suggest that GhPHOT2-1 is a conserved ortholog of PHOT2 that regulates the photo-relocation response in

cotton. Chloroplast mobility was hindered in *phot1 phot2*, however the 35S::GhPHOT2-1:GFP *phot1 phot2* transgenic lines were able to restore chloroplast accumulation and avoidance response.

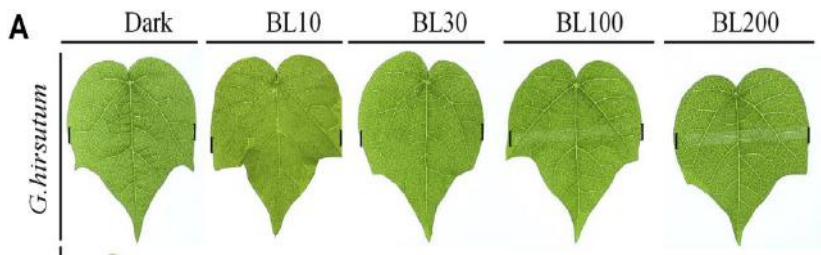


Fig 11a: Chloroplast avoidance response of cotton

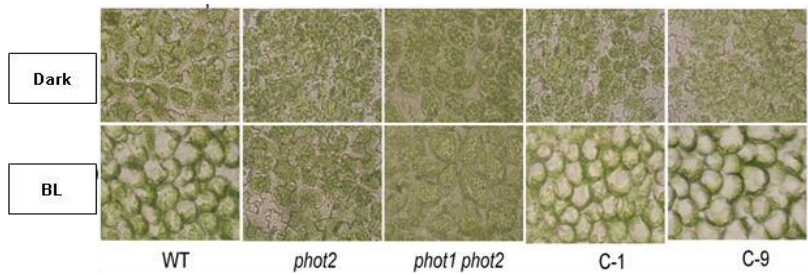


Fig 11b: Characteristics of chloroplast movement in (*phot2*, *phot1 phot2* mutant) in cotton

Conclusion

These photo sensory phosphoprotein photoreceptor involved in plant growth and development. Light-signaling components may offer potential as targets for genetic engineering in biotechnology aimed at improving crop plants. Genetic manipulation of these photoreceptors either alone or in concert with other photoreceptors and signaling pathways favorably places these genes as most suitable candidates for enhancing agronomic traits.

Future prospects

Gap in our knowledge exist regarding the regulation of photoreceptors signaling and the role of photoreceptors in plant development. Transcriptome profiling coupled with proteomic analyses at different stage of plant development under different environmental conditions can contribute to understanding of photoreceptor signaling. Information on the effect of transgenic photoreceptor expressions in controlled environments are available. However, performance under field conditions is still to be evaluated.

References

1. Fatini E, Sulli M, Zhang L, Aprea G, Jimenez-Gomez JM, Bendahmane A, et al. Pivotal role of Cryptochrome 1a and 2 in tomato development and physiology. *Plant Physiol.* 2019;179(2):732-748.
2. Franklin RA, Whitelam GC. Plants phytochromes and shade-avoidance responses in plants. *Ann Bot.* 2005;96(2):169-175.
3. Ganesan M, Han YJ, Bae TW, Hwang OJ, Chandrasekhar T, Shin AY, et al. *Planta.* 2012;236(19):1135-1150.
4. Garg AK, Sawers RJH, Wang H, Kim J, Walker JM, Brutnell TP, et al. Light-regulated overexpression of an Arabidopsis phytochrome A gene in rice alters plant architecture and increases grain yield. *Planta.* 2006;223:627-636.
5. Giliberto L, Perrotta G, Pallara P, Weller JL, Fraser PD, Bramley PM, et al. Tobacco phytochromes: genes, structure and expression. *Plant Cell Environ.* 1997;20(137):678-684.
6. Jie Z, Jin-Jun Z, Ying-Ying W, Jian-Wei G, Xian-Zhi X. Positive regulation of Phytochrome B on chlorophyll biosynthesis and chloroplast development in rice. *Rice Sci.* 2013;20(4):243-248.
7. Kwon CT, Song G, Kim SH, Han J, Yoo SC, An G, et al. Functional deficiency of phytochrome B improves salt tolerance in rice. *Environ Exp Bot.* 2017;148(1):100-108.
8. Liu H, Liu B, Zhao C, Pepper M, Lin C. The action mechanisms of plant cryptochromes. *Cell.* 2011;12(16):684-691.
9. Liu J, Zhang F, Zhou J, Chen F, Wang B, Xie X, et al. Phytochrome B control of total leaf area and stomatal density affects drought tolerance in rice. *Plant Mol Biol.* 2012;78(3):289-300.
10. Mawphlang OIL, Kharshiing EV. Photoreceptor mediated plant growth responses: implications for photoreceptor engineering toward improved performance in crops. *Front Plant Sci.* 2017;8:1181.
11. Podolec R, Ulm R. Photoreceptor-mediated regulation of the COP1/SAP E3 ubiquitin ligase. *Curr Opin Plant Biol.* 2018;45:18-25.
12. Shang B, Zang Y, Zhao X, Zhu J, Fan C, Gua X, et al. Functional characterization of GhPHOT2 in chloroplast avoidance of *Gossypium hirsutum*. *Plant Physiol Biochem.* 2018;135:51-60.
13. Taiz L, Zeiger E, Moller IM, Murphy AS. Plant physiology and development. Sinauer Associates, Inc. Publishing Co., U.S.; 2015. p. 375-377.

Chapter - 6

Illumina Next-Generation Sequencing

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Chapter - 6

Illumina Next-Generation Sequencing

Roshan K V Remesh and Priyadarsini G.

Abstract

Illumina sequencing, a cornerstone of Next-Generation Sequencing (NGS), has revolutionized genomic research by enabling rapid, high-throughput DNA analysis. This technology allows simultaneous sequencing of millions of DNA fragments, dramatically accelerating genetic studies. At its core are flow cells, microscopic glass slides where the sequencing magic happens. The process begins with DNA fragmentation and adapter ligation during library preparation. These fragments then undergo bridge amplification on the flow cell, creating dense clusters of identical DNA. The sequencing-by-synthesis method follows, where fluorescently labeled nucleotides are incorporated into growing DNA strands, emitting signals captured by high-resolution cameras. This massive parallel sequencing generates vast amounts of data, which sophisticated bioinformatics tools then align and analyze to reconstruct the original genome. Illumina sequencing's applications span numerous fields, from evolutionary biology to personalized medicine. It's instrumental in discovering new species, tracking disease outbreaks, unraveling genetic disorders, and guiding targeted cancer therapies. In microbiome research, it helps explore complex microbial communities, while in agriculture, it aids in developing resilient crop varieties. As the technology evolves, becoming faster and more cost-effective, it continues to unlock the secrets within our genetic code. From deciphering human evolution to advancing precision medicine, Illumina sequencing is not just reading DNA - it's reading the blueprint of life itself. This powerful tool offers insights that are transforming our understanding of biology and revolutionizing healthcare, standing as a testament to human ingenuity in our quest to understand the fundamental code shaping all living things.

Keywords: Illumina Sequencing, Next-Generation Sequencing (NGS), genomic research, sequencing-by-synthesis, bioinformatics analysis

Introduction to next-generation sequencing

The rapid advancement and widespread adoption of next-generation sequencing (NGS) technologies have revolutionized the landscape of genomic research, offering a powerful and cost-effective alternative to traditional sequencing methods. These new sequencing techniques have surpassed their original scope, providing researchers with a myriad of applications that allow for a more comprehensive analysis of microbial genomes and eukaryotic microorganisms ^[1, 2, 3]. The historical progression of sequencing technologies, from first-generation to second-generation and now the emergence of third-generation sequencing, has driven a dramatic shift in the scale and depth of genomic inquiries ^[3]. The need for higher throughput sequencing of large genomes at lower cost triggered the development of various second-generation or "next-gen" technologies, which can be broadly categorized into two major groups: sequencing by hybridization and sequencing by synthesis ^[4]. These new sequencing methods have surpassed their original scope and now provide a range of utility-based applications, which allow for a more comprehensive analysis of the structure and content of microbial genomes than was previously possible ^[1]. The challenge of ensuring comparability of data generated from different sequencing platforms remains an ongoing area of focus ^[3]. As the field of next-generation sequencing continues to evolve, the future holds exciting prospects for the integration of this technology with laboratory management and public health surveillance systems, moving towards the goal of performing sequencing directly from clinical samples and enabling a more personalized and targeted approach to disease prevention and treatment ^[3].

The development of these advanced sequencing techniques has been driven by a need to overcome the limitations of traditional Sanger sequencing, which was labor-intensive, time-consuming, and costly, particularly for large-scale projects such as whole-genome sequencing ^[2]. The emergence of next-generation sequencing (NGS) technologies, also known as high-throughput sequencing, has addressed these limitations by offering significantly higher throughput, speed, and scalability ^[1, 5]. These new methods have not only enabled the rapid and cost-effective sequencing of entire genomes, but have also opened up new avenues for exploring genetic diversity, gene expression, and other genomic features at an unprecedented level of detail ^[2, 6].

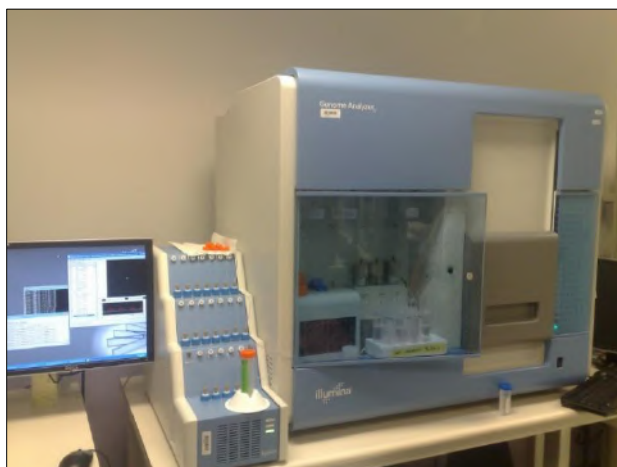
One of the key advantages of next-generation sequencing is its ability to simultaneously analyze multiple DNA or RNA samples in parallel, a process known as massively parallel sequencing ^[5]. This approach allows for the detection of all known and even previously unidentified viruses and other pathogens within a single sample, which has been particularly valuable in the

context of disease diagnosis and surveillance, such as the COVID-19 pandemic [5].

Next-Generation Sequencing (NGS) has revolutionized genomic research with various innovative techniques. These include Lynx therapeutics' massively parallel signature sequencing (MPSS), which allows for high-throughput analysis of gene expression. Polony sequencing, another NGS method, utilizes polymerase colonies for DNA amplification. Pyrosequencing, known for its real-time detection of nucleotide incorporation, has found applications in both research and clinical settings. Illumina (Solexa) sequencing has become widely adopted due to its high accuracy and scalability. The SOLiD (Sequencing by Oligonucleotide Ligation and Detection) system offers a unique approach using DNA ligase. DNA nanoball sequencing employs rolling circle replication to create nano-scale DNA balls for sequencing. Lastly, Helioscope single molecule sequencing allows for direct observation of individual DNA molecules, eliminating the need for DNA amplification. These diverse NGS technologies have significantly expanded our ability to analyze genetic material quickly and comprehensively.

Illumina sequencing

Illumina sequencing, is a type of next-generation sequencing (NGS), a powerful and widely-used technology that has revolutionized the field of genomics and molecular biology. This technology has enabled researchers to rapidly and cost-effectively sequence entire genomes, transcriptomes, and other genetic materials, opening up a wealth of opportunities for scientific discovery and medical advancements.



"Illumina Gene Sequencing machine" by [oncallas] is licensed under CC BY 2.0.

The development of next-generation sequencing techniques has been a significant milestone in the history of DNA sequencing ^[6]. For several decades, Sanger sequencing, a method developed in the 1970s, was the dominant approach for large-scale genome projects ^[2]. However, as the demand for higher throughput and lower costs increased, new sequencing technologies emerged to address these needs ^[3]. Illumina sequencing, in particular, has become a leading platform for next-generation sequencing, offering high accuracy, scalability, and a wide range of applications ^[1]. The technology is based on a cyclic-reversible termination method, where DNA fragments are amplified and sequenced simultaneously on a flow cell, generating millions or even billions of short read sequences in a single run ^[3].

The impact of Illumina sequencing on microbial genomics has been substantial, as it has enabled researchers to study the structure and content of microbial genomes in unprecedented detail ^[1]. From understanding the genetic diversity within a species to identifying new pathogens, Illumina sequencing has become an indispensable tool in the field of microbiology and public health ^[3]. Furthermore, the application of Illumina sequencing has extended beyond microbial genomics, with significant contributions to the understanding of human genetics and the development of personalized medicine ^[7]. As the cost of sequencing continues to decline and the technology becomes more accessible, the potential for Illumina sequencing to transform various fields of biology and medicine is immense.

The HiSeq X Ten Sequencing System, developed by Illumina, has been a game-changer in the industry, enabling the sequencing of an entire human genome in record time and at a cost below \$1000 ^[8]. This breakthrough has significantly expanded the scope of DNA sequencing, making it more accessible and feasible for population-level studies ^[8, 9].

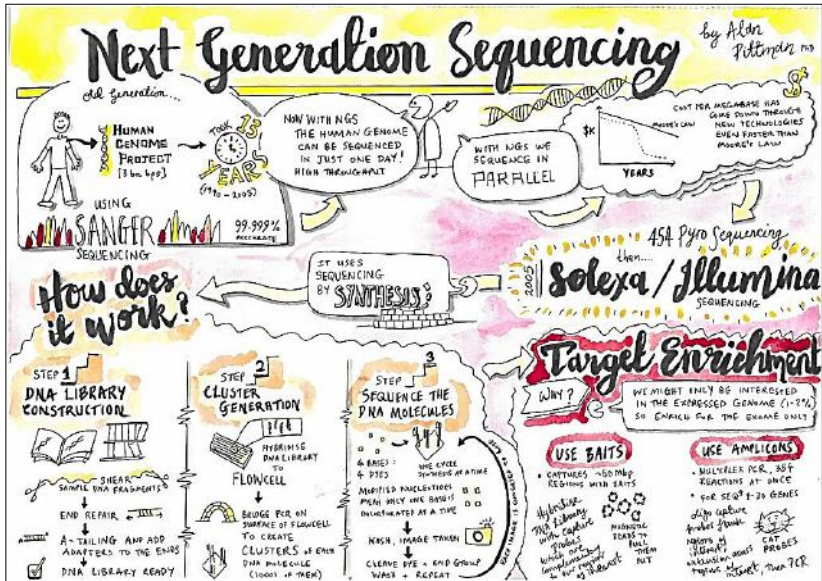
Illumina sequencing principle

Illumina sequencing, also known as sequencing by synthesis (SBS), is a widely adopted next-generation sequencing (NGS) technology that has revolutionized the field of genomics and molecular biology. This powerful technique enables the rapid and cost-effective sequencing of DNA and RNA molecules, providing researchers and clinicians with unprecedented access to genetic information.

The fundamental principle behind Illumina sequencing is the use of reversible terminator-based chemistry coupled with cyclic-array imaging ^[6]. During the sequencing process, DNA fragments are first attached to a solid surface, such as a flow cell, and then amplified through a process called bridge

amplification. This creates millions of identical DNA clusters, each representing a single DNA molecule [9]. The sequencing itself is carried out through a series of cycles, where fluorescently-labeled nucleotides are incorporated into the growing DNA strands. After each incorporation, the DNA strands are imaged, and the identity of the incorporated nucleotide is determined based on the emitted fluorescent signal [4].

However, the technology is not without its challenges, such as the potential for false positive and negative results, incomplete hybridization, and nonspecific binding. Nevertheless, Illumina sequencing has become a cornerstone of modern genomic research, enabling a wide range of applications, from whole-genome sequencing to targeted gene expression analysis and the detection of novel viral sequences.



"Next Generation Sequencing" by jennychamux is licensed under CC BY 2.0.

Illumina sequencing: Flow cells

The core of this technology is the flow cell, a crucial component that allows for the simultaneous sequencing of millions of DNA fragments. Which serves as both a solid support for DNA strands and a core sequencing reactor vessel. This flow cell contains lanes with nano wells, each equipped with numerous adapters on its surface. The flow cell's nanowells play a crucial role in managing DNA fragment distribution and preventing overcrowding.

These nanowells contain oligonucleotides that serve as anchoring points for adapters to attach. The flow cell's design allows it to function as a channel for adsorbing mobile DNA fragments, ensuring that all sequencing reactions take place within this specialized environment. As the DNA fragments from the sequencing library pass through the flow cell, they randomly attach to the lanes. Importantly, these lanes are theoretically independent, with no mutual influence between them.

Once the fragments are secured, cluster generation commences, involving bridge amplification PCR to create approximately a thousand copies of each DNA fragment. The process then moves to sequencing. Illumina flow cells are designed to create a dense, uniform array of DNA clusters, each containing thousands of copies of a single DNA fragment ^[9].

Illumina sequencing: An overview of the basic steps

The basic steps involved in the Illumina sequencing process, including DNA isolation, fragmentation, and analysis.

The Illumina sequencing process begins with the isolation of DNA samples from various sources, such as cells, tissues, or environmental samples ^[7]. The isolated DNA is then fragmented into smaller pieces, typically using mechanical, sonication or enzymatic methods, to prepare the samples for sequencing ^[3]. The fragmented DNA is then subjected to a series of steps, including adapter ligation, library preparation, and cluster generation, which ultimately enable the sequencing of the DNA fragments.

Library preparation of Illumina sequencing

The library preparation process, which involves the conversion of DNA or RNA samples into sequencing-ready libraries, is a crucial step in the Illumina sequencing workflow.

The library preparation process typically begins with the fragmentation of DNA or RNA samples, either through mechanical, transposon mediated, sonication or enzymatic means ^[3]. The fragmented nucleic acids are then subjected to a series of steps, including end-repair, A-tailing, and the ligation of Illumina-specific adapters. These adapters serve as the necessary priming sites for subsequent amplification and sequencing.

Following adapter ligation, the library undergoes PCR amplification to generate sufficient material for cluster generation and sequencing. The number of PCR cycles required can vary depending on the input material and the desired library size ^[10]. Illumina sequencing libraries are typically 400-500 base pairs in length, with the actual insert size ranging from 270-350 base pairs ^[10].

Illumina sequencing: The bridge amplification technique

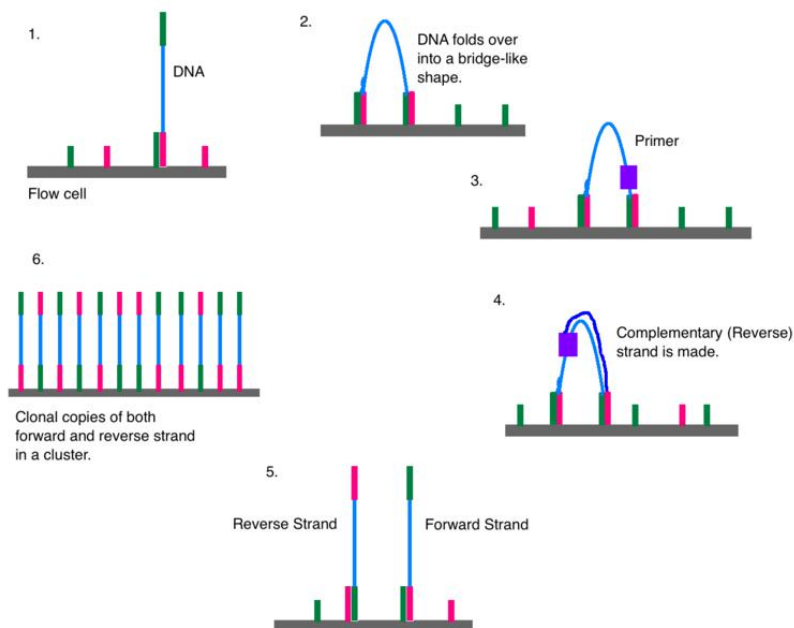
The bridge amplification technique begins with the attachment of adapter-ligated DNA fragments to a lawn of complementary oligonucleotides on a flow cell surface. These adapter sequences are designed to hybridize with the complementary sequences on the flow cell, effectively "bridging" the DNA fragments to the surface.

A crucial component of the Illumina sequencing process is the use of adaptors, which play a vital role in various steps, including library preparation, cluster generation, and sequencing-by-synthesis ^[11]. Illumina adaptors are designed to be compatible with the company's proprietary sequencing technology, which utilizes a reversible terminator-based method ^[12]. These adaptors typically consist of a universal sequence and sample-specific barcodes, enabling the multiplexing of multiple samples in a single sequencing run ^[10]. One of the key properties of Illumina adaptors is their ability to confer high accuracy during the sequencing process. The modular design of the adaptors allows for the optimization of various components, such as the linker sequence and the polymerase responsible for nucleotide binding, to enhance signal quality and reduce the likelihood of errors ^[12].

Bridge PCR is a crucial step in preparing DNA for sequencing. It starts when DNA fragments attach to adapters on a flow cell surface. The goal is to create hundreds of identical DNA strands, including both forward and reverse strands, which is why right and left adapters are used. During this process, DNA polymerase moves along each strand, creating its complementary strand. The original strand is then washed away. The remaining reverse strand bends and attaches to a matching oligo on the flow cell, forming a bridge-like shape. Polymerases then create a new complementary strand, which is identical to the original. This double-stranded DNA is separated, and each strand attaches to the flow cell separately. This process, called bridge amplification, happens thousands of times across the flow cell, creating clusters of identical DNA strands. These clusters amplify the signal intensity, making it strong enough for sequencing. Once cluster generation is complete, the DNA templates are ready for the next step in the sequencing process. This amplification step is crucial in Illumina sequencing, as it generates the necessary signal intensity for accurate base calling during the subsequent sequencing-by-synthesis (SBS) process.

Recent advancements in Illumina sequencing technology have addressed some of the challenges associated with this platform, such as false positive and negative results, incomplete hybridization, and nonspecific binding. The use of molecular beacons, single-stranded oligonucleotide hybridization

probes that form stem-loop structures, has been explored to enhance the specificity and sensitivity of Illumina-based assays ^[12]. Additionally, the combination of isothermal amplification and nanotechnology has opened new avenues for the development of complex nucleic acid nanostructures and hydrogels with diverse applications in biosensing, bioimaging, and nanomedicine ^[13, 14].



Ref: DMLapato

Paired sequences allow the DNA to cling to the flow cell. The strand forms a bridge when it bends over and joins with another section. The reverse strand is synthesized by a primer. The two threads uncurl and separate. Each creates a fresh bridge (amplification of a bridge). A cluster of clones of the forward and reverse strands of DNA is the end product.

Illumina sequencing: Cluster generation

The process of cluster generation as the resultant of bridge amplification is a fundamental aspect of Illumina sequencing, as it lays the foundation for the subsequent sequencing reactions. During this stage, single-stranded DNA molecules are amplified on a solid support, forming dense clusters of identical DNA fragments ^[11]. This amplification step is crucial, as it increases the signal-to-noise ratio, allowing for the detection of individual base incorporations with high accuracy ^[11].

Illumina sequencing: Clonal amplification technique

One of the critical steps in the Illumina Sequencing workflow is the process of clonal amplification, which is essential for obtaining a sufficient quantity of DNA templates for subsequent sequencing.

Clonal amplification in Illumina Sequencing involves the creation of multiple identical copies of each DNA fragment within a sample, a process that is crucial for improving the signal-to-noise ratio and ensuring accurate base calling during the sequencing reaction ^[9]. This technique overcomes the challenges associated with the low initial quantity of DNA obtained from samples, such as those derived from clinical or environmental sources ^[5].

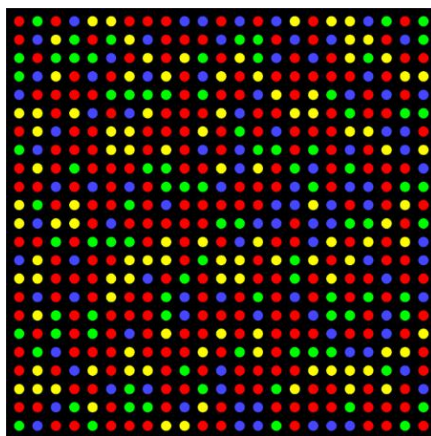
The bridge amplification step resulting in the formation of dense clusters of clonal DNA fragments, each representing a single starting DNA molecule ^[15]. The ability to generate clonal DNA clusters is a critical aspect of Illumina Sequencing, as it allows for the simultaneous sequencing of millions of DNA fragments in parallel, thereby increasing the throughput and efficiency of the overall sequencing process ^[9]. The optimization of the clonal amplification step, including factors such as the choice of library preparation method, DNA fragmentation, and amplification conditions, can have a significant impact on the quality and reliability of the final sequencing data ^[15].

Furthermore, the recent advancements in Illumina Sequencing technology, such as the development of small, scalable platforms with electrical detection of single-base incorporations, have the potential to further improve the accessibility and affordability of this powerful sequencing technique ^[11].

Illumina sequencing: Sequencing

In Illumina's sequencing process, after clonal amplification, only forward strands remain on the flow cell. A primer attaches to these strands, and a polymerase adds a single fluorescently tagged base. This base acts as a temporary blocker, allowing just one addition per round. The machine then uses lasers to excite the fluorescence and records the signal. In the four-color method, each base has a unique emission, while the two-color system uses red, green, a combination (orange), or no color to identify bases. After recording, the fluorescence is quenched and the blocker removed, enabling the next round. The computer converts these signals into sequence data. Once a strand is read, it's washed away, and index primers attach to add index sequences. The strand then forms a bridge, attaching to the flow cell again. A complementary strand is created, they separate, and the forward strand is removed. The process then repeats for the reverse strand, ensuring

comprehensive sequencing of the DNA fragment. One of the breakthrough innovations of Illumina's platform is the steady-state electrical detection of base incorporations ^[11].



"Sequencing flow cell (4 colour)" by Thomas Shafee is licensed under CC BY 4.0.

As the price of sequencing continues to decline, more and more individuals are having their genomes sequenced, which may facilitate better population-level disease subtyping and characterization, as well as individual-level diagnosis and personalized treatment and prevention plans ^[7].

The steady-state electrical detection of base incorporations, a unique feature of the Genapsys platform, has the potential to further streamline and simplify the sequencing process ^[11]. By automating all steps of the sequencing process within a single instrument, the Genapsys approach aims to drive down the cost and complexity associated with large-scale genomic studies ^[11]. As the cost of sequencing declines, the ability to explore the human genome on a population scale will enable comprehensive and sophisticated genetic studies, paving the way for advancements in personalized medicine and population-level disease prevention and treatment ^[8].

Illumina sequencing: Alignment and Data analysis

One of the key challenges in Illumina sequencing data analysis is the alignment of the generated sequence reads to a reference genome or transcriptome ^[16]. The vast amount of sequencing data produced requires sophisticated computational methods and algorithms to accurately map the short reads to their corresponding positions in the reference sequence ^[17]. Advances in bioinformatics tools and data analysis techniques have played a crucial role in unlocking the full potential of Illumina sequencing data ^[16].

These include the development of robust alignment algorithms, efficient data processing workflows, and comprehensive quality control measures to ensure the reliability and accuracy of the generated results ^[17, 18]. By employing parallel processing techniques and implementing robust quality control mechanisms, researchers are able to effectively transform the massive amounts of raw sequencing data into meaningful biological insights ^[17]. These advancements have enabled researchers to tackle a diverse range of applications, from identifying disease-associated genetic variants to unraveling the complexity of the human genome and transcriptome ^[16].

As the field of genomics continues to evolve, the integration of Illumina sequencing data with other high-throughput technologies and the advancement of computational methods will be essential in driving further discoveries and transforming our understanding of biological systems ^[19].

Applications of Illumina sequencing technology

The use of Illumina sequencing technology has been widely explored in various genomic applications ^[20] emphasized the importance of combining next and third-generation sequencing technologies to characterize bacterial model organisms at a detailed level ^[21] introduced a protocol for hybrid assembly of complete bacterial genomes using miniaturized multiplexed Illumina sequencing and PacBio sequencing, reducing the overall per-genome cost of sequencing ^[22]. Discussed the Illumina sequencing protocol and the NovaSeq 6000 system, highlighting the scalability of sequencing outputs for different applications ^[23, 24]. Focused on the accuracy and performance of short-read sequencing in clinical applications and benchmarking taxonomic classifiers for metagenomic diagnostic applications, respectively. They found that short-read sequencing is essential for clinical and basic research, with ^[24] reporting similar or superior performance of Illumina sequencing compared to Nanopore sequencing in certain scenarios ^[25]. Compared the application of long-read Nanopore sequencing to Illumina sequencing in searching for mutations in hypertrophic cardiomyopathy genes, noting a higher sequencing error rate with Nanopore technology ^[26] highlighted the superior resolution of Illumina short-read sequencing compared to ONT-based approaches in extracting MLVA profiles of *Vibrio cholerae* isolates. Overall, the literature review showcases the diverse applications of Illumina sequencing technology in genome assembly, metagenomics analysis, clinical diagnostics, and mutation detection, emphasizing its importance in various research fields.

References

1. Forde B, O'Toole PW. Next-generation sequencing technologies and their impact on microbial genomics. *FEMS Microbiol Lett.* 2013 Jan;12(5):440-53.
2. Nowrousian M. Next-Generation Sequencing Techniques for Eukaryotic Microorganisms: Sequencing-Based Solutions to Biological Problems. *Eukaryot Cell.* 2010 Sep;9(9):1300-10.
3. Besser JM, Carleton HA, Gerner-Smidt P, Lindsey RL, Trees E. Next-generation sequencing technologies and their application to the study and control of bacterial infections. *Clin Microbiol Infect.* 2018 Apr;24(4):335-41.
4. Slatko BE, Gardner AF, Ausubel FM. Overview of Next-Generation Sequencing Technologies. *Curr Protoc Mol Biol.*; 2018 Apr, 122(1).
5. Rahimi H, Salehiabar M, Barsbay M, Ghaffarlou M, Kavetsky T, Sharafi A, *et al.* CRISPR Systems for COVID-19 Diagnosis. *ACS Sens.* 2021 Jan;6(4):1430-45. Doi: 10.1021/acssensors.0c02312.
6. Nelson FK, Snyder M, Gardner AF, Hendrickson CL, Shendure J, Porreca GJ, *et al.* Introduction and Historical Overview of DNA Sequencing. *Curr. Protoc. Mol. Biol.*, 2011 Oct, 96(1).
7. Goldfeder RL, Wall DP, Khoury MJ, Ioannidis JPA, Ashley EA. Human Genome Sequencing at the Population Scale: A Primer on High-Throughput DNA Sequencing and Analysis. *Genet Med.* 2017 May10;186(8):1000-9.
8. Bahassi EM, Stambrook PJ. Next-generation sequencing technologies: breaking the sound barrier of human genetics. *Hum Mol. Genet.* 2014 Aug;29(5):303-10.
9. Shendure J, Aiden E. The expanding scope of DNA sequencing. *Nat Biotechnol.* 2012 Nov;30(11):1084-94. doi: 10.1038/nbt.2421.
10. Kalinin S, González-Prieto M, Scheiblich H, Lisi L, Kusumo H, Heneka MT, *et al.* Transcriptome analysis of alcohol-treated microglia reveals downregulation of beta amyloid phagocytosis. *J Neuroinflammation*, 2018 May, 15(1).
11. Esfandyarpour H, Parizi KB, Barmi MR, Rategh H, Wang L, Paliwal S, *et al.* High accuracy DNA sequencing on a small, scalable platform via electrical detection of single base incorporations. *Cold Spring Harb Symp Quant Biol.*; 2019 Apr.

12. Arslan S, García FJB, Guo M, Kellinger MW, Kruglyak S, Le Vieux JA, *et al.* Sequencing by avidity enables high accuracy with low reagent consumption. *Nat Biotechnol.* 2023 May;42(1):132-8.
13. Yan L, Zhou J, Zheng Y, Gamson AS, Roembke BT, Nakayama S, *et al.* Isothermal amplified detection of DNA and RNA. *Analyst.* 2014 Jan;10(5):970-970.
14. Srivastava PW, Prasad D. Isothermal nucleic acid amplification and its uses in modern diagnostic technologies. *Mol. Biol.;* 2023 May, 13(6).
15. Carretta C, Mallia S, Genovese E, Parenti S, Rontauoli S, Bianchi E, *et al.* Genomic Analysis of Hematopoietic Stem Cell at the Single-Cell Level: Optimization of Cell Fixation and Whole Genome Amplification (WGA) Protocol. *Int. J Mol. Sci.* 2020 Oct;21(19):7366-7366.
16. Koboldt DC, Chiang CWK, Larson DE, Wilson RK, Mardis ER. The Next-Generation Sequencing Revolution and Its Impact on Genomics. *Cell.* 2013 Sep;155(1):27-38.
17. Roy A, Diao Y, Mauceli E, Shen Y, Wu B. Massive genomic data processing and deep analysis. *ACM Trans Biol. Comput. Genomics.* 2012 Aug;5(12):1906-9.
18. El-Metwally S, Ouda O, Helmy M. Next Generation Sequencing Technologies and Challenges in Sequence Assembly. *Methods Mol. Biol.* 2014 Jan.
19. Soon WW, Hariharan M, Snyder M. High-throughput sequencing for biology and medicine. *Genome Biol.;* 2013 Jan, 9(1).
20. Qi W, Colarusso A, Olombrada M, Parrilli E, Patrignani A, Tutino ML, *et al.* New insights on *Pseudoalteromonas haloplanktis* TAC125 genome organization and benchmarks of genome assembly applications using next and third generation sequencing technologies. *Sci. Rep.;* 2019, 9.
21. Derakhshani H, Bernier SP, Marko VA, Surette MG. Completion of draft bacterial genomes by long-read sequencing of synthetic genomic pools. *BMC Genomics;* 2020, 21.
22. Modi A, Vai S, Caramelli D, Lari M. The Illumina sequencing protocol and the NovaSeq 6000 system. In: *Methods Mol. Biol. Humana Press;* 2021, Vol. XXXX.
23. Marin MG, Vargas R, Harris M, Jeffrey B, Epperson LE, Durbin D, *et al.* Genomic sequence characteristics and the empiric accuracy of short-read sequencing; 2021.

24. Govender KN, Eyre DW. Benchmarking taxonomic classifiers with Illumina and Nanopore sequence data for clinical metagenomic diagnostic applications. 2022.
25. Salakhov RR, Golubenko MV, Valiakhmetov NR, Pavlyukova EN, Zarubin AA, Babushkina NP, *et al.* Application of long-read Nanopore sequencing to the search for mutations in hypertrophic cardiomyopathy. *Int. J Mol. Sci.* 2022.
26. Ambroise J, Bearzatto B, Durant JF, Ireng L, Gala JL. On the ability to extract MLVA profiles of *Vibrio cholerae* isolates from WGS data generated with Oxford Nanopore Technologies; 2024.

Chapter - 7

SOLiD Sequencing: A Comprehensive Overview

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Chapter - 7

SOLiD Sequencing: A Comprehensive Overview

Roshan K.V. Remesh and Priyadarsini G.

Abstract

SOLiD (Sequencing by Oligonucleotide Ligation and Detection) is a next-generation sequencing technology that provides distinct advantages in DNA analysis through its unique sequencing-by-ligation approach and colour space encoding system. The methodology includes DNA sample preparation, adapter ligation, bead hybridization, emulsion PCR for clonal amplification, and bead deposition onto glass slides and sequencing. SOLiD's accuracy heavily relies on its adapter properties and ligation process. SOLiD presents specific strengths and limitations regarding read length, accuracy, and throughput compared to other sequencing technologies like Pyrosequencing and Illumina. Understanding these methodological aspects and comparative insights is crucial for researchers in selecting the appropriate sequencing technology for their applications.

Keywords: SOLiD sequencing, next-generation sequencing, DNA analysis, sequencing by ligation, comparative sequencing technologies

Introduction to SOLiD (Sequencing by Oligonucleotide Ligation and Detection) sequencing

SOLiD (Sequencing by Oligonucleotide Ligation and Detection) is a next-generation sequencing (NGS) technology developed by Applied Biosystems (now part of Thermo Fisher Scientific) ^[1, 2]. It is a sequencing-by-ligation (SBL) method that uses a unique approach to determine the sequence of DNA molecules ^[3]. During the sequencing process, a mixture of all four fluorescently labelled and 3'-blocked deoxynucleotides is used ^[3]. The bound label corresponds to two bases rather than one, so the signal derived from a SOLiD run is in a series of colours that represent overlapping dinucleotides, and each colour is directly correlated to a single base ^[3]. This colour-space information is then used to determine the underlying DNA sequence through a reference-based alignment ^[3]. SOLiD sequencing offers several advantages over other next-generation sequencing technologies, including high accuracy,

long read lengths, and the ability to detect rare sequence variants ^[4]. The ligation-based approach also allows for the use of optimized probes that can improve ligation efficiency and accuracy ^[5]. Overall, SOLiD sequencing is a powerful tool for various genomic applications, from molecular cloning to pathogenic gene identification and evolutionary studies ^[5].

Comparison of SOLiD Sequencing with Pyrosequencing and Illumina Sequencing

Among the leading platforms, the Roche/454 FLX, Illumina/Solexa Genome Analyzer, and Applied Biosystems SOLiD Analyzer have become dominant in the market ^[1]. One of the critical innovations in next-generation sequencing is the shift from the traditional Sanger sequencing method to alternative approaches, such as sequencing by hybridization and sequencing by synthesis (SBS) ^[2]. In contrast to the sequencing-by-synthesis methods used by Illumina and 454 platforms, the SOLiD system utilizes a ligation-based strategy, where DNA fragments are ligated to fluorescently labelled oligonucleotide probes ^[5]. This approach has been shown to provide high accuracy and throughput, making it a valuable tool for various genomic applications ^[5]. While the SOLiD system offers distinct advantages, it is important to compare its performance with other sequencing technologies, such as pyrosequencing and Illumina sequencing, to understand its capabilities and limitations ^[1, 6]. Pyrosequencing, pioneered by the Roche/454 platform, is characterized by its ability to generate longer read lengths compared to SOLiD and Illumina ^[1]. This feature can be beneficial for de novo genome assembly and the identification of structural variations ^[6]. On the other hand, the Illumina platform is known for its high-throughput and cost-effectiveness, making it a popular choice for various genomic applications, including microbial community analysis ^[6]. When it comes to the accuracy of the sequencing data, the SOLiD system is often considered to be one of the most accurate among the next-generation sequencing technologies ^[5]. This high accuracy is attributed to the ligation-based approach, which involves the detection of dinucleotide sequences rather than individual nucleotides ^[5]. This platform can generate an astounding 100 million to 1 billion small sequence reads simultaneously, utilizing a unique 2-base encoding method to interpret raw data. While distinct from "sequencing by synthesis" approaches like Roche-454 pyrosequencing and Illumina's Solexa system, this technology has contributed to a dramatic reduction in sequencing costs and a substantial increase in capacity. By 2009, expenses had plummeted to nearly \$0.0001 per base, while daily sequencing output per machine soared to over 5 billion bases.

Methodology

SOLiD sequencing is characterized by its two-base encoding approach, where each fluorescent signal corresponds to a combination of two nucleotides. This method uses DNA ligase to join oligonucleotide probes to the template DNA, enabling the detection of sequences in a "colour space" format. The process involves several key steps:

- 1) **Sample preparation:** Genomic DNA is fragmented using methods such as nebulization, sonication, or restriction enzyme digestion. Adapters are ligated to the ends of these fragments to facilitate amplification and sequencing.
- 2) **Amplification:** The fragments are amplified through emulsion PCR, where they are attached to beads that allow for high-density sequencing.
- 3) **Ligation and Imaging:** A library of octamer probes, each labelled with a different fluorescent dye, is hybridized to the target DNA. DNA ligase then joins the probes to the DNA, and the fluorescent signals are captured to determine the sequence.
- 4) **Data analysis:** The sequences are decoded based on the colours emitted from the fluorescent dyes, allowing for the reconstruction of the original DNA sequence (Garrido-Cardena, Next Gen SOLiD)

DNA Sample preparation methods in SOLiD sequencing

The advent of next-generation sequencing technologies, such as the SOLiD platform, has revolutionized the field of genomics and has provide researchers with powerful tools for DNA sequencing. One critical aspect of the SOLiD sequencing process is the fragmentation of DNA samples prior to library preparation and sequencing. DNA fragmentation is a crucial step in the SOLiD sequencing workflow, as it helps to generate smaller, manageable DNA fragments that can be efficiently amplified and sequenced. Several fragmentation methods have been employed in the context of SOLiD sequencing, each with its own advantages and considerations. Another fragmentation strategy is the use of enzymatic digestion, where specific restriction enzymes are employed to cleave the DNA at predetermined recognition sites ^[5]. This approach can provide a more controlled and reproducible fragmentation pattern, which is particularly important for subsequent library preparation and bioinformatic analysis. Nebulization, a process that utilizes high-velocity air or gas to break down DNA into smaller fragments, has garnered significant attention in the field of sequencing library preparation ^[2, 7]. This technique offers several advantages, including the ability

to generate fragments of desired lengths, the potential to minimize biases associated with enzymatic fragmentation, and the capacity to handle a wide range of input DNA quantities ^[5, 8]. In addition to these traditional methods, the emergence of next-generation sequencing platforms has also introduced new fragmentation techniques. These include methods such as sonication, which uses high-frequency sound waves to physically shear the DNA, and transposase-mediated fragmentation, where a specialized enzyme is used to randomly fragment the DNA ^[2]. One common fragmentation approach is pulsed-field gel electrophoresis, a technique that has been used since the 1970s ^[7]. This method utilizes an oscillating electric field to separate DNA fragments based on their size, allowing for the selection of desired fragment lengths. The development of automated DNA sequencers and fluorescence-based detection has further improved the speed and efficiency of DNA sequencing, including SOLiD-based approaches. The choice of fragmentation method for SOLiD sequencing depends on various factors, including the available sample material, the desired fragment size distribution, and the specific requirements of the downstream library preparation and sequencing workflows.

Adapter properties of solid sequencing

Solid sequencing is a next-generation sequencing technology that utilizes adapter-mediated amplification and ligation strategies to prepare DNA libraries for high-throughput sequencing ^[9]. The adapters used in Solid sequencing play a crucial role in the overall sequencing process and have several important properties. One key property of Solid sequencing adapters is their ability to facilitate emulsion PCR. These adapters contain sequences complementary to oligonucleotides on the surface of beads, allowing for clonal amplification of DNA fragments ^[10]. This process creates millions of beads, each carrying thousands of copies of a single DNA fragment. Another important feature of Solid adapters is their incorporation of specific sequences that enable base calling during the sequencing process. The adapters include a universal priming site for sequencing reactions and a unique barcode sequence for sample identification in multiplexed experiments ^[11]. The design of Solid adapters also allows for mate-pair sequencing, which can provide valuable information about the relative positions of sequenced fragments within a genome. This is achieved through the use of internal adapters that connect distant genomic regions ^[12]. Furthermore, the adapters used in Solid sequencing are engineered to minimize bias during library preparation and amplification steps. This is crucial for maintaining accurate representation of the original DNA sample in the sequencing output ^[13].

Efficient adapter ligation into DNA fragments

A crucial step in solid-phase sequencing is the ligation of adapter sequences to the DNA fragments, which enables the subsequent amplification and immobilization of the fragments on a solid support. The technique requires direct deposition and amplification of DNA fragments on a solid surface, eliminating the need for separate cloning and purification steps. Furthermore, the modular design of sequencing-by-avidity platforms, as discussed in ^[14], enables the optimization of individual components, including the adapter ligation step, to improve overall performance. Probe design and optimization, as highlighted in ^[5], can also play a crucial role in enhancing the ligation efficiency and accuracy of solid-phase sequencing.

Solid matrix utilization in solid-phase sequencing

SOLiD is a sequencing-by-ligation approach that relies on the use of beads to support the DNA library during the sequencing process ^[3]. In this technique, fragmented DNA is attached to beads, allowing for parallel processing and increased throughput compared to traditional Sanger sequencing ^[4]. The DNA library preparation for SOLiD sequencing involves the attachment of DNA fragments to oligos on the surface of the beads that can interact with the adaptors at one end of the DNA fragments. The concentration of the DNA fragment attaching to the beads may be limited in such a way that one DNA fragment attaches to one bead, followed by emulsion PCR to generate clonal amplification of the DNA on each bead ^[15]. This approach allows for the generation of a high density of DNA templates on the beads, which is essential for the high-throughput nature of SOLiD sequencing ^[3]. The beads serve as a solid support, enabling the immobilization of the DNA library and facilitating the successive ligation reactions that are at the core of the SOLiD sequencing method ^[16]. The beads provide several key properties that contribute to the platform's performance. Firstly, the attachment of DNA fragments to beads allows for parallel processing of a large number of samples, significantly increasing the throughput of the sequencing process ^[4]. Additionally, the miniaturized reaction volumes facilitated by bead-based systems help to reduce reagent consumption and lower the overall cost of sequencing ^[4].

To further enhance the capabilities of the SOLiD platform, researchers have explored strategies for the sequential attachment of multiple oligonucleotides to the beads ^[16]. This approach, known as "dual addressing," allows for the incorporation of two different address oligos on each bead, which can improve the decoding process and enable the use of larger numbers of bead types ^[16].

Hybridization to the Beads in SOLiD Next-Generation Sequencing

The SOLiD system employs a technique known as sequencing by ligation, which is distinctly different from the more common sequencing-by-synthesis approach used in other platforms such as Illumina and Ion Torrent ^[1]. In the SOLiD system, the DNA samples are first prepared by attaching them to beads, a process known as hybridization [4]. This hybridization step is crucial as it allows for the parallel processing of multiple DNA fragments, thereby increasing the throughput and efficiency of the sequencing process. The optimization of the probes used in the hybridization step is critical to the success of the SOLiD platform ^[5]. The accumulation of 3' terminal deoxyinosines in the sequencing primers can potentially reduce the fluorescence intensity and ligation efficiency, which is not desirable ^[5]. The incorporation of optimized probes can significantly improve the ligation efficiency and accuracy in sequencing by ligation ^[5]. Overall, the hybridization of DNA samples to beads and the optimization of probes used in the sequencing by ligation process are critical components of the SOLiD next-generation sequencing platform ^[3, 4, 5].

Emulsion PCR & clonal amplification in SOLiD next-generation sequencing

A key aspect of these technologies is the ability to generate clonal DNA amplicons through a process known as emulsion PCR (emPCR). ^[17] Emulsion PCR, also referred to as solid-phase PCR, involves the amplification of DNA templates that are immobilized on a solid surface, such as beads or a flow cell. ^[18] This approach enables the parallel amplification of millions of individual DNA molecules, significantly increasing the throughput of the sequencing stage ^[19]. In the context of SOLiD sequencing, emulsion PCR is a crucial step in the library preparation process. DNA samples are first fragmented and ligated with adapters, then encapsulated in water-in-oil emulsions along with PCR reagents and magnetic beads coated with oligonucleotides complementary to the adapters ^[20]. The emulsion is then subjected to thermal cycling, allowing for the amplification of the DNA templates on the bead surface. After the PCR process, the emulsion is broken, and the bead-bound, single-stranded DNA fragments are enriched and prepared for sequencing on the SOLiD platform ^[20]. The use of emulsion PCR in SOLiD sequencing, and other next-generation sequencing platforms, has several advantages over traditional cloning-based approaches. The ability to generate millions of clonal amplicons in parallel, combined with the high-throughput nature of the sequencing process, enables the rapid and cost-effective analysis of complex genomic samples, such as those found in environmental or clinical settings. ^[5]

Furthermore, the immobilization of the DNA templates on a solid surface, such as beads or a flow cell, allows for the efficient and accurate detection of the incorporated nucleotides during the sequencing-by-ligation process.

Efficient Removal of Empty Polystyrene Beads from Clonally Amplified Beads

Polystyrene beads have become an essential tool in the field of molecular biology, serving as a versatile platform for various applications, including clonal amplification and subsequent sequencing. However, the presence of empty beads, devoid of DNA, can significantly impact the efficiency and accuracy of downstream analysis. However, due to the Poisson distribution of DNA targets and beads, a significant proportion of the beads may remain empty, without any DNA fragments. To address this issue, developed a two-step approach that leverages the unique properties of Adaptor 1 and Adaptor 2. In the first step, Adaptor 1 is designed to selectively bind to the empty beads, leaving the beads containing clonally amplified DNA fragments unaffected. This step effectively enriches the population of beads with clonal DNA clusters, reducing the proportion of empty beads ^[15]. In the second step, Adaptor 2 is introduced, which is designed to recognize and bind to the clonal DNA fragments on the enriched beads. This step allows for the selective capture and recovery of the beads with clonally amplified DNA, while the remaining empty beads are discarded ^[20]. By implementing this two-step approach, a significant improvement in the efficiency of removing empty polystyrene beads from the pool of clonally amplified beads, ensuring a more accurate and reliable downstream analysis.

Bead Deposition into the Glass Slides

This process begins with the preparation of the glass slide surface to facilitate bead attachment, followed by the application of a bead solution containing millions of DNA-coated microspheres ^[10]. To ensure an even distribution of beads and optimal sequencing efficiency, centrifugation or other physical methods are often employed ^[21]. After deposition, excess beads and solution are carefully removed to prevent overcrowding and potential interference with sequencing signals ^[22]. The final step may involve fixing the beads in place to maintain their position throughout the sequencing process ^[23]. The precise execution of this bead deposition step is crucial for achieving high-quality sequencing data, as it directly impacts the density and distribution of DNA templates available for analysis on the SOLiD platform.

Sequencing process

Sequencing by Ligation

SOLiD sequencing technology operates differently from traditional methods by utilizing DNA ligase and short oligonucleotides instead of DNA polymerase and nucleotides^[21]. This technique involves multiple cycles where ligation, detection, and cleavage are repeatedly performed. The error-correcting properties of the SOLiD platform have been improved with the introduction of the Exact Call Chemistry in 2010^[24]. This new chemistry alters the way the sequence information is encoded, allowing for more accurate read generation by correcting for mistakes in the measurements.

Key steps in the ligation chemistry used in SOLiD (Sequencing by Oligonucleotide Ligation and Detection) next generation sequencing:

- 1) **Primer hybridization:** A universal primer is hybridized to the adapter sequence on the DNA template.
- 2) **Probe ligation:** A set of fluorescently labelled 8-mer probes compete for ligation to the primer. The first two bases are determined by the probe sequence, while the rest are degenerate.
- 3) **Detection:** The fluorescent signal from the successfully ligated probe is detected, identifying the two-base encoding at that position.
- 4) **Cleavage:** After imaging, the last three bases and the fluorescent dye are cleaved off, leaving a 5-mer on the growing strand.
- 5) **Cycle repetition:** Steps 2-4 are repeated multiple times to extend the sequenced length.

Probe design

SOLiD uses a set of fluorescently labelled 8-mer probes. The first two bases of these probes are known and determine the specificity of the ligation, allowing specific interaction with the target sequence (2 nucleotides). Following these, three nucleotides are universal and can bind with any of the four nucleotides. The last three bases are degenerate (can be any nucleotide), and bound with a fluorescent dye, and are cleaved off after ligation. Thus, the probe can attach to the target with five nucleotides, but specifically with only the first two. The colour of the dye attached at the last three nucleotides depends on the different permutations of the first two nucleotides. With the first two specific nucleotides, 16 different dinucleotide permutations are possible. That is, any one of the four nucleotides can occupy the first position, likely any of the four can occupy at the second position ($4 * 4 = 16$). Therefore, the colour of the fluorophore attached is dependent on the first two dinucleotides^[24].

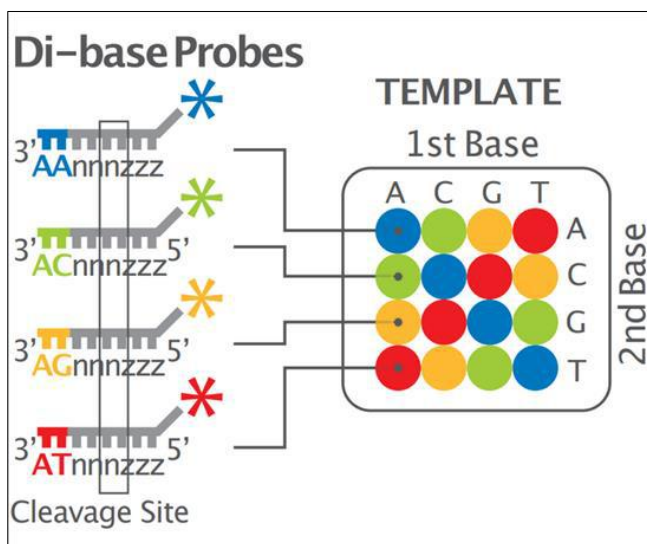


Fig: https://commons.wikimedia.org/wiki/File:Twobase_encoding_scheme.png

Probe ligation

In this sequencing process, a universal primer attaches to the adapter sequence on the template DNA. Multiple probes then compete for the opportunity to bind to this primer. The probe that forms a perfect match with the template is joined to the primer and later ligate through the action of DNA ligase. At this point, a fluorescent signal in the probe is observed and captured by a detector, revealing the identity of two specific bases in the sequence. Additionally, the highly efficient chemical cleavage and removal of the final three bases along with the fluorescent marker of the ligated probes is key to generating a 5' phosphate that can participate in the next round of ligation ^[25]. This cycle of events is repeated numerous times, gradually building up the complete DNA sequence. This method allows for efficient and accurate determination of the genetic code. The two-base encoding of the SOLiD probes, where each color represents a specific dinucleotide, differentiates it from other sequencing technologies that typically detect individual nucleotides ^[3]. The reference-based alignment of the color-space data to the base-space is the most efficient way to translate the SOLiD sequencing output ^[3].

Multiple rounds

After completing a round of ligation cycles, the synthesized strand is removed. A new primer, offset by one base (n-1) from the previous round, is then annealed. This process is repeated for a total of five rounds to ensure

every position in the sequence is covered^[26]. Five rounds of primer reset are completed for each sequence tag. Through the primer reset process, each base is interrogated in two independent ligation reactions by two different primers. For example, the base at read position 5 is assayed by primer number 2 in ligation cycle 2 and by primer number 3 in ligation cycle.

Colour space encoding

Instead of reading individual bases (A, T, G, C), SOLiD reads two adjacent bases at a time. This unique two-base encoding system called "colour space," where each color represents a combination of two adjacent bases, providing a built-in error-checking mechanism. Each two-base combination is assigned a specific colour. There are four colours used, typically represented as blue, green, yellow, and red.

Encoding scheme: The colour assignments follow a specific pattern:

- Blue: AA, CC, GG, TT
- Green: AC, CA, TG, GT
- Yellow: AG, GA, TC, CT
- Red: AT, TA, GC, CG



Sequencing process: As the sequencing proceeds, it generates a series of colours representing the sequence of two-base combinations along the DNA strand.

Error-checking mechanism

This system provides built-in error checking because each base is essentially interrogated twice. For example, if you have three consecutive bases "ACG", you would get two colour calls:

- A-C would give one colour
- C-G would give another colour

If there's an error in one of the colour calls, it will be inconsistent with the adjacent colour, making it easier to detect and potentially correct.^[27]

Decoding

To convert the colour space data back into the actual DNA sequence, you need to know the identity of the first base. From there, each subsequent colour in the sequence determines the next base.

Data analysis

Colour space data is converted to nucleotide sequence through "base calling"^[28]

SOLiD Software

Following SOLiD sequencing, the initial color-coded sequence is collected. Using a double-base coding matrix, the original base sequence can be decoded from the colour sequence, as the colour coding of each base affects the decoding of the next. Incorrect colour coding leads to decoding errors. BioScope, a SOLiD data analysis program, provides a framework for transcriptome analysis, ChIP-Seq, and resequencing. The reference base sequences are first converted into color-coded sequences for subsequent data analysis.

This two-base encoding system offers several advantages

The use of improved sequencing techniques allows for better discrimination between sequencing errors and true polymorphisms, thereby increasing the accuracy of genetic analysis. Each base is effectively sequenced twice, which boosts confidence in identifying true genetic variations. Additionally, aligning colour space data to a reference genome can be more straightforward and accurate than traditional base space alignment. However, this method presents challenges in data analysis, as many standard bioinformatics tools are designed for base space rather than colour space data.

Drawbacks of SOLiD Sequencing

One of the primary drawbacks of SOLiD sequencing is its reliance on ligation-based chemistry, which can be less efficient and more error-prone compared to the synthesis-based approaches used in other next-generation sequencing platforms^[29]. The ligation process can be sensitive to factors such as probe optimization, which can impact the accuracy and efficiency of the sequencing reactions^[5]. Additionally, the accumulation of 3' terminal deoxyinosines in the sequencing primers may reduce the fluorescence intensity and ligation efficiency, which can further contribute to inaccuracies

in the sequencing data ^[5]. Another limitation of SOLiD sequencing is its inherent difficulty in resolving homopolymeric regions, where a sequence of the same nucleotide is repeated ^[29]. This is a common issue across many next-generation sequencing technologies, as they tend to struggle with accurately determining the number of consecutive identical bases in such regions ^[2]. Furthermore, while the SOLiD platform was designed to improve accuracy through its unique colour-space encoding scheme, it can still generate sequence context-specific errors that cannot be entirely mitigated by increasing the number of reads ^[2]. Despite these drawbacks, SOLiD sequencing remains a valuable tool in the field of genomics and has found applications in a variety of settings, including molecular cloning, pathogenic gene identification, and evolutionary studies ^[5].

Applications of SOLiD Sequencing

SOLiD sequencing is a next-generation sequencing technology that has numerous applications in genomics and molecular biology. This method is characterized by its high accuracy and ability to generate large amounts of data, making it suitable for various research and clinical applications.

Whole genome sequencing

SOLiD sequencing is used for whole genome sequencing, allowing researchers to obtain comprehensive genetic information from various organisms. Its high throughput and accuracy make it ideal for sequencing complex genomes, including those of plants and animals, as well as in human genomics for personalized medicine ^[30].

Transcriptome analysis

This technology is employed in RNA sequencing (RNA-seq) to analyse gene expression profiles. By sequencing cDNA libraries, researchers can quantify RNA levels, identify novel transcripts, and understand alternative splicing events, which are crucial for studying gene regulation and function ^[31].

Targeted resequencing

SOLiD sequencing can be utilized for targeted resequencing of specific genomic regions, such as exomes or disease-associated genes. This application is particularly valuable in clinical settings for identifying mutations linked to genetic disorders and cancers ^[32].

Epigenomics

The technology is also applied in epigenomic studies to investigate DNA methylation patterns and histone modifications. Understanding these

epigenetic changes is essential for elucidating mechanisms of gene regulation and their implications in diseases ^[33].

Metagenomics

SOLiD sequencing facilitates metagenomic studies, allowing researchers to analyze the genetic material from environmental samples. This application is crucial for understanding microbial diversity, ecology, and the roles of microorganisms in various ecosystems ^[34].

Comparative genomics

Through comparative genomics, SOLiD sequencing enables the comparison of genetic material across different species. This can provide insights into evolutionary relationships and the genetic basis of phenotypic differences ^[35].

Cancer genomics

In cancer research, SOLiD sequencing is used to identify somatic mutations, copy number variations, and other genomic alterations that drive tumorigenesis. This information is vital for developing targeted therapies and understanding cancer biology. ^[36]

References

1. Zhang J, Chiodini R, Badr A, Zhang G. The impact of next-generation sequencing on genomics. Elsevier BV. 2011 Mar;38(3):95-109. <https://doi.org/10.1016/j.jgg.2011.02.003>
2. Slatko BE, Gardner AF, Ausubel FM. Overview of Next-Generation Sequencing Technologies. Wiley-Blackwell, 2018 Apr, 122(1). <https://doi.org/10.1002/cpmb.59>
3. Goodwin S, McPherson JD, McCombie WR. Coming of age: ten years of next-generation sequencing technologies. Nature Portfolio. 2016 May;17(6):333-51.
4. Ansorge W. Next-generation DNA sequencing techniques. Elsevier BV. 2009 Apr;25(4):195-203. <https://doi.org/10.1016/j.nbt.2008.12.009>
5. Pu D, Chen J, Qian X, Xiao P. Probe optimization for sequencing by ligation. Oxford University Press. 2014 Dec;157(5):357-64.
6. Degnan PH, Ochman H. Illumina-based analysis of microbial community diversity. Springer Nature. 2011 Jun;6(1):183-94.
7. Brassard É, Turmel C, Noolandi J. Pulsed field sequencing gel electrophoresis. Wiley. 1992 Jan;13(1):529-35. <https://doi.org/10.1002/elps.11501301109>

8. Rahimi H, Salehiabar M, Barsbay M, Ghaffarlou M, Kavetsky T, Sharafi A, *et al.* CRISPR Systems for COVID-19 Diagnosis. American Chemical Society. 2021 Jan;6(4):1430-45.
9. Mardis ER. Next-generation DNA sequencing methods. *Annu Rev Genomics Hum Genet.* 2008;9:387-402.
10. Shendure J, Ji H. Next-generation DNA sequencing. *Nat Biotechnol.* 2008;26(10):1135-45.
11. Metzker ML. Sequencing technologies-the next generation. *Nat Rev Genet.* 2010;11(1):31-46.
12. Van Dijk EL, Auger H, Jaszczyszyn Y, Thermes C. Ten years of next-generation sequencing technology. *Trends Genet.* 2014;30(9):418-26.
13. Quail MA, Smith M, Coupland P, Otto TD, Harris SR, Connor TR, *et al.* A tale of three next generation sequencing platforms: comparison of Ion Torrent, Pacific Biosciences and Illumina MiSeq sequencers. *BMC Genomics.* 2012;13(1):341.
14. Arslan S, García FJB, Guo M, Kellinger MW, Kruglyak S, Le Vieux JA, *et al.* Sequencing by avidity enables high accuracy with low reagent consumption. *Nature Portfolio.* 2023 May;42(1):132-8. <https://doi.org/10.1038/s41587-023-01750-7>
15. Kalinin S, González-Prieto M, Scheiblich H, Lisi L, Kusumo H, Heneka MT, *et al.* Transcriptome analysis of alcohol-treated microglia reveals downregulation of beta amyloid phagocytosis. *BMC Genomics.* 2018 May, 15(1).
16. Steinberg-Tatman G, Huynh M, Barker D, Zhao C. Synthetic Modification of Silica Beads That Allows for Sequential Attachment of Two Different Oligonucleotides. American Chemical Society. 2006 Apr;17(3):841-8.
17. Ma Z, Lee RW, Li B, Kenney P, Wang Y, Erikson J, *et al.* Isothermal amplification method for next-generation sequencing. *Proc. Natl. Acad. Sci. USA.* 2013 Aug;110(35):14320-3.
18. Palanisamy R, Connolly AR, Trau M. Considerations of Solid-Phase DNA Amplification. American Chemical Society. 2010 Mar;21(4):690-5.
19. Mitra RD. In situ localized amplification and contact replication of many individual DNA molecules. *Nucleic Acids Res.* 1999 Dec;27(24):34e-34.

20. Du L, Simons JF, Srinivasan M, Jarvie T, Taillon BE, Egholm M. 454 Sequencing: The Next Generation Tool for Functional Genomics. Wiley. 2008 Jun;447-76.
21. Valouev A, Ichikawa J, Tonthat T, Stuart J, Ranade S, Peckham H, *et al.* A high-resolution, nucleosome position map of *C. elegans* reveals a lack of universal sequence-dictated positioning. *Genome Res.* 2008 Jul;18(7):1051-63.
22. Rothberg JM, Leamon JH. The development and impact of 454 sequencing. *Nat Biotechnol.* 2008 Oct;26(10):1117-24.
23. McKernan KJ, Peckham HE, Costa GL, McLaughlin SF, Fu Y, Tsung EF, *et al.* Sequence and structural variation in a human genome uncovered by short-read, massively parallel ligation sequencing using two-base encoding. *Genome Res.* 2009 Sep;19(9):1527-41.
24. Massingham T, Goldman N. Error-correcting properties of the SOLiD Exact Call Chemistry. *BMC Bioinformatics*, 2012 Jun, 13(1). <https://doi.org/10.1186/1471-2105-13-145>
25. Pandey V, Nutter RC, Prediger E. Applied Biosystems SOLiD™ System: Ligation-Based Sequencing. In: *Next Generation Genome Sequencing: Towards Personalized Medicine*. Humana Press; 2008. p. 29-42.
26. Huang YF, Chen SC, Chiang YS, Chen TH, Chiu KP. Palindromic sequence impedes sequencing-by-ligation mechanism. *BMC Syst. Biol.*; 2012, 6(2).
27. Ondov BD, *et al.* An alignment algorithm for bisulfite sequencing using the Applied Biosystems SOLiD System. *Bioinformatics.* 2010 Aug;26(15):1901-2.
28. Sasson A, Michael TP. Filtering error from SOLiD Output. *Bioinformatics.* 2010 Mar;26(6):849-50.
29. Bruijns SR, Maesela M, Foushee JA, Wise R. The effect of prehospital transportation on outcomes in patients with traumatic brain injury: A systematic review and meta-analysis. *Prehosp Emerg Care.* 2018 Nov-Dec;22(6):750-7.
30. Koboldt DC, Steinberg KM, Larson DE, Wilson RK, Mardis ER. The Next-Generation Sequencing Revolution and Its Impact on Genomics. *Cell.* 2013 Sep;155(1):27-38. <https://doi.org/10.1016/j.cell.2013.09.006>
31. Korfhage C, Fricke E, Meier A, Geipel A, Baltes M, Krüger N, *et al.* Clonal rolling circle amplification for on-chip DNA cluster generation. *Nucleic Acids Res*; 2017 Jan, 2(1).

32. Garrido-Cardenas JA, Garcia-Maroto F, Alvarez-Bermejo JA, Manzano-Agugliaro F. DNA Sequencing Sensors: An Overview. *Sensors (Basel)*. 2017 Mar;17(3):588.
33. Next Gen SOLiD DNA Sequencing Method Explained. YouTube.
34. Morozova O, Marra MA. Applications of next-generation sequencing technologies in functional genomics. *Genomics*. 2008 Nov;92(5):255-64.
35. Ion Torrent and SOLiD Sequencing Techniques. Slideshare. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5375874/>

Chapter - 8

Biochemical Characterization of Maize (*Zea mays* L.) Genotypes under Different Irrigation Circumstances

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Chapter - 8

Biochemical Characterization of Maize (*Zea mays* L.) Genotypes under Different Irrigation Circumstances

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Abstract

Corn (*Zea mays* L.) is a substantial crop with extensive agricultural and economic reputation all over the world. With increasing anxieties over water insufficiency and climate change, understanding the responses of maize plants under water stress circumstances is crucial to evolve drought-tolerant cultivar (s). In current study, change in diverse biochemical traits, i.e., chlorophyll, malondialdehyde (MDA), hydrogen peroxide (H_2O_2), peroxidase, glutathione reductase (GR), and catalase activities, among 12 parental inbred lines along with 66 hybrids and two controls under irrigated and partial irrigated circumstances were investigated. Under irrigated situations, the chlorophyll content was highest in the parental inbred line IL8, intimately followed by IL6, while IL5 showed the lowest content. MDA levels were significantly higher in the parental line IL8 and hybrid IL1×IL6, whereas IL5 and IL3×IL11 revealed the lowest levels. H_2O_2 content was found to be higher in the parental line IL5 and hybrid IL8×IL12, whereas IL4 and IL2×IL5 displayed the lowest levels. Peroxidase activity was highest in IL7 and hybrid IL1×IL7, whilst IL6 and IL4×IL7 disclosed the lowest activity. Glutathione reductase action was found maximum in IL1 and IL9×IL12, whereas IL6 and hybrid IL1×IL3 displayed the lowest activity. Catalase activity was highest in IL8 and IL2×IL10, while IL4 and IL2×IL6 demonstrated the lowest activity under irrigated conditions. Under partial irrigated circumstances, almost similar trends were evident for the most of the parameters, with slight variations in the expression levels. Markedly, the drought-tolerant genotypes demonstrated higher chlorophyll content, peroxidase, glutathione reductase, and catalase expressions, while drought-sensitive genotypes unveiled elevated MDA levels and H_2O_2 content. Phylogenetic analysis exposed five major clusters, signifying significant variability in different biochemical profiles among the genotypes. The heat

map analysis supported the credentials of distinct expression patterns of biochemical traits, contributing to our understanding of the genotypic responses to varying irrigation conditions. These results deliver valuable visions for maize breeding programmes intended to breed drought-tolerant cultivar (s) with enhanced antioxidant defenses and tolerant to drought stress.

Keywords: Maize, chlorophyll, catalase, glutathione reductase (GR), hydrogen peroxide (H₂O₂), malondialdehyde (MDA), peroxidase

Introduction

Corn (*Zea mays*) classified under the family Poaceae, was initially described by Carl Linnaeus ^[1]. It is a diploid plant with a genome size of 2357 MB. It is a monoecious crop, as it has separate male and female flowers on the same plant and depend on cross-pollination for reproduction. The origin place is Mexico and firstly domesticated about 10000 years ago by the indigenous people of Mexico ^[2, 3]. It is also supposed to have originated in Northern Guatemala. Maize, being a major global crop, grips significant economic significance owing to its multipurpose uses in different areas, for instance food production, industrial materials, and animal feed ^[4-7]. Nevertheless, the growth and development of corn plants are suggestively prejudiced by an array of abiotic factors, including drought, salinity, and high temperature ^[8, 5, 6, 7]. Among these stresses, maize plants display heightened sensitivity and susceptibility to water stress compared to most other cereal crops ^[9]. The obtainability of water theatres a decisive part in defining the productivity and nutritional quality of maize, making drought stress a main anxiety ^[10].

Drought attitudes as a protruding abiotic stressor that deeply influences plant growth, development, and productivity ^[11-19]. With the predicted consequences of global warming and population growth, future situations encompass falling water resources and an expansion of arid and semiarid areas ^[20-21]. Subsequently, inspecting the mechanisms underlying plant adaptation and tolerance to drought, along with their capacity for post-water shortage retrieval, undertakes main significance in modern research activities ^[22-28].

Understanding the plant's capability to survive under drought situations requires an investigation into its different morpho-physiological ^[29, 5-6, 12, 14, 5-6] biochemical ^[30-37, 13, 17, 18, 21, 7] and molecular ^[38, 15, 12, 19, 20, 6] adaptations. Furthermore, exploring mechanisms related to both tolerance and recovery during rehydration is critical ^[11]. Plant tolerance to water deficit involves the

aptitude to uphold vital functions under uncomplimentary water circumstances and speedily restore water status and functions upon rewatering ^[12-20]. Current research highlights the significance of the recovery phase, as it deeply effects succeeding plant growth and development ^[39]. The assessment of plant tolerance to drought stress includes the identification of specific physiological features that are crucial for drought tolerance ^[40]. This pointer is commonly used in efforts to select drought-tolerant cultivars or decide tolerance levels ^[40].

Drought stress has been experimented to decrease chlorophyll content, leading to disruptions in photosynthesis ^[41]. Furthermore, it persuades oxidative stress, characterized by the generation of reactive oxygen species (ROS), which can cause membrane lipid peroxidation, protein degradation, DNA fragmentation, and eventually cell death ^[42]. These detrimental effects on cellular processes, including cell division, elongation, and differentiation, donate to restricted plant growth and reduced yields ^[43]. Maize displays numerous mechanisms to respond to drought stress, including redox regulation and osmotic regulation ^[44]. Redox regulation primarily comprises the activity of enzymes such as superoxide dismutase (SOD), peroxidase (POD), and catalase (CAT), which work to alleviate oxidative damage by clearing ROS and confirming the normal functioning of cells under drought stress circumstances ^[45].

Plants have established multifaceted antioxidant defense mechanisms to guard cells against the destructive influences of reactive oxygen species (ROS). Main enzymes involved in scavenging ROS and protecting plants from oxidative damage include catalase (CAT), superoxide dismutase (SOD), glutathione reductase (GR) and peroxidase (POD) ^[46-47]. These enzymes play critical roles in the detoxification and neutralization of ROS, contributing to the overall antioxidative capacity of plants. By proficiently removing ROS, these antioxidant enzymes help alleviate oxidative stress and uphold cellular homeostasis. Earlier investigations have verified that the actions of superoxide dismutase (SOD), peroxidase (POD), and catalase (CAT) display an increase under moderate drought situations, whereas they decrease under severe drought circumstances ^[48]. Remarkably, studies have stated that starch biosynthesis in maize leaves donates to sustaining leaf growth and photosynthesis under drought stress situations ^[49]. A decrease in membrane stability designates the incidence of lipid peroxidation instigated by reactive oxygen species (ROS) ^[50]. The content of chlorophyll (Chl.) in plants has a positive impact on photosynthetic rate ^[44]. Under water stress circumstances, a diminution in chlorophyll content assists as a suggestive

characteristic of oxidative stress and can be ascribed to chlorophyll photo-oxidation and degradation ^[51-52]. Shortfall irrigation meaningly diminishes chlorophyll a and b, total chlorophyll content, and the chlorophyll a/b ratio in plants ^[53]. The chlorophyll stability index (CSI), which is calculated as the ratio of chlorophyll levels under stress and normal circumstances, helps as a vital pointer for screening genotypes with abiotic stress tolerance. Substantial alterations and higher CSI values designate genotypes that display tolerance to abiotic stress ^[54]. The objective of current study was to assess the influence of water scarcity on the physiological and biochemical activities of maize inbred lines and their hybrids, concentrating on their characteristics and tolerance under deficit irrigation circumstances.

Materials and Method

Plant material and growth conditions

A total of 12 maize inbred lines were acquired from Sam Higgonbottom Agriculture Science and Technology University, Prayagraj, U.P., India (Table 1). A half diallel analysis technique as suggested by Jinks and Hayman ^[55], was used to develop 66 F₁ hybrids. A drought-tolerant HKI1105 and drought-susceptible HKI1128, cultivars were employed as control. A total of 80 genotypes, comprising of 12 parents, 66 F₁ hybrids, along with two controls, were raised in a randomized complete block design (RCBD) with two replications during Kharif 2020-21. Each genotype was planted in two rows measuring 4 meters in length, with a spacing of 60 cm between rows and 20 cm between plants. Drought stress was executed by withholding irrigation starting from 10 days preceding to flowering, and irrigation was resumed once the soil moisture grasped the momentary wilting point. At the reproductive phase after planting, leaf samples were collected from the second leaf from the top to analyze chlorophyll, malondialdehyde, H₂O₂, peroxidase, glutathione reductase, and catalase contents.

Table 1: List of inbred lines with their parentage utilized in investigation

S. No.	Lines	Parentage	Source
1.	IL-1	CM-13	SHUATS, Allahabad
2.	IL-2	CML-193	SHUATS, Allahabad
3.	IL-3	CML-439	SHUATS, Allahabad
4.	IL-4	NBPGR-36417	SHUATS, Allahabad
5.	IL-5	NBPGR-36417 X NBPGR-33000	SHUATS, Allahabad
6.	IL-6	(103) NBPGR-36548 × (97) NBPGR-36407	SHUATS, Allahabad
7.	IL-7	DMR-N 21 × NBPGR-32809	SHUATS, Allahabad
8.	IL-8	LM- 13 × NBPGR-31899	SHUATS, Allahabad

9.	IL-9	CML-224-1 × NBPGR-32809	SHUATS, Allahabad
10.	IL-10	NBPGR-36550 × NBPGR-36407	SHUATS, Allahabad
11.	IL-11	KL- 153237 × VL- 1016536	SHUATS, Allahabad
12.	IL-12	CML- 161 × VL- 1056	SHUATS, Allahabad

Biochemical estimation of maize genotypes

Chlorophyll content (mgcm⁻²)

Total chlorophyll was estimated as per method described by Arnon *et al.* [56]. Total chlorophyll content was assessed after 60 days after sowing. A fresh 100mg random leaf sample from each row from field after 60 days sowing was collected. Then leaf sample was crumpled finely in 10 ml 80% Acetone and transferred into falcon tube. Then centrifuged for 15 minutes at 10000 rpm and the green supernatant was transferred into fresh 15ml falcon tube. Readings were taken in a spectrophotometer at 645 nm, 663 nm and 470nm using a Spectrophotometer (UV-visible-160A, Shimadzu). Chlorophyll a and b concentrations was calculated by using the method of Arnon [56]. Total chlorophyll was the summation of chlorophyll a and b concentrations, and chlorophyll a/b ratio was the ratio between the concentration of chlorophyll a divided by the concentration of chlorophyll b. The chlorophyll stability index (CSI) was determined as described by Koleyoreas [57] using following equation:

$$\text{CSI \%} = [\text{Total chlorophyll content (stress)}/\text{Total chlorophyll content (control)}] \times 100$$

Malondialdehyde (MDA) test (Lipid peroxidation Assay)

The method described by Heath and Packer [58] was used for estimating the lipid peroxidation in terms of malondialdehyde (MDA) by thiobarbituric acid (TBA) content. Around 25 mg leaf sample was taken and crushed in fine powder. Then 500 microliters of 0.1% trichloro-acetic acid were added and vortexed and centrifuged at 10,000 rpm for 10 min. Around 100 microliters of supernatant were taken in an Eppendorf tube and 200 microliters 0.5% thiobarbituric acid was added. Then reaction mixture was heated for 95 °C for 30 minutes and rapidly kept at -80 °C for 2 minutes to stop the reaction. After 2 minutes contents come at room temperature and centrifuged at 10000 rpm for 10 min and supernatant was taken to take absorption at 532 nm in UV Spectrophotometer. The MDA content was expressed as $\mu\text{mol g}^{-1}$ FW.

Estimation of hydrogen peroxide (H₂O₂, mmol g⁻¹ FW)

Hydrogen peroxide was determined by using the protocol suggested by Alexieva *et al.* [59]. A 25.0 mg leaf sample was crushed into a fine powder.

Around 500 microliters of 0.1% trichloro-acetic acid were added and vortexed and centrifuged at 10000 rpm for 10 min. Hundred microliters of supernatant were taken in an Eppendorf tube and 200 microliters of 0.5% TBA was added. The reaction mixture was heated at 95 °C for 30 minutes and rapidly kept at -80 °C for 2 minutes to stop the reaction. After 2 minutes allowed to come at room temperature and centrifuged it at 10000 rpm for 10 min and supernatant was taken for reading at 532 nm absorption.

Peroxidase activity (units/mg protein)

Peroxidase activity assay was estimated as per the protocol described by Kar and Mishra ^[60]. Enzyme extraction was performed by homogenizing 100 mg leaf sample of normal and water-logged plants in 5.0 ml, 0.1M phosphate buffer (pH 6.4). The crude extract was centrifuged at 10,000 rpm for 20 minutes at 4 °C. Supernatant was stored at 4 °C till the enzymatic action was accomplished. Reaction mixture was prepared by adding 4.6 ml 0.1M phosphate buffer (pH 6.4), 0.2 ml pyrogallol (50 µM) and 0.1 ml 50 µM H₂O₂ and 0.1 ml enzyme extract. Mixture was incubated at 25 °C for 5 minutes. Then 0.5 ml 5.0 per cent H₂SO₄ was added to terminate the reaction. Absorbance was measured at 420 nm with the help of spectrophotometer.

Glutathione reductase activity (GR)

Glutathione reductase activity was measured following the method of Smith *et al.* ^[61]. The reagents used were 50 mM potassium phosphate buffer (pH 7.6) comprehending 1 mM EDTA, 5 mM NADPH, 6 mM 5,5'-dithio-bis (2-nitrobenzoic acid) [DTNB] and 0.2 mM oxidized glutathione (GSSG). To make the reaction mixture to determine GR activity, diluted enzyme extract (25 p1), DTNB (250 p1), 1mM EDTA (20 p1) and 0.2 mM GSSG (100 p1) were added in 175 p1 of 50 mM potassium phosphate buffer (pH 7.6). To initiate GR activity 50 p1 of 5 mM NADPH added to the reaction mixture. Absorbance was taken using spectrophotometer at 412 nm was followed at 15 s interval up to 2 min.

Catalase activity (units/mg protein)

Catalase activity was determined according to the protocol given by Aebi ^[62]. The 100 mg leaf sample was collected from normal and water-logged plants and samples were homogenized in 5.0 ml of 0.1M phosphate buffer (pH 6.4). The crude extract was centrifuged at 10,000 rpm for 20 minutes at 4 °C. The enzyme extract was stored at low temperature until completion of enzyme assay. The enzymatic activity was assayed by taking 2.6 ml, 0.1M phosphate buffer (pH 6.4), 0.1 ml enzyme extract and 0.1 ml

and 1.0 percent H_2O_2 . The reaction mixture was mixed speedily at room temperature. A blank was prepared also in which 0.1M phosphate buffer (pH 6.4) was added in reaction mixture in place of enzyme extract. The absorbance of the reaction mixture was read instantly at 230 nm with the UV Spectrophotometer at an interval of 15 second were noted for 2 minutes.

Statistical analysis of biochemical traits of maize genotypes

Statistical analysis was performed to analyze biochemical parameters employing the NTSYS pc software (version 2.02) ^[63]. The dendrogram and heat map were generated to envisage the relationships between the diverse biochemical traits. To construct the dendrogram, the Unweighted Pair Group Method with Arithmetic Mean (UPGMA) algorithm was used ^[64].

Results and Discussion

Biochemical analysis of the inbred lines and their hybrids

The mean performances of drought tolerance indexes viz., chlorophyll, malondialdehyde (MDA), and H_2O_2 content along with peroxidase, glutathione reductase and catalase activities were assessed in elite eighty maize genotypes subjected to drought treatment at the reproductive stage. Wide discrepancy was observed for all investigated biochemical traits during drought stress. The descriptive statistics for the phenotypes related to drought stress are presented in the Table 2.

Table 2: Biochemical parameters of the inbred lines, their hybrids and controls under irrigated and partial irrigated circumstances

Genotype	Chlorophyll Content		Malondialdehyde Content		H ₂ O ₂ Content		Peroxidase		Glutathione		Catalase	
	I	PI	I	PI	I	PI	I	PI	I	PI	I	PI
IL1	0.74	0.71	12.5	9.7	64.4	42.4	25.8	14.1	0.37	0.36	10.4	7.20
IL2	0.75	0.70	13.5	8.4	57.5	38.8	29.8	19.3	0.32	0.30	9.80	7.10
IL3	0.77	0.70	12.7	9.7	64.3	47.6	28.4	16.3	0.28	0.24	9.80	6.50
IL4	0.75	0.69	15.8	10.2	38.2	33.9	29.2	14.8	0.30	0.27	8.80	6.90
IL5	0.69	0.62	11.6	8.5	68.2	45.4	22.7	14.4	0.33	0.27	9.30	7.80
IL6	0.88	0.80	15.9	9.4	56.5	44.3	19.1	17.1	0.22	0.19	9.60	7.30
IL7	0.76	0.71	15.5	10.3	50.7	43.5	31.9	13.2	0.32	0.28	10.70	7.60
IL8	0.81	0.70	16.4	9.8	62.3	46.5	26.7	15.5	0.25	0.20	11.90	7.10
IL9	0.84	0.79	14.5	10.4	45.4	38.6	29.4	14.7	0.34	0.29	10.70	6.50
IL10	0.89	0.80	12.5	9.7	58.7	45.5	24.2	15.1	0.31	0.28	11.20	6.90
IL11	0.76	0.75	13.5	8.4	54.3	39.6	31.9	15.9	0.29	0.24	10.80	7.80
IL12	0.77	0.71	12.7	9.7	56.4	40.5	25.5	13.9	0.33	0.29	9.60	7.30
IL1 × IL2	0.80	0.76	12.9	10.3	42.1	38.5	21.3	18.9	0.30	0.26	10.70	7.80
IL1 × IL3	0.82	0.79	15.2	9.5	53.3	37.9	28.9	15.7	0.21	0.18	11.20	6.30
IL1 × IL4	0.87	0.80	13.5	10.7	56.2	49.3	29.6	20.1	0.26	0.22	10.80	7.80
IL1 × IL5	0.78	0.71	11.7	9.2	63.6	38.6	30.9	15.9	0.26	0.21	9.60	6.50
IL1 × IL6	0.76	0.73	17.4	10.3	56.2	36.7	28.9	16.4	0.29	0.22	10.90	7.90
IL1 × IL7	0.77	0.72	15.1	10.4	59.4	41.1	31.8	18.7	0.32	0.28	10.20	6.30
IL1 × IL8	0.73	0.69	15.8	14.3	55.3	53.5	30.4	19.6	0.30	0.27	10.20	8.30

IL1 × IL9	0.78	0.71	16.6	14.6	55.7	46.9	27.6	17.2	0.28	0.21	10.80	7.80
IL1 × IL10	0.76	0.71	15.7	10.3	69.6	45.3	29.4	17.3	0.30	0.26	10.30	7.10
IL1 × IL11	0.74	0.69	12.8	9.6	66.4	44.4	27.8	19.1	0.27	0.21	11.7	7.70
IL1 × IL12	0.75	0.69	15.9	10.9	55.7	47.7	31.4	16.7	0.33	0.28	10.30	7.20
IL2 × IL3	0.77	0.70	14.6	10.2	55.7	49.4	27.3	17.2	0.34	0.29	10.40	7.10
IL2 × IL4	0.75	0.72	15.3	10.2	60.7	47.3	30.6	19.4	0.28	0.24	9.80	6.50
IL2 × IL5	0.69	0.60	12.7	9.6	53.7	45.6	21.8	16.6	0.36	0.31	9.80	6.90
IL2 × IL6	0.88	0.80	10.9	7.8	48.9	46.8	25.8	17.3	0.29	0.22	8.80	7.80
IL2 × IL7	0.76	0.75	10.8	9.9	53.9	47.1	24.2	16.5	0.25	0.20	9.3	7.3
IL2 × IL8	0.81	0.75	11.2	8.5	51.6	46.9	28.3	17.7	0.28	0.22	9.6	7.6
IL2 × IL9	0.84	0.79	12.9	8.2	52.5	47.9	26.9	19.7	0.32	0.29	10.9	8.2
IL2 × IL10	0.89	0.82	12.2	9.4	59.5	42.1	31.4	16.9	0.29	0.22	11.9	7.8
IL2 × IL11	0.76	0.71	13.3	7.9	48.3	40.7	26.5	16.4	0.31	0.28	10.7	6.5
IL2 × IL12	0.77	0.7	12.5	8.4	49.8	46.2	24.8	15.3	0.28	0.22	11.2	7.9
IL3 × IL4	0.76	0.71	13.7	7.2	66.4	40.1	26.4	14.4	0.30	0.27	10.8	6.3
IL3 × IL5	0.77	0.7	13.2	9.9	46.9	39.8	27.1	17.8	0.30	0.26	9.6	8.3
IL3 × IL6	0.71	0.69	12.8	7.4	55.7	50.2	20.5	17.5	0.26	0.22	10.9	7.8
IL3 × IL7	0.78	0.72	12.7	8.3	50.5	42.7	25.1	17.1	0.24	0.19	10.2	7.1
IL3 × IL8	0.76	0.7	13.8	7.3	65.5	47.3	20.3	18.5	0.23	0.19	10.2	7.7
IL3 × IL9	0.74	0.69	12.6	8.9	48.1	42.4	23.1	19.6	0.23	0.21	10.8	7.2
IL3 × IL10	0.75	0.7	10.9	7.5	64.8	39.1	30.9	16.3	0.28	0.23	10.3	7.1
IL3 × IL11	0.77	0.72	10.3	7.3	54.6	35.7	26.6	16.7	0.26	0.22	11.7	6.5
IL3 × IL12	0.75	0.73	10.5	8.2	57.8	36.3	23.7	16.6	0.24	0.20	10.3	6.9

IL4 × IL5	0.69	0.62	12.2	9.1	54.3	34.9	23.3	16.2	0.23	0.19	10.4	7.8
IL4 × IL6	0.88	0.81	13.6	8.4	53.1	38.1	24.9	14.1	0.29	0.21	9.8	7.3
IL4 × IL7	0.76	0.72	12.7	7.2	57.9	42.2	18.4	15.9	0.32	0.27	9.8	7.6
IL4 × IL8	0.81	0.78	17.4	10.3	52.5	43.8	25.7	17.2	0.27	0.22	8.8	6.5
IL4 × IL9	0.84	0.8	15.1	10.4	54.2	45.5	24.2	18.6	0.27	0.21	9.3	7.9
IL4 × IL10	0.80	0.76	15.8	14.3	68.5	51.8	24.2	16.9	0.25	0.21	9.6	6.3
IL4 × IL11	0.82	0.79	16.6	14.6	61.4	39.8	21.6	16.6	0.30	0.25	10.9	8.3
IL4 × IL12	0.87	0.82	15.7	10.3	52.9	46.6	29.2	16.9	0.24	0.20	11.9	7.8
IL5 × IL6	0.78	0.74	12.8	9.6	55.4	48.1	23.7	16.6	0.23	0.18	10.7	7.1
IL5 × IL7	0.76	0.71	15.9	10.9	51.9	42.1	23.9	17.5	0.26	0.21	11.2	7.7
IL5 × IL8	0.77	0.72	14.6	10.2	57.1	50.5	21.5	16.3	0.34	0.30	10.8	7.2
IL5 × IL9	0.73	0.69	15.3	10.2	52.4	40.8	23.2	17.5	0.26	0.21	9.6	7.1
IL5 × IL10	0.78	0.71	12.7	9.6	50.4	42.9	23.8	17.1	0.29	0.22	10.9	6.5
IL5 × IL11	0.76	0.74	10.9	7.8	60.8	48.1	24.4	16.6	0.32	0.28	10.2	6.9
IL5 × IL12	0.74	0.69	10.8	9.9	63.4	44.6	20.1	17.3	0.28	0.22	10.2	7.8
IL6 × IL7	0.75	0.71	11.2	8.5	59.7	36.8	20.6	15.3	0.34	0.30	10.8	7.3
IL6 × IL8	0.77	0.72	12.9	8.2	61.3	45.3	23.1	15.5	0.30	0.27	10.3	7.6
IL6 × IL9	0.75	0.71	12.2	9.4	68.4	49.3	24.9	16.9	0.30	0.26	11.7	6.5
IL6 × IL10	0.69	0.62	13.3	7.9	51.9	44.7	22.1	17.3	0.26	0.22	10.3	7.9
IL6 × IL11	0.87	0.8	12.5	8.4	60.7	42.2	19.9	12.5	0.28	0.21	10.4	6.3
IL6 × IL12	0.76	0.72	13.7	7.2	67.7	45.9	22.3	15.4	0.37	0.32	9.8	8.3
IL7 × IL8	0.81	0.76	13.2	9.9	59.6	45.5	24.5	16.6	0.35	0.30	9.8	7.8
IL7 × IL9	0.84	0.8	12.8	7.4	61.4	38.1	22.4	15.7	0.31	0.25	8.8	7.1

IL7 × IL10	0.89	0.81	12.7	8.3	66.2	38.7	19.1	17.1	0.31	0.26	9.3	7.7
IL7 × IL11	0.76	0.71	13.8	7.3	60.7	47.8	24.9	16.4	0.26	0.21	9.6	7.2
IL7 × IL12	0.77	0.7	12.6	8.9	70.8	42.9	23.8	16.2	0.24	0.20	10.9	7.1
IL8 × IL9	0.76	0.72	10.9	7.5	59.4	46.5	25.3	15.6	0.26	0.21	11.9	6.5
IL8 × IL10	0.77	0.73	10.3	7.3	62.6	45.3	22.5	16.4	0.30	0.28	10.7	6.9
IL8 × IL11	0.71	0.68	10.5	8.2	70.1	41.1	22.5	15.7	0.29	0.24	11.2	7.8
IL8 × IL12	0.80	0.76	12.2	9.1	76.9	47.7	30.5	16.1	0.28	0.22	10.8	7.3
IL9 × IL10	0.82	0.75	17.4	10.3	73.1	46.1	22.2	16.7	0.27	0.21	9.6	7.6
IL9 × IL11	0.87	0.81	15.1	10.4	44.2	38.3	20.1	14.2	0.26	0.22	10.9	6.5
IL9 × IL12	0.78	0.74	15.8	14.3	58.3	41.7	22.2	16.4	0.40	0.35	10.2	7.9
IL10 × IL11	0.76	0.72	16.6	14.6	50.5	44.3	23.1	15.8	0.37	0.31	10.2	6.3
IL10 × IL12	0.77	0.71	15.7	10.3	60.8	43.3	28.2	15.8	0.34	0.29	10.8	8.3
IL11 × IL12	0.73	0.69	15.5	10.3	57.8	43.4	23.5	13.4	0.34	0.28	10.3	7.8
HKI 1105	0.78	0.70	16.4	9.8	55.7	45.4	28.5	14.5	0.35	0.30	11.7	7.1
HKI 1128	0.76	0.71	14.5	10.4	58.7	46.5	38.7	15.7	0.26	0.24	10.3	7.7
SEm	1.023	1.659	1.081	1.236	1.728	1.965	2.68	2.541	0.196	1.235	1.641	1.234
CV	6.02	5.23	7.5	8.56	6.32	3.45	5.36	4.53	3.36	5.23	4.23	6.32

Chlorophyll content

Chlorophylls have been precisely selected as “pigments of life” due to their central role in living systems accountable for garnering sunlight and converting its energy into biochemical energy important for life on earth. It is one of the main chloroplast components in photosynthesis and had a positive correlation with the photosynthetic rate. Reductions in chlorophyll contents under deficit irrigation circumstances can be considered a typical symptom of oxidative stress that causes pigment photo-oxidation and chlorophyll degradation. The chlorophyll contents *i.e.*, Chl. a, Chl. b, and total Chl are very viable parameters that indicate water stress conditions ^[44]. Deficit irrigation leads to the inhibition of photosynthesis by damaging the photosynthetic device ^[65], which leads to a decrease in photosynthetic pigments and a reduction in the ingesting of energy and carbon required for chlorophyll synthesis inside the plants ^[66].

The parental inbred line IL10 confirmed the highest chlorophyll content (0.89 ± 0.02) closely tracked by IL6 (0.88 ± 0.02), IL9 (0.84 ± 0.02), and IL8 (0.81 ± 0.02). Contrariwise, IL5 showed the lowest chlorophyll content (0.69 ± 0.02). While, genotypes *viz.*, IL1 (0.74 ± 0.02), IL2 (0.75 ± 0.02), and IL4 (0.75 ± 0.02) displayed intermediate chlorophyll contents under irrigated circumstances. Likewise, under partial irrigated situations, inbred IL10 upheld the highest chlorophyll content (0.80 ± 0.02), followed by IL6 (0.80 ± 0.02) and IL9 (0.79 ± 0.02). In contrast, IL5 displayed the lowest chlorophyll content (0.62 ± 0.02), whereas IL4 (0.69 ± 0.02), IL2, IL3 and IL8 (0.70 ± 0.02) disclosed somewhat higher but analogous levels.

Regarding hybrids under irrigated circumstances, the IL2 $\times$ IL10 hybrid demonstrated a remarkably high chlorophyll content (0.89 ± 0.02), along with IL7 $\times$ IL10 (0.89 ± 0.02), IL2 $\times$ IL6 (0.88 ± 0.02), and IL4 $\times$ IL6 (0.88 ± 0.02). Whereas, the hybrid IL4 $\times$ IL5 displayed the lowest chlorophyll content (0.69 ± 0.02), and likewise, IL6 $\times$ IL10 (0.69 ± 0.02), IL3 $\times$ IL6 (0.71 ± 0.02), and IL8 $\times$ IL11 (0.71 ± 0.02) showed comparatively lower chlorophyll levels. Similarly, under partial irrigated situations, the hybrid IL4 $\times$ IL12 and IL2 $\times$ IL10 displayed a relatively higher chlorophyll content (0.82 ± 0.02), chased by IL4 $\times$ IL6 (0.81 ± 0.02) and IL1 $\times$ IL11 (0.81 ± 0.02). While hybrid IL2 $\times$ IL5 presented a relatively low chlorophyll content (0.60 ± 0.02) followed by IL4 $\times$ IL5, IL6 $\times$ IL10 (0.62 ± 0.02) (Table 2). These results deliver scientific indication concerning the variation in chlorophyll content among diverse parental inbred lines and their hybrids under both irrigation conditions. In present research, water shortage stress instigated a substantial decrease in the chlorophyll contents in the inbred

lines and their hybrids (Fig. 1 and Fig. 2). For chlorophyll content, Kumari *et al.*^[67] and Manasa *et al.*^[68] described that under drought circumstances, there is decrease in chlorophyll content. Likewise, Anjum *et al.*^[69] and Naghizadeh *et al.*^[51] conveyed that deficit irrigation triggered a momentous reduction in the chlorophyll contents in maize hybrids. The drought-tolerant genotypes have higher Chl a, Chl b, and total Chl, compared to drought-sensitive maize hybrids.

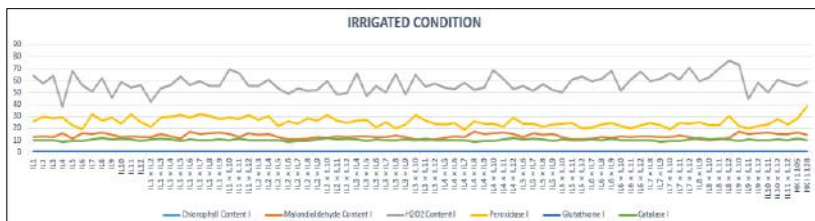


Fig 1: Line diagram of biochemical parameters investigated in 80 maize genotypes under irrigated circumstances

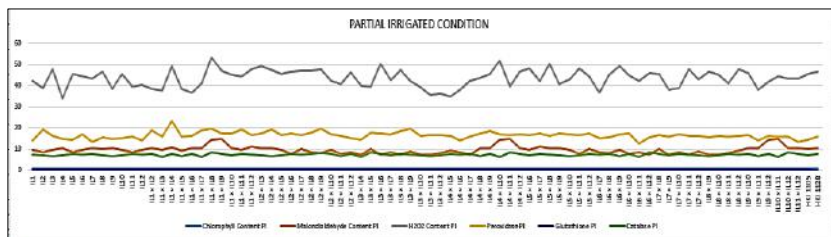


Fig 2: Line diagram of biochemical parameters investigated in 80 maize genotypes under partial irrigated circumstances

Malondialdehyde content (MDA TEST)

The MDA content is a pointer of oxidative grievance and acts as an indicator for membrane lipid peroxidation because of divulging plants to stress ^[66]. Drought-encouraged overproduction of ROS increases MDA, which consequences in a disrupted cell bilayer structure and leads to the porosity of the cell membrane, with a reduction in membrane stability shimmering the level of lipid peroxidation triggered by ROS ^[50].

Under irrigated conditions, among the parental inbred lines, IL8 displayed the highest MDA content (16.4 ± 0.2 nmol/g FW), followed by IL6 (15.9 ± 0.2 nmol/g FW), and IL4 (15.8 ± 0.2 nmol/g FW). Conversely, IL5 displayed the lowest MDA content (11.6 ± 0.2 nmol/g FW), while IL1 and IL10 showed somewhat higher but comparable levels (12.5 ± 0.2 nmol/g FW). Under partial irrigated conditions, IL9 demonstrated the highest MDA content (10.4 ± 0.2 nmol/g FW), followed by IL7 (10.3 ± 0.2 nmol/g FW),

and IL4 (10.2 ± 0.2 nmol/g FW). However, the lowest MDA content was observed in IL5 (8.5 ± 0.2 nmol/g FW).

Regarding hybrids under irrigated circumstances, the hybrids IL1 $\times$ IL6 and IL9 $\times$ IL10 displayed considerably higher MDA content (17.4 ± 0.2 nmol/g FW), tracked by IL1 $\times$ IL9, IL4 $\times$ IL11 (16.6 ± 0.2 nmol/g FW), and IL5 $\times$ IL7 (15.9 ± 0.2 nmol/g FW). Alternatively, the hybrid IL3 $\times$ IL11, IL8 $\times$ IL10 exhibited the lowest MDA content (10.3 ± 0.2 nmol/g FW), tracked by IL3 $\times$ IL12, IL8 $\times$ IL11, (10.5 ± 0.2 nmol/g FW), and IL2 $\times$ IL7 (10.8 ± 0.2 nmol/g FW). While under partial irrigated surroundings, the hybrids viz., IL1 $\times$ IL9, IL10 $\times$ IL11, IL4 $\times$ IL11 showed remarkably high MDA content (14.6 ± 0.2 nmol/g FW), along with IL1 $\times$ IL8, IL9 $\times$ IL12, IL4 $\times$ IL10 (14.3 ± 0.2 nmol/g FW). Correspondingly, the hybrids IL4 $\times$ IL7, IL6 $\times$ IL12 displayed the lowest MDA content (7.2 ± 0.2 nmol/g FW), followed by IL3 $\times$ IL6, IL3 $\times$ IL8, IL3 $\times$ IL11, and IL8 $\times$ IL10 (7.3 ± 0.2 nmol/g FW) under partial irrigated circumstances (Table 2, Fig. 1 and Fig. 2).

The content of MDA is frequently used as a pointer of lipid peroxidation in plant tissues, resulting from oxidative stress induced by different abiotic stresses. Recently, an increase in MDA content under drought stress has been described in leaves of drought-sensitive genotype of maize, while no alteration is experimented in tolerant genotypes [70-71].

H₂O₂ content

Hydrogen peroxide is a reactive oxygen species (ROS) that can play a role in several physiological processes in plants, together with responses to stress and defense mechanisms. Nevertheless, its levels can differ contingent to environmental factors, plant health, and other circumstances. The studies explored the H₂O₂ (hydrogen peroxide) content in diverse maize parental inbred lines and hybrids under changing irrigation circumstances, revealing distinguishing patterns of H₂O₂ accumulation. Under irrigated surroundings, the parental inbred line IL5 displayed the highest H₂O₂ content (68.2 ± 0.2 nmol/g FW), tracked by IL1 (64.4 ± 0.2 nmol/g FW) and IL3 (64.3 ± 0.2 nmol/g FW). Inversely, the parental inbred line IL4 (38.2 ± 0.2 nmol/g FW) exhibited the lowest H₂O₂ content, along with IL9 (45.4 ± 0.2 nmol/g FW) and IL7 (50.7 ± 0.2 nmol/g FW) displayed higher levels of H₂O₂ under the similar irrigation situations. Moreover, IL3 (47.6 ± 0.2 nmol/g FW) confirmed the highest H₂O₂ content, tracked by IL8 (46.5 ± 0.2 nmol/g FW) and IL10 (45.5 ± 0.2 nmol/g FW) under partial irrigated condition. Whereas, IL4 (33.9 ± 0.2 nmol/g FW) exposed the lowest H₂O₂ content followed by IL2 (38.8 ± 0.2 nmol/g FW) and IL9 (38.6 ± 0.2 nmol/g FW) under partial irrigated condition (Table 2).

Among the hybrids, under irrigated circumstances, the hybrid IL8 × IL12 exhibited the highest H₂O₂ content (76.9 ± 0.2 nmol/g FW), tracked by IL9 × IL10 (73.1 ± 0.2 nmol/g FW), IL7 × IL12 (70.8 ± 0.2 nmol/g FW), and IL8 × IL11 (70.1 ± 0.2 nmol/g FW). While, the hybrid IL2 × IL5 displayed the lowest H₂O₂ content (42.1 ± 0.2 nmol/g FW), followed by IL9 × IL11, IL10 × IL12, IL11 × IL12 (all three 43.2 ± 0.2 nmol/g FW) and IL3 × IL5 (46.9 ± 0.2 nmol/g FW). Whereas under partial irrigated circumstances, the hybrid IL1 × IL8 confirmed comparatively low H₂O₂ content (53.5 ± 0.2 nmol/g FW), lagged by IL4 × IL10 (51.8 ± 0.2 nmol/g FW), IL5 × IL8 (50.5 ± 0.2 nmol/g FW), and IL3 × IL6 (50.2 ± 0.2 nmol/g FW). Contradictory, the hybrid IL4 × IL5 showed the lowest H₂O₂ content (34.9 ± 0.2 nmol/g FW), followed by IL3 × IL11 (35.7 ± 0.2 nmol/g FW), IL1 × IL6 (36.7 ± 0.2 nmol/g FW), and IL3 × IL12 (36.3 ± 0.2 nmol/g FW) (Table 2).

These results shed light on the varying responses of maize genotypes to H₂O₂ accumulation under diverse irrigation circumstances, providing valuable visions into their oxidative stress tolerance machineries. The findings donate to the understanding of plant stress responses and may have inferences for the development of stress-tolerant maize cultivars in agricultural practices. Tolerant genotype (s) did not accumulate more H₂O₂ under drought stress, demonstrating less severe oxidative damage. Accretion of higher H₂O₂ content in leaf tissue of sensitive genotype of maize, compared to tolerant ones under drought stress induced by polyethylene glycol has been reported ^[72, 71]. Higher H₂O₂ content in the leaves of drought sensitive maize under drought stress compared to drought tolerant maize has also been investigated ^[70, 71].

Peroxidase activity

Peroxidase is most important enzyme in the decomposition of H₂O₂ into water and oxygen in the cytosol and chloroplast. Alteration in peroxidase action have been frequently correlated to the response of tolerance or susceptibility of plants to stresses ^[73]. The data evidently displays that activity of peroxidase increased with increase in stress in roots and shoots of all the genotypes.

Peroxidase activity, an essential enzyme involved in plant defense against oxidative stress, was examined in several maize parental inbred lines and hybrids under diverse irrigation circumstances. The research exposed discrete designs of peroxidase activity across the genotypes. Under irrigated situations, the parental inbred lines IL7 and IL11 showed the highest peroxidase activity (31.9 ± 0.2 units/mg protein), closely followed by IL2 (29.8 ± 0.2 units/mg protein) and IL9 (29.4 ± 0.2 units/mg protein). Instead,

IL6 (19.1 ± 0.2 units/mg protein) and IL5 (22.7 ± 0.2 units/mg protein) exhibited the lowest peroxidase activity, with IL10 (24.2 ± 0.2 units/mg protein) showing an intermediate level. Furthermore, under the partial irrigated circumstances, IL2 (19.3 ± 0.2 units/mg protein) exhibited the highest peroxidase activity, followed by IL6 (17.1 ± 0.2 units/mg protein) and IL3 (16.3 ± 0.2 units/mg protein). Likewise, IL5 (14.4 ± 0.2 units/mg protein) confirmed the lowest peroxidase activity, tracked by IL1 (14.1 ± 0.2 units/mg protein) and IL12 (13.9 ± 0.2 units/mg protein).

Moreover, under irrigated circumstances, the hybrid IL1×IL7 (31.8 ± 0.2 units/mg protein) showed the highest peroxidase activity, tracked by IL2 × IL10, (31.4 ± 0.2 units/mg protein) and IL3 × IL9 (30.9 ± 0.2 units/mg protein). In contrast, the hybrid IL4 × IL7 (18.4 ± 0.2 units/mg protein) showed relatively lower peroxidase activity, along with IL7 × IL10 (19.1 ± 0.2 units/mg protein) and IL6 × IL10 (19.9 ± 0.2 units/mg protein) under the similar situations. Besides, under partial irrigated circumstances, the hybrid IL1 × IL4 (23.1 ± 0.2 units/mg protein) established the highest peroxidase activity, tracked by IL5 × IL12 (20.1 ± 0.2 units/mg protein) and IL6 × IL11 (19.9 ± 0.2 units/mg protein), while the hybrid IL6 × IL11 (12.5 ± 0.2 units/mg protein) confirmed the lowest peroxidase activity tracked by IL11×IL12 (13.4 ± 0.2 units/mg protein) and IL4 × IL6 (14.1 ± 0.2 units/mg protein).

Like with our findings, an increase in peroxidase activity in drought tolerant and sensitive genotypes of maize under water stress has also been described by Kolarovic *et al.* ^[74]. Such harmonies were found in other crop species including cotton, rice and wheat ^[75-80]. Abedi and Pakniyat ^[81] recognized that under water stress circumstances, the peroxidase activity increased in the oilseed rape plants. Comparable findings were also addressed by Chugh *et al.* ^[71], Moharramnejad *et al.* ^[82] and Xie *et al.* ^[83] in maize.

Glutathione reductase activity

Glutathione reductase (GR) is a crucial enzyme that supports to reduced glutathione (GSH) from its oxidized form (GSSG). Glutathione is a tripeptide composed of three amino acids viz., glutamate, cysteine, and glycine ^[18]. It plays a significant role in protecting plant cells from oxidative stress by rummaging destructive reactive oxygen species (ROS) and protecting cellular components from damage ^[17]. In plants, including maize, glutathione reductase activity is closely linked to the antioxidant defense system, as it supports to uphold an adequate pool of abridged glutathione. The balance between GSH and GSSG is indispensable for cellular redox signaling and stress responses ^[23].

Under irrigated circumstances, the parental inbred line viz., IL1 showed the highest GR activity (0.37 ± 0.02), tracked by IL9 (0.34 ± 0.02), and IL2 and IL7 (0.32 ± 0.02). Contrarywise, the lowest GR activity was evidenced in IL6 (0.22 ± 0.02) tracked by IL8 (0.25 ± 0.02) and IL3 (0.28 ± 0.02) congruently, under partial irrigated situations, IL1 displayed the highest GR activity (0.36 ± 0.02), followed by IL9, IL7 (both 0.29 ± 0.02), Nevertheless, the lowest GR activity was detected in IL6 (0.19 ± 0.02), tracked by IL8 (0.20 ± 0.02) and IL3, IL11 (both 0.24 ± 0.02).

For hybrids, under irrigated circumstances, the hybrid IL9 $\times$ IL12 showed the highest GR activity (0.40 ± 0.02), followed by IL6 $\times$ IL12, IL10 $\times$ IL11 (0.37 ± 0.02), and IL2 $\times$ IL5 (0.36 ± 0.02). Alternatively, the hybrid IL1 $\times$ IL3 displayed the lowest GR activity (0.21 ± 0.02), followed by IL3 $\times$ IL8, IL3 $\times$ IL9, IL4 $\times$ IL5, IL3 $\times$ IL12 and IL5 $\times$ IL6 (0.23 ± 0.02 each). While, under partial irrigated situations, the hybrids IL2 $\times$ IL5 confirmed the highest GR activity (0.36 ± 0.02), tracked by IL9 $\times$ IL12, IL2 $\times$ IL3, IL5 $\times$ IL8 (0.35 ± 0.02 each), Contrariwise, the hybrids IL1 $\times$ IL3, IL5 $\times$ IL6 (0.18 ± 0.02) followed by IL3 $\times$ IL7 (0.19 ± 0.02) displayed the lowest GR activity. The findings indicate noteworthy disparities in GR activity among diverse maize inbred lines and their hybrids under changing irrigation circumstances. These results donate to our knowledge of the antioxidative defense apparatuses in maize and may have suggestions for breeding programmes aimed to develop drought-tolerant maize cultivars. Glutathione reductase (GR) activity was increased under drought in plant species such as maize ^[84], wheat ^[85], rice ^[86-87]. GR activity was increased with short-term drought treatment in leaves in wheat ^[88]. Total GR activity was augmented in the drought-tolerant sugarcane genotype under severe water stress but not mild stress, but was amplified even under mild stress in non-drought-tolerant cultivars ^[89]. GR activity was increased in a drought-tolerant wheat cultivar subjected to 100% oxygen and water stress ^[90]. Nevertheless, under incessant drought, dual-targeted GR transcripts were upregulated in the drought-sensitive cultivar but downregulated in the tolerant genotype ^[91].

Catalase activity

CAT decomposes H_2O_2 into water and oxygen at diverse cellular sites ^[46]. A decline in CAT activity is considered as a common response to many stresses ^[92]. During stress circumstances, CAT activity is allegedly reduced owing to the inhibition of enzyme synthesis or alteration in the assembly of enzyme subunits. It may also be related with degradation instigated by persuaded peroxisomal proteases or owing to the photo-inactivation of the enzyme ^[81].

Catalase activity, a main enzyme involved in the detoxification of hydrogen peroxide, was judged in several maize parental inbred lines and hybrids under both irrigation circumstances. The investigation exposed different outlines of catalase activity across the genotypes. Under irrigated circumstances, the parental inbred line IL8 showed the highest catalase activity (11.9 ± 0.2 units/mg protein), tracked by IL10 (11.2 ± 0.2 units/mg protein) and IL11 (10.8 ± 0.2 units/mg protein). Inversely, IL7 and IL12 showed the lowest catalase activity (9.6 ± 0.2 units/mg protein). Furthermore, under the partial irrigated situations, IL5 and IL11 confirmed the highest catalase activity (7.8 ± 0.2 units/mg protein), tracked by IL7 (7.6 ± 0.2 units/mg protein) and IL6 and IL12 (7.3 ± 0.2 units/mg protein). Moreover, under partial irrigated situations, the parental inbred line IL4 and IL10 displayed the lowest catalase activity (6.9 ± 0.2 units/mg protein).

In respect to hybrids under irrigated situations, the hybrid IL2 $\times$ IL10 and IL4 $\times$ IL12 verified the highest catalase activity (11.9 ± 0.2 units/mg protein), followed by IL1 $\times$ IL11, IL3 $\times$ IL11, and IL6 $\times$ IL9 (11.7 ± 0.2 units/mg protein). In contrast, the hybrid IL7 $\times$ IL9 and IL2 $\times$ IL6, IL4 $\times$ IL8 exhibited relatively lower catalase activity (8.8 ± 0.2 units/mg protein), along with IL4 $\times$ IL9 (9.3 ± 0.2 units/mg protein) under the similar situations. Contrarywise, under partial irrigated circumstances, the hybrids *i.e.*, IL1 $\times$ IL8, IL3 $\times$ IL5, IL4 $\times$ IL11, IL6 $\times$ IL12, and IL10 $\times$ IL12 displayed the highest catalase activity (8.3 ± 0.2 units/mg protein), followed by IL2 $\times$ IL9 (8.2 ± 0.2 units/mg protein). While the hybrids *viz.*, IL1 $\times$ IL3, IL1 $\times$ IL7, IL10 $\times$ IL11, IL3 $\times$ IL4 and IL4 $\times$ IL10 exhibited the lowest catalase activity (6.3 ± 0.2 units/mg protein) tracked by IL8 $\times$ IL9, IL5 $\times$ IL10, IL2 $\times$ IL11, and IL2 $\times$ IL14 (6.5 ± 0.2 units/mg protein).

In the absence of natural scavengers for instance CAT and POD, high level of H₂O₂ accrues in tissues. Catalase is heterogeneous in nature under drought stress. It might be amplified and continue unaffected or decreased on exposure to water stress ^[93]. Like our findings, Anjum *et al.* ^[94] also stated that catalase activity reduced owing to water deficiency compared to well-watered circumstances in maize hybrids. Xie *et al.* ^[83] confirmed a decay in the catalase enzyme activity in maize hybrids under drought stress circumstances.

Diversity and expression analysis among different biochemical traits

A heat map was made on a set of 12 inbred lines along with 66 hybrids, and two controls to investigate their biochemical profiles. The analysis resulted in the formation of different clusters based on a map (Fig. 3). These

clusters comprehended a total of 80 genotypes, which were principally classified into five major clusters. Cluster I, the largest group, contained 39 genotypes. Cluster II confined four genotypes, while cluster III comprised seven genotypes. Cluster IV encompassed 23 genotypes, and Cluster V had five genotypes. Besides, additional subdivisions were investigated within these clusters. It is notable that this clustering pattern designates substantial variations in the biochemical, and antioxidant profiles among the genotypes under investigation.

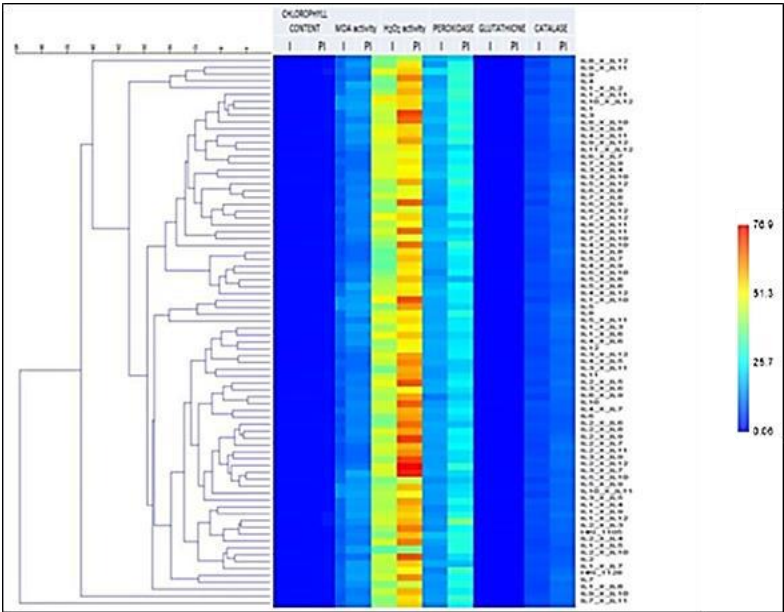


Fig 3: Dendrogram and heat map analysis of different biochemical traits of maize genotypes under irrigated (I) and partial irrigated conditions (PI)

Heat maps are frequently used in expression analysis investigations to visually represent data and enable quality control. In this study, heat map analysis was employed to inspect 80 genotypes based on their expression levels of six biochemical parameters viz., chlorophyll, MDA and H₂O₂ contents along with peroxidase, glutathione reductase, and catalase activities. The analysis was conducted under two different irrigation circumstances viz., irrigated and partially irrigated. The heat map visualization used a color key, where the colour blue was utilized to represent a range of values from 0.69 to 76.9. By employing this tactic, the heat map enabled the identification of outlines and disparities in the expression levels of the diverse biochemical parameters among the genotypes under diverse irrigation (irrigated and partially irrigated) situations.

Based on the heat map analysis conducted under irrigated circumstances, certain observations can be made regarding the expression levels of diverse biochemical traits in the genotypes investigated. The expression of chlorophyll content was found to be highest in the parent IL10 and the hybrid IL2 $\times$ IL10, while it was lowest in IL5 and IL6 $\times$ IL10. In respect to the MDA test, the expression was highest in IL9 and IL1 $\times$ IL9 for the parent and hybrid, correspondingly, whereas it was lowest in IL2 and IL6 $\times$ IL12. Regarding H₂O₂ content, IL3 and IL1 $\times$ IL8 displayed the highest expression, while IL4 and IL4 $\times$ IL5 exhibited the lowest expression. For peroxidase activity, IL2 and IL1 $\times$ IL4 demonstrated the highest expression, while IL7 and IL6 $\times$ IL11 showed the lowest expression. Glutathione reductase activity was found to be highest in IL1 and IL9 $\times$ IL12, whereas it was lowest in IL6 and HKI 1128 for the parent and a control. Finally, catalase activity displayed maximum expression in IL5 and IL1 $\times$ IL8, and minimum in genotypes IL9 and IL10 $\times$ IL11 for the parent and hybrid.

Under partial irrigated circumstances, the expression of chlorophyll content was highest in the genotypes IL10 and IL2 $\times$ IL10, while it was lowest in IL5 and IL6 $\times$ IL10. In respect to the MDA test, expression was highest in the genotypes IL8 and IL1 $\times$ IL6, while it was lowest in IL5 and IL8 $\times$ IL10. Concerning H₂O₂ content, IL5 and IL8 $\times$ IL12 displayed the highest expression level, while IL4 and IL1 $\times$ IL2 showed the lowest expression. For peroxidase activity, IL7 and HKI 1128 exhibited the highest expression for the parent and control correspondingly, while IL6 and IL4 $\times$ IL7 displayed the lowest expression. Glutathione reductase activity was found to be highest in IL1 and IL9 $\times$ IL12, whereas it was lowest in IL6 and HKI 1128 for the parent and control cultivar. Finally, catalase activity exhibited maximum expression level in genotypes IL8 and IL2 $\times$ IL10, and minimum for the genotypes IL4 and IL7 $\times$ IL9.

According Santos *et al.* ^[95], genotypes from the same group are genetically similar and their combinations may cause inferior variability when compared with the other groups. The inbred lines in distant groups, is suggestive of being genetic divergence and may be considered promising in artificial crosses. Moreover, the divergence and genetic relations investigations regarding physiological quality and biochemical composition support the selection strategies, aiming at the quality of seeds ^[96-100].

Conclusion

Present research explored the variation in diverse biochemical traits associated to antioxidant defense mechanisms in maize parental inbred lines

and their hybrids under diverse irrigation circumstances. The findings exposed substantial alterations in chlorophyll, MDA, and H₂O₂ contents, along with peroxidase, glutathione reductase, and catalase activities among the genotypes investigated. Overall, the phylogenetic analysis and heat map visualization highlighted the diversity in the antioxidant defense mechanisms among the maize genotypes, signifying potential variations in stress response and antioxidant activities. These findings may have important implications for maize breeding programmes meant to develop drought-tolerant cultivars with improved antioxidant capability and stress flexibility.

References

1. Gondim FA, Miranda RDS, Gomes-Filho E, Prisco JT. Enhanced salt tolerance in maize plants induced by H₂O₂ leaf spraying is associated with improved gas exchange rather than with non-enzymatic antioxidant system. *Theor. Exp. Plant Physiol.* 2013;25(4):251-60. Doi: 10.1590/S2197-00252013000400003.
2. Benz BF. On the origin, evolution, and dispersal of maize. In: *Pacific Latin America in Prehistory: The Evolution of Archaic and Formative Cultures*. Pullman: Washington State University Press; 1999. p. 25-38.
3. Yadav PK, Tripathi MK, Tiwari S, Chauhan S, Tripathi N, Sikarwar RS, *et al.* Genetic components and variability assessment for grain yield and its accrediting traits in maize (*Zea mays* L.). *Int. J Environ Climate Change*. 2023;13(9):772-84. doi: 10.9734/ijec/2023/v13i92298.
4. Huang L, Tan H, Zhang C, Li Q, Liu Q. Starch biosynthesis in cereal endosperms: An updated review over the last decade. *Plant Commun.* 2021;2(5):100-237. Doi: 10.1016/j.xplc.2021.100237. PubMed: 34746765.
5. Yadav PK, Singh AK, Tripathi MK, Tiwari S, Rathore J. Morpho-physiological characterization of maize (*Zea mays* L.) genotypes against drought. *Biol. Forum*. 2022;14(2):0975-1130.
6. Yadav PK, Singh AK, Tripathi MK, Tiwari S, Yadav SK, Tripathi N. Morpho-physiological and molecular characterization of maize (*Zea mays* L.) genotypes for drought tolerance. *Eur. J Appl. Sci.* 2022;10(6):65-87.
7. Yadav PK, Singh AK, Tripathi MK, Tiwari S, Yadav SK, Solanki RS, *et al.* Assessment of maize (*Zea mays* L.) genotypes on the basis of biochemical contents in respect to drought. *Pharma Innov J.* 2022;11(6):1996-2002.

8. Li P, Cao W, Fang H, Xu S, Yin S, Zhang Y, *et al.* Transcriptomic profiling of the maize (*Zea mays* L.) leaf response to abiotic stresses at the seedling stage. *Front Plant Sci.* 2017;8:290. doi: 10.3389/fpls.2017.00290. PubMed: 28298920.
9. Hernández VAG, Cruz EL, Onofre LEM, Varela AS, Espinosa MAG, García FZ. Maize (*Zea mays* L.) landraces classified by drought stress tolerance at the seedling stage. *Emir J Food Agric.* 2021;33:29-36.
10. Zhang X, Mi Y, Mao H, Liu S, Chen L, Qin F. Genetic variation in ZmTIP1 contributes to root hair elongation and drought tolerance in maize. *Plant Biotechnol J.* 2020;18(5):1271-83. Doi: 10.1111/pbi.13290. PubMed: 31692165.
11. Asati R, Tripathi MK, Tiwari S, Yadav RK, Tripathi N. Molecular breeding and drought tolerance in chickpea. *Life (Basel).* 2022;12(11):18-46. doi: 10.3390/life12111846.
12. Choudhary ML, Tripathi MK, Tiwari S, Pandya RK, Gupta N, Tripathi N, *et al.* Screening of pearl millet [*Pennisetum glaucum* (L.) R. Br.] germplasm lines for drought tolerance based on morpho-physiological traits and SSR markers. *Curr. J Appl. Sci. Technol.* 2021;40(5):46-63.
13. Choudhary ML, Tripathi MK, Gupta N, Tiwari S, Tripathi N, Parihar P, *et al.* Screening of pearl millet [*Pennisetum glaucum* (L.) R. Br.] germplasm lines against drought tolerance based on biochemical traits. *Curr. J Appl. Sci. Technol.* 2021;40(23):1-12.
14. Mishra N, Tripathi MK, Tiwari S, Tripathi N, Gupta N, Sharma A. Morphological and physiological performance of Indian soybean [*Glycine max* (L.) Merrill] genotypes in respect to drought. *Legume Res;* 2021. doi: 10.18805/LR-4550.
15. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A. Validation of drought tolerance gene-linked microsatellite markers and their efficiency for diversity assessment in a set of soybean genotypes. *Curr. J Appl. Sci. Technol.* 2021;40:48-57.
16. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A, *et al.* Cell suspension culture and in vitro screening for drought tolerance in soybean using polyethylene glycol. *Plants (Basel).* 2021;10(3):517. Doi: 10.3390/plants10030517.
17. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A, *et al.* Changes in biochemical and antioxidant enzymes activities play significant role in drought tolerance in soybean. *Int. J Agric. Technol.* 2021;17:1425-46.

18. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A, *et al.* Role of biochemical and antioxidant enzymes activities in drought tolerance in soybean: A recent study. In: Current Topics in Agricultural Sciences. 2021;3:102-19. Doi: 10.9734/bpi/ctas/v3/2117C.
19. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A, *et al.* Screening of soybean genotypes against drought on the basis of gene-linked microsatellite markers. In: Innovations in Science and Technology. 2022;3:49-61. Doi: 10.9734/bpi/ist/v3/2454C.
20. Paliwal S, Tripathi MK, Tiwari S, Tripathi N, Payasi DK, Tiwari PN, *et al.* Molecular advances to combat different biotic and abiotic stresses in linseed (*Linum usitatissimum* L.): A comprehensive review. Genes (Basel). 2023;14(7):1461. Doi: 10.3390/genes14071461.
21. Rajput S, Jain S, Tiwari S, Barela A, Chauhan S, Tiwari PN, *et al.* Biochemical characterization of chickpea (*Cicer arietinum* L.) genotypes. Plant Cell Biotechnol Mol. Biol. 2023;24(3-4):1-9.
22. Sinay H, Karuwal RL. Proline and total soluble sugar content at the vegetative phase of six corn cultivars from Kisar Island Maluku, grown under drought stress conditions. Int. J Adv. Agric. Res. 2014;5(2):77-82.
23. Sharma A, Tripathi MK, Tiwari S, Gupta N, Tripathi N, Mishra N. Evaluation of soybean (*Glycine max* L.) genotypes on the basis of biochemical contents and anti-oxidant enzyme activities. Legume Res; 2021, 44.
24. Sahu VK, Tiwari S, Gupta N, Tripathi MK, Yasin M. Evaluation of physiological and biochemical contents in Desi and Kabuli chickpea. Legume Res. 2020. doi: 10.18805/LR-4265.
25. Sahu VK, Tiwari S, Tripathi MK, Gupta N, Tomar RS, Yasin M. Morpho-physiological and biochemical traits analysis for Fusarium wilt disease using gene-based markers in Desi and Kabuli genotypes of chickpea (*Cicer arietinum* L.). Indian J Genet. 2020;80:16.
26. Tiwari PN, Tiwari S, Sapre S, Babbar A, Tripathi N, Tiwari S, *et al.* Screening and selection of drought-tolerant high-yielding chickpea genotypes based on physio-biochemical selection indices and yield trials. Life (Basel). 2023;13(6):1405. doi: 10.3390/life13061405.
27. Tripathi MK, Tripathi N, Tiwari S, Mishra N, Sharma A, Tiwari S, *et al.* Identification of Indian soybean (*Glycine max* [L.] Merr.) Genotypes for drought tolerance and genetic diversity analysis using SSR markers. Sci. 2023;3(3):31-46.

28. Yadav RK, Tripathi MK, Tiwari S, Tripathi N, Asati R, Chauhan S, *et al.* Genome editing and improvement of abiotic stress tolerance in crop plants. *Life (Basel)*. 2023;13(7):1456. Doi: 10.3390/life13071456.
29. Tiwari S, Yadav SK, Sahu VK, Tripathi MK. Current status and future prospects of marker assisted breeding for genetic improvement of minor millets. *Int. J Curr. Microbiol. App Sci*. 2018;7(12):2587-90.
30. Gupta N, Tiwari S, Tripathi MK, Bhagyawant SS. Antinutritional and protein-based profiling of diverse desi and wild chickpea accessions. *Curr. J Appl. Sci. Technol*. 2021;40(6):7-18.
31. Yadav PK, Tripathi MK, Tiwari S, Chauhan S, Tripathi N, Sikarwar RS, *et al.* Biochemical characterization of parental inbred lines and hybrids of maize (*Zea mays* L.) under different irrigation conditions. *Int. J Plant Soil Sci*. 2023;35(18):1743-62.
32. Rathore MS, Tiwari S, Tripathi MK, Gupta N, Yadav S, Singh S, *et al.* Genetic diversity analysis of groundnut germplasm lines in respect to early and late leaf spot diseases and biochemical traits. *Legume Res*; 2022. Doi: 10.18805/LR-4833.
33. Sharma N, Tiwari S, Tripathi MK, Gupta N, Singh S, Sikarwar RS. Selection of groundnut germplasm lines based on yield, nutritional, anti-nutritional and antioxidant parameters. *Sci. Int*. 2023;2(1):96-105. Doi: 10.5281/zenodo.7536324.
34. Shyam C, Tripathi MK, Tripathi N, Tiwari S, Sikarwar RS. Genetic variations in fatty acids and oil compositions among 188 Indian mustard *Brassica juncea* (Linn.) Czern & Coss genotypes. *Curr. J Appl. Sci. Technol*. 2022;40(46):9-28. doi: 10.9734/CJAST/2021/v40i4631629.
35. Shyam C, Tripathi MK, Tripathi N, Tiwari S, Sikarwar RS. Analysis of genetic differences in fatty acids and oil contents among *Brassica juncea* (Linn.) Czern & Coss genotypes. In: *Research Developments in Science and Technology*. 2022;1:127-49. Doi: 10.9734/bpi/rdst/v1/6010F.
36. Tomar YS, Tiwari S, Tripathi MK, Singh S, Gupta N. Genetic diversity, population structure and biochemical parameters estimations driving variations in groundnut germplasm. *Legume Res*; 2022. Doi: 10.18805/LR-4965.
37. Yadav S, Tiwari S, Tripathi MK, Tripathi N, Gupta N, Tiwari S. Evaluation of high oleic acid content in a set of 96 genotypes of *Arachis hypogaea* L. *Scientist*. 2023;2:132-43.

38. Tripathi N, Tripathi MK, Tiwari S, Payasi DK. Molecular breeding to overcome biotic stresses in soybean: update. *Plants (Basel)*. 2022;11(15):19-67.
39. Kosová K, Vítámvás P, Urban MO, Prášil IT, Renaut J. Plant abiotic stress proteomics: the major factors determining alterations in cellular proteome. *Front Plant Sci*. 2018;9:122. Doi: 10.3389/fpls.2018.00122. PubMed: 29472941.
40. Ai NS, Banyo Y. The concentration of leaf chlorophyll as water-deficit indicator in plants. *J Ilmiah Sains*. 2011;11(2):168-73.
41. Naeem M, Naeem MS, Ahmad R, Ahmad R, Ashraf MY, Ihsan MZ, *et al*. Improving drought tolerance in maize by foliar application of boron: water status, anti-oxidative defense and photosynthetic capacity. *Arch Agron Soil Sci*. 2018;64(5):626-39. Doi: 10.1080/03650340.2017.1370541.
42. Cui G, Zhao X, Liu S, Sun F, Zhang C, Xi Y. Beneficial effects of melatonin in overcoming drought stress in wheat seedlings. *Plant Physiol Biochem*. 2017;118:138-49. Doi: 10.1016/j.plaphy.2017.06.014. PubMed: 28633086.
43. Kamran M, Wennan S, Ahmad I, Xiangping M, Wenwen C, Xudong Z, *et al*. Application of paclobutrazol affects maize grain yield by regulating root morphological and physiological characteristics under a semi-arid region. *Sci. Rep*. 2018;8(1):4818. Doi: 10.1038/s41598-018-23166-z. PubMed: 29556086.
44. Farooq M, Wahid A, Kobayashi N, Fujita D, Basra SMA. Plant drought stress: effects, mechanisms and management. *Agron Sustain Dev*. 2009; 29(1):185-212. Doi: 10.1051/agro:2008021.
45. Horváth E, Szalai G, Janda T. Induction of abiotic stress tolerance by salicylic acid signaling. *J Plant Growth Regul*. 2007;26(3):290-300. Doi: 10.1007/s00344-007-9017-4.
46. Mittler R, Vanderauwera S, Gollery M, Van Breusegem F. Reactive oxygen gene network of plants. *Trends Plant Sci*. 2004;9(10):490-8. Doi: 10.1016/j.tplants.2004.08.009. PubMed: 15465684.
47. Foyer CH, Noctor G. Redox homeostasis and antioxidant signaling: a metabolic interface between stress perception and physiological responses. *Plant Cell*. 2005;17(7):1866-75. Doi: 10.1105/tpc.105.033589. PubMed: 15987996.

48. Zhang L. Superoxide dismutase, peroxidase, and catalase activities in response to drought stress of Kentucky bluegrass. *J Plant Physiol.* 2011;135(3):185-92.
49. AbdElgawad H, Avramova V, Baggerman G, Van Raemdonck G, Valkenburg D, Van Ostade X, *et al.* Starch biosynthesis contributes to the maintenance of photosynthesis and leaf growth under drought stress in maize. *Plant Cell Environ.* 2020;43(9):2254-71. Doi: 10.1111/pce.13813. PubMed: 32488892.
50. Yang F, Miao LF. Adaptive responses to progressive drought stress in two poplar species originating from different altitudes. *Silva Fenn.* 2010;44(1):23-37. Doi: 10.14214/sf.160.
51. Naghizadeh M, Kabiri R, Hatami A, Oloumi H, Nasibi F, Tahmasebi Z. Exogenous application of melatonin mitigates the adverse effects of drought stress on morphophysiological traits and secondary metabolites in Moldavian balm (*Dracocephalum moldavica*). *Physiol. Mol. Biol. Plants.* 2019;25(4):881-94. doi: 10.1007/s12298-019-00674-4. PubMed: 31402815.
52. Salim BBM, Abou El-Yazied A, Salama YAM, Raza A, Osman HS. Impact of silicon foliar application in enhancing antioxidants, growth, flowering and yield of squash plants under deficit irrigation condition. *Ann Agric. Sci.* 2021;66(2):176-83. Doi: 10.1016/j.aosas.2021.12.003.
53. El-Mageed TAA, Mekdad AAA, Rady MOA, Abdelbaky AS, Saudy HS, Shaaban A. Physio-biochemical and agronomic changes of two sugar beet cultivars grown in saline soil as influenced by potassium fertilizer. *J Soil Sci Plant Nutr.* 2022;22(3):3636-54. Doi: 10.1007/s42729-022-00916-7.
54. Madhan MM, Lakshmi NS, Ibrahim SM. Chlorophyll stability index (CSI): its impact on salt tolerance in rice. *Int. Rice Res Notes.* 2000;25:38-9.
55. Jinks JL, Hayman BI. The analysis of diallel crosses. *Maize Genet Coop News Lett.* 1954;27:48-54.
56. Arnon D. Copper enzymes isolated chloroplasts, polyphenol oxidase in *Beta vulgaris*. *Plant Physiol.* 1949;24:1-15.
57. Koleyoreas SA. A new method for determining drought resistance. *Plant Physiol.* 1958;33:232-3.
58. Heath RL, Packer L. Photoperoxidation in isolated chloroplasts I. Kinetics and stoichiometry of fatty acid peroxidation. *Arch Biochem*

- Biophys. 1968;125(1):189-98. Doi: 10.1016/0003-9861(68)90654-1. PubMed: 5655425.
59. Alexieva MS, Bhagwat AA, Cregan PB. Length polymorphisms of simple sequence repeat DNA in soybean. *Genetics*. 2001;132(4):1131-9.
 60. Kar M, Mishra D. Catalase, peroxidase, and polyphenol oxidase activities during rice leaf senescence. *Plant Physiol*. 1976;57(2):315-9. Doi: 10.1104/pp.57.2.315. PubMed: 16659474.
 61. Smith IK, Vierheller TL, Thorne CA. Properties and functions of glutathione. 1989.
 62. Aebi H. Catalase *in vitro*. *Methods Enzymol*. 1984;105:121-6. doi: 10.1016/s0076-6879(84)05016-3. PubMed: 6727660.
 63. Rohlf FJ. NTSYS-pc: numerical taxonomy and multivariate analysis system. Version 2.02. Exeter Software; 2000.
 64. Sokal RR, Michener CD. A statistical method for evaluating systematic relationships. *Univ Kans Sci Bull*. 1958;38(2):1409-38.
 65. Ahmad P, Jamsheed S, Hameed A, Rasool S, Sharma I, Azooz MM, Hasanuzzaman M. Drought stress induced oxidative damage and antioxidants in plants. In: Ahmad P, editor. *Oxidative damage to plants: antioxidant networks and signaling*. Academic Press; 2014. p. 345-67.
 66. Moharramnejad S, Sofalian O, Valizadeh M, Asghari A, Shiri MR, Ashraf M. Response of maize to field drought stress: oxidative defense system, osmolytes' accumulation and photosynthetic pigments. *Pak J Bot*. 2019;51(3):1-9. doi: 10.30848/PJB2019-3(1).
 67. Kumari P, Saunginetti MC, Liu C, Li Y, Wang TY, Giuliani S, *et al*. Root ABA1 QTL affects root lodging, grain yield and other agronomic traits in maize grown under well-watered and water-stressed conditions. *J Exp Bot*. 2004;58:319-26.
 68. Manasa MJ, Ribaut J, Vargas M, Crossa J, Van EF. A multi-trait multi-environment QTL mixed model with an application to drought and nitrogen stress trials in maize (*Zea mays* L.). *Euphytica*. 2014;161:241-57.
 69. Anjum SA, Xie X, Wang L, Saleem MF, Man C, Lei W. Morphological, physiological and biochemical responses of plants to drought stress. *Afr J Agric Res*. 2011;6(9):2026-32.

70. Moussa HR, Abdel-Aziz SM. Comparative response of drought tolerant and drought-sensitive maize genotypes to water stress. *Aust J Crop Sci.* 2008;1(1):31-6.
71. Chugh V, Kaur N, Gupta AK. Evaluation of oxidative stress tolerance in maize (*Zea mays* L.) seedlings in response to drought. *Indian J Biochem Biophys.* 2011;48(1):47-53. PubMed: 21469602.
72. Kellos T, Tímár I, Szilágyi V, Szalai G, Galiba G, Kocsy G. Stress hormones and abiotic stresses have different effects of antioxidants in maize lines with different sensitivity. *Plant Biol (Stuttg).* 2008;10(5):563-72. doi: 10.1111/j.1438-8677.2008.00071.x. PubMed: 18761495.
73. Zoz T, Steiner F, Guimaraes VF, Castagnara DD, Meinerz CC, Fey R. Peroxidase activity as an indicator of water deficit tolerance in soybean cultivars. *Biosci J.* 2013;29(5):1664-71.
74. Kolarovič L, Valentovič P, Luxová M, Gasparikova O. Changes in antioxidants and cell damage in heterotrophic maize seedlings differing in drought sensitivity after exposure to short-term osmotic stress. *Plant Growth Regul.* 2009;59(1):21-6. doi: 10.1007/s10725-009-9384-x.
75. Zafar SA, Hameed A, Nawaz MA, Wei MA, Noor MA, Hussain M, Rahman M. Mechanisms and molecular approaches for heat tolerance in rice (*Oryza sativa* L.) under climate change scenario. *J Integr Agric.* 2017;16(0):60345-7.
76. Zahra S, Hussain M, Zulfiqar S, Ishfaq S, Shaheen T, Akhtar M, Rahman M. EMS-based mutants are useful for enhancing drought tolerance in spring wheat. *Cereal Res Commun.* 2021;50(4):767-78.
77. Zahra S, Shaheen T, Hussain M, Zulfiqar S, Rahman M. Multivariate analysis of mutant wheat (*Triticum aestivum* L.) lines under drought stress. *Turk J Agric For.* 2021;45(5):617-33.
78. Sarwar MKS, Ashraf MY, Rahman M, Zafar Y. Genetic variability in different biochemical traits and their relationship with yield and yield parameters of cotton cultivars grown under water stress conditions. *Pak J Bot.* 2012;44(2):515-20.
79. Rahman M, Ullah I, Ashraf M, Zafar Y. A study of genotypic variation for drought tolerance in cotton. *Agron Sustain Dev.* 2008;28:439-47.
80. Ullah I, Rahman M, Ashraf M, Zafar Y. Genotypic variation for physiological and productivity traits in cotton (*Gossypium hirsutum* L.) under two contrasting water regimes. *Flora.* 2008;203:105-15.

81. Abedi T, Pakniyat H. Antioxidant enzyme changes in response to drought stress in ten cultivars of oilseed rape (*Brassica napus* L.). Czech J Genet Plant Breed. 2010;46(1):27-34. doi: 10.17221/67/2009-CJGPB.
82. Moharramnejad S, Sofalian O, Valizadeh M, Asgari A, Shiri M. Response of antioxidant defense system to osmotic stress in maize seedlings. Fresen Environ Bull. 2016;25(3):805-11.
83. Xie T, Gu W, Zhang L, Li L, Qu D, Li C, *et al.* Modulating the antioxidant system by exogenous 2-(3,4-dichlorophenoxy) triethylamine in maize seedlings exposed to polyethylene glycol-simulated drought stress. PLoS One. 2018;13(9):1-22.
84. Jiang M, Zhang J. Water stress-induced abscisic acid accumulation triggers the increased generation of reactive oxygen species and up-regulates the activities of antioxidant enzymes in maize leaves. J Exp Bot. 2002;53(379):2401-10. doi: 10.1093/jxb/erf090. PubMed: 12432032.
85. Chen KM, Gong HJ, Chen GC, Wang SM, Zhang CL. Gradual drought under field conditions influences the glutathione metabolism, redox balance and energy supply in spring wheat. J Plant Growth Regul. 2004;23(1):20-8. doi: 10.1007/s00344-003-0053-4.
86. Selote DS, Khanna-Chopra R. Drought-induced spikelet sterility is associated with an inefficient antioxidant defense in rice panicles. Physiol Plant. 2004;121(3):462-71. doi: 10.1111/j.1399-3054.2004.00341.x.
87. Sharma P, Dubey RS. Drought induces oxidative stress and enhances the activities of antioxidant enzymes in growing rice seedlings. Plant Growth Regul. 2005;46(3):209-21. doi: 10.1007/s10725-005-0002-2.
88. Bartoli CG, Simontacchi M, Tambussi E, Beltrano J, Montaldi E, Puntarulo S. Drought and watering-dependent oxidative stress: effect on antioxidant content in *Triticum aestivum* L. leaves. J Exp Bot. 1999;50(332):375-83. doi: 10.1093/jxb/50.332.375.
89. Boaretto LF, Carvalho G, Borgo L, Creste S, Landell MG, Mazzafera P, *et al.* Water stress reveals differential antioxidant responses of tolerant and non-tolerant sugarcane genotypes. Plant Physiol. Biochem. 2014;74:165-75. doi: 10.1016/j.plaphy.2013.11.016. PubMed: 24308986.
90. Pastori GM, Trippi VS. Oxidative stress induces high rate of glutathione reductase synthesis in a drought-resistant maize strain. Plant Cell Physiol. 1992;33(7):957-61.

91. Contour-Ansel D, Torres-Franklin ML, Cruz DE, Carvalho MH, D'Arcy-Lameta A, Zuily-Fodil Y. Glutathione reductase in leaves of cowpea: cloning of two cDNAs, expression and enzymatic activity under progressive drought stress, desiccation and abscisic acid treatment. *Ann Bot.* 2006;98(6):1279-87. doi: 10.1093/aob/mcl217. PubMed: 17008354.
92. Liu J, Xie X, Du J, Sun J, Bai X. Effects of simultaneous drought and heat stress on Kentucky bluegrass. *Sci. Hortic.* 2008;115(2):190-5. doi: 10.1016/j.scienta.2007.08.003.
93. Cruz de Carvalho MH. Drought stress and reactive oxygen species: production, scavenging and signaling. *Plant Signal Behav.* 2008;3(3):156-65. doi: 10.4161/psb.3.3.5536. PubMed: 19513210.
94. Anjum SA, Ashraf U, Tanveer M, Khan I, Hussain S, Shahzad B, *et al.* Drought induced changes in growth, osmolyte accumulation and antioxidant metabolism of three maize hybrids. *Front Plant Sci.* 2017;8:1-12.
95. Santos ER, Barros HB, Capone A, Melo AV, Cella AJS, Santos WR. *Divergência genética* entre genotypes de soja com base na qualidade de sementes. *Agraria.* 2012;7(2):247-54. doi: 10.5039/agraria.v7i2a1560.
96. Sistu R, Tiwari S, Tripathi MK, Singh S, Gupta N, Tripathi N, *et al.* Effect of different biochemical parameters and antioxidant enzymes activities on drought indices in chickpea (*Cicer arietinum* L.). *Legume Res.* 2023. doi: 10.18805/LR-5204.
97. Rana SS, Tiwari S, Gupta N, Tripathi MK, Tripathi N, Singh S, *et al.* Validating the nutraceutical significance of minor millets by employing nutritional-antinutritional profiling. *Life.* 2023;13(9):1918. doi: 10.3390/life13091918.
98. Paliwal S, Tripathi MK, Sikarwar RS, Yadav RK, Sonaniya P. Evaluation of biochemical parameters in linseed (*Linum usitatissimum*). *Int. J Adv. Biochem. Res.* 2024;8(4):260-7. doi: 10.33545/26174693.2024.v8.i4d.947.
99. Tiwari PN, Tiwari S, Sapre S, Babbar A, Tripathi N, Tiwari S, *et al.* Prioritization of microsatellite markers linked with drought tolerance associated traits in chickpea (*Cicer arietinum* L.). *Legume Res;* 2023. doi: 10.18805/LR-5191.
100. Tiwari PN, Tiwari S, Sapre S, Tripathi N, Payasi DK, Singh M, *et al.* Prioritization of physio-biochemical selection indices and yield-

attributing traits toward the acquisition of drought tolerance in chickpea (*Cicer arietinum* L.). *Plants*. 2023;12:31-75. doi: 10.3390/plants12183175.

Chapter - 9
**Decoding the Genome: An In-Depth Analysis of
454/Roche GS FLX Sequencing**

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Chapter - 9

Decoding the Genome: An In-Depth Analysis of 454/Roche GS FLX Sequencing

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Abstract

454/Roche GS FLX Sequencing, also known as pyrosequencing, was a groundbreaking next-generation sequencing technology that revolutionized genomic research in the early 2000s. Developed by 454 Life Sciences and later acquired by Roche, this method employed a unique combination of emulsion PCR and pyrosequencing chemistry to achieve high-throughput DNA sequencing. The process began with DNA fragmentation and adapter ligation, followed by amplification on microscopic beads within water-in-oil emulsion droplets. These DNA-coated beads were then deposited into picotiter plates, where sequencing-by-synthesis occurred through the sequential addition of nucleotides. As each nucleotide was incorporated, a chemiluminescent signal was produced and detected by a high-resolution camera. This technology offered several advantages over traditional Sanger sequencing, including longer read lengths (up to 700 base pairs), increased throughput, and reduced costs. The GS FLX system found applications in various fields, such as de novo genome sequencing, metagenomics, and transcriptomics. However, it faced challenges in homopolymer regions and had higher error rates compared to some competing technologies. Despite its eventual discontinuation in 2016, 454 sequencing played a crucial role in advancing genomic research and paved the way for subsequent developments in high-throughput sequencing technologies.

Keywords: 454/Roche GS FLX sequencing, pyrosequencing, high-throughput DNA sequencing, chemiluminescent signal, de novo genome sequencing

Introduction to 454/Roche GS FLX sequencing

The 454/Roche GS FLX sequencing platform, developed by 454 Life Sciences later acquired by Roche, revolutionized the field of DNA sequencing by introducing a novel approach known as pyro-sequencing. This technique

relies on the detection of light emission during the incorporation of nucleotides into a growing DNA strand, allowing for rapid and high-throughput sequencing. The 454/Roche GS FLX system is a notable advancement in next-generation sequencing technologies, offering several key advantages over traditional Sanger sequencing methods. One of the primary strengths of this platform is its ability to generate long read lengths, with the latest GS FLX Titanium chemistry capable of producing reads up to 1,000 base pairs in length ^[1]. This feature is particularly beneficial for applications such as de novo genome assembly and metagenomics. Where long reads can facilitate the resolution of repetitive genomic regions and improve the quality of the final assembly ^[2].

The 454/Roche GS FLX sequencing platform, introduced in the early 2000s, ^[1, 3]. It has a lower throughput and higher cost per base ^[4]. While the 454/Roche GS FLX system offers longer read lengths compared to other next-generation sequencing platforms, such as Illumina, Nevertheless, the platform has found widespread use in various fields, including environmental and microbial genomics, where the longer reads can provide valuable insights into complex microbial communities ^[4]. In addition to the 454/Roche GS FLX system, the field of next-generation sequencing has witnessed the emergence of other innovative platforms, such as the Ion Torrent system from Ion Torrent, which utilizes semiconductor technology for sequencing signal detection ^[5]. These advancements in sequencing technologies have led to significant improvements in terms of speed, cost, and accessibility, broadening the applications of high-throughput DNA sequencing in various areas of research and clinical practice ^[3].

The principle of 454/Roche GS FLX sequencing

The 454/Roche GS FLX system operates on the principle of pyrosequencing, a sequencing-by-synthesis approach that relies on the detection of inorganic pyrophosphate released during the incorporation of nucleotides by DNA polymerase ^[1]. This method eliminates the need for bacterial cloning of DNA fragments, a requirement in traditional Sanger sequencing, and instead utilizes a cell-free library preparation system ^[5].

The 454/Roche GS FLX Sequencing process begins with the fragmentation of DNA samples, followed by the attachment of adapter sequences to the fragments. These adapter-ligated fragments are then immobilized on beads, and each bead is isolated in a separate oil-water emulsion PCR reaction, resulting in the amplification of a single DNA fragment per bead ^[1]. The beads are then loaded into a picotitre plate, where each bead is positioned in a well, and the sequencing reaction takes place.

During the sequencing process, the release of inorganic pyrophosphate caused by nucleotide incorporation in a template-directed primer extension reaction triggers the production of luminescence, which is detected by a charge-coupled device camera ^[1]. This method of sequencing, known as "pyrosequencing" allows for the real-time detection of nucleotide incorporation, resulting in the generation of long read lengths.

Library preparation for 454/Roche GS FLX sequencing

The advent of high-throughput sequencing technologies has revolutionized genomics, with platforms like the 454/Roche GS FLX playing a pivotal role. Library preparation is a crucial step in the sequencing workflow, significantly influencing the quality and reliability of the sequencing data. This essay delves into the detailed process of library preparation for 454/Roche GS FLX sequencing, encompassing DNA fragmentation, adapter ligation, amplification, and quality control, while providing insights into best practices and potential challenges.

DNA Fragmentation

The first step in library preparation is DNA fragmentation. For 454/Roche GS FLX sequencing, genomic DNA is typically sheared into smaller fragments to facilitate efficient sequencing. Mechanical shearing methods such as nebulization or sonication are commonly employed. These methods randomly break the DNA into fragments of varying sizes, which are then size-selected to ensure they fall within the optimal range for sequencing, typically 300-800 base pairs (bp) ^[6]. Proper fragmentation is essential as it impacts the coverage and accuracy of the sequencing results.

End repair and adapter ligation

Once the DNA is fragmented, the ends of these fragments are polished to create blunt ends, ensuring they are suitable for the subsequent ligation of adapters. This polishing step typically involves the use of DNA polymerase to fill in overhangs and exonucleases to remove any remaining single-stranded regions and phosphorylate the 5' ends of the DNA fragments.

The next step involves the ligation of specific oligonucleotide adapters to both ends of the DNA fragments. These adapters serve multiple purposes: they provide a priming site for the polymerase chain reaction (PCR) amplification, facilitate the attachment of the fragments to sequencing beads, and enable the identification of different samples through barcoding if multiplexing is required ^[6].

Following adapter ligation, the DNA-adapter constructs are mixed with beads under conditions that favour the attachment of a single DNA fragment per bead. Each bead carries adapters complementary to those on the DNA fragments, allowing for their capture. This mixture is then emulsified in a water-in-oil emulsion, creating millions of microreactors. In each microreactor, the DNA fragment on the bead is amplified by emulsion PCR (emPCR), resulting in each bead carrying around 10 million copies of the original DNA fragment ^[7].

Amplification and Library enrichment

After adapter ligation, the DNA library undergoes amplification to enrich fragments that have successfully incorporated adapters. This is typically achieved through emulsion PCR (emPCR), where individual DNA fragments are encapsulated within aqueous droplets in an oil phase, each containing PCR reagents. EmPCR ensures clonal amplification of each fragment, producing multiple copies of each fragment, which is essential for generating sufficient signal during sequencing ^[7]. The quality of emPCR can significantly affect sequencing performance, necessitating careful optimization of reaction conditions.

Quality control in the library preparation process

Quality control (QC) is an integral part of the library preparation process, ensuring that the final library meets the standards required for high-quality sequencing. Several QC steps are implemented throughout the preparation process:

Fragment size distribution: Analyzing the size distribution of fragmented DNA using gel electrophoresis or capillary electrophoresis ensures that fragments fall within the desired size range.

Adapter ligation efficiency: Quantifying the ligated DNA to check the efficiency of adapter ligation using methods such as quantitative PCR (qPCR) or fluorometric assays.

Library quantification: Accurate quantification of the final library concentration using qPCR or fluorometry ensures that the appropriate amount of library is used for sequencing ^[8].

Potential Challenges and Solutions during library preparation

Several challenges can arise during library preparation for 454/Roche GS FLX sequencing:

Next, the enriched beads are loaded into a picotiter plate, a high-density array with millions of wells, each capable of holding a single bead. Beads are loaded such that ideally, each well contains one bead with a single DNA fragment, and small enzyme beads and packing beads are also added to ensure each well contains the necessary reagents for the sequencing reaction ^[6]. The plate is then placed in the sequencing instrument, which cycles through the addition of reagents: the four nucleotides (A, T, C, G) and enzymes. DNA polymerase extends the DNA strand by adding nucleotides, ATP sulfurylase converts released pyrophosphate (PPi) into ATP, and luciferase uses ATP to produce light, while apyrase degrades unincorporated nucleotides and ATP, preparing the well for the next cycle. The machine systematically flows each nucleotide type across the wells. When the correct nucleotide is present, it is incorporated into the growing DNA strand, releasing PPi ^[9].

Light emission occurs when a nucleotide is incorporated, with the light intensity being proportional to the number of nucleotides added. A CCD camera captures the emitted light as a series of peaks (flowgrams), where each peak corresponds to the incorporation event, with the height indicating the number of nucleotides added in that particular cycle ^[6].

Detailed data analysis process

Once sequencing is complete, the raw data undergoes a series of computational steps to generate meaningful sequence information. Raw flowgrams (light intensity signals) are first processed using base-calling algorithms, converting light intensity data into nucleotide sequences and assigning quality scores to each base call to reflect confidence in the call, with low-quality reads filtered out ^[10]. The individual sequence reads are then aligned and assembled into longer contiguous sequences (contigs). If a reference genome is available, reads are aligned to this genome using alignment algorithms such as BLAST or BWA. In the absence of a reference genome, de novo assembly algorithms like Newbler or Velvet are used to assemble reads into contigs based on overlapping sequences, employing sophisticated overlap-layout-consensus (OLC) algorithms to reconstruct the original sequence by identifying overlaps between reads and constructing contigs ^[8].

Error correction algorithms address sequencing errors, especially homopolymer errors, which occur in regions with repeated nucleotides (e.g., "AAAA"). Statistical models and error correction algorithms compare reads and correct discrepancies, ensuring accuracy in applications requiring precise sequence data such as variant calling and structural variant detection ^[8]. Assembled sequences are annotated using bioinformatics tools to identify

genes, exons, introns, regulatory elements, and other functional regions. Databases such as GenBank, RefSeq, and Ensembl are used to map known genes and predict functions of newly discovered sequences. Functional analysis includes predicting protein-coding regions, identifying potential mutations, and assessing their impact on gene function and disease ^[9].

The final stage of data analysis involves visualizing sequence data and generating reports. Visualization tools like IGV or the UCSC Genome Browser allow researchers to explore and visualize sequence data, aiding in identifying regions of interest, verifying assembly accuracy, and presenting findings. Reports summarize sequencing results, including metrics like read length distribution, coverage depth, and identified variants ^[10].

Advanced considerations in sequencing and data analysis

Homopolymer regions are a significant challenge due to the difficulty in accurately counting identical nucleotides. Algorithms specifically designed to handle homopolymer regions, such as the use of correction factors based on known sequence patterns and flowgram characteristics, are employed to mitigate these errors ^[8]. Data integration with other omics data (transcriptomics, proteomics) provides a comprehensive view of biological systems, helping in understanding gene expression, protein interactions, and metabolic pathways. Accurate variant calling (SNPs, indels) is crucial for clinical applications and understanding genetic diversity, while structural variant detection (e.g., large deletions, duplications) is facilitated by long-read capabilities and advanced algorithms ^[9].

Advantages

The 454/Roche GS FLX System, also known as the 454 Sequencing System, was a pioneering next-generation sequencing (NGS) platform that significantly advanced genomics research when it was introduced. One of its main advantages was the ability to generate longer read lengths compared to other NGS technologies available at the time, such as Illumina. These longer reads facilitated better assembly of genomes, especially for complex or repetitive regions, which were challenging to sequence with shorter-read technologies ^[6]. Additionally, the 454 system produced high-quality reads with relatively low error rates, making it suitable for a wide range of applications including whole-genome sequencing, transcriptome analysis, and metagenomics ^[7]. Its capability to deliver large amounts of sequence data quickly was another notable advantage, enabling researchers to complete time-sensitive projects more efficiently.

Limitations

Despite these benefits, the 454 system had several limitations. The cost was relatively high, both in terms of the initial investment for the platform and the per-base sequencing cost, which made it less accessible for some laboratories ^[7]. Another significant limitation was the system's susceptibility to errors in homopolymer regions (regions with a sequence of identical bases). These errors could affect the overall accuracy of the sequencing data ^[6]. Furthermore, compared to later NGS platforms like Illumina and Ion Torrent, the 454 system had lower throughput, meaning it generated fewer bases of sequence data per run, which limited its efficiency for large-scale projects ^[11].

The introduction of third-generation sequencing technologies, such as Pacific Biosciences (PacBio) and Oxford Nanopore Technologies, further highlighted the limitations of the 454 system. These newer technologies offered even longer read lengths and higher throughput, making the 454 system less competitive ^[12]. Eventually, Roche discontinued the 454 platform and stopped supporting it, leading to a decline in its usage and the availability of reagents and support for existing systems ^[13]. Despite its eventual obsolescence, the 454/Roche GS FLX System played a crucial role in the development of NGS technologies and significantly contributed to the advancement of genomics research.

Applications

The 454/Roche GS FLX Sequencing platform, utilizing pyrosequencing technology, has had a profound impact across multiple research disciplines due to its high throughput and long-read capabilities. In microbial ecology, this platform has been pivotal in elucidating the complexity of microbial communities by offering comprehensive profiles of microbial diversity and functional potential. Studies have leveraged 454 sequencing to explore microbial populations in environments ranging from soil to the human gut, revealing insights into microbial interactions and ecosystem functions ^[6, 15]. In clinical research, the platform's ability to detect low-frequency mutations has enhanced diagnostic precision for genetic disorders and cancers, leading to more effective and personalized treatment strategies ^[7, 9]. Additionally, 454 sequencing has made significant contributions to genomics by facilitating the assembly of complex genomes and identifying structural variations that are difficult to resolve with shorter-read technologies. This capability has been particularly useful in plant and animal genomics, where long reads aid in the resolution of repetitive regions and large structural variants, thus advancing our understanding of genomic architecture and evolution ^[10, 14]. The platform's versatility has also enabled large-scale studies in metagenomics and

epigenomics, further underscoring its role in advancing both basic and applied research in the biological sciences ^[9, 14].

References

1. Xuan J, Yu Y, Qing T, Guo L, Shi L. Next-generation sequencing in the clinic: Promises and challenges. *Cancer Lett.* 2013 Nov;340(2):284-95. DOI: 10.1016/j.canlet.2012.11.025.
2. Nowrousian M. Next-Generation Sequencing Techniques for Eukaryotic Microorganisms: Sequencing-Based Solutions to Biological Problems. *Eukaryot Cell.* 2010 Sep;9(9):1300-10. doi: 10.1128/EC.00123-10.
3. Zhang J, Chiodini R, Badr A, Zhang G. The impact of next-generation sequencing on genomics. *J Genet Genomics.* 2011 Mar;38(3):95-109. doi: 10.1016/j.jgg.2011.02.003.
4. Degnan PH, Ochman H. Illumina-based analysis of microbial community diversity. *ISME J.* 2011 Jun;6(1):183-94. doi: 10.1038/ismej.2011.74.
5. Yang Y, Xie B, Yan J. Application of Next-generation Sequencing Technology in Forensic Science. *Genomics Proteomics Bioinformatics.* 2014 Oct;12(5):190-7. doi: 10.1016/j.gpb.2014.09.001.
6. Margulies M, Egholm M, Altman WE, Attiya S, Bader JS, Bembien LA, *et al.* Genome sequencing in micro-fabricated high-density picolitre reactors. *Nature.* 2005 Jul;437(7057):376-80. doi: 10.1038/nature03959.
7. Rothberg JM, Leamon JH. The development and impact of 454 sequencing. *Nat Biotechnol.* 2008 Oct;26(10):1117-24.
8. Quail MA, Kozarewa I, Smith F, *et al.* A large genome center's improvements to the Illumina sequencing system. *Nat Methods.* 2008 Dec;5(12):1005-10.
9. Mardis ER. Next-generation DNA sequencing methods. *Annu Rev Genomics Hum Genet.* 2008;9:387-402.
10. Metzker ML. Sequencing technologies-the next generation. *Nat Rev Genet.* 2010 Jan;11(1):31-46. Doi: 10.1038/nrg2626.
11. Liu L, Li Y, Li S, Hu N, He Y, Pong R, *et al.* Comparison of next-generation sequencing systems. *J Biomed Biotechnol.*; 2012. p. 251-364. Doi: 10.1155/2012/251364.
12. Rhoads A, Au KF. PacBio sequencing and its applications. *Genomics Proteomics Bioinformatics.* 2015 Oct;13(5):278-89. doi: 10.1016/j.gpb.2015.08.002.

13. Roche. Roche to discontinue 454 sequencing platform. Genome Web. 2013. Available from: <https://www.genomeweb.com/roche-discontinue-454-sequencing-platform>.
14. Schuster SC, Morell V, Davidson EH. Advances in genome sequencing and assembly. *Annu Rev Genomics Hum Genet.* 2008;9:403-29.
15. Sogin ML, Morrison HG, Huber JA, Mark Welch D, Daugherty SC, Emerald MA, *et al.* Microbial diversity in the deep sea and the underexplored "rare biosphere". *Proc. Natl. Acad. Sci. USA.* 2006 Aug;103(32):12115-20. Doi: 10.1073/pnas.0605127103.

Chapter - 10

Alteration in Proline and Sugar Content in Maize (*Zea mays* L.) Genotypes under drought Condition

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Chapter - 10

Alteration in Proline and Sugar Content in Maize (*Zea mays* L.) Genotypes under drought Condition

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Abstract

Maize is a cereal crop that has an imperative part of trades as well as a principal nutriment in several nations all over the globe. Its growth, production and productivity extremely influenced by drought stress. A study was conducted to examine the influence of drought condition on sugar and proline content in 80 genotypes including parents raised in half diallel crosses under irrigated and partial irrigated circumstances in RBD with two replications. Proline and sugar content was estimated to evaluate the potential of genotypes against drought. Upsurge in the content of sugar and proline among genotypes was investigated highly significant at 1% probability level. Among eighty maize genotypes, genotypes viz., IL1XIL4, IL3XIL5, IL2XIL5, IL2XIL11, IL7XIL11, IL9XIL10, IL8, IL5XIL11, IL1XIL7 and IL1XIL8 showed enhanced levels of sugar and proline content. Henceforth, these genotypes may be drought tolerant and could be used in varietal development programme to breed drought tolerant cultivar (s).

Keywords: Maize, biochemical parameters, drought, proline and sugar content

Introduction

Maize is a crop that has been used as a staple food in several nations all over the world ^[1-4]. Environmental factors that could influence growth, production and productivity of many crop plants including maize is drought ^[5-22]. This is comprehensible because water is the chief essential part of living organisms, including maize. Drought is one of the greatest momentous abiotic stresses distressing plant growth, development and productivity. When tested with convinced environmental circumstances for instance drought, some plants have the skill to familiarize to the situation where they are cultivated ^[23]. In upcoming days, worldwide heating and the expansion of human population can lead to lessening of water capitals and an upsurge in

arid and semiarid zones. Consequently, the investigation of mechanisms of plant adaptation and tolerance to drought, along with the ability to convalesce after water discrepancy is an imperative job for present investigation ^[24].

To explicate plant competence to endure under drought, it is necessary to investigate the diverse physio- morphological ^[25-26] and biochemical ^[27-37] adaptation and tolerance along with the devices of recovery under rehydration. Plant tolerance to water shortage needs the ability to sustain roles under uncomplimentary water circumstances and to convalesce water availability and functions speedily after rewatering. Current investigations displayed that repossession stage is as imperative as the stress treatment as an effectual retrieval influences added development and growth ^[38-39].

Assessment of tolerance against drought stress may be performed by recognizing the physiological features essential to tolerance ^[40-43]. This is a pointer used frequently to choose tolerant genotypes or tolerance to drought ^[44-46, 40]. Physiological activities that may be utilized is organic matter accretion in the cytosol, like osmolyte and to uphold turgor pressure inside the cell. The alteration in carbohydrate metabolism under drought circumstances are meticulously associated to photosynthesis and transpiration and are of prime significance for maintaining water balance of crop plants ^[47-50, 44, 24]. Earlier, a shrill upsurge in the content of reducing sugars and proline concurrently with substantial decrease in the rate of photosynthesis and transpiration throughout the acclimatization of corn saplings against drought ^[51] has been described. Gathering of osmolytes in the cells is recognized to lead to the development of concentration gradient between the inside and outside cell cubicles. This concentration gradient might produce auspicious circumstances for the transfer of osmolytes from the photosynthesizing cells into apoplast.

The objective of current study was to succeed a theory that the accretion of soluble sugars and proline under drought circumstances possibly facilitate to a depletion in ψ_{wa} (water potential of mesophyll cells' apoplast in substomatal cavity) and, the conflicting, the decrease of osmolyte content after rewatering consequences in an upsurge in ψ_{wa} together with refurbishment of the leaf water situation. Considerate physiological response of maize genotypes under drought circumstances may consequences in envisaging drought tolerant cultivars of maize. Considering these realities in attention, this study was intended to assess the reply of maize genotypes to proline and total soluble sugars contents in the maize leaves, at the foliar growth stage under irrigated and partly irrigated circumstances.

Materials and Method

Experimental material and growth circumstances

The seeds of 12 corn inbred lines were collected from Sam Higginbottom Agriculture Science and Technology University, Prayagraj, U.P., India (Table 1) and hybridization was performed in half diallel design as recommended by Jinks and Hayman^[52] and 66 F₁ hybrids were obtained. These hybrids were experimented accompanied by two controls *i.e.*, HKI1105 (drought tolerant) and HKI1128 (drought susceptible). Finally, a total of 80 genotypes together with 12 parents, 66 F₁ hybrids and two controls were experimented under irrigated and partially irrigated circumstances in a RBD (randomized block design) in two replications. Each genotype was cultivated in 2 rows of 4 meters with an arrangement of 60 cm between rows and 20 cm between plant to plant. Drought condition was created from 10 days before flowering by preventing irrigation. The irrigation was continued when soil moisture grasps impermanent wilting point. The assemblage of maize leaves for sugar and proline content examination was carried out on reproductive phase after sowing. The second leaf from the top leaf were utilized for sugar and proline analyses.

Table 1: List of inbred lines with their parentage used in study

S. No.	Lines/ genotypes	Parentage	Source
1.	IL-1	CM-13	SHUATS, Allahabad
2.	IL-2	CML-193	SHUATS, Allahabad
3.	IL-3	CML-439	SHUATS, Allahabad
4.	IL-4	NBPGR-36417	SHUATS, Allahabad
5.	IL-5	NBPGR-36417 X NBPGR-33000	SHUATS, Allahabad
6.	IL-6	(103) NBPGR-36548 X (97) NBPGR-36407	SHUATS, Allahabad
7.	IL-7	DMR-N 21 X NBPGR-32809	SHUATS, Allahabad
8.	IL-8	LM- 13 X NBPGR-31899	SHUATS, Allahabad
9.	IL-9	CML-224-1 X NBPGR-32809	SHUATS, Allahabad
10.	IL-10	NBPGR-36550 X NBPGR-36407	SHUATS, Allahabad
11.	IL-11	KL- 153237 X VL- 1016536	SHUATS, Allahabad
12.	IL-12	CML- 161 X VL- 1056	SHUATS, Allahabad
13.	DS	HKI 1128	
14.	DT	HKI 1105	

DS: Drought stress, **DT:** Drought tolerance.

Sugar content estimation

Total sugar content was estimated using Anthrone tactic as described by Irigoyen *et al.* [53]. A 0.5 gram of the leaf was crumpled and 5.0 ml of 80% hot ethanol was supplemented to it. The combination was centrifuged at 6000 rpm for 15 min. The resultant supernatant was unglued into another test tube and 12.5 ml of 80% ethanol was mixed with it. A 1.0 ml of the solution was engaged and supplemented with 1.0 ml of 0.2% anthrone. The combination was animated in a water bath at 100 °C for 10 min. The reaction was terminated by incubating the mixture on ice for 5 min. Total soluble sugar content was determined employing a spectrophotometer at 620 nm wavelength. Calculation of the total soluble sugar content was done by creating a standard curve using a standard glucose and was denoted as μgg^{-1} fresh weight (μgg^{-1} FW).

Analysis of proline content

Proline content was investigated following the method of Bates [54-55]. A 1.0 gram of the leaf was wrinkled and was regimented with 5.0 ml of 3% sulfosalicylic acid. The homogenate was centrifuged at 6,000 rpm for 15 min. A 1.0 ml of the supernatant was engaged and 1.0 ml each of nynthidrin and acetic acid were supplemented to it. This was heated in water bath for 1 h and was hatched in ice for 5 min. A 2.0 ml of the solution was engaged and removed with 2.0 ml of toluene and quickly stunned with a vortex until chromoporm was shaped. The upper layer of the chromoporm was acquired and the absorbance was documented with a spectrophotometer at 520 nm wavelength. To decide the proline content, a standard curve was drawn employing pure proline. The proline content was determined in μmolg^{-1} fresh weight (μmolg^{-1} FW) unit.

Analysis of data

The acquired data were calculated employing ANOVA to predict the influence of drought stress on the maize genotypes. An average assessment was made employing the DMRT (Duncan's multiple range test) at 5% probability level. ANOVA and DMRT were accomplished employing software R [56-57].

Results and Discussion

Sugar and protein content of 80 genotypes were analyzed and presented in Table 2, Fig. 1 and Fig. 2. The analysis of variance evidently designated presence of considerable extent of disparities among 80 corn genotypes encompassed in the current study for both of the biochemical traits

investigated and displayed in Table 3. Generally, the enhanced levels of proline and total soluble sugar was inspected under the partial irrigated condition while the lowest under the irrigated situation.

Table 2: Level of sugar and proline content of the maize leaf under Irrigated (I) and Partial irrigated (PI) conditions

Sr. No.	Genotype	Sugar content ($\mu\text{mol/g FW}$)		Proline content ($\mu\text{mol/g FW}$)	
		I	PI	I	PI
1	IL1 X IL2	1.14	1.3	32.33	42.91
2	IL1 X IL3	1.35	1.6	35.1	38.11
3	IL1 X IL4	1.41	1.3	39.45	42.76
4	IL1 X IL5	1.3	1.41	34.9	42.45
5	IL1 X IL6	0.85	1.02	32.86	33.76
6	IL1 X IL7	0.87	1.17	41.9	45.11
7	IL1 X IL8	0.94	1.24	45.1	38.86
8	IL1 X IL9	1.3	1.32	38.35	42.71
9	IL1 X IL10	1.15	1.48	38.7	41.08
10	IL1 X IL11	1.35	1.56	32.66	34.36
11	IL1 X IL12	1.33	1.57	36.46	39.01
12	IL2 X IL3	0.96	1.26	37.61	40.21
13	IL2 X IL4	0.96	1.39	37.1	40.06
14	IL2 X IL5	0.94	2.4	36.25	42.81
15	IL2 X IL6	0.99	1.11	37.41	39.01
16	IL2 X IL7	0.93	1.18	35.01	36.76
17	IL2 X IL8	0.9	0.99	35.36	37.51
18	IL2 X IL9	0.82	0.91	34.3	38.26
19	IL2 X IL10	0.87	1.02	37.95	40.27
20	IL2 X IL11	0.94	2.37	37.81	40.96
21	IL2 X IL12	1.14	1.3	31.46	33.46
22	IL3 X IL4	1.35	1.59	20.05	29.62
23	IL3 X IL5	1.41	1.6	39.4	40.51
24	IL3 X IL6	1.3	1.62	36.95	40.81
25	IL3 X IL7	0.85	1.3	33.86	36.01
26	IL3 X IL8	0.87	1.05	34.61	36.31
27	IL3 X IL9	0.94	1.18	35.36	36.61
28	IL3 X IL10	1.3	1.48	20.9	23.61
29	IL3 X IL11	1.15	1.47	27.36	30.76
30	IL3 X IL12	1.35	1.54	35.26	39.76
31	IL4 X IL5	1.33	1.6	38.46	42.21

32	IL4 X IL6	0.96	1.15	29.26	30.46
33	IL4 X IL7	0.96	1.14	33.01	34.36
34	IL4 X IL8	0.94	1.44	16.4	23.71
35	IL4 X IL9	0.99	1.41	21.8	28.56
36	IL4 X IL10	0.93	1.14	31.56	35.86
37	IL4 X IL11	0.9	1.44	23.6	29.56
38	IL4 X IL12	0.82	1.41	22.9	28.96
39	IL5 X IL6	0.87	1.17	24.45	27.91
40	IL5 X IL7	0.94	1.33	16.4	22.65
41	IL5 X IL8	0.94	1.45	21.3	25.94
42	IL5 X IL9	0.87	1.84	33.81	38.41
43	IL5 X IL10	0.88	1.81	34.06	38.56
44	IL5 X IL11	0.9	1.17	46.46	50.41
45	IL5 X IL12	0.99	1.33	17.1	31.81
46	IL6 X IL7	1.14	1.38	25.86	30.82
47	IL6 X IL8	1.35	1.51	18.05	32.11
48	IL6 X IL9	1.41	1.8	17.6	37.36
49	IL6 X IL10	1.3	1.93	21.45	22.72
50	IL6 X IL11	0.85	1.81	22.25	40.36
51	IL6 X IL12	0.87	1.51	20.9	34.21
52	IL7 X IL8	0.94	1.8	21.85	30.76
53	IL7 X IL9	1.3	1.93	25.7	34.81
54	IL7 X IL10	1.15	1.81	24.95	42.96
55	IL7 X IL11	1.35	1.51	53.8	63.41
56	IL7 X IL12	1.33	1.8	21.65	31.21
57	IL8 X IL9	0.96	1.93	36.85	39.01
58	IL8 X IL10	0.96	1.81	28.06	31.06
59	IL8 X IL11	0.94	1.51	29.61	33.31
60	IL8 X IL12	0.99	1.8	19.7	22.41
61	IL9 X IL10	0.93	1.93	25.55	56.16
62	IL9 X IL11	0.9	1.81	23.3	31.61
63	IL9 X IL12	0.82	1.14	26.41	31.38
64	IL10 X IL11	0.87	1.35	33.16	36.31
65	IL10 X IL12	0.94	1.41	29.21	31.21
66	IL11 X IL12	1.14	1.3	26.11	28.41
67	HKI 1105	1.35	1.14	37.61	39.91
68	HKI 1128	1.41	1.35	29.26	31.51
69	IL1	1.3	1.41	29.96	31.51

70	IL2	0.85	1.3	35.6	41.71
71	IL3	0.87	1.35	27.56	29.56
72	IL4	0.94	1.54	29.71	31.36
73	IL5	1.3	1.71	38.95	42.61
74	IL6	1.15	1.45	37.21	39.61
75	IL7	1.35	1.3	28.06	28.96
76	IL8	1.33	2.49	45.4	53.71
77	IL9	0.96	1.12	24.7	21.01
78	IL10	0.96	1.72	37	41.26
79	IL11	0.94	1.78	25.35	37.21
80	IL12	0.99	1.84	32.56	33.91
Mean		1.06	1.48	30.92	35.99
CV		8.74	10.10	11.85	5.70
CD_{0.05}		0.185	0.298	7.290	4.080

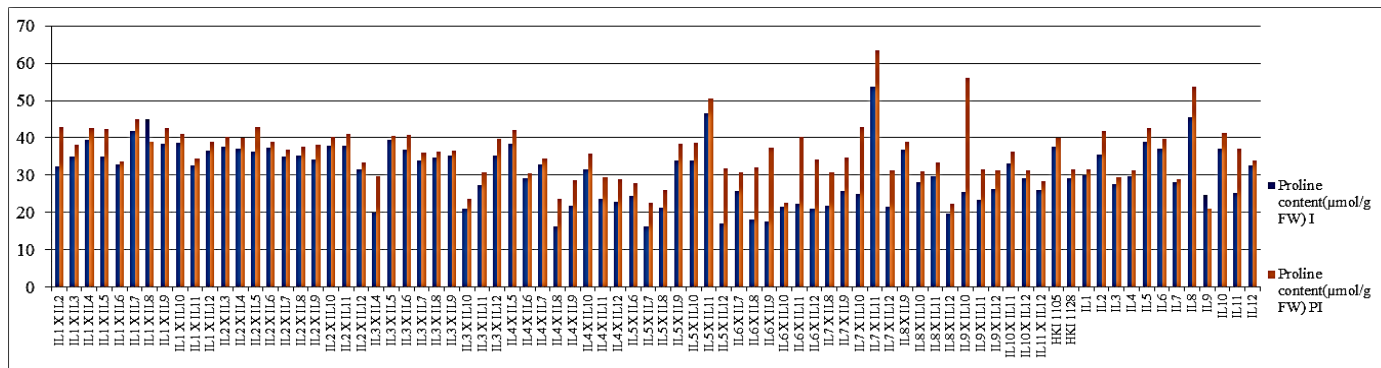
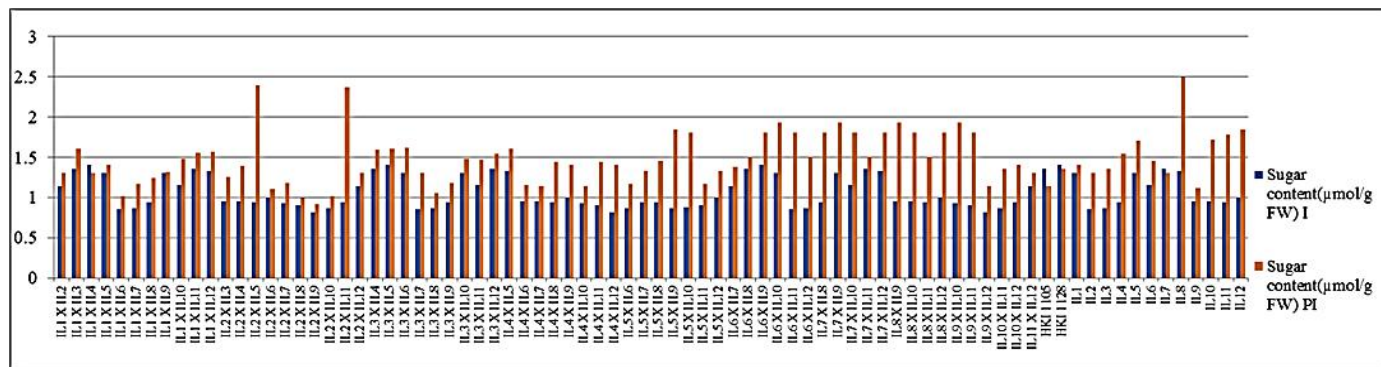
Table 3: Analysis of variance for proline and sugar content

Level of variance	Df		SS		MSS		F value		Probability (>F)	
	PC	SC	PC	SC	PC	SC	PC	SC	PC	SC
Replication	1	1	184.3	2.659	184.35	2.659	20.13	105.22	0.00 ***	0.00 ***
Treatment (A)	79	79	16057.9	13.300	203.26	0.168	22.20	6.66	0.00 ***	0.00 ***
Drought stress(B)	1	1	2059.1	14.074	2059.13	14.074	224.86	556.89	0.00 ***	0.00 ***
Interaction (AxB)	79	79	2221.3	8.792	28.12	0.111	3.07	4.40	0.00 ***	0.00 ***
Errors	159	159	1456.1	4.018	9.16	0.025				

***Significantly different at $\alpha < 0.0001$

PC=Proline content

SC=Sugar content



Soluble sugars are the main osmotic alteration ingredients and hence are substantial indicators of tolerance. Under irrigated situation, among 80 corn genotypes, sugar content arrayed between $0.82\mu\text{molg}^{-1}$ FW to $1.41\mu\text{molg}^{-1}$ FW with an average worth of $1.06\mu\text{molg}^{-1}\text{FW}$ (Table 2; Fig.1). While in irrigated situation, the maximum sugar content was investigated in genotype IL1XIL4 (1.41) tracked by IL3XIL5 (1.41), IL6XIL9 (1.41), HK11128 (1.41) and IL1XIL3 (1.35) whereas the minimum in genotype IL2XIL9 (0.82). Nevertheless, under partial irrigated situation, it speckled between $0.91\mu\text{molg}^{-1}$ FW to $2.49\mu\text{molg}^{-1}$ FW with an average worth of $1.48\mu\text{molg}^{-1}$ FW. While under partial irrigated circumstances, the maximum count was evident in genotype IL8 (2.49) pursued by IL2XIL5 (2.4), IL2XIL11 (2.37), IL7XIL9 (1.93) and IL6XIL10 (1.93), whereas the lowest worth was observed in genotype IL2XIL9 (0.91). Genotypes chasing higher sugar content might be drought tolerant as testified previous by Irigoyen *et al.* [53], Kachare *et al.* [58], Choudhary *et al.* [10], Mishra *et al.* [16] and Sharma *et al.* [20] as they detailed that the sugar content in leaves of the plant enhanced under drought circumstances. Arabzadeh [59] stated that the sugar melts compatible metabolites and absorption enhances with enlarged drought stress and abridged soil water content. Furthermore, in another way, plants usage to endure drought stress is by modifiable osmotic potential of the cell, particularly if drought stress upsurges progressively from slight stress to severe one [8, 23, 2].

Once plants are challenged by drought pressure, the osmotic pressure of the plant cell adjusts several progressions by means of the gathering of non-toxic solutes privileged the cell [8, 23]. This osmotic gathering ensues since the cell water potential diminutions thus snowballing the concentration of liquified substances to sanctuary turgidness of the cell. Additionally, compatible solutes avert interface between the ions and cellular machineries by substituting the water molecules around the component, thereby preventing destabilization during drought. Osmolyte accumulation is also due to increased biosynthesis without degradation [23, 24]. Osmotic cell potential can be accustomed by increasing the concentration of sugar which can decrease water potential of cells without inhibiting the function of the enzyme and does not reduce turgidity of the cell. Sugar accumulation in drought stress conditions helps to uphold the stability of the membrane, prevent and protect membrane fusion and; keep protein so as to remain functional [60, 59, 23, 24, 2].

Proline is also believed as a convincing drought tolerance indicator. As like under irrigated ailment, proline content varied between $16.4\mu\text{molg}^{-1}$ FW

to 53.8 μmolg^{-1} FW with a mean worth of 30.92 μmolg^{-1} FW (Table 2; Fig.2). The maximum proline level investigated in genotype IL7XIL11 (53.8 μmolg^{-1} FW) followed by genotypes IL5XIL11 (46.46 μmolg^{-1} FW), IL8 (45.4 μmolg^{-1} FW), IL1XIL8 (45.1 μmolg^{-1} FW) and IL1XIL7 (41.9 μmolg^{-1} FW) whereas the minimum count was found in genotype IL4XIL8 (16.4 μmolg^{-1} FW). Nevertheless, under partial irrigated situation, proline level was ranged between 21.01 μmolg^{-1} FW to 63.41 μmolg^{-1} FW with an average worth of 35.99 μmolg^{-1} FW. Under partial irrigated circumstances the maximum proline content was investigated in genotype IL7XIL11 (63.41 μmolg^{-1} FW) intimately followed by IL9XIL10 (56.16 μmolg^{-1} FW), IL8 (53.71 μmolg^{-1} FW), IL5XIL11 (50.41 μmolg^{-1} FW) and IL1XIL7 (45.11 μmolg^{-1} FW), whereas the minimum count was recorded in genotype IL9 (21.01 μmolg^{-1} FW). Genotypes displaying higher proline level may be putative drought tolerance as recommended by Kachare *et al.* [58], Tripathi *et al.* [17] and Sharma *et al.* [20] as they also determined in their studies on screening of soybean genotypes. It may be perchance owing to improved proline levels sustains cell water level under drought [9]. Umezawa *et al.* [61] and Krasensky and Jonak [7] reported that plants have the aptitude to accrue non-toxic amalgams for instance proline that defends cells damage because of low water potential of cells, which is a means of acclimatization of plants against drought.

Corn is very subtle to water stress [62]. The variation in the responses against drought among 80 corn genotypes displayed that each cultivar has diverse aptitude to synthesis proline and total soluble sugar with an upsurge with drought stress action. On the basis of these findings, it is vibrant that the proline and sugar content in the leaves was suggestively amplified owing to the upsurge in the level of drought stress. Comparable findings were also reported by Sinay *et al.* [24].

Conclusion

In this study, eighty maize genotypes were assessed based on sugar and proline content. Amplified level of sugar and proline content were investigated with upsurge under drought condition. The results advise that, genotypes IL1XIL4, IL3XIL5, IL2XIL5, IL2XIL11, IL7XIL11, IL9XIL10, IL8, IL5XIL11, IL1XIL7 and IL1XIL8 may be utilized for future crop enhancement scheme to develop drought tolerant varieties as augmented level of sugar and proline was explored in such genotypes.

Conflict of Interest: None.

References

1. Khayatnezhad M, Hasanuzzaman M, Gholamin R. Assessment of yield and yield components and drought tolerance at end-of season drought condition on corn hybrids (*Zea mays* L.). Aust. J. Crop Sci. 2011;5(12):1493-1500
2. Naghavi MR, Aboughadareh AP, Khalili M. Evaluation of drought tolerance indices for screening some of corn (*Zea mays* L.) cultivars under environmental conditions. Notulae Scientia Biologicae. 2013;5(3):388-393
3. Yadav PK, Singh AK, Tripathi MK, Tiwari S, Rathore J. Morpho-physiological characterization of maize (*Zea mays* L.) genotypes against drought. Biological Forum. 2022a;14(2):0975-1130.
4. Yadav PK, Tripathi MK, Tiwari S, Chauhan S, Tripathi N, Sikarwar RS, et al. Genetic components and variability assessment for grain yield and its accrediting traits in maize (*Zea mays* L.). International Journal of Environment and Climate Change. 2023a;13(9):772-784.
5. Bahar B, Yildirim M. Heat and drought resistances criteria in spring bread wheat: Drought resistance parameters. Sci. Res. Essays. 2010;5(13):1742-1745
6. Batlang U. Identification of drought-responsive genes and validation for drought resistance in rice. Crop and Soil Environmental Science. Doctoral Dissertations; 2010.
7. Krasensky J, Jonak C. Drought, salt and temperature stress induced metabolic rearrangements and regulatory networks. J Exp. Bot; 2012. p. 1-16
8. Lipiec J, Doussan C, Nosalewicz A, Kondracka K. Effect of drought and heat stresses on plant growth and yield. A review. Int. Agrophys. 2013;27:463-477.
9. Choudhary ML, Tripathi MK, Tiwari S, Pandya RK, Gupta N, Tripathi N, et al. Screening of pearl millet [*Pennisetum glaucum* (L) R Br] germplasm lines for drought tolerance based on morpho-physiological traits and SSR markers. Curr. J Appl. Sci. Technol. 2021a;40(5):46-63.
10. Choudhary ML, Tripathi MK, Gupta N, Tiwari S, Tripathi N, Parihar P, et al. Screening of pearl millet [*Pennisetum glaucum* [L] R Br] germplasm lines against drought tolerance based on biochemical traits. Current Journal of Applied Science & Technology. 2021b;40(23):1-12.

11. Asati R, Tripathi MK, Tiwari S, Yadav RK, Tripathi N. Molecular breeding and drought tolerance in chickpea. *Life*. 2022;12:18-46. <https://doi.org/10.3390/life12111846>
12. Mishra N, Tripathi MK, Tiwari S, Tripathi N, Gupta N, Sharma A. Morphological and physiological performance of Indian soybean [*Glycine max* (L.) Merrill] genotypes in respect to drought. *Legume Research*; 2021a. DOI: 10.18805/LR-4550
13. Mishra N, Tripathi MK, Tiwari S, Tripathi N, Ahuja A, Sapre S, *et al.* Cell suspension culture and *in vitro* screening for drought tolerance in soybean using poly-ethylene glycol. *Plants*. 2021b;10(3):517-536.
14. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A. Validation of drought tolerance gene-linked microsatellite markers and their efficiency for diversity assessment in a set of soybean genotypes. *Current Journal of Applied Science & Technology*. 2021c;40(25):48-57.
15. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A, *et al.* Changes in biochemical and antioxidant enzymes activities play significant role in drought tolerance in soybean. *International Journal of Agricultural Technology*. 2021d;17(4):1425-1446.
16. Mishra N, Tripathi MK, Tripathi N, Tiwari S, Gupta N, Sharma A, *et al.* Role of biochemical and antioxidant enzymes activities in drought tolerance in soybean: A recent study. In book: *Current Topics in Agricultural Sciences*. 2021e;3(8):102-119.
17. Tripathi MK, Tripathi N, Tiwari S, Mishra N, Sharma A, Tiwari S, *et al.* Identification of Indian soybean (*Glycine max* [L.] Merr.) genotypes for drought tolerance and genetic diversity analysis using SSR markers. *Scientist*. 2023;3(3):31-46.
18. Tiwari PN, Tiwari S, Sapre S, Babbar A, Tripathi N, Tiwari S, *et al.* Screening and selection of drought-tolerant high-yielding chickpea genotypes based on physio-biochemical selection indices and yield trials. *Life*. 2023a;13:1405.
19. Yadav PK, Singh AK, Tripathi MK, Tiwari S, Yadav SK, Solanki RS, *et al.* Assessment of maize (*Zea mays* L.) genotypes on the basis of biochemical contents in respect to drought. *Pharma Innovation*. 2022b;11(6):1996-2002.
20. Sharma A, Tripathi MK, Tiwari S, Gupta N, Tripathi N, Mishra N. Evaluation of soybean (*Glycine max* L.) genotypes on the basis of biochemical contents and anti-oxidant enzyme activities. *Legume Res*. 2021;44:LR-467.

21. Yadav PK, Singh AK, Tripathi MK, Tiwari S, Yadav SK, Tripathi N. Morphophysiological and molecular characterization of maize (*Zea mays* L.) genotypes for drought tolerance. *European Journal of Applied Sciences*. 2022c;10(6):65-87.
22. Yadav PK, Tripathi MK, Tiwari S, Chauhan S, Tripathi N, Sikarwar RS, *et al.* Biochemical characterization of parental inbred lines and hybrids of maize (*Zea mays* L.) under different irrigation conditions. *International Journal of Plant & Soil Science*. 2023b;35(18):1743-1762.
23. Lisar SYS, Motafakkerazad R, Hossain MM, Rahman IMM. Water stress in plants: Causes, effects and responses. *Water stress*, Prof. Ismail Md. Mofizur Rahman (Ed.); 2012. In Tech, Available from: <http://www.intechopen.com/books/water-stress/water-stress-inplants-causes-effects-and-responses>.
24. Sinay H, Karuwal RL. Proline and total soluble sugar content at the vegetative phase of six corn cultivars from Kisar Island Maluku, grown under drought stress conditions. *Int. J Adv. Agric. Res*. 2014;2:77-82.
25. Asati R, Tripathi MK, Yadav RK, Tiwari S, Chauhan S, Tripathi N, *et al.* Morphological description of chickpea (*Cicer arietanum* L.) genotypes using DUS characterization. *International Journal of Environment and Climate Change*. 2023;13(9):1321-1341.
26. Rajpoot P, Tripathi MK, Solanki RS, Tiwari S, Tripathi N, Chauhan S, *et al.* Genetic variability and multivariate analysis in pearl millet (*Pennisetum glaucum* (L.) R. Br.) germplasm lines. *Pharm Innovation*, 12(4), 216-226.
27. Sahu VK, Tiwari S, Tripathi MK, Gupta N, Tomar RS, Yasin M. Morpho-physiological and biochemical traits analysis for *Fusarium* wilt disease using gene-based markers in desi and Kabuli genotypes of chickpea (*Cicer arietinum* L.). *Indian J. Genet*. 2020a;80(2):163-172.
28. Sahu VK, Tiwari S, Gupta N, Tripathi MK, Yasin M. Evaluation of physiological and biochemical contents in desi and Kabuli chickpea *Legume Research*; 2020b. DOI:10.18805/LR-4265
29. Shyam C, Tripathi MK. Biochemical studies in Indian mustard [*Brassica juncea* (Linn) Czern & Coss] for fatty acid profiling. *Int. J Chem. Stud*. 2019;7(4):338-343.
30. Shyam C, Tripathi MK, Tripathi N, Tiwari S, Sikarwar RS. Genetic variations in fatty acids and oil compositions among 188 Indian Mustard *Brassica juncea* (Linn.) Czern & Coss genotypes. *Curr J Appl Sci Technol*. 2021;40(46):9-28 DOI: 10.9734/CJAST/2021/v40i4631629.

31. Gupta N, Tiwari S, Tripathi MK, Bhagyawant SS. Anti-nutritional and protein-based profiling of diverse desi and wild chickpea accessions. *Current Journal of Applied Science and Technology*. 2021;40(6):7-18. DOI: 10.9734/cjast/2021/v40i631312
32. Rajput S, Jain S, Tiwari S, Barela A, Chauhan S, Tiwari PN, *et al.* Biochemical characterization of chickpea (*Cicer arietinum* L.) genotypes. *Plant Cell Biotechnol. Mol. Biol.* 2023;24:1-9.
33. Sistu R, Tiwari S, Tripathi MK, Singh S, Gupta N, Tripathi N, *et al.* Effect of different biochemical parameters and antioxidant enzymes activities on drought indices in chickpea (*Cicer arietinum* L.). *Leg. Res.*; 2023. DOI: 10.18805/LR-5204.
34. Tomar YS, Tiwari S, Tripathi MK, Singh S, Gupta N. Genetic diversity, population structure and biochemical parameters estimations driving variations in groundnut germplasm. *Legume Research*; 2022. DOI: 10.18805/LR-4965.
35. Rana SS, Tiwari S, Gupta N, Tripathi MK, Tripathi N, Singh S, *et al.* Validating the nutraceutical significance of minor millets by employing nutritional-antinutritional profiling. *Life*. 2023;13(9):19-18. Doi: <https://doi.org/10.3390/life13091918>
36. Rathore MS, Tiwari S, Tripathi MK, Gupta N, Yadav S, Singh S, *et al.* Genetic diversity analysis of groundnut germplasm lines in respect to early and late leaf spot diseases and biochemical traits. *Legume Research*; 2022. DOI: 10.18805/LR-4833
37. Paliwal S, Tripathi MK, Sikarwar RS, Yadav RK, Sonaniya P. Evaluation of biochemical parameters in linseed (*Linum usitatissimum*). *International Journal of Advanced Biochemistry Research*. 2024;8(4):260-267. DOI: 10.33545/26174693.2024.v8.i4d.947
38. Chen D, Wang S, Cao B, *et al.* Genotypic variation in growth and physiological response to drought stress and re-watering reveals the critical role of recovery in drought adaptation in maize seedlings. *Front. Plant Sci.* 2016;6:12-41.
39. Kosová K, Vítámvás P, Urban MO, *et al.* Plant abiotic stress proteomics: The major factors determining alterations in cellular proteome. *Front. Plant Sci.* 2018;9:122.
40. Nio Song A, Banyo Y. Concentration of leaf chlorophyll as an indicator to water deficit in plant. *Sci. J.* 2011;11(2):166-173.

41. Tiwari PN, Tiwari S, Sapre S, Babbar A, Tripathi N, Tiwari S, *et al.* Prioritization of microsatellite markers linked with drought tolerance associated traits in chickpea (*Cicer arietinum* L.). Legume Research; 2023b. DOI:10.18805/LR-5191
42. Tiwari PN, Tiwari S, Sapre S, Tripathi N, Payasi DK, Singh M, *et al.* Prioritization of physio-biochemical selection indices and yield-attributing traits toward the acquisition of drought tolerance in chickpea (*Cicer arietinum* L.). Plants. 2023c;12:31-75. <https://doi.org/10.3390/plants12183175>
43. Paliwal S, Tripathi MK, Tiwari S, Tripathi N, Payasi DK, Tiwari PN, *et al.* Molecular advances to combat different biotic and abiotic stresses in linseed (*Linum usitatissimum* L.): A comprehensive review. Genes. 2023;14(7):14-61. Available: <https://doi.org/10.3390/genes14071461>
44. Chaves MM, Maroco JP, Pereira JS. Understanding plant responses to drought: From genes to whole plant. Func. Plant Biol. 2003;30:239-264.
45. Bartles D, Chandler JW. Drought: Avoidance and adaptation. Encyclopedia of Water Science 2nd Edition; 2007.
46. McCann SE, Huang B. Evaluation of drought tolerance and avoidance traits for six creeping bentgrass cultivar. Hort. Sci. 2008;43(2):519-524.
47. Hare PD, Cress WA, Van Staden J. Dissecting the role of osmolyte accumulation during stress. Plant Cell Environ. 1998;21:535-553.
48. Tarchevsky IA, Grechkin AN. Metabolism of plants under stress. Selected papers; 2001. p. 9-102.
49. Reddy AR, Chaitanya KV, Vivekanandan M. Drought-induced responses of photosynthesis and antioxidant metabolism in higher plants. J Plant Physiol. 2004;161:1189-1202.
50. Chaves MM, Flexas J, Pinheiro C. Photosynthesis under drought and salt stress: regulation mechanisms from whole plant to cell. Ann. Bot. London. 2009;103:551-560.
51. Nikolaeva MK, Maevskaya SN, Voronin Pyu. Photosynthetic CO₂/H₂O gas exchange and dynamics of carbohydrates content in maize leaves under drought. Russ. J Plant Physiol. 2017;64:536-542.
52. Jinks JL, Hayman BI. The analysis of diallel crosses, maize genetics coop. News Lett. 1953;27:48-54.

53. Irigoyen JJ, Emerich DW, Sanchez-Diaz M. Water stress induced changes in concentrations of proline and total soluble sugars in nodulated alfalfa (*Medicago sativa*) plants. *Physiol. Plant.* 1992;84:55-60.
54. Bates LS, Waldren RP, Teare ID. Rapid determination of free proline for water-stress studies. *Plant Soil.* 1973;39:205-207.
55. Mohammadkhani N, Heidari R. Drought-induced accumulation of soluble sugars and proline in two maize varieties. *World App. Sci. J.* 2008;3(3):448-453.
56. R Core Team. A language and environment for statistical computing. R Foundation for Statistical Computing Vienna, Austria; 2020. <http://www.R-project.org>.
57. De Mendiburu F. Una herramienta de análisis estadístico para la investigación agrícola Universidad Nacional de Ingeniería (UNI). Lima, Peru; 2009.
58. Kachare S, Tiwari S, Tripathi N, Thakur VV. Assessment of genetic diversity of soybean [*Glycine max* (L.) Merr.] genotypes using qualitative traits and microsatellite makers. *Agricultural Research*; 2019.
59. Arabzadeh N. The effect of drought stress on soluble carbohydrates (Sugars) in two species of *Haloxylon persicum* and *Haloxylon aphyllum*. *Asian J Plant Sci.* 2012;11(1):44-51.
60. Xonostle-Cazares B, Ramirez-Ortega FA, Flores-Elenes L. Ruiz-Medrano R. Drought tolerance in crop plants. *Am. J Plant Physiol.*; 2011. p. 1-16.
61. Umezawa T, Fujita M, Fujita Y, Yamaguchi SK, Shinozaki K. Engineering drought tolerance in plants: discovering and tailoring genes to unlock the future. *Current Opinion in Biotechnology.* 2006;17:113-122.
62. Shaddad MAK, Hamdia M, El-Samad A, Mohammed HT. Interactive effects of drought stress and phytohormones or polyamines on growth and yield of two maize (*Zea mays* L.) genotypes. *Am. J Plant Sci.* 2011;2:790-807.

Chapter - 11

Surface Plasmon Resonance (SPR)

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Chapter - 11

Surface Plasmon Resonance (SPR)

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Abstract

The advancement of new era technologies lead to the development of innovative detection techniques in science and medical research such as Surface plasmon resonance. SPR is a label-free detection method which emerged during the last two decades as a suitable and reliable platform in clinical analysis for biomolecular interactions. SPR is a phenomenon that occurs when electrons in a thin metal sheet become excited by light that is directed to the sheet with a particular angle of incidence and then travel parallel to the sheet. Assuming a constant light source wavelength and that the metal sheet is thin, the angle of incidence that triggers SPR is related to the refractive index of the material and even a small change in the refractive index will cause SPR. The technique makes it possible to measure interactions in real-time with high sensitivity and without the need of labels. This review book chapter discusses about the detailed view of SPR and its wide range of applications in optical-based sensors using either surface plasmon resonance (SPR) or Surface Plasmon Resonance Imaging (SPRI). Here I summarize the principles and provide working methods, examples and demonstrate the utility of SPR and SPRI through applications from the biomedical, proteomics, genomics and bioengineering fields. In addition, SPR signal amplification strategies and surface functionalization are covered in this chapter.

Keywords: Surface plasmon resonance, surface plasmon resonance imaging, biomolecular interactions, electrons, signal amplification

Introduction

Advancement of Science and Medical research, there are numerous strategies for protein labelling have been developed that allow the characterization of proteins regarding their structure, folding or interaction with other proteins ^[1, 2]. Surface plasmon resonance (SPR) is a label free detection method which has arisen during the last few decades. Labelling strategies are used to bring about the covalent attachment of reporter tags, such

as biotin, radioisotopes, fluorophores, or enzymes to the target biomolecules (proteins and nucleotides) to quantitatively assess binding among biomolecules [3, 4]. In addition, the use of molecular labels can cause steric hindrance or change in structural configurations, affecting the labelled molecules' affinities for their target biomolecules, which is a major challenge. Label-free detection eliminates the need for specialized tags or dyes, thereby allowing the sensitive measurement of target analytes and enabling the use of native biomolecules suitable for biologically relevant approaches. SPR is one of the suitable and reliable platforms is used in clinical analysis for biomolecular interactions. These techniques make it possible to measure interactions in real time with high sensitivity and without the need of labels [5, 6, 7].

In the theoretical physics the free electrons in a metal are treated as a liquid, entirely composed of electrons, that has a very high density (plasma). The fluctuations of density that appear on the surface of this material are called plasmons or surface plasmons. Each plasmon represents the quantization's of classically oscillating plasma waves. The plasmons represent discreet values of an oscillating plasma wave, therefore most of their properties can be directly derived from Maxwell's equations. Surface plasmon resonance refers to the electromagnetic response that occurs when plasmons are oscillating with the same frequency on the surface of a material. As these plasmons oscillate at specific resonant frequencies, they move with periodic driving forces that can become large amplitude oscillations when they interact. This phenomenon is stimulated by a light source. The frequency of the incidence of light must be equal to the natural frequency of the material or resonance will not occur. These oscillations travel on the surface between the material and air and travel in the direction of the negative dielectric material surface. Because these plasmons are on this boundary, they are very sensitive to a change in external stimuli such as the absorption of energy into the material. In order to excite the surface plasmons in a manner so that they can resonate, there are two popular configurations used to excited surface plasmon waves currently, the Otto and Kretschmann configurations.

One of the important properties of nanoparticles is that they exhibit SPR. When electric fields of light are directed at nanoparticles, the surface plasmons become excited and begin to resonate. This electric field also creates a separation of charge, which can be seen in Figure 3, that then forms a dipole oscillation in the same direction as the electric field of light. Due to the fact that the frequencies are the same, the SPR allows a strong absorption of the incidence light while also allowing some scattering of light (Fig. 1) these can

be measured using a UV-Vis Spectrometer. The SPR band intensity and wavelength is dependent on the properties of the particle, including the shape, structure, metal type, size, and dielectric material surrounding the medium which can include air. The band intensity is strongest for metals Ag and Au, however, other metals like Cu have been used before. Au is most commonly used to observe this phenomenon due to the fact that it is biologically compatible and inert even though Ag exhibits the strongest bands and sharpest peaks. At the nanoscale, SPR works the most efficiently and is capable of being observed through spectrometry [8, 9].

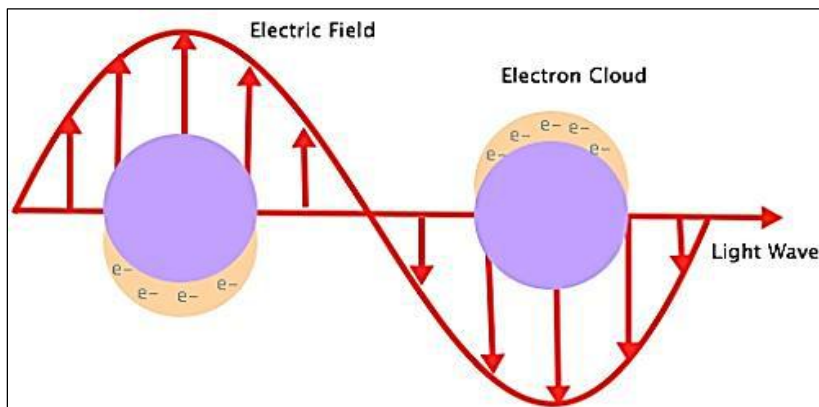


Fig 1: A light wave passes through a material, the induced electric field creates a charge separation in the atoms, creating an electron cloud that then allows for electrons to move freely [8, 9]

Plasmon Resonance Energy Transfer (PRET) and SPR Imaging

Plasmon Resonance Energy Transfer occurs when nanoparticles are connected to molecular chromophores (an atom or molecule whose presence is responsible for the color of the compound), then the plasmon resonance energy can be transferred to the molecular chromophore. The transfer of this energy paired with the natural frequencies of the biomolecules causes an overlap of resonant energy peak positions. The overlap of these two frequencies can cause spectral quenching dips on the Rayleigh scattering spectrum of a single nanoparticle. This quenching allows for ultrasensitive nanoscopic absorption spectroscopy, which is much more specific as well as faster and more efficient than optical absorption spectroscopy (Fig. 2.).

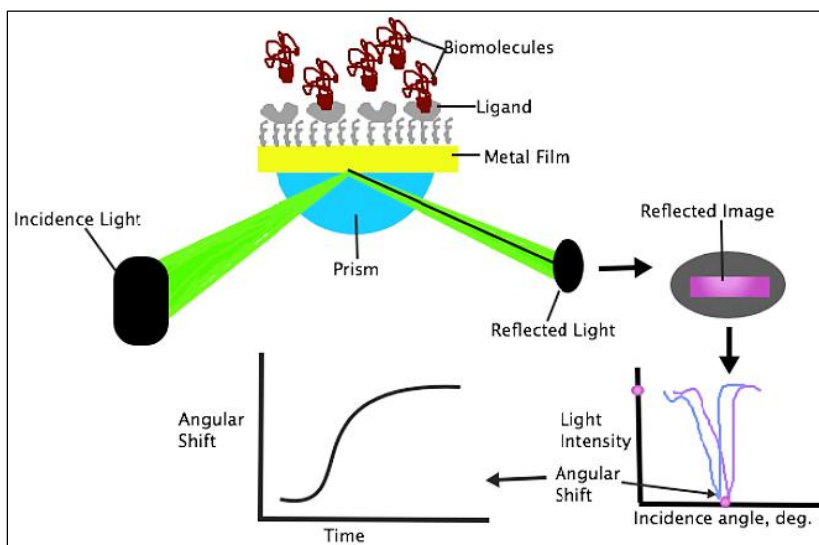


Fig 2: A simple schematic representation of how light enters the prism, goes through the thin metal film where SPR is occurring on the side of the film touching the ligands and biomolecules. The SPR allows certain frequencies of light through, allowing for an image of how the biomolecules are attached to the ligands

In early 1990s it was first introduced and SPR has been proven to be one of the most powerful technologies to determine specificity, affinity and kinetic parameters during the binding of macromolecules in many bonds types, including protein-protein ^[10, 11] protein-DNA ^[12, 13], enzyme-substrate or inhibitor ^[14, 15] receptor-drug ^[16, 17] lipid membrane-protein ^[18, 19], protein-polysaccharide ^[20] cell or virus-protein ^[21, 22, 23], among others. This optical technique measures the refractive index changes in the vicinity of thin metal layers (i.e., gold, silver, or aluminium films) in response to biomolecular interactions. Before a sample solution flows across the SPR surface, capturing agents, such as antibodies, enzymes, peptides and DNAs are immobilized on the surface. The changes in the SPR angle, which is the angle of minimum reflectivity, can be determined by varying the incidence angle and recording the reflected light intensity during the biological binding reactions between various biomolecules. So far, numerous studies have advanced the potential of SPR sensors by increasing the effectiveness of the technique.

The theory

The optical phenomenon of Surface Plasmon Resonance (SPR) used by Biacore systems enables the detection and quantification of protein-protein interactions in real time, without the use of labels. Biacore systems are widely

used for characterizing the interactions of proteins with other proteins, peptides, nucleic acids, lipids, and small molecules. When Biacore is used to measure protein interactions, one of the interactants is immobilized onto a sensor chip surface and the other interactant is passed over that surface in solution via an integrated microfluidic flow system. The immobilized interactant is referred to as the ligand and the injected interactant in solution is referred to as the analyte. Binding responses are measured in resonance units (RU) and are proportional to the molecular mass on the sensor chip surface and consequently to the number of molecules on the surface. Surface Plasmon Resonance (SPR) (Fig.3) was first shown to be amenable for the label-free study of interactions between biomolecules over 20 years ago ^[24]. Biacore pioneered commercial SPR biosensors offering a unique technology for collecting high quality, information-rich data from biomolecular binding events. Biacore's optical biosensors is to characterize binding events with samples ranging from proteins, nucleic acids, small molecules to complex mixtures, lipid vesicles, viruses, bacteria and eukaryotic cells.

Biacore

Biacore's optical biosensors are designed around three core technologies:

- An optical detector system that monitors the changes in SPR signal brought about by binding events in real time.
- An exchangeable sensor chip upon which one of the interacting biomolecules is immobilized or captured. The resulting biospecific surface is the site where biomolecular interactions occur.
- A microfluidic and liquid handling system that precisely controls the flow of buffer and sample over the sensor surface.

These systems are contained within the processing unit that communicates with a computer equipped with control and data evaluation software.

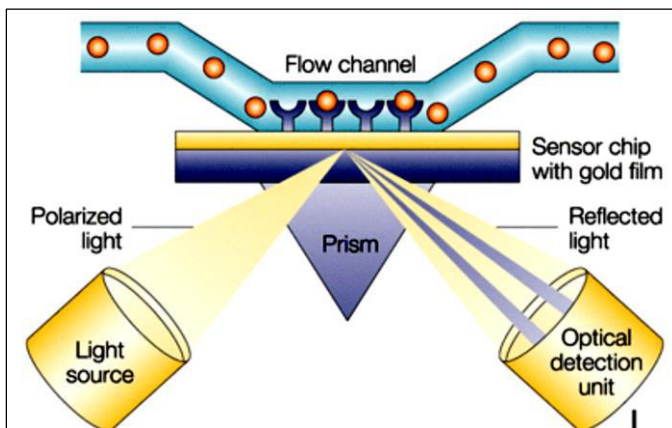


Fig 3: Surface Plasmon Resonance principle

The SPR detection system

Surface Plasmon Resonance (SPR) is a phenomenon that occurs in thin conducting films placed at the interface between two media of different refractive indices. In Biacore systems, a 50-nm layer of gold on the sensor chip is sandwiched between the glass layer of a sensor chip and the sample solution flowing through the microfluidic cartridge.

Plane polarized light from a near-infrared LED is focused on the back of the sensor chip under conditions of total internal reflection and a diode array detector monitors the intensity of the reflected light (Fig.1). Under these conditions, the light leaks an electromagnetic component called an evanescent wave across the gold interface into the sample/buffer solution. At a certain angle of incident light, the evanescent wave field excites electrons in the gold film resulting in the formation of surface plasmons (electron charge density waves) within the gold film with a concomitant drop in the intensity of the reflected light at this angle (SPR angle). When a change in mass occurs near the sensor chip surface, e.g., as a result of a binding event, the angle of light at which SPR occurs shifts due to a change in refractive index near the sensor chip surface.

A sensorgram depicts changes in SPR angle in real time, with responses measured in resonance units (RU). One RU corresponds to 0.0001° shift in SPR angle, or to 1pg/mm^2 of molecule on the surface. Since only the evanescent wave penetrates the sample, measurements can be carried out with colored, turbid, or even opaque solutions without interference from conventional light absorption or light scattering.

A sensorgram is a plot of the binding response in resonance units versus time in seconds, which is displayed and recorded as a change in mass occurs on the sensor chip surface (Fig. 2). The sensorgram provides essentially two kinds of information that are relevant to different types of applications:

- 1) The rate of interaction (association, dissociation, or both), which provides information on kinetic rate constants and analyte concentration.
- 2) The binding level, which provides information on affinity constants and can be used for qualitative or semi-quantitative applications (Fig 4a).

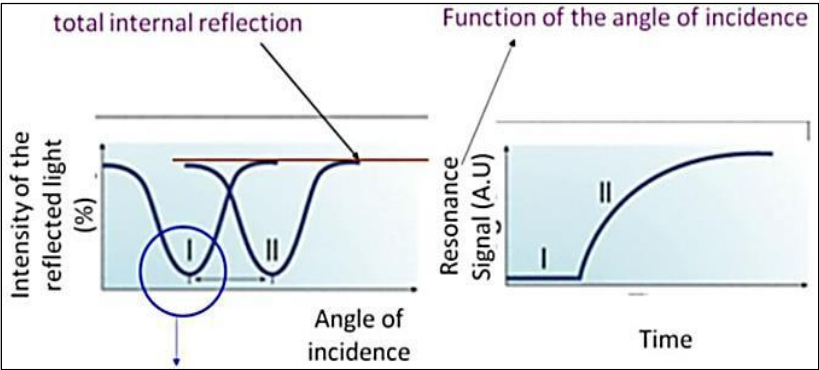
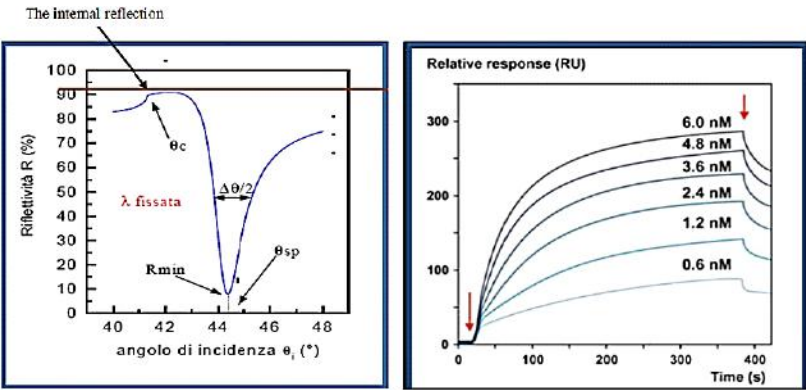


Fig 4a): The SPR wave absorbs energy and the reflected light decreases in the intensity



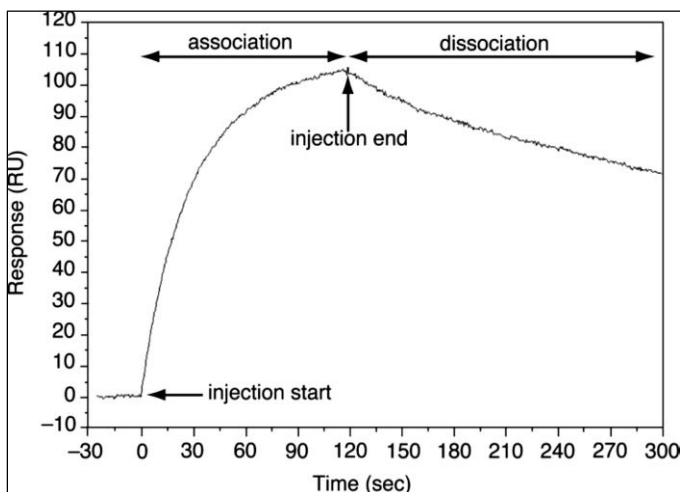


Fig 4(b)

Figure 4 (a). The sensorgram is a plot of response in resonance units (RU) versus time in seconds, which is presented continuously in real time. Upon injection of an analyte, if a binding interaction occurs, then an increase in mass occurs on the sensor chip surface and association is measured. (b). At the end of the analyte injection, as the complex decays, a decrease in mass occurs and dissociation is measured. Association and dissociation are measured as changes in response.

The microfluidic system

Biacore systems make use of an integrated microfluidic cartridge (IFC) to deliver samples and buffer to the sensor surface. The IFC consists of a series of micro-channels encased in a plastic housing. In automated systems, samples are transferred through a needle into the IFC. Flow cells are created through pressure contact between molded channels in the IFC and the sensor chip surface. A series of computer-controlled pneumatic valves meters precise access to the channels and the flow cells. The design of the IFC, as well as the number and configuration of flow cells, is specific to each instrument platform. Pulse-free syringe pumps ensure a continuous flow of buffer over the sensor surface whenever samples are not being injected.

There are two flow cells in the Biacore X instrument. Two different molecules can be immobilized on one sensor chip or one flow cell can be used as a reference and the other immobilized with a molecule of interest. Dual-channel detection of the SPR signal and serial flow allow simultaneous monitoring of both flow cells.

The instrument control software can be set to automatic in-line reference subtraction of the control surface signal from the same sample injection. Sample bound to the sensor chip surface can be recovered for complementary analysis and/or identification, e.g., using mass spectrometry. Biacore X is well suited for laboratory environments with low-sample throughput where many users handle different types of samples.

Surface preparation

a) Sensor chips

Although SPR can be generated in thin films made from conducting metals, all Biacore sensor chips are coated with a thin, uniform gold layer. Gold has a number of advantages in that it results in a well-defined reflectance minimum when a visible light source is used to generate the SPR signal, gold is amenable to covalent attachment of surface matrix layers and, in physiological buffer conditions, gold is mostly inert.

A variety of sensor surfaces are available to give a range of alternative possibilities for different experimental situations:

Table 1: Different types of sensor chips

	Sensor Chip CM5	Sensor Chip CM4	Sensor Chip CM3	Sensor Chip C1	Sensor Chip SA	Sensor Chip NTA	Sensor Chip L1	Sensor Chip NTR
Molecule to be immobilized								
Proteins	The first choice for immobilization of proteins via any, the CM5 chip (C200)	Useful if contaminants have a high positive charge	If partner in solution is very large e.g. a molecular complex	If partner in solution is multivalent or very large e.g. a molecular complex	When biotinylated			
Tagged proteins	For GST tags surface is generated with anti-GST antibody						For His tags	
UHR molecules, typically <1000 Da								
Membrane-associated molecules						Incorporate molecule into a lipid monolayer	Incorporate molecule into a lipid bilayer	
Nucleic acids	When modified with an amine group	Useful if contaminants have a high positive charge	If partner in solution is very large e.g. a molecular complex	If partner in solution is multivalent or very large e.g. a molecular complex	When biotinylated			
Carbohydrates	When modified with an aldehyde group	Useful if contaminants have a high positive charge	If partner in solution is very large e.g. a molecular complex	If partner in solution is multivalent or very large e.g. a molecular complex	When biotinylated			
Viruses or intact cells			Keep large particles close to the surface to maintain sensitivity	Keep large particles close to the surface to maintain sensitivity				
- Recommended choice - Good alternative								

Table 1. The most popular type of sensor chip has a layer of carboxymethylated (CM) dextran on top of the gold layer carboxyl groups allow covalent attachment of ligands or capturing molecules and also provide a hydrophilic environment for the interaction. A range of techniques can be used for ligand immobilization to a CM surface. Covalent coupling of the ligand can be performed by a variety of methods via free amino groups, thiol-disulfide exchange, or aldehyde groups.

An alternative to covalent immobilization is ligand capture, in which a molecule with high affinity for the ligand is chemically coupled to the sensor

chip. The ligand is then adsorbed to the capturing molecule from a solution in a separate step.

Sensor chips are reusable since noncovalently bound material can be removed from the Sensor Chip surface with an injection of a suitable regeneration solution.

Immobilization chemistries

A critical step in the development of reliable SPR assays is the selection of the most suitable immobilization technique such that ligand activity is maintained and binding sites are available to interacting partners. Commonly utilized immobilization strategies are outlined here. Guidelines and a detailed review of immobilization chemistries have been published recently [26].

Direct immobilization chemistries

A number of covalent immobilization techniques are available to covalently attach proteins or other biomolecules to the dextran on the sensor chip surface. All types of immobilization can be performed directly in the Biacore system, and a typical immobilization reaction usually takes <30 min to complete. Immobilization is usually carried out at low flow rates (5 to 10 $\mu\text{l}/\text{min}$) and the concentration of ligand used for immobilization is typically in the range of 1 to 100 $\mu\text{g}/\text{ml}$. To ensure specificity, it is recommended that the ligand purity exceed 95%.

a) Amine coupling

Amine coupling is the most generally applicable covalent coupling chemistry used to immobilize protein ligands (Fig. 5). Immobilization is via free primary amine groups such as lysine residues that are abundant in most proteins or the N-terminus of proteins and peptides. The dextran matrix on the sensor chip surface is first activated with a 1:1 mixture of 0.4 M 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC) and 0.1 M N-hydroxysuccinimide (NHS) to create reactive succinimide esters. The standard activation time is 7 min; this can be varied from 1 to 10 min to create fewer or more, respectively, reactive groups on the sensor chip surface depending on the immobilization level required. The ligand is then injected in low salt buffer lacking primary amines at a pH below the isoelectric point (pI) of the protein. Under these conditions, the ligand acquires a positive charge and is effectively preconcentrated into the negatively charged carboxymethyl dextran matrix. High local concentrations of ligand obtained through electrostatic preconcentration maximize the efficiency of amine coupling. Finally, unreacted esters are blocked with ethanolamine. The volume or concentration

of ligand injected may be varied to adjust the immobilization level. The ligand may be prevented from attaching to the surface via sites important to the interaction under study by having analyte present during immobilization. Stabilization of ligand activity during immobilization was recently reported by [27] where conformationally sensitive protein kinases, p38 α and GSK3 β , were immobilized on Sensor Chip CM5 by amine coupling in the presence of a specific reversible inhibitor. This treatment resulted in a more stable surface with much higher specific binding activity. In cases where amine coupling interferes with the binding site on the ligand or in the case of very acidic proteins, the ligand can be attached using alternative coupling chemistries or a high-affinity capture approach.

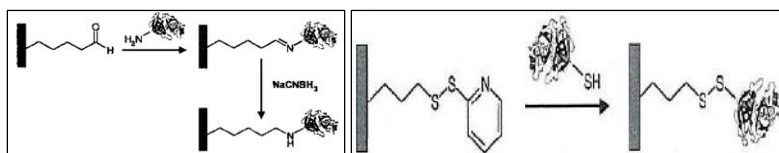


Fig 5: Ammine coupling and Thiol coupling

b) Thiol coupling

Thiol coupling is based on exchange reactions between thiol and active disulfide groups. In the case where the ligand has a free thiol group (typically cysteine residues), a surface thiol approach can be used where a disulfide group is introduced on the sensor chip dextran by first activating the surface with NHS and EDC to amine couple 2-(2-pyridinyldithio) ethaneamine (PDEA). Injection of the ligand results in thiol-disulfide exchange and excess PDEA groups are inactivated with cysteine- HCl. If the ligand lacks a free thiol group, a reactive disulfide (PDEA) can be linked to carboxyl groups on the ligand. The pyridyldisulfide groups linked to the ligand are then coupled to thiol groups on the sensor surface that has been derivatized through injection of NHS and EDC, cystamine and subsequent reduction with DTE or DTT. Modification of carboxyl groups results in an increase in the isoelectric point of the proteins, which is of additional benefit in the case of acidic proteins.

c) Maleimide coupling

Maleimide coupling is an alternative form of thiol coupling, which involves the formation of a stable thioether bond between reactive maleimido groups on the sensor surface and the thiol groups of the ligand molecule. Such surfaces have the capacity to withstand basic pH (>9.5) as well as reducing agents such as heterobifunctional reagents are available for introduction of

reactive maleimido groups to the sensor surface, including sulfo-MBS (m-maleimidobenzoyl-N-hydroxysulfosuccinimide ester), sulfo-SMCC (sulfosuccinimidyl-4-(N-maleimidomethyl)-cyclohexane-1-carboxylate), and GMBS (N-(g-maleimidobutyryloxy)sulfosuccinimide ester).

d) Aldehyde coupling

Aldehyde coupling involves the formation of a hydrazone bond via condensation of hydrazide groups on the sensor surface with aldehyde groups on the ligand molecule. These aldehyde moieties may be native to the protein or introduced through mild oxidation of cis-diols present in the ligand molecule. Aldehyde coupling is particularly useful for site directed immobilization of glyco-conjugates, glyco-proteins, and polysaccharides, and may also be useful for orientation-specific immobilization of proteins containing functional groups that may be converted to aldehyde moieties.

Indirect (capture) immobilization

a) General capture methods

Capture approaches provide an alternative to covalent immobilization and take advantage of tags commonly used for ligand purification. This technique involves high-affinity capture of the ligand onto a capturing molecule that has been covalently immobilized using one of the techniques described earlier (Fig.6). The requirement for ligand purity is less stringent for capture approaches than for covalent immobilization since the capture step can also provide a ligand purification step. Another benefit of capture approaches is the creation of a homogenous surface since all ligands are similarly oriented through a common site on the ligand (the tag). The affinity of the ligand for the capture molecule should be sufficiently high to ensure that little or no ligand dissociates from the surface for the duration of an analysis cycle. Monoclonal antibodies are frequently used as capture molecules, e.g., anti-GST antibodies can be immobilized and used to capture GST-tagged molecules. In general, regeneration of the surface removes both the ligand and the analyte at the end of an assay cycle such that fresh ligand must be captured for a new cycle.

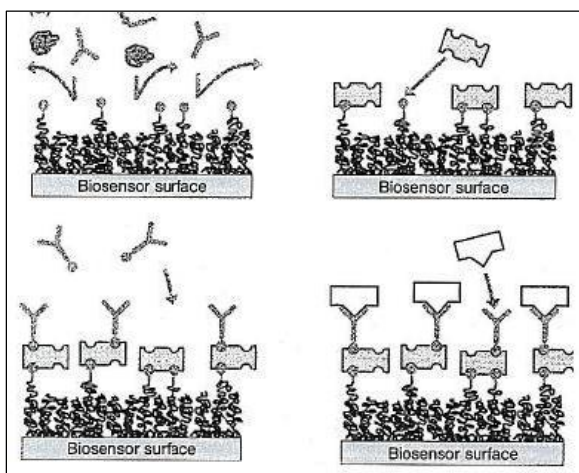


Fig 6: Bioconjugation

Binding specificity

Biacore is well suited to carry out qualitative studies to confirm the specificity of interactions as well as quantitative measurements for affinity, kinetics, and concentration determination. A small volume of analyte can be tested easily for selective binding to 2 to 400 targets simultaneously, depending on the instrument platform chosen. Furthermore, analyte activity is not compromised since interactants do not need to be labelled. It is possible to monitor a number of sequential binding events since each yields a concomitant increase in mass on the sensor chip surface and all stages in the binding process can thus be monitored. Examples of specificity assays include identification of binding sites ^[28] monitoring steps involved in complex formation ^[29-31] and assessing cofactor requirements for an interaction to occur (e.g., Ca^{2+}) ^[30].

SPR technology is increasingly being used to monitor immune responses either to an immunotherapeutic protein, vaccine, or even whole virus in research and preclinical environments ^[31-34].

One simple assay requiring small sample volumes can provide information regarding antibody isotype and active or relative concentration in serum. Another advantage of using Biacore for immunogenicity studies is the ability of the technique to detect both high- and low affinity antibodies, whereas traditional endpoint assays (e.g., ELISA) often fail to detect fast-dissociating antibodies ^[35, 36].

Kinetics and Affinity Analysis

A hallmark of Biacore's SPR biosensors is the ability to derive kinetic association and dissociation rate constants from real-time measurement of binding interactions, thereby providing valuable information regarding complex formation and complex stability that is not revealed through affinity measurements. Rate constants can provide a link between protein function and structure, e.g., through the evaluation of the impact of amino acid substitutions on an interaction. Affinity values can be derived either from interactions that have reached equilibrium or from the ratio of the dissociation and association rate constants in cases where the system does not reach steady state during the time frame of the experiment. The typical working range for affinity measurements with Biacore is picomolar to high micromolar for KD (M). Association rate constants can be measured ranging from 10^3 to $10^8 \text{ m}^{-1} \text{ sec}^{-1}$ and dissociation rate constants from 10^{-5} to 1 sec^{-1} [37].

To measure rate constants accurately, proper experimental design is critical. As with all surface-based analysis methods, the phenomenon of mass transport should be considered. For analyte molecules to bind to a ligand on the sensor surface, they must be transported from the bulk analyte solution to the surface. Under laminar flow conditions used in Biacore, the rate of transport of the analyte to the surface is proportional to the cube root of the flow rate, and is also influenced by the dimensions of the flow cell and the diffusion properties of the analyte. If the rate of transport of analyte to the surface is much faster than the rate of analyte association with the ligand, the observed binding will be determined by the kinetic rate constants. However, if mass transport is much slower than association, the binding interaction will be limited by the rate of analyte transport and there will be partial or no kinetic information in the binding data (mass transport limitation). Optimal assay conditions to minimize mass transport limitations to measure rate constants are a combination of high flow rates and low surface binding capacity. High flow rates minimize the diffusion distance from the bulk flow to the surface, while low ligand densities reduce analyte consumption in the surface layer.

In practice, this translates to using ligand densities that result in a maximum analyte binding response no greater than 50 to 150 RU and flow rates $>30 \text{ } \mu\text{l/min}$. It is important to note that mass transport is a well-understood physical property of the system and partial mass transport limitations can be accounted for during data analysis [38, 39]. Reliable detailed kinetic analysis requires data from four to six analyte concentrations, spanning the range of 0.1 to 10 times KD. Analyte concentrations must be accurately known to determine correct association rate constants. Analytes should be in

the same buffer as the continuous flow buffer to minimize bulk refractive index differences to avoid the so-called bulk effect, so that can lead to low signal-to-noise ratios. This is often most easily achieved through dilution of a concentrated analyte stock into running buffer. Samples containing high refractive index solutions, such as high salt, glycerol, or DMSO, should either be exchanged into the running buffer or the concentration of the high refractive index component should be matched precisely in the continuous flow running buffer.

Kinetic assays should include a series of start-up cycles using buffer as analyte to equilibrate the surface as well as cycles with zero concentration of analytes as part of the concentration series for the purposes of double-referencing^[40] during data analysis.

Although it is not necessary to reach equilibrium, it is recommended that the association times used be sufficient for at least one analyte concentration to reach steady state. To accurately determine dissociation rate constants, a measurable decrease in signal should occur during the dissociation period. If possible, kinetic experiments should be designed such that the data are described by the simplest interaction model. For example, in the case of an antibody-antigen interaction, the antibody should be immobilized or captured on the surface and the antigen used as analyte to avoid avidity effects resulting from the bivalent nature of the antibody. Avidity refers to the ability of an antibody to bind to two antigen molecules simultaneously, thus, the antibody may not dissociate from the antigen immobilized on the chip surface before binding another antigen molecule. Avidity effects will slow down the dissociation rate yielding enhanced affinity values compared to those measured from a 1:1 interaction.

It is also important that both the ligand and analyte be as homogeneous as possible. Impurities from partially purified material can complicate the results by affecting the accurate determination of analyte concentration or introducing nonspecific binding (Fig. 7).

Lastly, analyte should be injected over both a reference surface and an active ligand surface. Reference surfaces are necessary to subtract bulk refractive index responses from the specific binding signal as well as to ensure that there is no nonspecific interaction with the sensor chip surface. Several excellent reviews on the topic provide detailed guidelines on experimental setup and interpretation of results^[41-45].

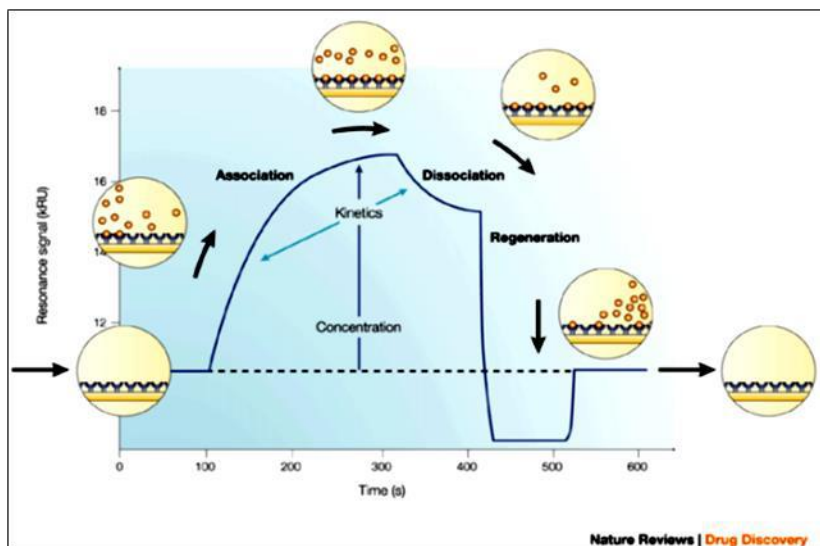


Fig 7: Binding kinetics

1. Receptor molecules are immobilized on the sensor surface.
2. At $t = 0$ s, buffer is contacted with the receptor through a microfluidic flow cell or through a cuvette.
3. At $t = 100$ s, a solution of analyte in the running buffer is passed over the receptor. As the analyte binds to the surface, the refractive index of the medium adjacent to the sensor surface increases, which leads to an increase in the resonance signal. Analysis of this part of the binding curve gives the observed association rate (k_{obs}). If the concentration of the analyte is known, then the association rate constant of the interaction (k_{ass}) can be determined. At equilibrium, by definition, the amount of analyte that is associating and dissociating with the receptor is equal. The response level at equilibrium is related to the concentration of active analyte in the sample.
4. At $t = 320$ s, the analyte solution is replaced by buffer, and the receptor-analyte complex is allowed to dissociate. Analysis of these data gives the dissociation rate constant (k_{diss}) for the interaction.
5. Many complexes in biology have considerable half-lives, so a pulse of a regeneration solution (for example, high salt or low pH) is used at $t = 420$ s to disrupt binding and regenerate the free receptor.
6. The entire binding cycle is normally repeated several times at varying concentrations of analyte to generate a robust data set for global

fitting to an appropriate binding algorithm. The affinity of the interaction can be calculated from the ratio of the rate constants ($KD = k_{\text{diss}}/k_{\text{ass}}$) or by a linear or nonlinear fitting of the response at equilibrium at varying concentrations of analyte. In addition to determining the interaction affinities and kinetics, a thermodynamic analysis of a biomolecular interaction is also possible.

Concentration analysis

Most chemical and spectroscopic methods used to quantify proteins measure total protein content, do not distinguish active from inactive molecules, and cannot be used with unpurified samples. Since SPR is a noninvasive technology (i.e., no light penetrates the sample), it is possible to measure sub-femtomole amounts of analyte bound to the sensor chip surface from complex matrices such as food products, serum, and cell extracts, to name a few ^[46]. Instrument automation decreases operator involvement thereby leading to highly reproducible measurements. Various assay formats are possible. For analytes >5000 Da, a direct binding assay format can be used with the optional response enhancement from a secondary detecting molecule. Enhancement not only increases the dynamic range of the assay but can also improve assay specificity. An enhancement step can also be used to determine the isotype of antibodies in serum that are generated in response to a protein therapeutic or vaccine. Unlike many other immunoassays, concentration analysis with Biacore requires no separation and washing steps and, since binding responses are monitored continuously, it is possible to quantify fast-dissociating, low-affinity interactants. The point at which analyte concentration is measured can be chosen, giving flexibility in the assay design, which is not available with standard end-point assays.

Inhibition or competition assay formats are well suited for quantification of low-molecular-weight molecules (<5000 Da). Inhibition assays rely on mixing the sample with a known concentration of a molecule that binds with high affinity, typically an antibody, then injecting the mixture over the target molecule immobilized on the sensor surface to determine the concentration of the high-molecular-weight binder in the mixture.

Biacore evaluation software provides a direct readout of analyte concentration from a calibration curve generated using standard analyte concentrations. In a competition assay, the low-molecular-weight analyte competes with a fixed concentration of a high-molecular-weight molecule that shares the same binding site. Generally, concentration analysis is carried out at high ligand densities and slow flow rates. Efficient regeneration of the

surface while maintaining ligand activity is paramount for successful concentration analysis measurements.

Thermodynamics

By studying temperature dependence of rate and affinity constants it is possible to determine thermodynamic parameters for a binding interaction. Not only can the equilibrium values for changes in enthalpy (ΔH) and entropy (ΔS) associated with complex formation be determined, but transition state energetics can also be evaluated ^[47].

SPR Imaging principle

In high-throughput screenings and multiplex analyses, a combination of protein arrays and the SPR technique would provide an excellent alternative and label-free method in comparison with the existing methods that require kinetic data. However, standard SPR biosensor instrumentation, with 3-4 flow cells on a single sensor chip, acts more as a hindrance than an aid to high-throughput screening (HTS) applications. To overcome this obstacle, a modified version of SPR design, called SPR imaging (SPRI), has been developed to simultaneously process hundreds or thousands of samples. SPR imaging systems using rapid optical arrays have potential applications in the high-throughput screening of drugs and biomarkers, as well as clinical diagnosis through their use in multi-array detection systems. SPR imaging technology represents an important step forward to SPR analysis. Not eliminating the sensitive label-free method, SPRI's additional value lies in its capability to visualize a whole biochip via a CCD camera. The improved design enables biochips to be prepared in an array format, in which each array spot can provide a wealth of SPR information simultaneously. As shown in Figure 2, in an SPRI system, a coherent polarized light beam is used instead of polychromatic light. This change helps expand the light cover on a larger area of the sensing surface. The reflected light is captured by a charge-coupled device (CCD) camera for further imaging analysis (Fig. 8). The high-resolution CCD camera provides images across the array format in real-time with up to hundreds of active spots. Captured images that show local changes at the chip surface can provide detailed information on molecular binding, interactions or kinetic processes. Unlike conventional SPR, the measurement conducted by SPRI is stringently performed at a constant wavelength and a constant angle (48). Thus, changes in reflected light intensity are proportional to any variation of the refractive index near the metal surface ^[49].

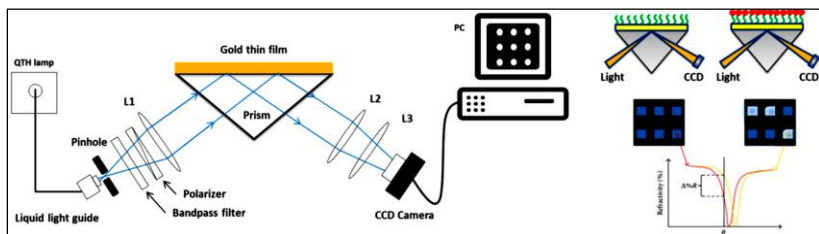


Fig 8: General principle of surface plasmon resonance imaging (SPRI)

(Left) The instrumentation of an SPR imaging system: The light source is a quartz tungsten-halogen lamp; the light is delivered through a liquid light guide to a goniometer arm, collimated by lenses, and passed through a narrow interference filter and a polarizer. A *p*-polarized and monochromatic light beam is then focused directly onto a prism coupler. The reflected light from the gold surface is captured by a monochromatic CCD camera. L2, L3 are lenses positioned in front of CCD for higher quality images. The images could be digitally stored using a B/W frame grabber and further analyzed using photography software; (Right) The analyte-ligand interaction shifts the SPR curve towards a higher angle (red to orange). Due to the measurement confinements (fixed wavelength and angle of incidence θ), changes in the reflectivity ($\Delta\%R$) at a single spot of the array can be simultaneously detected. Adapted from ^[50].

Applications of SPR-based biosensors

a) Biomedical applications

SPR biosensing appears to be one of the most powerful approach for monitoring of affinity binding of biomolecules, and primary screening of druggable molecules. SPR-type sensors are increasingly used to study a variety of biological entities, such as DNAs, RNAs, proteins, carbohydrates, lipids, and cells in the field of biomedical research. In this subsection, several examples of biomedical applications of SPR technology including interaction analyses, conformational change studies, and mutation detection are described.

b) Interaction analyses

A wide range of applications has been developed for the use of SPR biosensors in the biomedical field. First of all, SPR has been used as a powerful tool to study interactions between biomolecules based on affinity binding analysis of a variety of bonds, including antibody-antigen ^[51], ligand-receptor kinetics ^[52-57], enzyme-substrate reaction ^[58, 59] and epitope mapping. Real-time monitoring of DNA manipulation such as hybridization kinetics,

enzymatic modifications, and DNA strand separation using SPR biosensors was earlier introduced in 1995 by Nilsson *et al.* [60]. The advent of click chemistry has allowed scientists to design DNA analogs with novel properties unseen in nature as well as improved stability, functionality and binding characteristics that can be utilized to develop innovative therapeutic agents or new tools for diagnostics. Artificial nucleosides with unusual structural features, such as peptide nucleic acid (PNA), locked nucleic acid (LNA), hexitol nucleic acid (HNA) and phosphoramidates morpholino (MORF) oligomers have proven advantages over functional nucleic acids (aptamers and DNazymes) in terms of denaturation and biodegradation stability in body fluids. In SPR studies, aptamers are considered promising recognition elements with good chemical stability, high selectivity and high affinity toward their targets, and they are easily chemically modified. Aptamers offer more advantages than antibodies. SPR detection was employed in the selection of an RNA aptamer for human influenza [61] and aptamer-based SPR analyses were successfully applied in the detection of human IgE [62] C-reactive protein (CRP) [63] and the HIV-1-trans-activating (Tat) protein [64] and RBP4 (retinol binding protein 4), a diabetes biomarker [65]. There is another branch of DNA analogs that are designed to target single-stranded DNA and RNA with high affinity and specificity; they are conformationally restricted DNA analogs, such as PNA, LNA, HNA and MORF. These molecules have great uses in radiopharmaceutical applications. Many researchers have utilized these artificial molecules to study DNA hybridization [66, 67] pathogen DNA detection, single-nucleotide polymorphisms (SNPs) [68] and miRNA detection. A thorough review of the use of these DNA analogs as recognition elements in SPR-based sensing can be found elsewhere [68] Conformational Change Studies In addition, the SPR signal intensity has been shown to be strongly affected by optical thickness changes in the sensor metal film, as well as by refraction index changes taking place near the metal surface (~200 nm). As a protein molecule undergoes a structural change, those optical indicators are also affected and can be monitored by SPR biosensors. Nevertheless, the SPR technique is often used as a complementary method to verify conformational changes study rather than as a primary technique. This application of the SPR technique has been used to monitor structural transition in protein-small molecule interactions [69] proteins under diverse environmental conditions [70, 71] or impacts on apoptosis inducers [72]. In an attempt to detect protein conformational changes, in 2005, Kim *et al.* developed an antibody chip with conformational specificity to the Bax protein, a pro-apoptotic member of the Bcl-2 family of proteins, which plays a pivotal role in the mitochondrial pathway for apoptosis [73]. Bax conformational change was first induced by

the administration of an apoptosis inducer, TNF-related apoptosis-inducing ligand (TRAIL) and then measured by SPRI. The results indicated that only structurally altered Bax gave visible SPR images, while intact Bax seldom showed any data.

c) Mutation detection

Another extension of SPR-based detection applications is its use in point mutation detection by combining SPR with other conventional techniques. For example, an SPR biosensor was utilized for the detection of point mutation using polymerization extension reaction ^[74]. In this experiment, the capture DNA and probe complementary DNA were coated by a natural complement, and PCR reaction was carried out directly on-chip. Only wild type DNA showed signal boosting by PCR, while mutant DNA showed no SPR signal amplification. Mutation in protein molecules has also been studied using the SPR technique ^[75]. The DNA-binding capability of tumor protein p53 was evaluated. This protein is the “master switch” for the control of cell proliferation, whose mutation causes genetic alterations in human cancers. DNA was immobilized on a BIACORE CM5 chip for the protein binding experiment. Purified wild p53 and mutant p53 (R248W) were injected at a concentration of 100 nM and a flow rate of 5 $\mu\text{L} \cdot \text{min}^{-1}$ for 6 min. The SPR data showed that the RU value of the mutant p53 protein was 4.7 times lower than that of wild p53 protein in response to 100 nM of each protein. Based on the fact that the SPR response is directly proportional to the amount of p53 protein that interacted with the consensus DNA coated on the gold thin film, the results obtained from SPR sensorgrams demonstrated that the wild p53 protein could bind to the cognate DNA sequence while mutant p53 protein could not.

d) High-Throughput Screening (HTS)

The application extensions of SPR biosensors are not only limited to ligand-receptor interaction kinetics dynamic analyses; they are also used for drug discovery and drug development. There are several different formats of SPR biosensors, including the array format, multi-channel unit format, and SPR imaging format, which allow simultaneous and continuous detection to analyze the performance of hundreds to thousands of affinity binding events on a chip surface ^[76, 77]. In SPR imaging, the incidence angle remains fixed, and the binding of biomolecules on a gold surface is measured as the change in reflectivity (or reflectance) in relation to the incident ray intensity, unlike SPR sensors that depend on the measurement of the absorption dip in the SPR angle or SPR wavelength. Despite the excellent benefits inherent in SPR

technology, conventional propagating SPR biosensors have a serious limitation in their inability to support multiplex analysis, as less than four analyses with conventional SPR instruments make such parallel operations feasible. In contrast, SPR imaging technology using a multi-analyte biosensor permits a high-throughput approach, and it achieves a similar degree of sensitivity that achieved by conventional SPR biosensors (Figure 3) ^[78]. Thus, SPR imaging systems without any labeling requirements are more suitable for high-throughput screening (HTS), particularly in drug discovery, than any other optics-based detection techniques (Fig 9) ^[79].

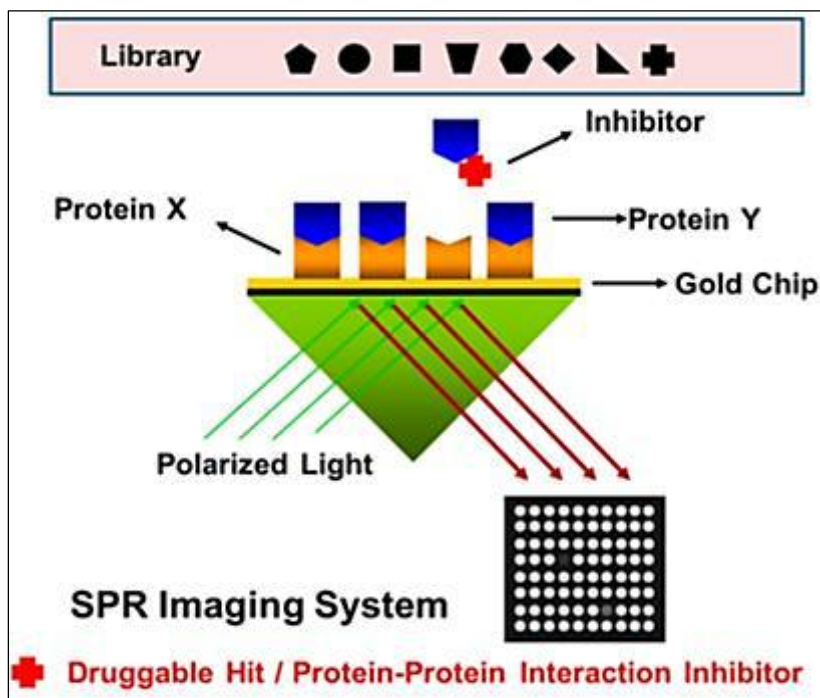


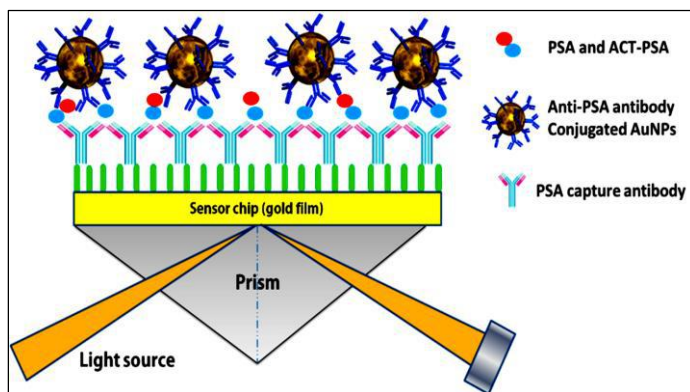
Fig 9: High-throughput drug screening using an SPR imaging protein chip system

The bright image indicates protein-protein interaction on a gold surface. Upon the binding of an inhibitor to the target protein, protein-protein interactions are disrupted, resulting in changes in SPR imaging signal intensity and a darker image. Adapted from ^[80] Krishnamoorthy *et al.* ^[81] used a commercial SPR imaging instrument for the screening of kinetics and affinity constants of multiple biomolecular interactions simultaneously on a 6×4 microarray chip ^[82]. Ligands were serially diluted and spotted on the array following by an analyte sample injection. The SPR signal responses of 24 microarray spots were collected and used as data for the rates and affinity

constant calculations. This method significantly reduces experiment time and cost in comparison with traditional approaches. In 2004, Maillart *et al.* [83] described an SPR imaging system-based HTS approach for multiple analyses of interactions between the p53 transcription factor and its corresponding cis-acting DNA elements [84]. In 2007, [85] developed a method capable of detecting up to 9216 target-ligand interactions on a single array [86].

e) Disease-related protein biomarkers

In contrast to genetic markers, protein markers enable rigorous evaluation for clinical applications therefore, protein-based assays are being developed. Because biomarkers in real blood samples exist in small concentration, this hinders the use of SPR biosensors. To overcome this, the aid of nanoparticles is needed for signal amplification. With the assistance of nanoparticles, researchers have been able to employ SPR in the detection of important biomarkers, such as total prostate-specific antigen (tPSA) [87] carbohydrate antigen 15-3 (CA15-3) [88] carcinoembryonic antigen (CEA) (89) C-reactive protein (CRP) [90] human epidermal growth factor receptor 2 (HER2) [91] estrogen receptor (ER) progesterone receptor (PR) [92]. Figure 4 shows the schematic representation of PSA sandwich assay using PSA detection antibody-modified AuNPs [93]. By using 40 nm nanoparticles conjugated with the PSA antibody, a tPSA assay was performed on 75% human serum at a detection limit of $0.29 \text{ ng}\cdot\text{mL}^{-1}$ (8.5 pM); tPSA was obtained in the 75% serum, which corresponds to $0.39 \text{ ng}\cdot\text{mL}^{-1}$ in whole serum with the SPR sensor. As this detection limit is lower than the threshold value for prostate cancer detection ($2.5 \text{ ng}\cdot\text{mL}^{-1}$), it is possible to clinically diagnose prostate cancer with the developed immunoassay. Short assay time, repeated usability of the same sensor chip, ability to detect PSA in high serum concentration, and real-time testing enable the application of the optimized assay as a point-of-care device for clinical prostate cancer diagnosis and prognosis.



Schematic representation of the PSA immunosensor based on PSA detection antibody-modified Au nanoparticles using surface plasmon resonance (SPR) (Fig 10). For the PSA sandwich analysis, following PSA/tPSA binding to the SPR sensor layer, $1.5 \mu\text{g}\cdot\text{mL}^{-1}$ PSA detection antibody or Anti-PSA antibody conjugated AuNPs (20 nm, 5.1×10^{10} nanoparticles/mL or 40 nm, 5.1×10^{10} nanoparticles/mL) was injected on the surface. Adapted from CEA is the human cancer-associated antigen and serologic tumor biomarkers for diagnosis of colorectal, gastric and pancreatic cancers. The CEA level is a commonly used index in treatment after breast cancer surgery to identify recurrences. The normal CEA level is in the range of 3-5 ng/mL for a healthy woman. In 2006, Tang *et al.* used an SPR-based immunosensor that used protein A conjugated on a gold surface. Next, a monoclonal anti-CEA antibody was coupled to protein A and prepared for CEA injections. The detection limit of 0.5 ng/mL was obtained for the purified CEA. C-reactive protein (CRP) is a principal blood serum biomarker for conventional inflammation and low-grade inflammation diagnosis. CRP is produced by the liver and rises in level whenever inflammation occurs throughout the body. Jung and coworkers used the CRP monoclonal antibody and an amide-linked (AL) N-hydroxysuccinimide (NHS)-dextran functionalized gold array surface in a spectral SPR system to detect C-reactive protein (CRP) in human sera with up to 120 human sera.

Conclusions

SPR-based biosensor applications extend beyond the scope of this book chapter paper. The SPR technique has been proven to be one of the most versatile frameworks for biosensors applications in the most concerning medical, biology, environment, and food safety areas. Hence, SPR based biosensors provide an excellent analytical performance in terms of high sensitivity, fast response, LOD and reproducibility with its label-free, real-time approach. However, there are challenges for SPR technology. Furthermore, the kinetic rate and equilibrium constants data for biomolecular interactions obtained by SPR biosensors have low significance in POC analysis. SPR technology is one of the most promising tools because it is continuously progressing and modified in order to keep its efficiency keep constant in the diagnosis area. Clearly, a portable SPR system offers a better way in onsite analysis in order to check effect of pollutants and effect of contamination *e.g.* bioterrorism. In the coming years, more research on the development of innovative chip chemistry and antifouling strategies in combination with amplification schemes and miniaturization are needed to make SPR an irreplaceable tool for routine clinical analysis and it may open a new channel in SPR application.

References

1. Stephanopoulos N, Francis MB. Choosing an effective protein bioconjugation strategy. *Acc Chem Biol*. 2011;7:876-84.
2. Tugarinov V, Kanelis V, Kay LE. Isotope labeling strategies for the study of high-molecular-weight proteins by solution NMR spectroscopy. *Nat Protoc*. 2006;1:749-54.
3. Phelan ML, Nock S. Generation of bioreagents for protein chips. *Proteomics*. 2003;3:2123-34.
4. Fng CC, Lai WP, Leung YC, Lo SCL, Wong MS, Yang M. Study of substrate-enzyme interaction between immobilized pyridoxamine and recombinant porcine pyridoxal kinase using surface plasmon resonance biosensor. *Biochim Biophys Acta*. 2002;1596:95-107.
5. Nelson BP, Grimsrud TE, Liles MR, Goodman RM, Corn RM. Surface plasmon resonance imaging measurements of DNA and RNA hybridization adsorption onto DNA microarrays. *Anal Chem*. 2001;73:1-7.
6. Caruso F, Rodda E, Furlong DN, Niikura K, Okahata Y. Quartz crystal microbalance study of DNA immobilization and hybridization for nucleic acid sensor development. *Anal Chem*. 1997;69:2043-9.
7. Arwin H, Poksinski M, Johansen K. Total internal reflection ellipsometry: principles and applications. *Appl Opt*. 2004;43:30-28.
8. Garcia G, Runnerstrom E, Llordes A, Anders A, Millron DJ, Richardson TJ, *et al*. Surface plasmon resonance sensors. *Nano Lett*. 2011;11:3056-62.
9. Kim M, Park K, Jeong EJ, Shin YB, Chung BH. Surface plasmon resonance imaging analysis of protein-protein interactions using on-chip-expressed capture protein. *Anal Biochem*. 2006;351:298-304.
10. Madeira A, Vikeved E, Nilsson A, Sjögren B, Andrén PE, Sveningsson P. Identification of protein-protein interactions by surface plasmon resonance followed by mass spectrometry. *Curr Protoc Protein Sci*. 2011;65:19.21.1-19.21.23.
11. Majka J, Speck C. Analysis of protein-DNA interactions using surface plasmon resonance. *Adv Biochem Eng Biotechnol*. 2007;104:13-36.
12. Teh HF, Peh WYX, Su X, Thomsen JS. Characterization of protein-DNA interactions using surface plasmon resonance spectroscopy with various assay schemes. *Biochemistry*. 2007;46:2127-35.

13. Geitmann M, Danielson UH. Studies of substrate-induced conformational changes in human cytomegalovirus protease using optical biosensor technology. *Anal Biochem.* 2004;332:203-14.
14. Salamon Z, Cowell S, Varga E, Yamamura HI, Hruby VJ, Tollin G. Plasmon resonance studies of agonist/antagonist binding to the human delta-opioid receptor: new structural insights into receptor-ligand interactions. *Biophys J.* 2000;79:2463-74.
15. Rich RL, Hoth LR, Geoghegan KF, Brown TA, LeMotte PK, Simons SP, *et al.* Kinetic analysis of estrogen receptor/ligand interactions. *Proc Natl Acad Sci USA.* 2002;99:8562-7.
16. Baron OL, Pauron D, Antipolis S. Protein-lipid interaction analysis by surface plasmon resonance (SPR). *Bio-Protocol.* 2014;4:1-8.
17. Erb EM, Chen X, Allen S, Roberts CJ, Tendler SJ, Davies MC, *et al.* Characterization of the surfaces generated by liposome binding to the modified dextran matrix of a surface plasmon resonance sensor chip. *Anal Biochem.* 2000;280:29-35.
18. Zhang H, Yang L, Zhou B, Wang X, Liu G, Liu W, *et al.* Investigation of biological cell-protein interactions using SPR sensor through laser scanning confocal imaging-surface plasmon resonance system. *Spectrochim Acta A Mol Biomol Spectrosc.* 2014;121:381-6.
19. Besenicar M, Macek P, Lakey JH, Anderluh G. Surface plasmon resonance in protein-membrane interactions. *Chem Phys Lipids.* 2006;141:169-78.
20. Miyoshi H, Suehiro N, Tomoo K, Muto S, Takahashi T, Tsukamoto T, *et al.* Binding analyses for the interaction between plant virus genome-linked protein (VPg) and plant translational initiation factors. *Biochimie.* 2006;88:329-40.
21. Liedberg B, Nylander C, Lundstrom I. Surface plasmon resonance for gas detection and biosensing. *Sens Actuators B Chem.* 1983;4:299-304.
22. Karlsson R, Larsson A. Affinity measurement using surface plasmon resonance. *Methods Mol Biol.* 2004;248:389-415.
23. Casper D, Bukhtiyarova M, Springman EB. A Biacore biosensor method for detailed kinetic binding analysis of small molecule inhibitors of p38alpha mitogen-activated protein kinase. *Anal Biochem.* 2004;325:126-36.

24. Jokiranta TS, Hellwage J, Koistinen V, Zipfel PF, Meri S. Each of the three binding sites on complement factor H interacts with a distinct site on C3b. *Protein Struct Funct Genet.* 2000;275(36):27657-62.
25. Schuster SC, Swanson RV, Alex LA, Bourret RB, Simon MI. Assembly and function of a quaternary signal transduction complex monitored by surface plasmon resonance. *Nature.* 1993;365:343-7.
26. Clark WA, Jian X, Chen L, Northup JK. Independent and synergistic interaction of retinal G-protein subunits with bovine rhodopsin measured by surface plasmon resonance. *Biochem J.* 2001;358(2):389-97.
27. Thai C-T, Ogata RT. Recombinant C345C and Factor I modules of complement components C5 and C7 inhibit C7 incorporation into the complement membrane attack complex. *J Immunol.* 2005;174(10):6227-32.
28. Schlattner U, Dolder M, Wallimann T, Tokarska-Schlattner M. Mitochondrial creatine kinase and mitochondrial outer membrane porin show a direct interaction that is modulated by calcium. *Metabolism.* 2001;276(51):48027-30.
29. Alaedini A, Latov N. A surface plasmon resonance biosensor assay for measurement of anti-GM1 antibodies in neuropathy. *Neurology.* 2001;56(7):855-60.
30. Swanson NL, Billard BD. Optimization of extinction from surface plasmon resonances of gold nanoparticles. *Nanotechnology.* 2003;14:353.
31. Thorpe R, Swanson SJ. Current methods for detecting antibodies against erythropoietin and other recombinant proteins. *Clin Vaccine Immunol.* 2005, 12(1).
32. Karlsson OP, Löfås S. Flow-mediated on-surface reconstitution of G-protein coupled receptors for applications in surface plasmon resonance biosensors. *Anal Biochem.* 1999;300(2):132-8.
33. Myszka DG, Wood SJ, Biere AL. Analysis of fibril elongation using surface plasmon resonance biosensors. *Anal Biochem.* 1999;309:386-402.
34. Karlsson R, Fält A. Experimental design for kinetic analysis of protein-protein interactions with surface plasmon resonance biosensors. *J Immunol Methods.* 1997;200(1-2):121-33.

35. Englebienne P, Van Hoonacker A, Verhas M. Surface plasmon resonance: principles, methods and applications in biomedical sciences. *Spectroscopy*. 2003;17:255-73.
36. Karlsson R, Larsson A. Affinity measurement using surface plasmon resonance. In: *Protocols for Antibody Engineering*. 2004;389-415.
37. Brockman JM, Nelson BP, Corn RM. Surface plasmon resonance imaging measurements of ultrathin organic films. *J Am Chem Soc*. 2000;51:41-63.
38. Strandh M, Persson B, Roos H, Ohlson S. Studies of interactions with weak affinities and low-molecular-weight compounds using surface plasmon resonance technology. *J Mol Recognit*. 1999;10:135-8.
39. Yu X, Xu D, Cheng Q. Label-free detection methods for protein microarrays. *Proteomics*. 2006;6:5493-503.
40. Shumaker-Parry JS, Campbell CT. Quantitative methods for spatially resolved adsorption/desorption measurements in real time by surface plasmon resonance microscopy. *Anal Chem*. 2004;76:907-17.
41. Zeder-Lutz G, Zuber E, Witz J, Van Regenmortel MH. Thermodynamic analysis of antigen-antibody binding using biosensor measurements at different temperatures. *Anal Biochem*. 1997;246:123-32.
42. Evans SV, Roger MacKenzie C. Characterization of protein-glycolipid recognition at the membrane bilayer. *J Mol Recognit*. 1999;12:155-68.
43. Cooper MA, Hansson A, Löfås S, Williams DH. A vesicle capture sensor chip for kinetic analysis of interactions with membrane-bound receptors. *Anal Biochem*. 2000;277:196-205.
44. Cooper MA, Williams DH. Kinetic analysis of antibody-antigen interactions at a supported lipid monolayer. *Anal Biochem*. 1999;276:36-47.
45. Cooper MA, Try AC, Carroll J, Ellar DJ, Williams DH. Surface plasmon resonance analysis at a supported lipid monolayer. *Biochim Biophys Acta-Biomembr*. 1998;1373:101-11.
46. Wegner GJ, Wark AW, Lee HJ, Codner E, Saeki T, Fang S, *et al*. Real-time surface plasmon resonance imaging measurements for the multiplexed determination of protein adsorption/desorption kinetics and surface enzymatic reactions on peptide microarrays. *Anal Chem*. 2004;76:5677-84.

47. Sibille P, Strosberg AD. A FIV epitope defined by a phage peptide library screened with a monoclonal anti-FIV antibody. *Immunol Lett.* 1997;59:133-7.
48. Fong CC, Lai WP, Leung YC, Lo SC, Wong MS, Yang M. Study of substrate-enzyme interaction between immobilized pyridoxamine and recombinant porcine pyridoxal kinase using surface plasmon resonance biosensor. *Biochim Biophys Acta.* 2002;1596:95-107.
49. Nilsson P, Persson B, Uhlén M, Nygren PA. Real-time monitoring of DNA manipulations using biosensor technology. *Anal Biochem.* 1995;224:400-8.
50. Misono TS, Kumar PKR. Selection of RNA aptamers against human influenza virus hemagglutinin using surface plasmon resonance. *Anal Biochem.* 2005;342:312-7.
51. Kim YH, Kim JP, Han SJ, Sim SJ. Aptamer biosensor for label-free detection of human immunoglobulin E based on surface plasmon resonance. *Sens Actuators B Chem.* 2009;139:471-5.
52. Bini A, Centi S, Tombelli S, Minunni M, Mascini M. Development of an optical RNA-based aptasensor for C-reactive protein. *Anal Bioanal Chem.* 2008;390:1077-86.
53. Tombelli S, Minunni M, Luzi E, Mascini M. Aptamer-based biosensors for the detection of HIV-1 Tat protein. *Bioelectrochemistry.* 2005;67:135-41.
54. Lee SJ, Youn BS, Park JW, Niazi JH, Kim YS, Gu MB. ssDNA aptamer-based surface plasmon resonance biosensor for the detection of retinol binding protein 4 for the early diagnosis of type 2 diabetes. *Anal Chem.* 2008;80:2867-73.
55. Lao AIK, Su X, Aung KMM. SPR study of DNA hybridization with DNA and PNA probes under stringent conditions. *Biosens Bioelectron.* 2009;24:1717-22.
56. Ratilainen T, Holmén A, Tuite E, Nielsen PE, Nordén B. Thermodynamics of sequence-specific binding of PNA to DNA. *Biochemistry.* 2000;39:7781-91.
57. Wang J, Rivas G, Cai X, Chicharro M, Parrado C, Dontha N, *et al.* Detection of point mutation in the p53 gene using a peptide nucleic acid biosensor. *Anal Chim Acta.* 1997;344:111-8.

58. D'Agata R, Spoto G. Artificial DNA and surface plasmon resonance. *Artif DNA PNA XNA*. 2012;3:45-52.
59. Geitmann M, Danielson UH. Studies of substrate-induced conformational changes in human cytomegalovirus protease using optical biosensor technology. *Anal Biochem*. 2004;332:203-14.
60. Sota H, Hasegawa Y, Iwakura M. Detection of conformational changes in an immobilized protein using surface plasmon resonance. *Anal Chem*. 1998;70:2019-24.
61. Mannen T, Yamaguchi S, Honda J, Sugimoto S, Kitayama A, Nagamune T. Observation of charge state and conformational change in immobilized protein using surface plasmon resonance sensor. *Anal Biochem*. 2001;293:185-95.
62. Kim M, Jung SO, Park K, Jeong EJ, Joung HA, Kim TH, *et al*. Detection of Bax protein conformational change using a surface plasmon resonance imaging-based antibody chip. *Biochem Biophys Res Commun*. 2005;338:1834-8.
63. Li Y, Yan Y, Lei Y, Zhao D, Yuan T, Zhang D, *et al*. Surface plasmon resonance biosensor for label-free and highly sensitive detection of point mutation using polymerization extension reaction. *Colloids Surf B Biointerfaces*. 2014;120:15-20.
64. Han SH, Kim SK, Park K, Yi SY, Park HJ, Lyu HK, *et al*. Detection of mutant p53 using field-effect transistor biosensor. *Anal Chim Acta*. 2010;665:79-83.
65. Smith EA, Corn RM. Surface plasmon resonance imaging as a tool to monitor biomolecular interactions in an array-based format. *Appl Spectrosc*. 2003;57:320A-332A.
66. Steiner G. Surface plasmon resonance imaging. *Anal Bioanal Chem*. 2004;379:328-331.
67. Jung SO, Ro HS, Kho BH, Shin YB, Kim MG, Chung BH. Surface plasmon resonance imaging-based protein arrays for high-throughput screening of protein-protein interaction inhibitors. *Proteomics*. 2005;5:4427-4431.
68. Kim M, Han SH, Shin Y. Surface plasmon resonance biosensor chips. *Biochip J*. 2007;1:81-89.
69. Krishnamoorthy G, Beusink JB, Schasfoort RBM. High-throughput surface plasmon resonance imaging-based biomolecular kinetic screening analysis. *Anal Methods*. 2010;2:10-20.

70. Maillart E, Brengel-Pesce K, Capela D, Roget A, Livache T, Canva M, *et al.* Versatile analysis of multiple macromolecular interactions by SPR imaging: Application to p53 and DNA interaction. *Oncogene*. 2004;23:5543-5550.
71. Neuman WM, Mullett EP, Lai EPC, Yeung JM. Surface plasmon resonance-based immunoassays. *Methods*. 2000;22(1):77-91.
72. Uludag Y, Tothill IE. Cancer biomarker detection in serum samples using surface plasmon resonance and quartz crystal microbalance sensors with nanoparticle signal amplification. *Anal Chem*. 2012;84:5898-5904.
73. Chang CC, Chiu NF, Lin DS, Chu-Su Y, Liang YH, Lin CW. High-sensitivity detection of carbohydrate antigen 15-3 using a gold/zinc oxide thin film surface plasmon resonance-based biosensor. *Anal Chem*. 2010;82:1207-1212.
74. Tang DP, Yuan R, Chai YQ. Novel immunoassay for carcinoembryonic antigen based on protein A-conjugated immunosensor chip by surface plasmon resonance and cyclic voltammetry. *Bioprocess Biosyst Eng*. 2006;28:315-321.
75. Jung SH, Jung JW, Suh IB, Yuk JS, Kim WJ, Choi EY, *et al.* Analysis of C-reactive protein on amide-linked N-hydroxysuccinimide-dextran arrays with a spectral surface plasmon resonance biosensor for serodiagnosis. *Anal Chem*. 2007;79:5703-5710.
76. Martin VS, Sullivan BA, Walker K, Hawk H, Sullivan BP, Noe LJ. Surface plasmon resonance investigations of human epidermal growth factor receptor 2. *Appl Spectrosc*. 2006;60:994-1003.
77. Yuan J, Oliver R, Li J, Lee J, Aguilar M, Wu Y. Sensitivity enhancement of SPR assay of progesterone based on mixed self-assembled monolayers using nanogold particles. *Biosens Bioelectron*. 2007;23:144-148.